

1. What is Sql Server?

Ans. SQL Server is a Microsoft product. We can call it as Microsoft SQL Server or MS SQL Server.

Sql Server is a RDBMS (Relational database management system) software.

Before RDBMS, there is DBMS. Both DBMS and RDMS are useful to way of storing the data. But DBMS is working very slow while retrieve the big data. So, the father of RDBMS is "EF. CORD" introduced RDMS. RDMS useful to create small small tables and we can create relationship b/w the tables.

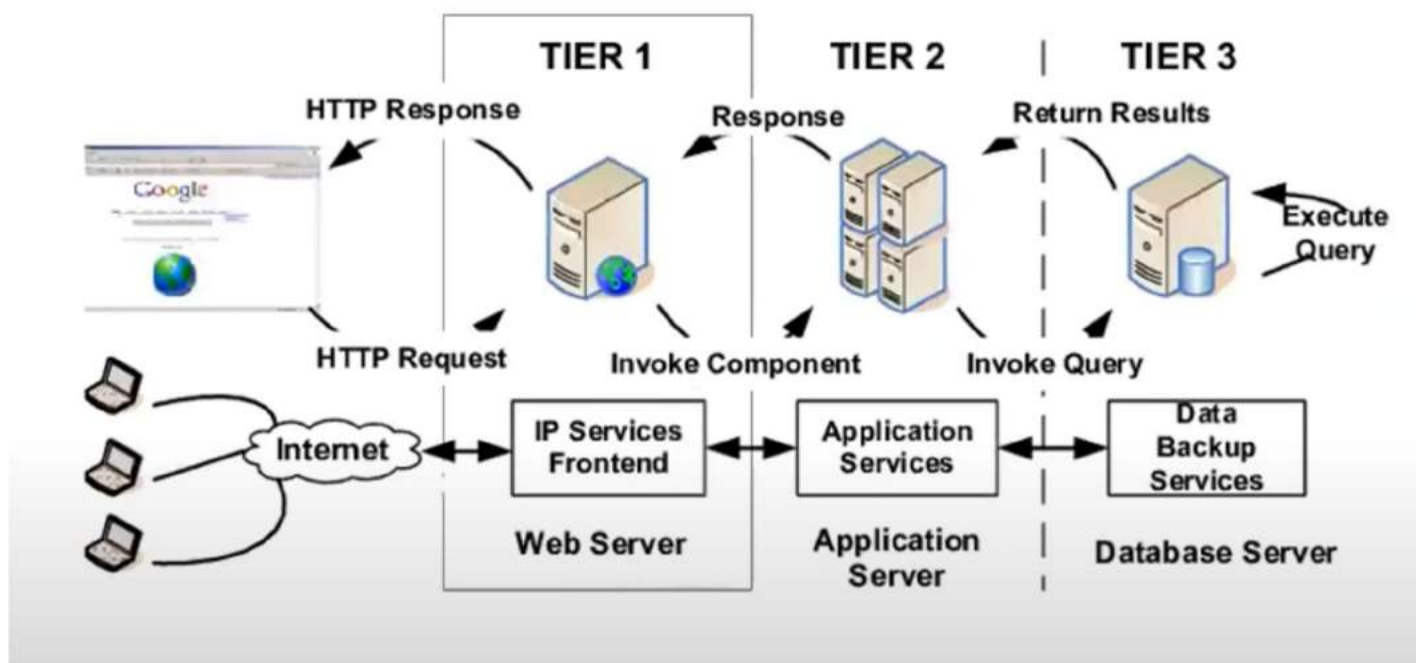
Sql Server is database software, which is used for storing, processing & retrieving the data."

Every application needs a database in the backend.

Like SQL Server, we have other software's also Oracle, DB2, MySQL etc.

In the market, based on the client budget, compatibility, requirement they will choose the database software.

Every application has 3 tier architectures as below,



Tier 1 – Front End

Tier 2 – Application

Tier 3 – Database (backend)

2. What are the duties & activities of DBA?

Ans. Installing and Maintaining the SQL Servers. Creating Logins & giving permissions.

Taking Backups & doing restores, database refresh tasks, backup failures.

Creating and scheduling jobs and maintenance plans.

Configuring high availabilities. Fixing and troubleshooting the issues. Tuning the Sql server.

3.What are the environments in organization?

Ans. There are two environments

1. Production (Live Application- it has the live data)
2. NON production (developers used to test new logic in the non-production application. It's not a live application. Used to test before applying to the production. Below are the non-production environments.
 - a. Dev
 - b. Stage
 - c. QA
 - d. UAT
 - e. TEST

Not all the companies invest on non-production servers.

For example, college Websites not getting any revenue. Only its useful to know the college information. So, for colleges NON production environment is not required.

3. What is database refresh task or database refresh activity?

Ans. To test the any new logic in NON production, we don't have any data to test. So, DBA will take the backup from live application(production) and restore it in the NON production. Now, developer can test their logic in the NON production.

4. In what environments you have worked in previous organization? (Interview question)

Ans. I have worked in production and non-production. In non-production, worked in Dev, Stage, QA, UAT & TEST.

5. What is configuring highly availability or HADR?

Ans. Used to keep the server highly available to the users or customers, even at the time of maintenance.

Below are the 5 HADR features helps to configure the highly availability of the server.

- a. Log Shipping
- b. Mirroring
- c. Replication
- d. Clustering (Windows server failure cluster (WSFS) and SQL server failure cluster (SSFS))
- e. Always on

These features are called High availability disaster recovery (HADR).

These features help as secondary server to the databases and it helps when the primary server down, immediately secondary server helps to work on the databases.

As a DBA, we have to configure, maintain & troubleshoot for all these HADR options.

6. Types of application?

Ans. a. External Application – used by the public.

b. Internal application – used by the organizations or banks or any institutions.

Whether internal or external application, will have a database

7. What is tuning the SQL server?

Ans. at the time of server response is very slow. We have to troubleshoot to improve the performance of SQL Server.

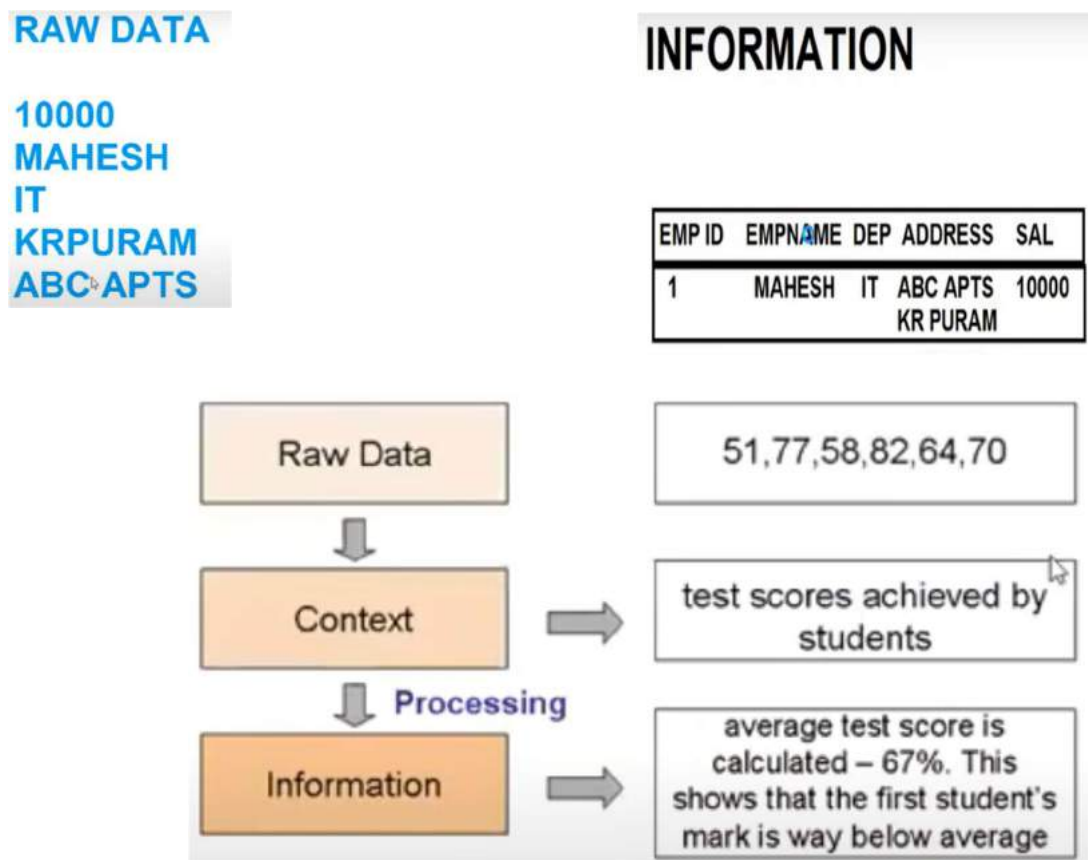
Ex: User given the select query and taking lot of time to execute the command.

8. What is difference b/w Data & Information?

Ans. Data is a Raw formatted information. We can't understand raw information clearly. By using database software, we will convert the raw data in to information. Now, we can understand meaning clearly.

Raw data don't have any meaning. Where, Information has a meaning.

When we add the attributes to the raw data it will be converted into information.



9. In which format data will be stored in the database software's?

Ans. In the form of Rows and columns called it as Table. Tables are very easy to represent, process, understand and retrieve the database. What is a table?

What is a table?

The data in an RDBMS is stored in database objects which are called as **tables**. This table is basically a collection of related data entries and it consists of numerous columns and rows.

Remember, a table is the most common and simplest form of data storage in a relational database. The following program is an example of a CUSTOMERS table:

ID	NAME	AGE	ADDRESS	SALARY
1	Ramesh	32	Ahmedabad	2000.00
2	Khilan	25	Delhi	1500.00
3	kaushik	23	Kota	2000.00
4	Chaitali	25	Mumbai	6500.00
5	Hardik	27	Bhopal	8500.00
6	Komal	22	MP	4500.00
7	Muffy	24	Indore	10000.00

1	Ramesh	32	Ahmedabad	2000.00
---	--------	----	-----------	---------

A record is a horizontal entity in a table.

What is a column?

A column is a vertical entity in a table that contains all information associated with a specific field in a table.

For example, a column in the CUSTOMERS table is ADDRESS, which represents location description and would be as shown below:

ADDRESS
Ahmedabad
Delhi
Kota

10. What are the versions of SQL Server software's? (V.V.IMP interview Question)

Ans. SQL Server introduced in the year 1997. From SQL Server 2000 version onwards it's become popular.

Below are the versions of SQL Server and release year with Compatibility. Each Version of the SQL server is valid for 10 years only.

SQL Server Name	Build Number	Compatibility
sql server 97	7.0	70
sql server 2000	8.0	80
sql server 2005	9.0	90
sql server 2008	10.0	100
sql server 2008R2	10.50	100
sql server 2012	11.0	110
sql server 2014	12.0	120
sql server 2016	13.0	130
Sql Server 2017	14.0	140
Sql Server 2019	15.0	150

Up to SQL server 2016, SQL Server is platform dependent only on Microsoft operating systems. It Means, SQL Server will work only on Microsoft operating systems not in any other OS

There are 2 types of Microsoft OS,

- Desktop OS (using for personal purpose like widows 7, windows XP, windows 10)
- Server OS (Using in the organizations - windows server operating system)

In the organizations, we are not going to install Sql server in the personal laptops. We will be having virtual machine or we have to create virtual machines on the organization, where we are going to install SQL Server on the top of that we are going to instal windows server operating system. It means if you want to install a SQL server in a virtual machine, we should install windows server OS first.

From SQL Server 2017 & 2019 are independent, can be installed in LINUX OS also.

If you want to install in any other OS, we have to install a virtual machine software like Oracle VM Virtual box, create a virtual machine, install windows operating system and install SQL server.

11. What are the Windows Server Versions or server operating system versions?

Ans. In windows server operating systems also, From the Windows server 2000 version it becomes popular.

Windows server 2019 also came in to the market and it is the latest version.

In organizations Windows servers called as Boxes. Example: Please reboot the box.

Windows Server Versions

```
Windows NT 3.0
Windows NT 4.0
Windows Server 2000
Windows Server 2003
Windows Server 2008
Windows Server 2008R2
Windows Server 2012
Windows Server 2012R2
Windows Server 2016
```

The comfort Zone for installing SQL Server in the Server operating systems is -7 to +7.

For Example: If we want to install SQL Server 2000,

Then $2007 - 7 = 1998$ and $2007 + 7 = 2012$

So, SQL Server 2000 can be installed from any of the Windows server operating system, which was invented b/w 1998 to 2012. It can be installed in the below windows servers.

```
Windows Server 2000
Windows Server 2003
Windows Server 2008
Windows Server 2008R2
```

12. what is Build number?

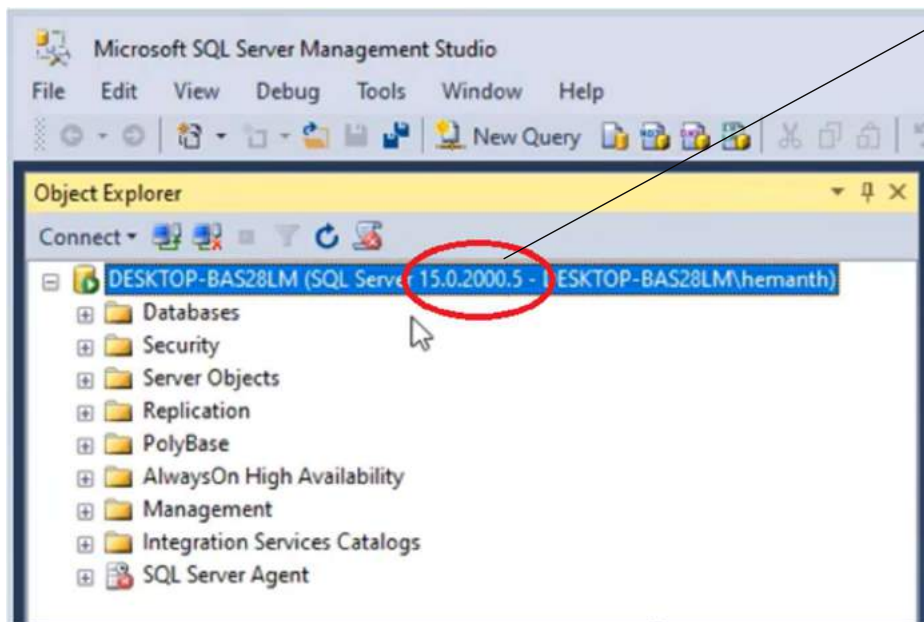
Ans. Build number helps to find out the version of the SQL Server.

SQL Server Name	Build Number
sql server 97	7.0
sql server 2000	8.0
sql server 2005	9.0
sql server 2008	10.0
sql server 2008R2	10.50
sql server 2012	11.0
sql server 2014	12.0
sql server 2016	13.0
Sql Server 2017	14.0
Sql Server 2019	15.0

We can find the build number in the SSMS (SQL Server management studio), Below highlighted is the Build number and match that with the above SQL Server Versions. Below is the 2019 version.

Build number contains four parts, majorversion.minorversion.Buildnumber.revisionhistory(15.0.2000.5).

Here 15 is the major version,0 is the minor version,2000 is the build number and 5 is the revision history.



By using query, we can find out the build number with version together,

Query: **Select @@ Version;** and it represents like below



13. What is CTP, RC, RTM & SERVICE PACK?

Ans. CTP – Community technology Preview. It is also a SQL Server software, which is used for testing purpose. any new software is ready, first Microsoft release the software to the community people to test the software. It's a test phase. Here users will work on the software and give the feedback to the Microsoft. Now Microsoft will rectify the bugs and errors.

RC – Release Candidate. It is also a SQL Server software. Once CTP Completed, Microsoft will release one more test copy to Release candidate. Here also users will work on the software and give the feedback to the Microsoft. Now Microsoft will rectify the bugs and errors.

RTM – Released to market. After completing the tests in the CTP and RC. Microsoft release the software to the market. This is the base version or first copy of software is called RTM Software.

EX: Client is asking to install SQL Server 2016 RTM. It means we have to install first copy of SQL Server 2016.

SP – Service Pack – It contains the updates of the RTM Server. If we want to install any updated, we use service packs.

CTP and RC software are free versions, which can be installed by anyone in the desktop OS.

RTM is the paid version, which can be installed in server OS.

14. What are the editions in the SQL Server?

Ans. Limitations to the features of the software is called Edition.

Below are the famous editions from Microsoft SQL Server,

- a. Express Edition- It's a free edition. This edition won't allow to create a database more than 10GB. There is no SQL Server Agent in this version.
- b. Evaluation Edition- it is also a free edition but for limited time period that is 180 days only. In this edition also, so many features locked to access.
- c. Standard Edition – it's not a free edition. This will be used in the Production environment. Not all features unlocked.
- d. Developer Edition – From SQL Server 2017 Version onwards, It's a free edition. This edition can't be installed in production environment.
- e. Enterprise Edition- This is the highest edition. It's not a free edition. This will be used by the Production environment. All features unlocked in this edition.

95% of the organizations are using Enterprise edition only.

Paid Editions are Standard and Enterprise

Free editions are Express, Evaluation & Developer Edition.

15. What are the main components or features or applications or ADD ONs in the SQL Server?

Ans. Below are the main features of the SQL Server.

- a. Database engine Services- This is the brain and heart of the SQL Server. SQL Server is running based on the Database engine services.
- b. Business intelligence development studio (BIDS)
- c. Reporting Services – To generate reports
- d. Analysis Services
- e. Integration Services – To import data into SQL Server from external applications

When the client is requested to install the BIDS or Analysis, Integration services then only we have to install these features in the SQL Server. Why because these are the ADD ON, Client has pay extra money to access these features.

Before installing any SQL Server, we have to ask the client- Please provide the build sheet. In the build sheet, they will mention everything like Which version of SQL Server, Windows Server OS, RAM, Storage, ADDONS etc.

15. What are the types of machines?

Ans. There are two types.

- a. Physical Machine – We can touch the machine
- b. Virtual Machine – We can't touch the machine but we can feel it.

In a single physical machine, we can create number of virtual machines based on the capacity (RAM, Storage) of the physical machine. If physical machine is down, we can't access the virtual machine.

Every physical Machine have three important components to install the any application those are OS, RAM, Hard disk, CPU.

In the same way to create the virtual machine in the physical machine, below are the few software to create the virtual machine. These softwares are called Virtualization Softwares.

- a. Oracle Virtual Box
- b. VM Ware
- c. Hyper V (It's a Microsoft product) etc

16. How many physical and virtual machines is there in your previous organization? (Interview Question)

Ans. 80-90 % are virtual machines and 10-15% are Physical machines

17. What is the process in the organization to install SQL Server or any software? (Interview Question).

Ans. 1. we have to Raise the Change request or Ticketing for installing SQL Server.

2. VM Team creates the Virtual machine

3. Windows team will install the Windows server OS

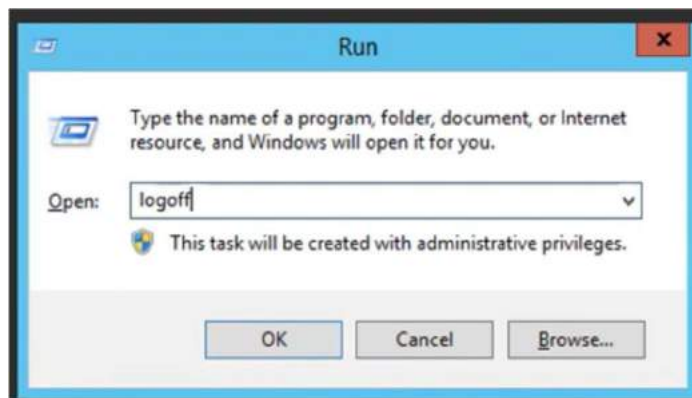
4. Network team will install network drives

5. Now it comes to DBA team to install SQL Server.

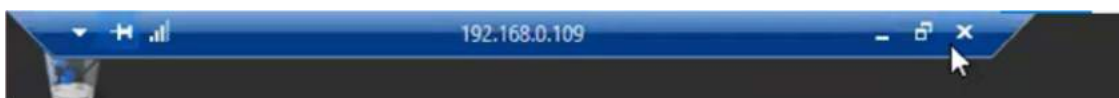
DBA performance can be track with all the Change requested and incident requests in the organization.

Note: Never shutdown the any server because server should run 24/7. If you have any problem restart the server. If your work is completed logoff from the server. DBA never shutdowns the server.

To log off from the server, open Run and type Log off.



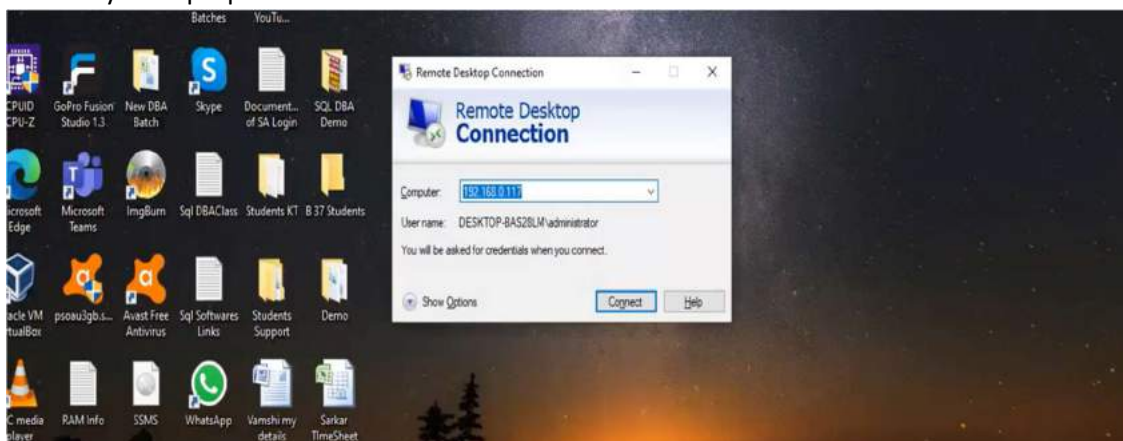
If there is any pending work in the server like backup, restore, query handling, jobs running etc, don't log off from the server. Just click on the "X" button of the server, now your session only will disconnect and pending work will be continued in the server.



18. How to connect to the server remotely?

Ans. In the organization, we will already be having the IP address or server name to login to server.

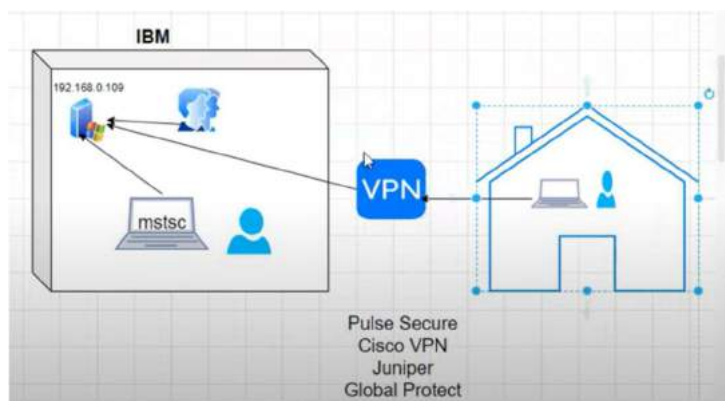
a. Open the Remote desktop option by typing in the search button "remote" or "RDP" or type "mstsc" in the RUN on your laptop



- b. Type the IP address of the Virtual Machine in the above Remote desktop connection and connect.
- c. After that give the user name and password to login server.
- d. Now you can access the virtual machine on your laptop.



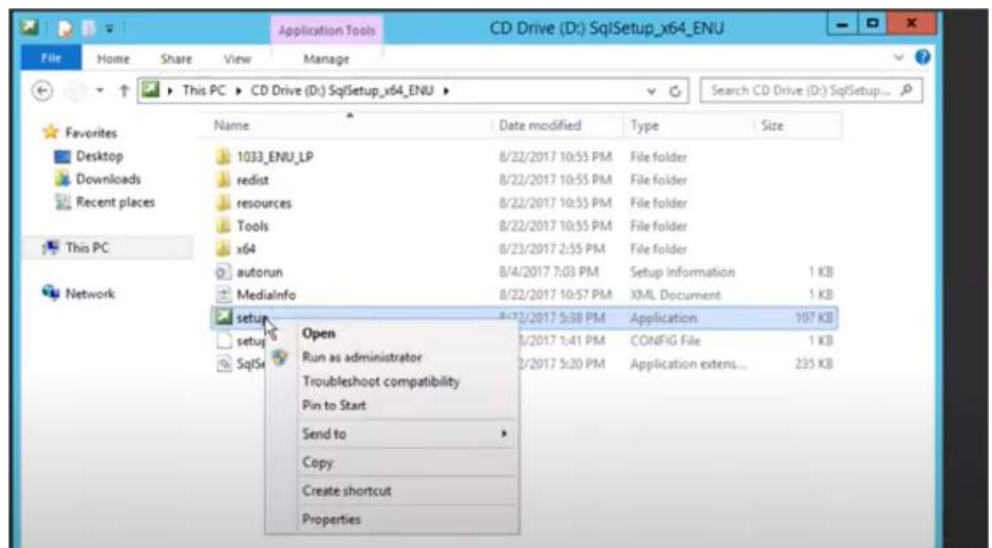
- e. Now login to the server by using the User name and password. User name is always administrator
- f. Windows server allow only 1 admin and 2 users to access the server at a time. If you want to access the same server, then you to kill the 1 user access. But before killing the user access ask them and then do it. Without their permission don't kill the any user access." Ask the user, I want to connect to the server, please disconnect your connection from the server".
- g. If you're working from home, to connect to the organization virtual machines use the VPN



19. How to install SQL Server or Instance?

Ans. One instance is called One SQL Server software. We can install 50 standalone instances in a single Windows Server. Below are the steps to install SQL Server. (Interview Question). Prerequisites will be installed by windows team or application team. If any prerequisite is missing, we should ask them to install.

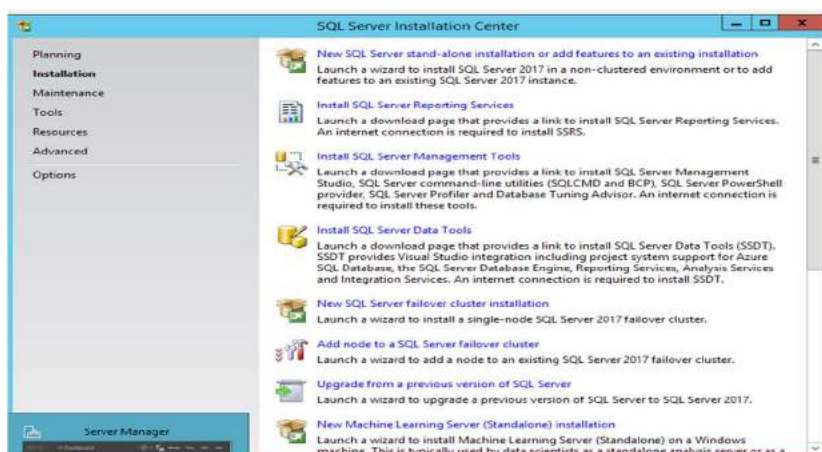
- a. First copy the SQL Server setup from the Shared paths to the virtual machine.
- b. Open the Setup folder and run the setup file as administrator.



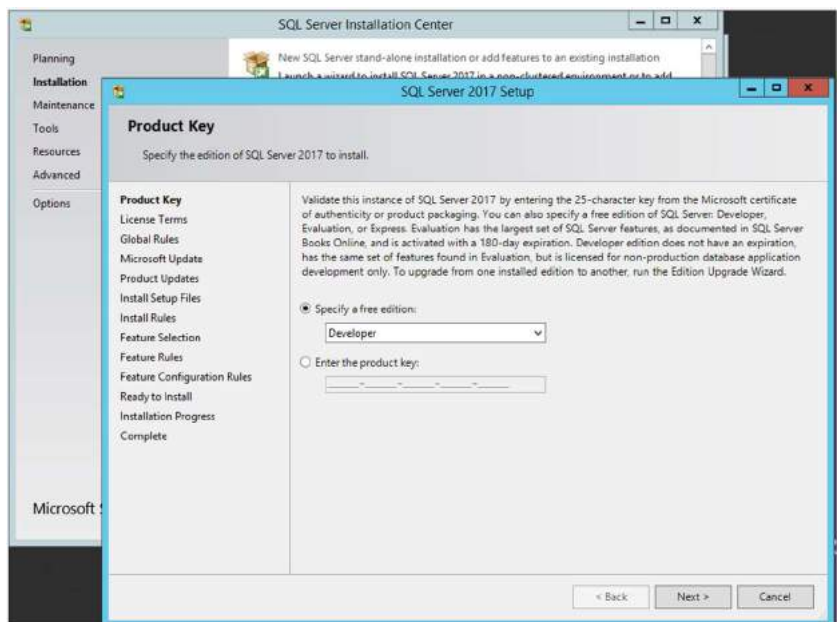
- c. Now SQL Server installation centre will be opened like below and go to installation page.



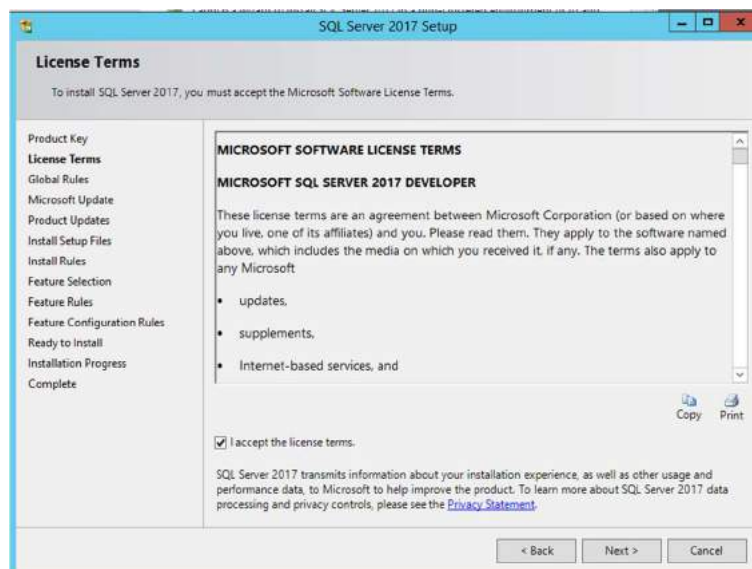
- d. If you're installing the SQL Server in the single windows server or single machine it's called stand-alone installation.



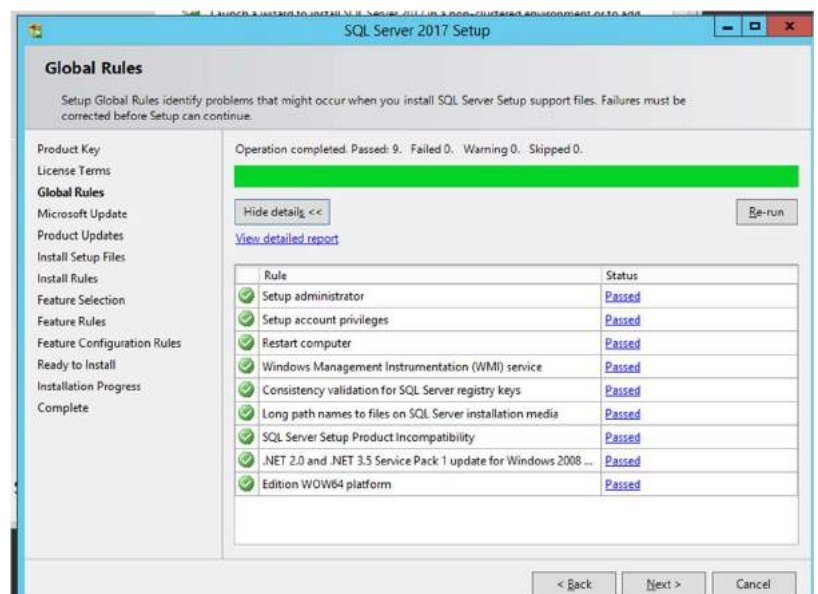
- e. Click on the new SQL server stand-alone installation for installing a fresh SQL Server or to add any additional features to the existing SQL Server Like Reporting services, Integration services etc. after that below screen appears.



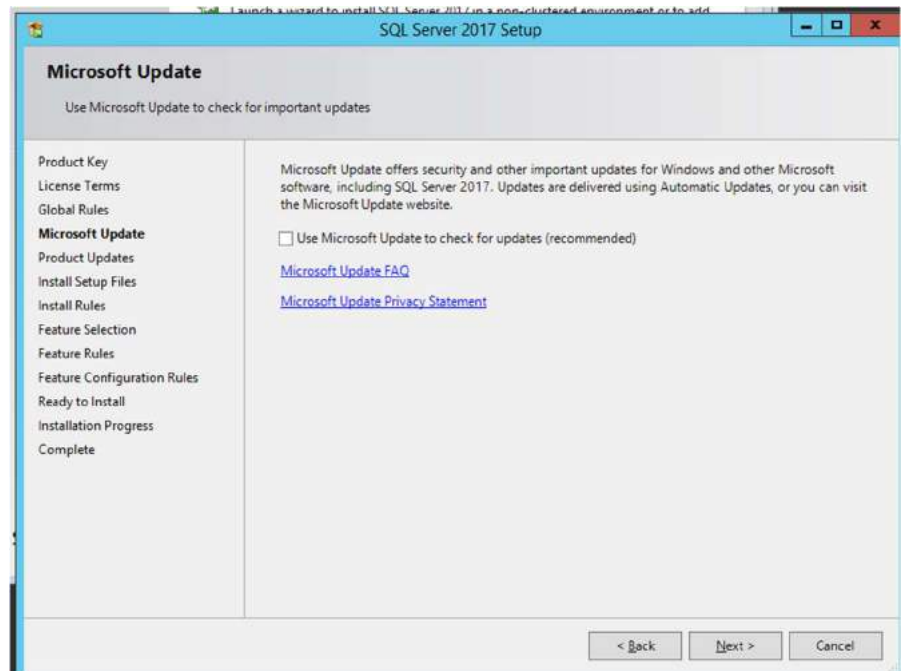
- f. Select the edition type. In the organization's product keys available. Click Next. After that accept the license acceptance click next



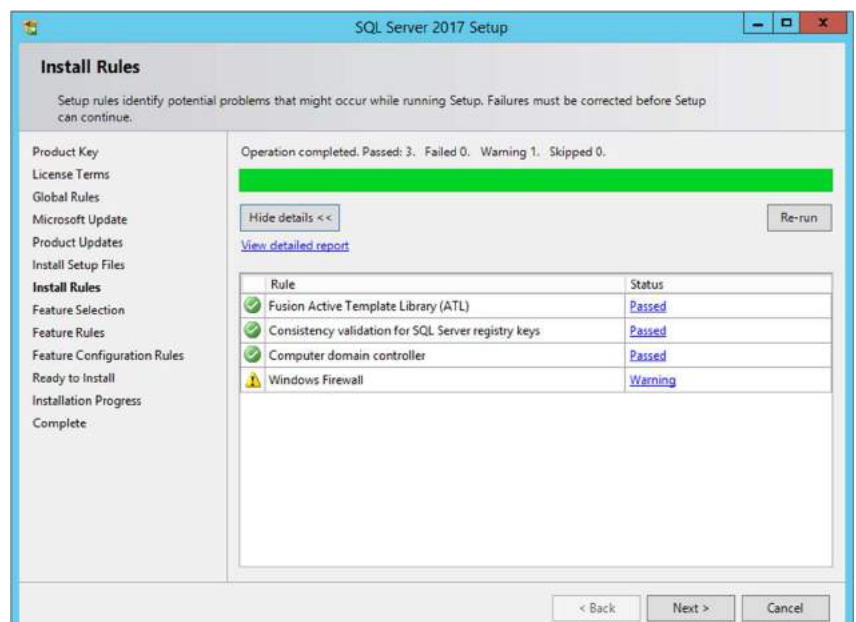
- g. Now the page goes to check, whether prerequisites of SQL Server installation. If any check is failed, Next Option won't work in the setup. we have to install that software first and then we have to continue the SQL Server setup.



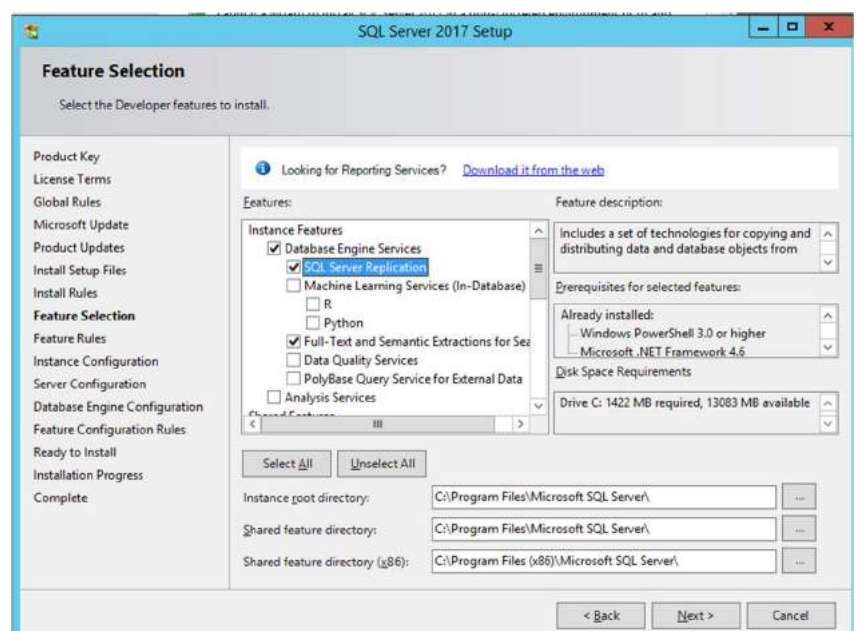
- h. Now Microsoft update screen will come. In the organizations, Servers won't have internet connections. So, don't tick the Microsoft update option and click next.



- i. Now, it will install setup files. After it will do the checks like below, if there is any check failed, installation won't continue.



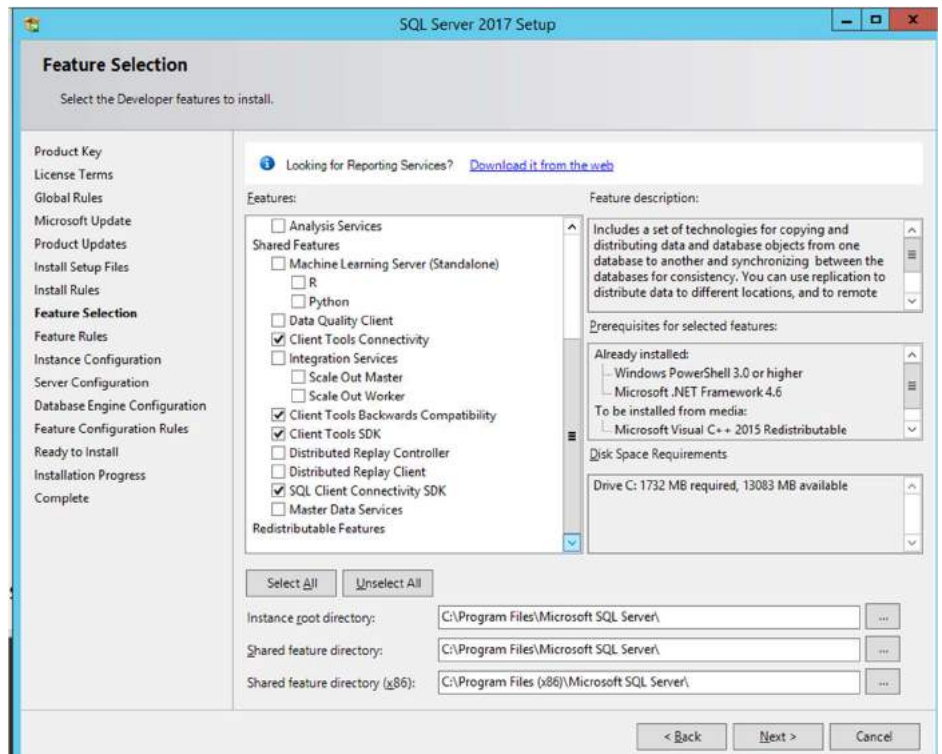
- j. Install Rules and Selecting Components or Features. **Instance Features or services - These features, we have to select for each and every instance or Sql server.** Select database engine services, SQL Server replication & Full-Text semantic Extraction. If any other services required by client, then only select the other features. These features are also called as specific features or instance aware services



Shared features or services-

Selected features in this column will be applied to all the SQL Servers. It means, when you want install second instance, shared instances have been selected already. Only any other features are required we have selected in this column again these features will be applied to all the SQL Servers or instances. These features are also called as instance unaware services.

Select
Client tools connectivity
Client tools backwards compatibility
Client tools sdk
Sql client connectivity SDK



If you go to services.msc, the shared services will be used for all the SQL Servers in a single virtual machine. But, in the specific services- separate copy of services will be created for each SQL Server in a virtual machine. To identify, which services of SQL Services in services .msc is Specific or shared- At the end of the service instance name is there means it's a specific service otherwise shared service.

For example: if you already installed the first instance in the windows server with 3 specific services and 2 shared services of sql server, then for the second instance only specific services only created and already created shared services will be used for the first and second instance. Now total 6 specific and 2 shared will be running in the services.msc of SQL Server.

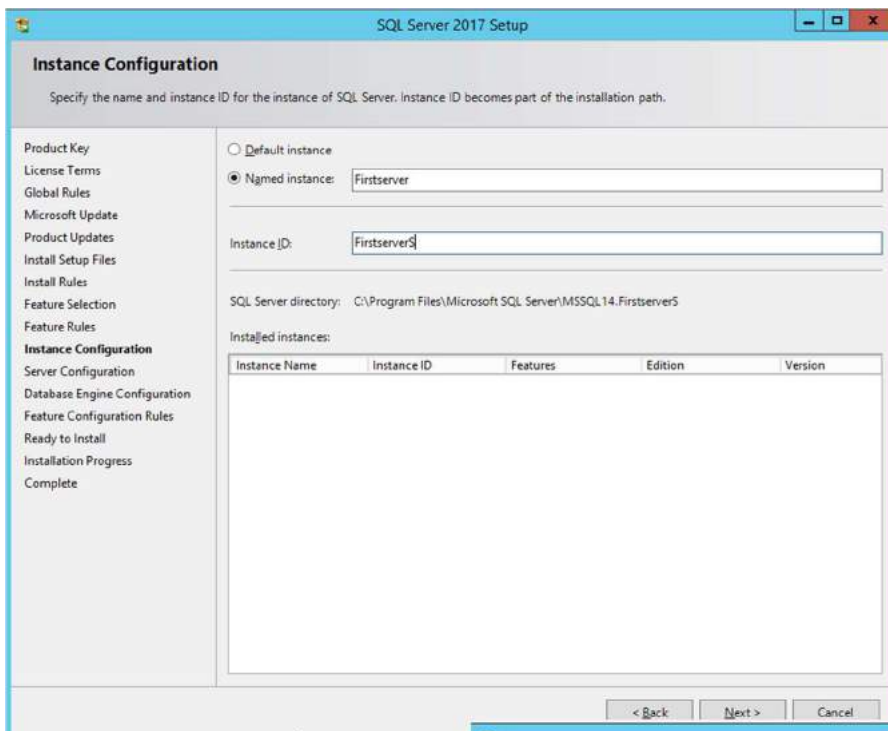
Service Name	Service Description	Status	Startup Type	Path
SQL Full-text Filter Daemon Launcher (MSSQLSERVER)	Service to la...	Running	Manual	NT Service\MSSQLFDLauncher
SQL Server (MSSQLSERVER)	Provides sto...	Running	Automatic	NT Service\MSSQLSERVER
SQL Server Agent (MSSQLSERVER)	Executes jo...	Running	Automatic	NT Service\SQLSERVERAGENT
SQL Server Browser	Provides SQ...	Stopped	Disabled	Local Service
SQL Server CEIP service (MSSQLSERVER)	CEIP service...	Running	Automatic	NT Service\SQLTELEMETRY
SQL Server VSS Writer	Provides th...	Running	Automatic	Local System

- k. Up to SQL Server 2016, Reporting services are available in the feature selection. After that onwards, this feature has been separated and we have installed separately this feature when there is a client requirement.
- l. Instance Configuration.

There are 2 types of instances.

Both will act as same only the difference is the way of connecting to the server is different. We can choose any one.

- Default Instance – To connect to the default instance, Will use “Computer name” to login the server. Only one default instance can be created for one Windows server. By default, it takes the computer name as Instance name. No need to give any name of the instance
- Named Instance - To connect to the Named instance, will use” Computer name\Instance name” together. Named instance will allow maximum of 16 characters for the name. Instance name should start with alphabet only.
Special characters not allowed in the named instance name, it allows only “_” and “\$” only.



Instance ID – it will create the folder for the instances with the name whatever we mentioned.

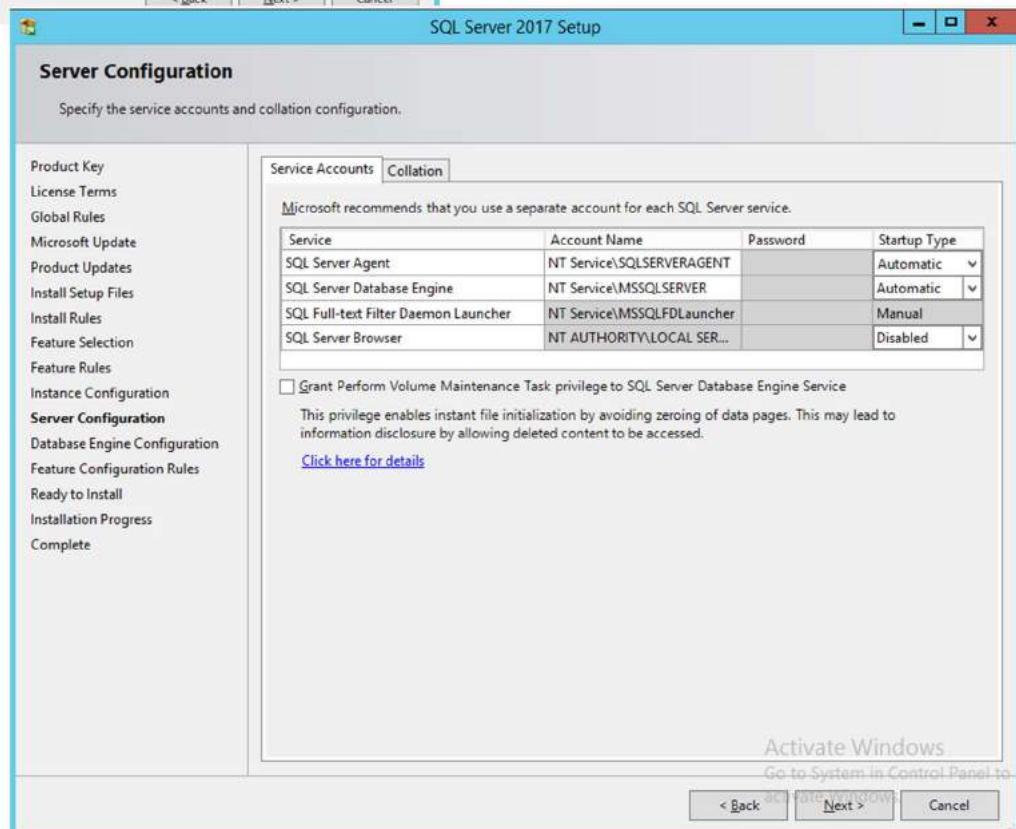
*) we can see already installed instances also in this screen

m. Service Accounts –

Any machine has 3 service accounts.

Below services will do the work for the all the softwares, based on their selected service accounts.

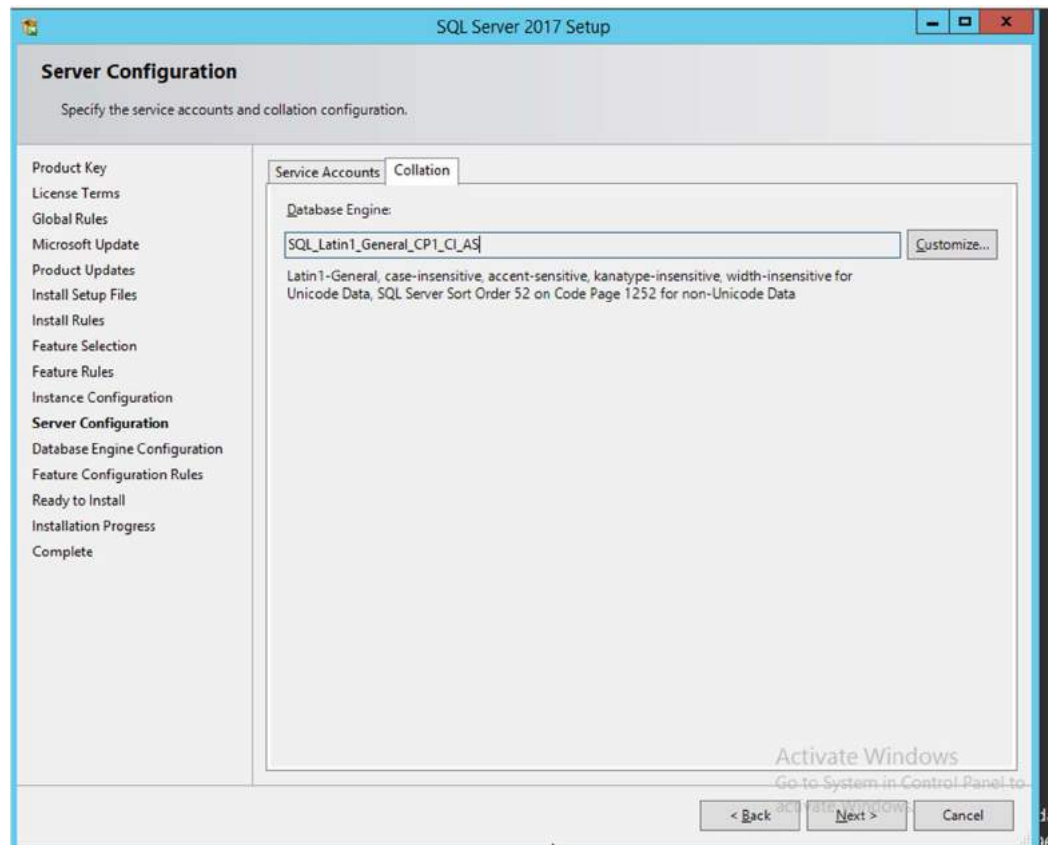
- Local service
- Local System – Very IMP software runs under the Local System. Read, write operations happened in this account.



- Network Service

By default, windows give a service account for SQL server. We can use the same or else we can select the Local system as Sql server service account or we can create customized services accounts for the Sql server. Start-up type for SQL Server agent & Sql server database engine change to automatic.

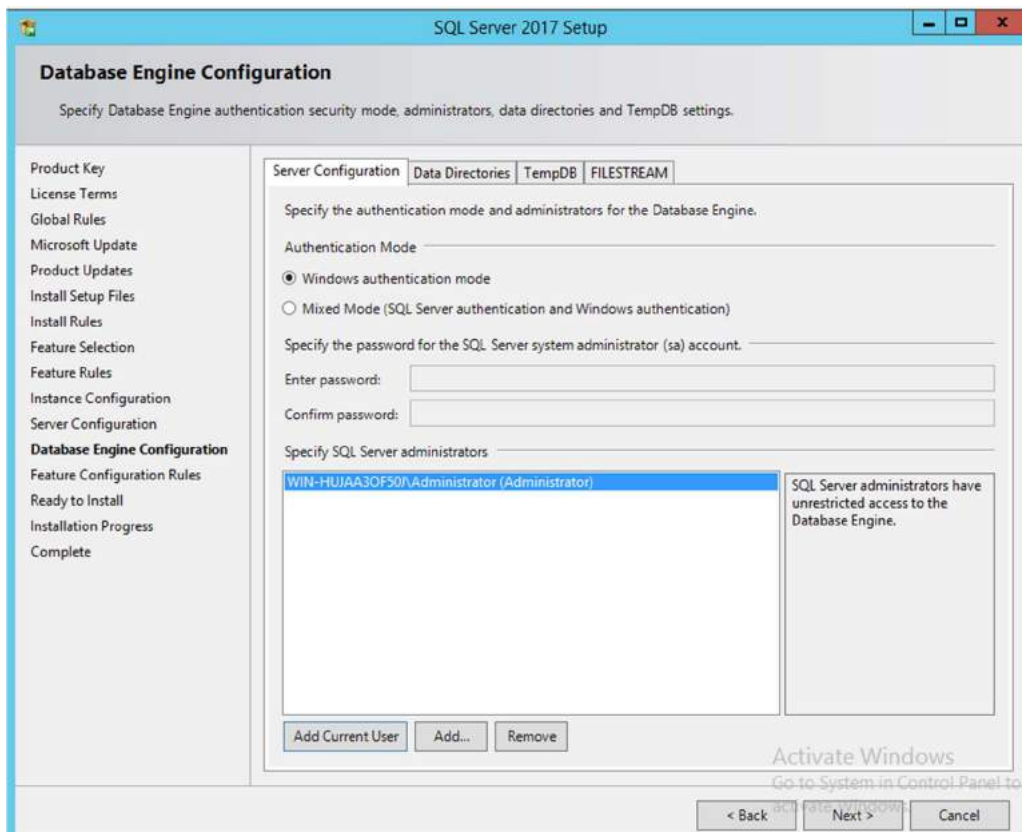
In the same screen, Collation tab is there- Collation means language setup for the SQL databases. By default, "SQL Latin 1" will be there- It's an English language. We can change it as per client requirement.



n. Database engine Configuration.

There are 2 authenticate modes in the SQL Server.

- Windows mode- By using the windows user
- Mixed mode – by using windows user or SQL Server user login details.



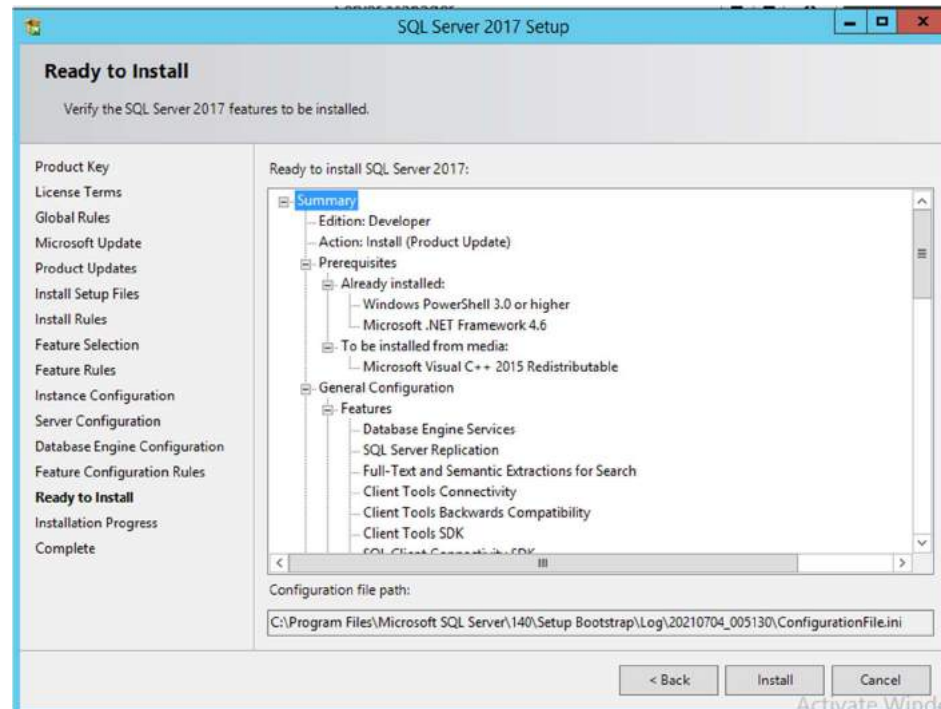
We have to add the list of the users we want give access to the SQL Server by using add current user and add option for other users. We can add the others users after the installion also. But at least to login after the installation of Sql server, we should add the current user. If we won't add any users at this step, we can't access the Sql server after the installation. Normally in the organization, we will add the all-DBA users, Customized Service accounts at this option.

Next go to data directories and give the location to store the data directories.

Next go to TempDB and configure the database

Next go to File stream, if you want to store movie content enable it.

- o. Finally, Ready to install -All the above GUI options stored in a "configurationfile.ini" file used for the installation.
GUI is only to understand the process to the user. Everything inside the application happening in command form only.



20. What are the regular activities of DBA?

Ans.

- Installing SQL Server
- Configuring SQL Server
- Taking backups
- Restore backups
- Troubleshooting the backup & restore failures
- Drive space monitoring
- Jobs
- Patching
- Sql Server upgradation or migration
- User Deployments
- HADR Options (Log Shipping, Mirroring, Replication, clustering, always on)
- Performance tuning of the server

21. How to install Sql Server management studio (SSMS)?

Ans. From SQL Server 2016 onwards, Microsoft separated SSMS feature in the SQL Server software.

We have to install SSMS Separately from the SQL Server 2016. SSMS is tool to connect to SQL Server.

We can connect lower version sql server instances in higher version management studio.

We can connect higher version sql server instances in lower version management studio. But We won't get higher version features of SSMS

Copy the SSME Software from the Shared folders to the Windows server and then install it. Don't install from the shared folder directly. It's not a good habit. Always copy any software to the virtual machine first and then start the installation. Open the installation file with Run as administrator.

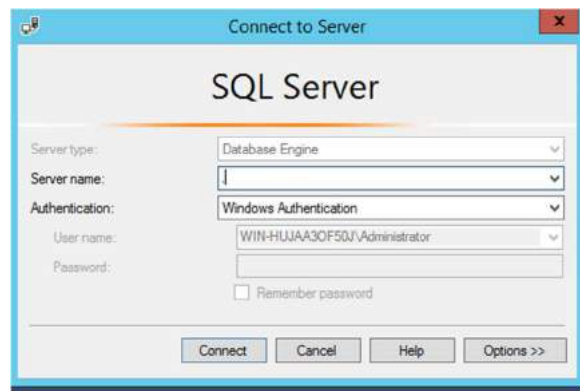
No need to install separate SSMS for all the instances. With one ssms we can connect to all the sql servers.

Click on install.

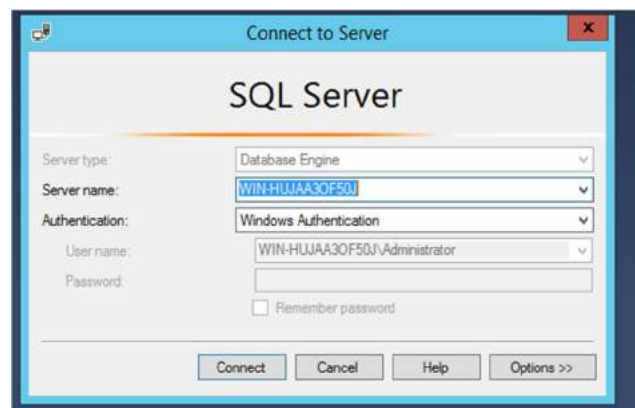
22. What are the ways to connect SQL Server by using default instance?

Ans. There are 5 options.

- By giving “.” (dot) in server name



- BY giving “Computer name” in the server name

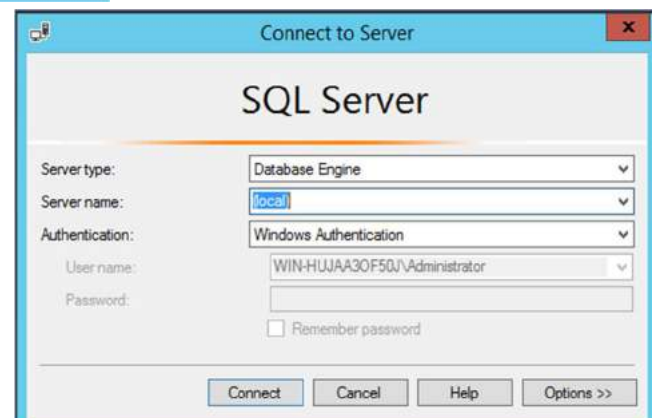


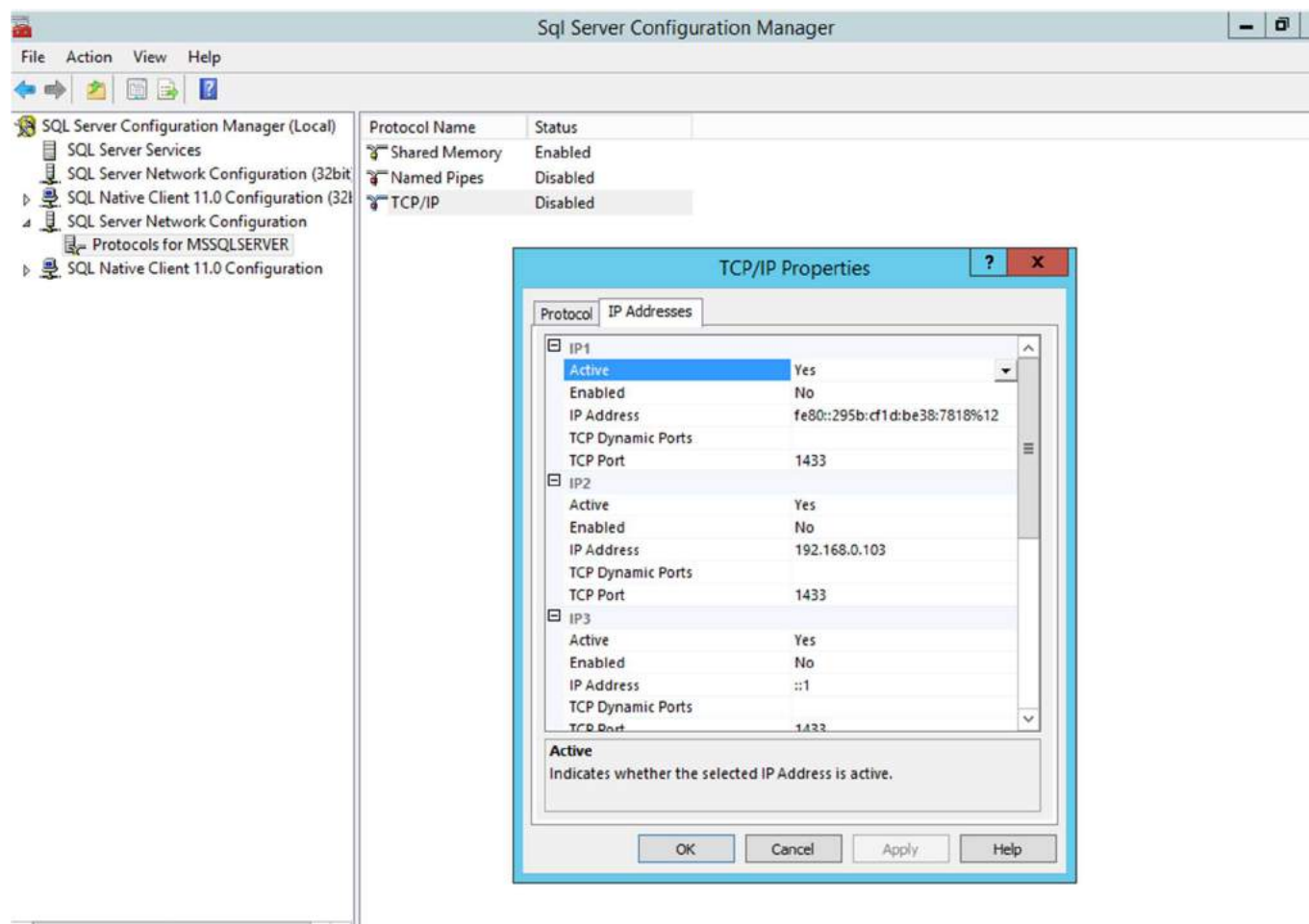
- giving “localhost” in the server name.



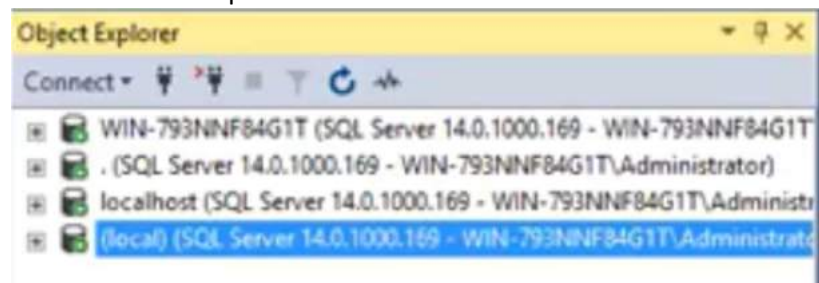
- by giving “(local)” in the brackets in server name
- By giving “IP address and Port ID” in the server name

Remember default Port ID for default instance is 1433. (Interview question) or you can see the port id in the sql server configuration manager.





Below is the example where default instance connected with four different ways.



23. What is Parser?

Ans. It helps to identify the Syntax & Symantec errors. Short cut is Cntrl+F5.

24. How to select the database while running the query itself.

Ans. Syntax: select * from Table name databasename. schema. tablename

Example: select * from employee Testdb.dbo.employee

25 Which database is hidden in the System databases?

Ans. Resource Database.

25. What are the ways to stop & start SQL Server or instance?

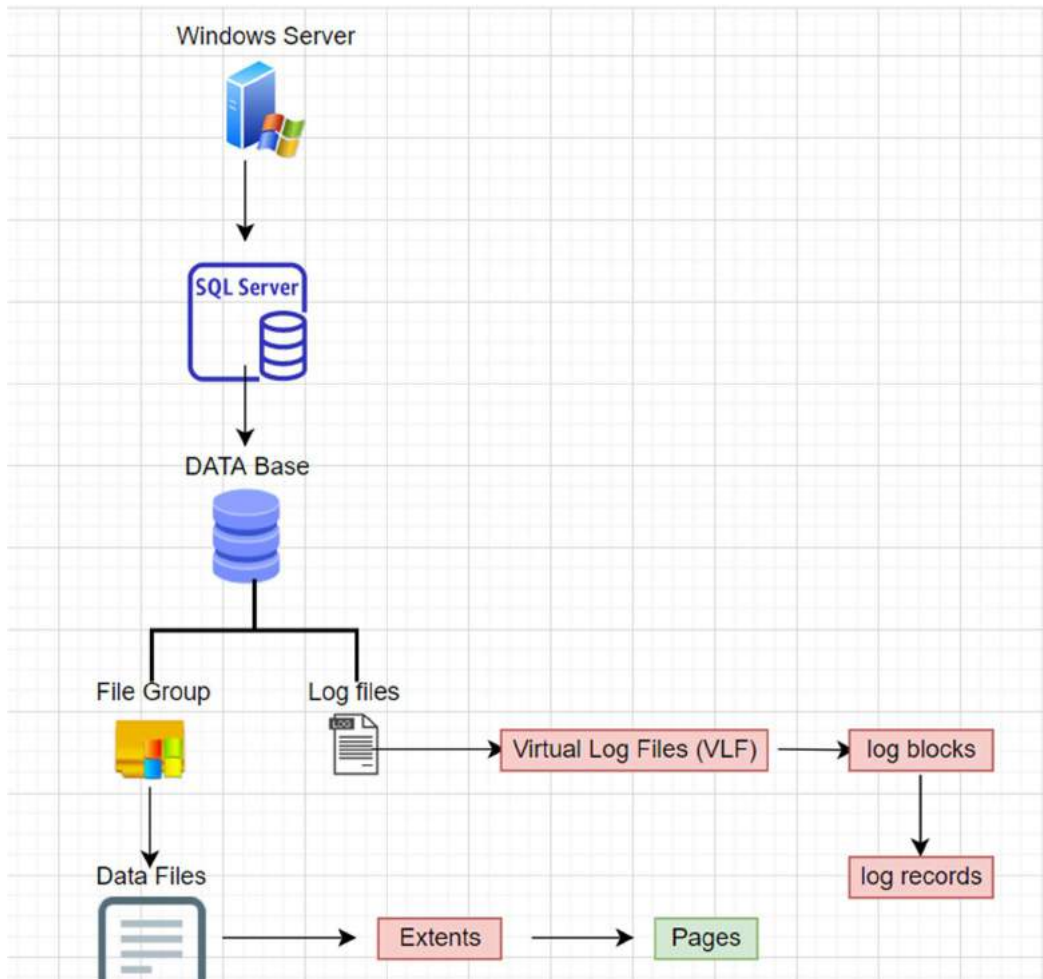
Ans. There are different ways.

- Open services.msc, select the sql server instance and stop it
- Open sql server configuration manager, select the instance and stop it
- Open the command prompt, type “net stop instance name”

We can start the instance from the above options, from command prompt type “net start instance name”

26. Physical hierarchy or architecture of physical server?

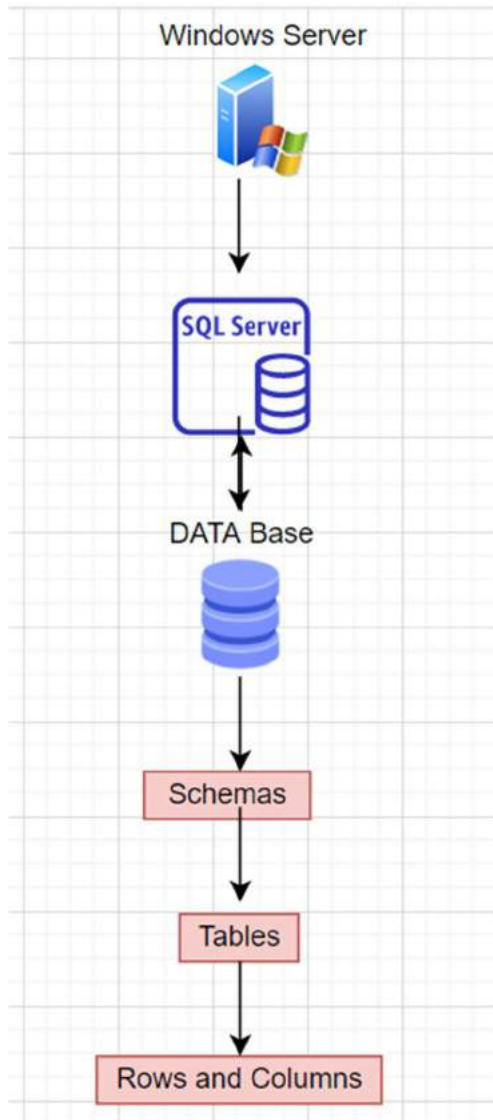
Ans.



- Data base is a collection of data files and log file.
- Data files created under the file groups. By default, data files created under primary group.
- In the organization, datafiles stores in separate drive, log files stored in separate drive & Temp DB in separate drive.
- Organizations maintains separate drives for backups.
- Separate purpose, there is separate drive in the organization.

27. Logical hierarchy of SQL Server?

Ans.



- Schema- A collection of tables. It's a container.

28. What are the files in a database?

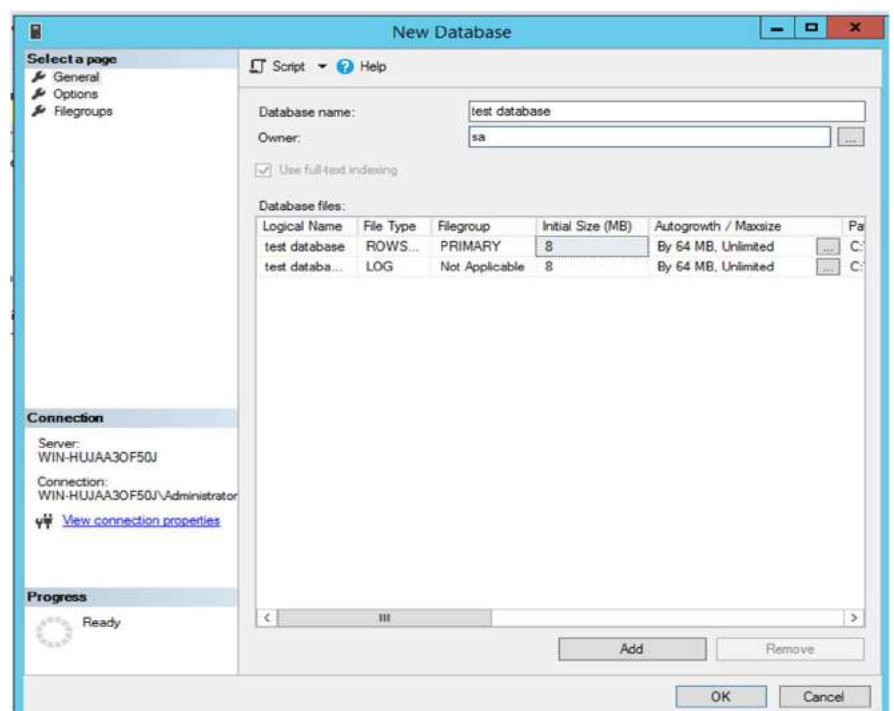
Ans. Datafile and log file.

Data file Extension is mdf and log file extension is ldf

Note: While creating any object in the database, change the owner name "default" to "sa".

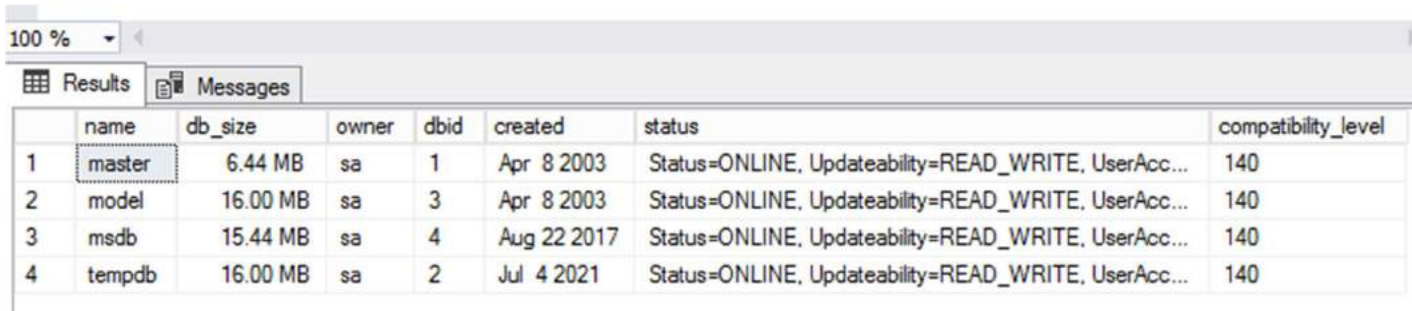
We can change the owner of the already created data bases also, by rgt click in database – open properties-go to files option – change the owner to sa

- Give the path for storing the data file and log file.



29. How to know all the databases names, owners , size of the database?

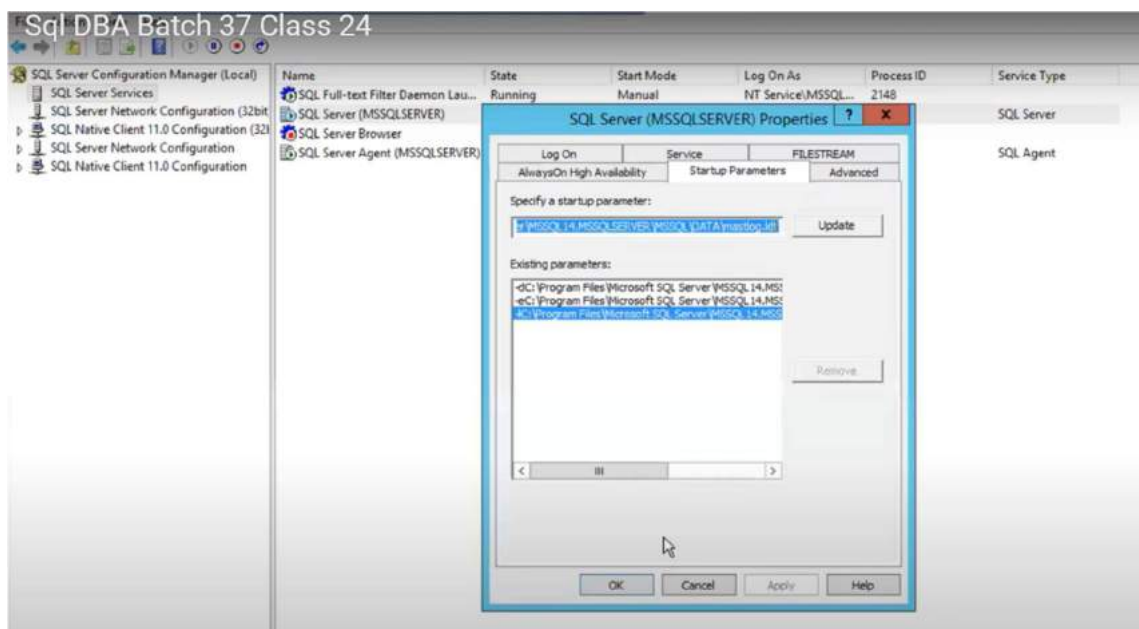
Ans. Query is SP_HELPDB and executes like below



	name	db_size	owner	dbid	created	status	compatibility_level
1	master	6.44 MB	sa	1	Apr 8 2003	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	140
2	model	16.00 MB	sa	3	Apr 8 2003	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	140
3	msdb	15.44 MB	sa	4	Aug 22 2017	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	140
4	tempdb	16.00 MB	sa	2	Jul 4 2021	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	140

30. What are the system databases?

Ans. When we install SQL Server, we will get 5 system data bases. Only for sql server 2000, we will get 4 system databases. Without the system data bases, Sql server won't work. To start the sql server, there should be 3 files, Master data file, master log file & error file. These 3 files are called start up parameters. We can see the start-up parameters in the Sql server configuration, right click on main server and go to start-up parameters option. Here we can see the path of start-up parameters of sql server.



Below are the System data bases

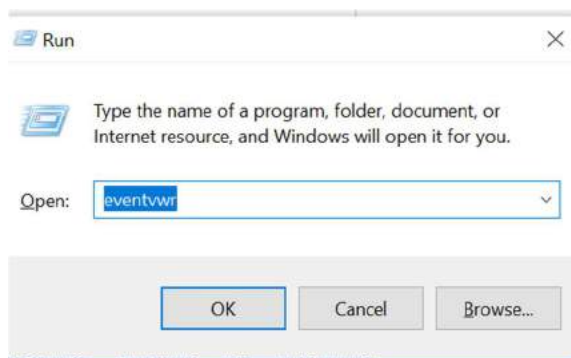
- 1 **Master:** This is the main database, which controls the sql server. The meta data of each object will be stored in the master data base. META data base means, data about the data (It means how many columns, rows, storage, path, logins, size etc of all the objects). To create any objects like database, logins, tables, records etc created by the master only. It stores the information of the other system databases and user databases. We can't start sql server, if master data file & log file corrupted. Always master data first comes online in the sql server, because of it contains all the paths of the other databases. Instance level configurations also stored in master data base only.
- 2 **Model:** Model database will act as a template for newly created databases. All the new databases will follow the default settings of the model database to create the database. Even model database is corrupted, we can't start SQL Server.
- 3 **MSDB:** All the automation information (like job, backups, restore, maintenance activities information) will be stored in this data base. If MSDB is corrupted, SQL server can start but automation of jobs or maintenance activities will stop. It's an optional to start the SQL Server.SSIS packages information stored in the MSDB.
- 4 **TEMPDB:** All the users will store the data in TEMP DB for temporary. It's better to maintain a separate drive for TEMP db. We can't take backup of TEMP DB , because whenever we restart the server new

TEMP DB will be created. If temp db is corrupted, sql server will be down. Index rebuild and sorting of data will be done in TempDB. It acts as a RAM to the SQL Server. We can't start the sql server if there is no TempDB. New temp db also depends on the template of model database. That's why without model database we can't create temp db, so without TempDB we can't run sql server. So, both model and TempDB also required to start the sql server.

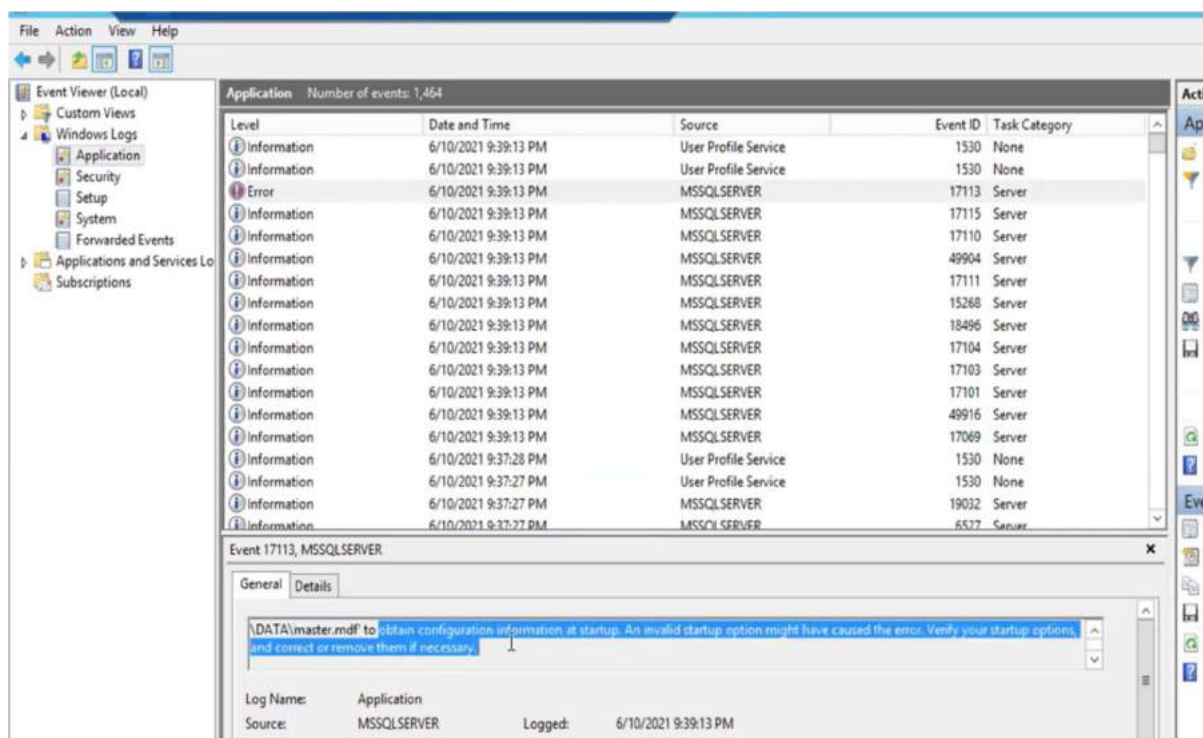
- 5 Resource: It's a hidden data base. Not visible in the SSMS. In this database, all the objects stored physically. Whatever data stored in the master; the same data will be stored in resource db also but only in the physically. In master data stored in logically. Before SQL Server 2005, there is no resource data base. In SQL Server 2005, we can find the resource db in data folder of SQL instance. From SQL Server 2008 onwards, it's a hidden data base. We can find this data base in root directory \binn folder.

31. Where to check the error, if the Sql Server not started?

Ans. Open the RUN and type "Eventvwr" (Event Viewer). - Its stores all the applications errors.



Go to Windows logs option, and select the application, where you can see the error details.



32. What is the order of system data bases when we start the SQL Server?

Ans.

- a. Master
- b. Resource
- c. Model
- d. TempDB
- e. Msdb
- f. User data bases

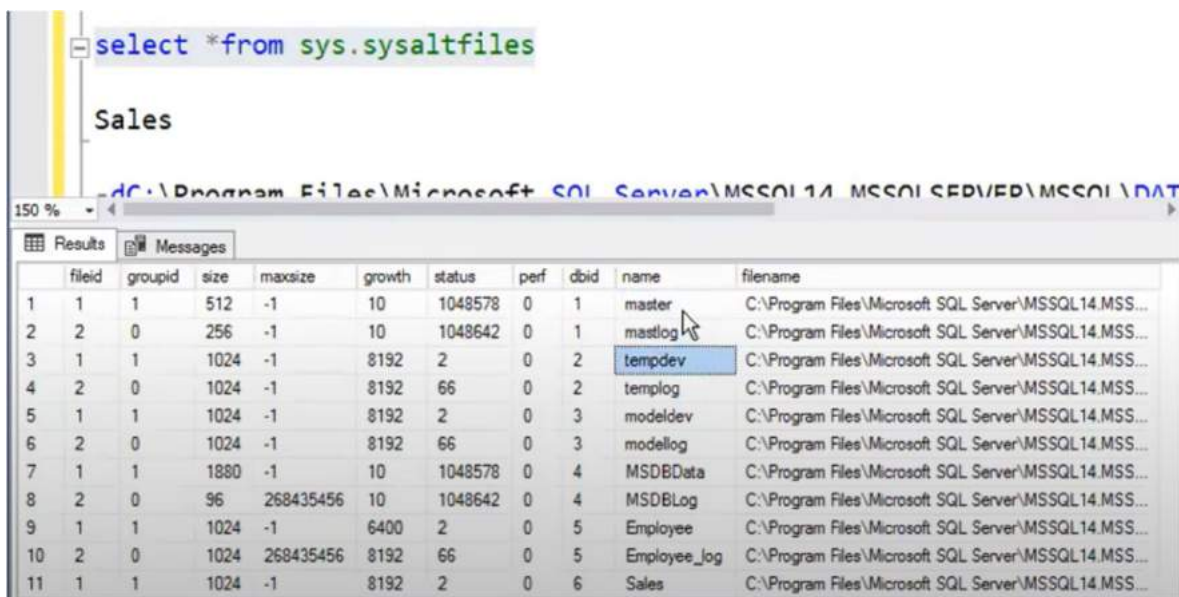
33. What are the data base id for the System data bases?

Ans.

Database Ids	
Master	1
TempDB	2
Model	3
MSDB	4
ResourceDB	32767

34. What is the query to see all the data files and log files?

Ans. select * from sys. sysaltfiles



The screenshot shows a SQL Server Enterprise Manager interface. At the top, a query window displays the command `select * from sys.sysaltfiles`. Below the query, a table titled 'Sales' is visible. The main part of the screenshot is a 'Results' pane showing a table with 11 rows and 10 columns. The columns are: fileid, groupid, size, maxsize, growth, status, perf, dbid, name, and filename. The rows list various system databases and their files, including master, mastlog, tempdev, templog, modeldev, modellog, MSDBData, MSDBLog, Employee, Employee_log, and Sales.

fileid	groupid	size	maxsize	growth	status	perf	dbid	name	filename
1	1	512	-1	10	1048578	0	1	master	C:\Program Files\Microsoft SQL Server\MSSQL14.MSS...
2	2	0	256	-1	1048642	0	1	mastlog	C:\Program Files\Microsoft SQL Server\MSSQL14.MSS...
3	1	1024	-1	8192	2	0	2	tempdev	C:\Program Files\Microsoft SQL Server\MSSQL14.MSS...
4	2	0	1024	-1	8192	66	0	templog	C:\Program Files\Microsoft SQL Server\MSSQL14.MSS...
5	1	1024	-1	8192	2	0	3	modeldev	C:\Program Files\Microsoft SQL Server\MSSQL14.MSS...
6	2	0	1024	-1	8192	66	0	modellog	C:\Program Files\Microsoft SQL Server\MSSQL14.MSS...
7	1	1880	-1	10	1048578	0	4	MSDBData	C:\Program Files\Microsoft SQL Server\MSSQL14.MSS...
8	2	0	96	268435456	1048642	0	4	MSDBLog	C:\Program Files\Microsoft SQL Server\MSSQL14.MSS...
9	1	1024	-1	6400	2	0	5	Employee	C:\Program Files\Microsoft SQL Server\MSSQL14.MSS...
10	2	0	1024	268435456	8192	66	5	Employee_log	C:\Program Files\Microsoft SQL Server\MSSQL14.MSS...
11	1	1024	-1	8192	2	0	6	Sales	C:\Program Files\Microsoft SQL Server\MSSQL14.MSS...

35. What are the types of data files?

Ans. There are 2 types.

- a. Master data file(mdf) – It stores the meta data of each data base. Already discussed in the above pages. Default file group is primary.
- b. New\next data file(ndf) – if you want store the data file in .NDF, then create a new file group.

36. How many databases or logins or users or files or file groups etc we can create?

Ans. 32767

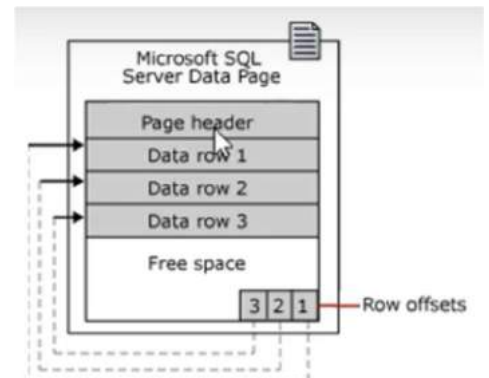
37. What are pages?

Ans. Pages are nothing but the basic storage unit of data file. All the data files of the SQL Server are stored in pages only.

Each page is 8 KB. We can't change the page size. It's fixed. A page is used for only one object. Each object will have separate pages. But a single table can be stored in multiple pages.

Suppose we have a database size of 1 GB = 1024MB = 1048576 kb = $1048576/8 = 131072$ pages are the size of this database. A page contains 3 parts,

- Page header – it stores the meta data like Page id's, object Id's, how much space in this page etc.
- Body (Actual data) – Data or records will be stored in row-by-row manner.
- Row offset – It works as an index. It has the address of each data or record



Page header uses 96 bytes of space,

Row offsets use 36 bytes of space,

Balance size of 8060 bytes used for body. Total 8192 bytes = 8 kb = 1 page

1MB – 128 pages

38. What are Extents?

Ans. Extent is a collection of 8 pages. It's a basic allocation unit of sql server data file.

1 page = 8kb = $8 \times 8 = 64$ kb = 1 extent. So, size of each extent is 64 Kb. It's a fixed size we can't change.

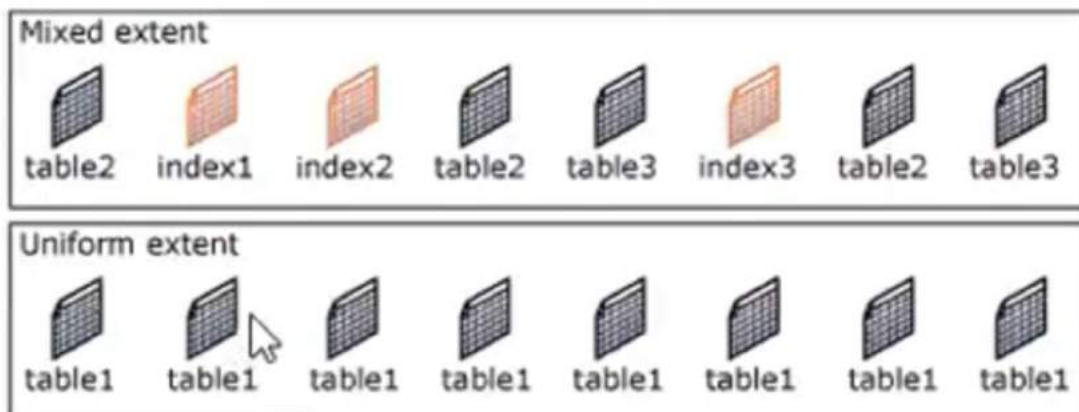
We have two types of extents,

- Mixed Extent – all the 8 pages can be used for multiple objects
- Uniform Extent – all the 8 pages can be used for only one object.

For example, whenever we create a table, sql server first creates the one mixed extent with 1 page to save the storage. When the data is increasing in the data file, then all the subsequent pages of the same datafile will be converted into uniform extent.

Remember, while creating the object- sql server chooses the mixed extents only.

Depends on the data, SQL Server will allocate Mixed extents and uniform extents to the data file. Mostly it allocates mixed extent.



39. What are the types of pages?

Ans. Mainly we have 2 pages,

1. Data pages - 3 types

All the data will be stores in the any of the below 3 types pages only.

- Data page – it will store the small data types data except large data types like text, ntext, image, nvarchar(max), varchar(max), varbinary(max) and xml data.
- Index page – it stores the indexes.
- Text/image it will store the large data type data like text, ntext, image, nvarchar(max), varchar(max), varbinary(max) and xml data.

2. System pages - 6 types

- Gam/SGAM (Global allocation map\Shared global allocation map)-

GAM: It tracks the status of extents, which extent is free and which extent is allocated. It tracks the status in zeros and ones.

Bit 1 = Free extent

Bit 0 = allocated extent

SGAM: It tracks all the mixed extents, which is having at least one unused page or space. If the bit is 1 then extent is used as mixed extent with free page.

- PFS (Page free space): it will record the information about page allocation and free space available on pages. It will calculate in percentages.
- IAM (Index allocation map): it will track information about extents used by a table or index per allocation unit. It will track all the tables and in which extent table is present.
- BCM (Bulk changed map): information about all the extents, which are modified by the bulk operations. Like bulk insert, bulk updates etc. So, when we did the bulk operations like inserting, data will be stored in any pages and extents will be changed. These modified extents will be tracked by BCM.
- DCM (Differential Changed map): information about extents that have changed since the last full backup database statement per allocation using all the changes that we can do to an extent.

To see on what pages data is stores, use the query,

DONE USE ALL THE BELOW QUERIES IN PRODUCTION

(DBCC- DATABASE CONSOLE COMMANDS)

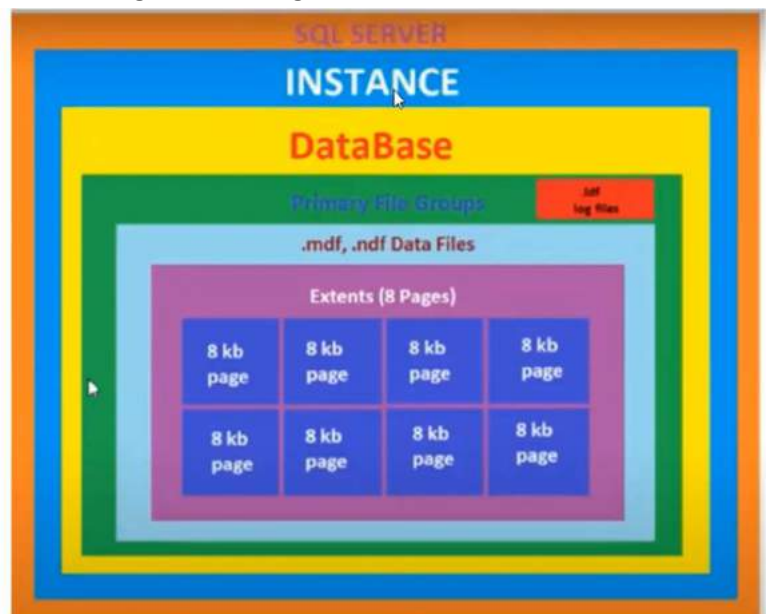
“DBCC IND (DB NAME, TABLE NAME, -1)”

-1 is used for display option

To see the page internals,

First this query “dbcc traceon (3604)”

Second query “Dbcc page (db name, file id, page id,1)” (1 IS TO DISPLAY)



```
dbcc traceon(3604)
DBCC PAGE(A,1,309,1)
```

100 % Messages

PAGE: (1:309)

BUFFER:

BUF @0x00000002668AD2C0

bpage = 0x000000025200A000 bhash = 0x0000000000000000 bpageno = (1:309)
bdbid = 12 breferences = 0 bcpu ticks = 0
bsampleCount = 0 buse1 = 31939 bstat = 0x9
blog = 0x15a0215a bnxt = 0x0000000000000000

PAGE HEADER:

IndexID	PartitionNumber	PartitionID	iam_chain_type	PageType	IndexLevel
0	1	72057594040614912	In-row data	10	NULL
0	1	72057594040614912	In-row data	1	0

40. What are the types of transactions in SQL Server?

Ans. A query is called a transaction.

There are 2 types

- a. Implicit Transaction: All the implicit transactions are auto committed. It means without Begin Tran and Commit

EX: insert in to table values(1,"name");

- b. Explicit Transaction: All the explicit transactions are not auto committed. We have to mention commit. If we won't mention commit, data will be stored temporarily only.

Ex: begin Tran

insert in to table values(1,"name")

Commit

41. Explain Log File architecture?

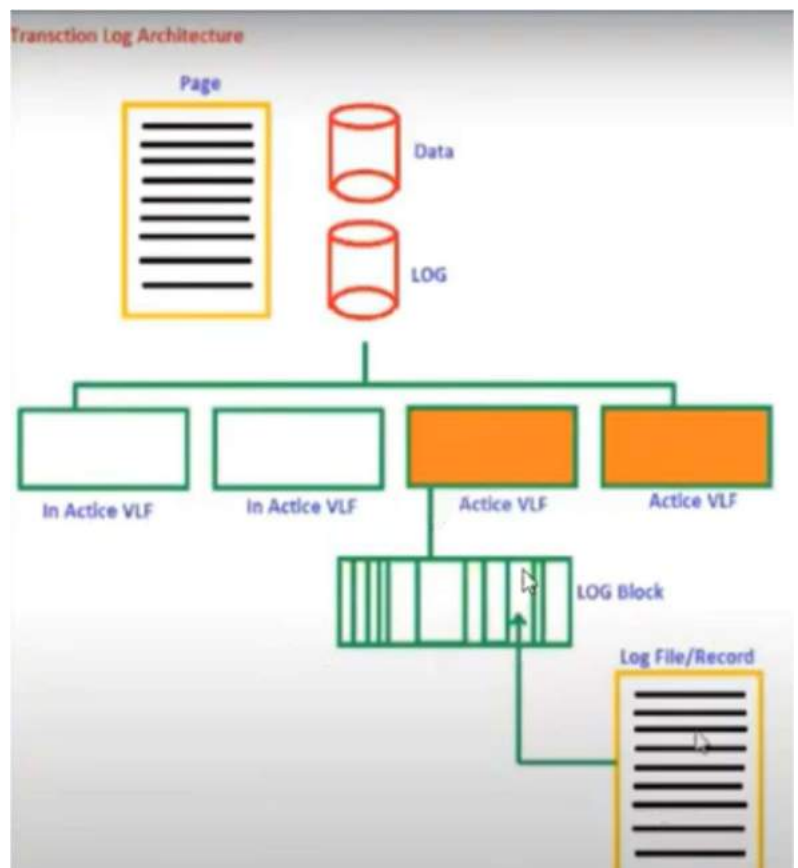
Ans. log file stores the transactions.log file contains virtual log files (VLF).

There are 2 types of log files in architecture,

- a) Active VLF: Currently which is holding the transaction is called Active VLF. Active VLF contains Log blocks. Each log block contains Log file/Record. One record is one transaction.
- b) Inactive VLF: Log file, which doesn't contain any transactions is called inactive VLF.

To see how many VLF's is there for each data base log file

Use the query "dbcc loginfo"



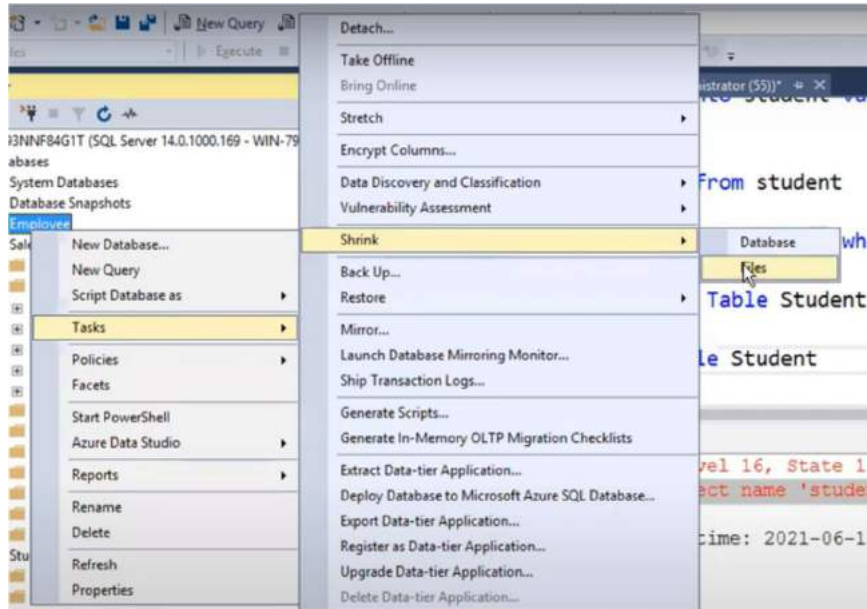
dbcc loginfo

	RecoveryUnitId	Field	FileSize	StartOffset	FSeqNo	Status	Party	CreateLSN
1	0	2	2031616	8152	36	0	64	0
2	0	2	2031616	203808	37	0	64	0
3	0	2	2031616	4071424	38	0	64	0
4	0	2	2265568	6103040	39	0	64	0
5	0	2	16777216	8388608	40	0	64	39000000397200001
6	0	2	16777216	25165824	41	0	64	39000000397200001
7	0	2	16777216	41943040	42	2	64	39000000397200001
8	0	2	16777216	58720256	0	0	0	39000000397200001

42. How to shrink the log file or how to release the unused space of the log file?

Ans.

a) right click on database, select the tasks, choose the shrink option, select files like below

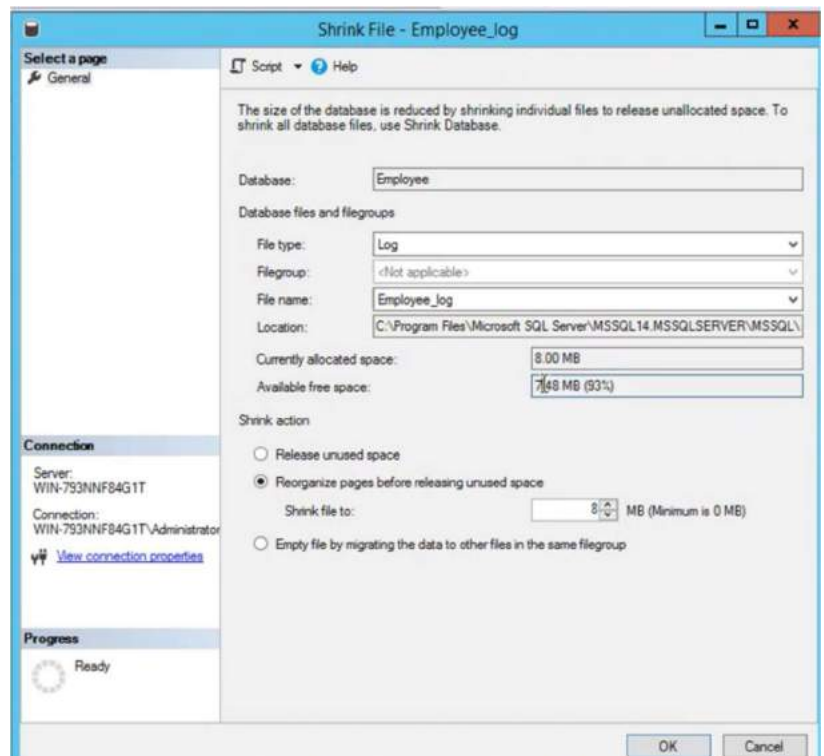


b) Choose the file type as log, and here we can see, how much space allocated to log file and how much is free space available.

a. Here we have 2 actions, one is releasing unused space, two is reorganise pages before releasing unused space.

Choose the 2 nd option to shrink the log file size. Like screenshot provided.

In the organizations, we won't do the shrinking for 1 gb or 2 gb. We do for bigger sizes.



43. To Which system data bases, we can create a new file groups and data files?

Ans. Only MSDB allows to create new file group and files.

44. What is service packs?

Ans. Service packs are major updates of the SQL Server. Within a year or 2 years once, Microsoft release the service packs. Once service pack released, Microsoft send the emails to all the customers. After that, service pack update will be download by managers or team leaders from Microsoft website and kept in the shared folder. Now, DBA will patch the service packs to sql server. We can install any service pack, it's not mandatory to install in order. We can install the latest one sp4 alone. Whenever we uninstall any service pack, SQL server will go to the previous update of service pack sp1 or sp2 or sp3. If there are no previous service pack updates, SQL Server will go starting level RTM Copy (Release to market.)

		SP1	SP2	SP3	SP4
SQL Server 2016 SQL Server 13 Support end date: 2021-07-13 Ext. end date: 2026-07-14	13.0.1601.5 + CU9	13.0.4001.0 or 13.1.4001.0 + CU15	13.0.5026.0 or 13.2.5026.0 + CU17		
SQL Server 2014 SQL Server 12 Support end date: 2019-07-09 Ext. end date: 2024-07-09	12.0.2000.8 + CU14	12.0.4100.1 or 12.1.4100.1 + CU13	12.0.5000.0 or 12.2.5000.0 + CU18	12.0.6024.0 or 12.3.6024.0 + CU4	
SQL Server 2012 SQL Server 11 codename: Denali Support end date: 2017-07-11 Ext. end date: 2022-07-12	11.0.2100.60 + CU11	11.0.3000.0 or 11.1.3000.0 + CU16	11.0.5058.0 or 11.2.5058.0 + CU16	11.0.6020.0 or 11.3.6020.0 + CU10	11.0.7001.0 or 11.4.7001.0

Microsoft release up to four service packs only.

From SQL Server 2017 version on wards there is no Service packs.

45. What are cumulative updates?

Ans. Cumulative updates are minor updates of RTM or service packs. Within a month or every 2 months, Microsoft release the cumulative updates. In cumulative updated, there are 2 types,

These are called hot fixes.

- GDR (General distribution release): it's a security update to safeguard from the viruses, threats.
- QFE (Quick fix engineering): This update helps to fix the bugs.

On RTM Copy also Cumulative updates will be there. All the service packs also will have Cumulative updates. If you want to install any cumulative update of service packs, you must patch the service pack first and then you can patch cumulative update.

Sql Server 2012				
RTM	SP1	SP2	SP3	SP4
CU1	CU1	CU1	CU1	CU1
CU2	CU2	CU2	CU2	CU2
CU3	CU3	CU3	CU3	CU3
CU4	CU4	CU4	CU4	CU4
CU5	CU5	CU5	CU5	CU5
CU6	CU6	CU6	CU6	CU6
CU7	CU7	CU7	CU7	CU7
CU8	CU8	CU8	CU8	CU8
CU9	CU9	CU9	CU9	CU9
CU10	CU10	CU10	CU10	CU10

From SQL Server 2017 version on wards there is no Service packs.

46. What are the precautions before patching the service packs or CU to SQL Server or production server?

Ans. Patching is a downtime activity. Whenever we want to patch, we have to restart windows server first. We should not restart the windows server directly; we have to take proper approval. Patching will take 45 mins to 1 hour time. During the patching, SQL Server restarts multiple times automatically. That's why, we can't patch the sql server during business hours. Below are the precautions before patching.

- a. First find the application owner.
- b. Inform to owner about downtime and ask for the timings to do the patching.
- c. Timings, which are provided by the owner is called “patching Window”.
- d. Raise a change request with start time and end time of patching.
- e. CAB (Change authority board) will discuss about the request
- f. CAB will approve the change request.
- g. Now DBA start the patching as per Patch window

47. What are the precautions we have to take when you are going to start patching & after finishing patching?

Ans.

- a. Check how many SQL Servers or Instances are running in the same Windows Server. Because when you start the patching windows server needs to restart. At this time, other instances also go off line.
- b. Take the screenshot of SQL Server Configuration manager before starting the patching. Because some of the servers are already offline and some servers are online. We don't know which server is online or which server is offline after patching. So, to keep the servers in the same online or offline position keep the screen shot of configuration manager with you.
- c. Take the screen shot of database status before patching. Few databases already in offline or restoring mode like that. After patching, screenshot helps to check the databases are in the same status or not.
- d. Make sure the name of the server in the object explorer should match with running the query “select @@servername”. If both are not matching, patching will be failed.
- e. Check the build numbers before patching in the object explorer and Query “select @@version”. Both should be same.
- f. Make sure you have a proper free space in the drive at least 3 GB. If you have less space, ask the windows team to extend the space.
- g. request the windows team to take the Virtual machine snapshot (it means back up of entire virtual machine) – it helps if SQL Server not starting after patch-up. Windows team can restore the entire virtual machine.
- h. Check whether machine is physical machine or virtual machine. (We can check it, by typing “system information” in the windows search box, there you can see system model tells us it's a VirtualBox or vm ware or hyper wiser etc for virtual machines. If it's a physical machine, system model tells us HP or Lenovo or dell or ASUS etc). if it's a physical machine, try to patch on week days instead of weekends. Why because, if after patching physical machine is not coming online- in week days no one will be there in office to turn the physical machine online.
- i. Send an email to whole DBA Team.
- j. After all the above precautions start the patching process.

48. What is the query to know the server name & Version of the sql server?

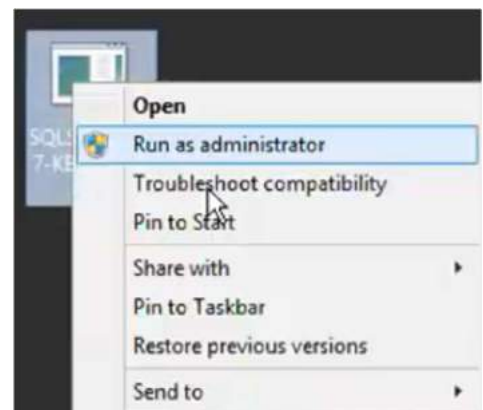
Ans. select @@servername – for server name

Select @@ Version – for version

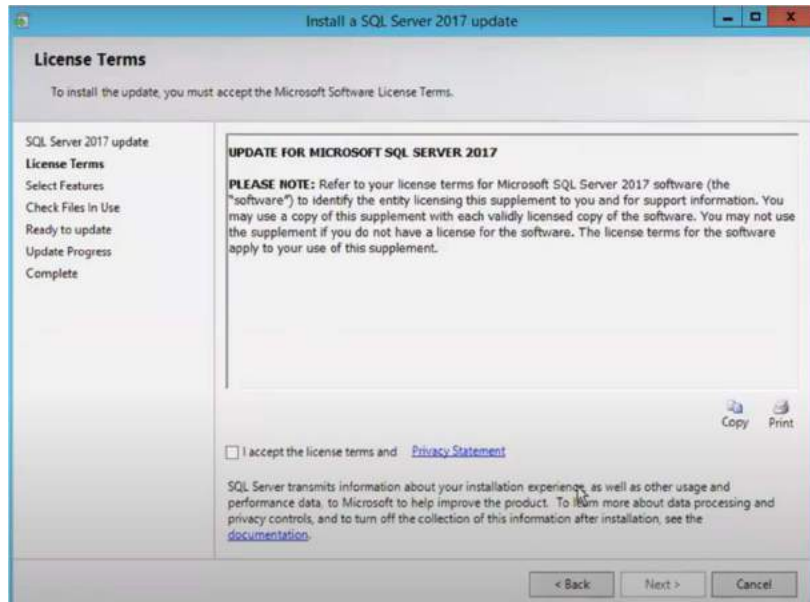
49. What are steps to install Service Pack or Cumulative update?

Ans. After installed the cumulative update or service pack how will know whether patching done successfully. After update, the build no. of the sql number will change. There will come to know patched successfully.

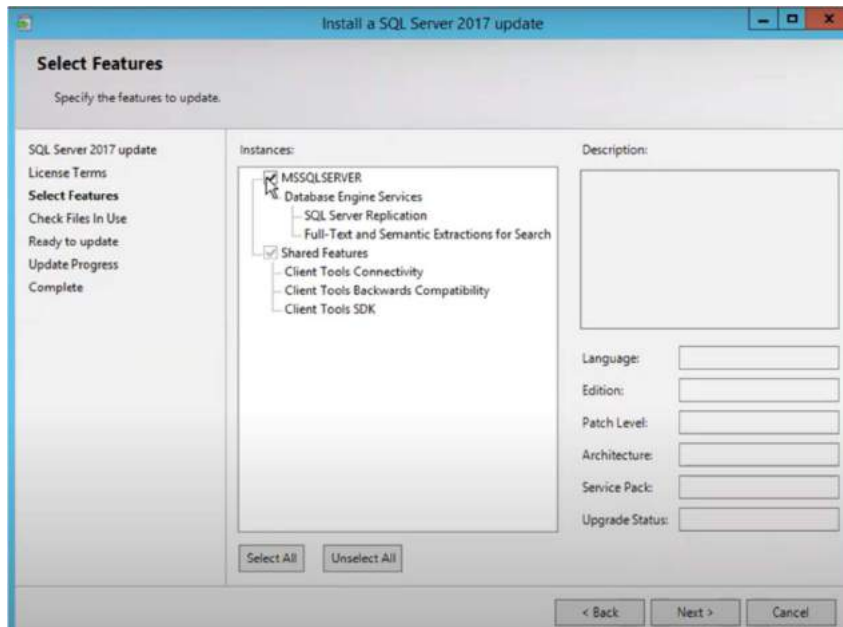
- a. Copy the service pack or CU from shared path to the windows server. (Which is already downloaded and kept in the shared folder by team leader or project managers)
- b. Right click – run as administrator



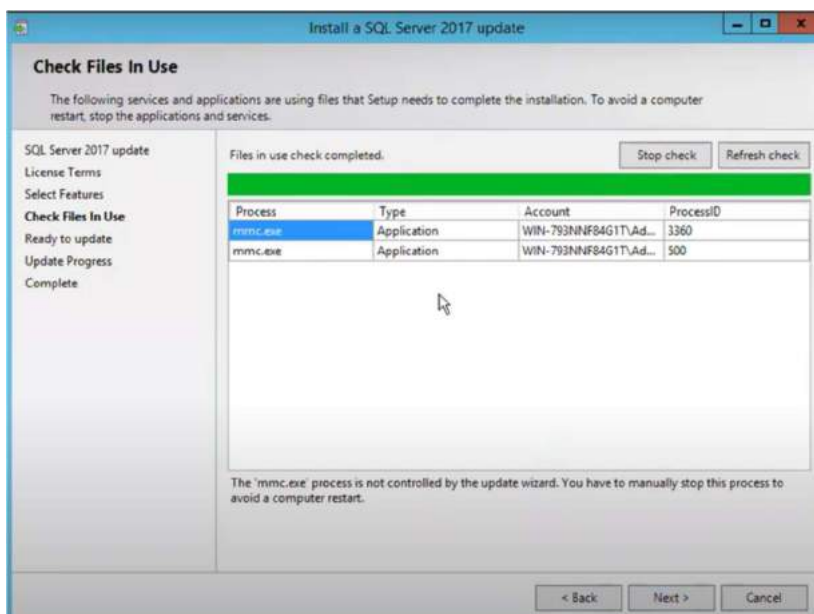
- c. Accept the license terms, click next



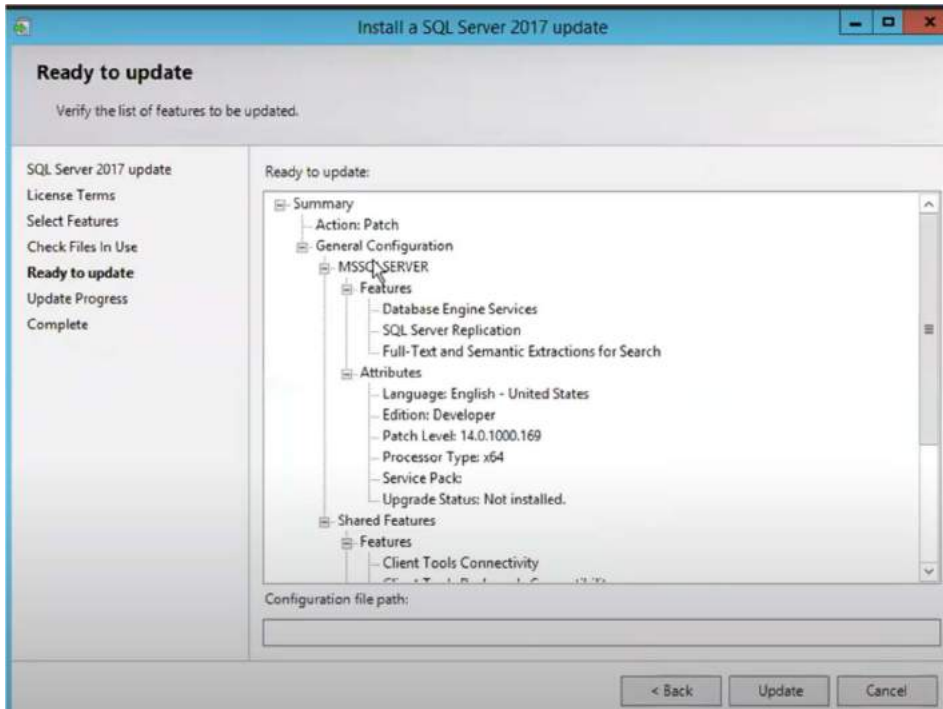
- d. Select the which server or which instance, you want to patch, click next



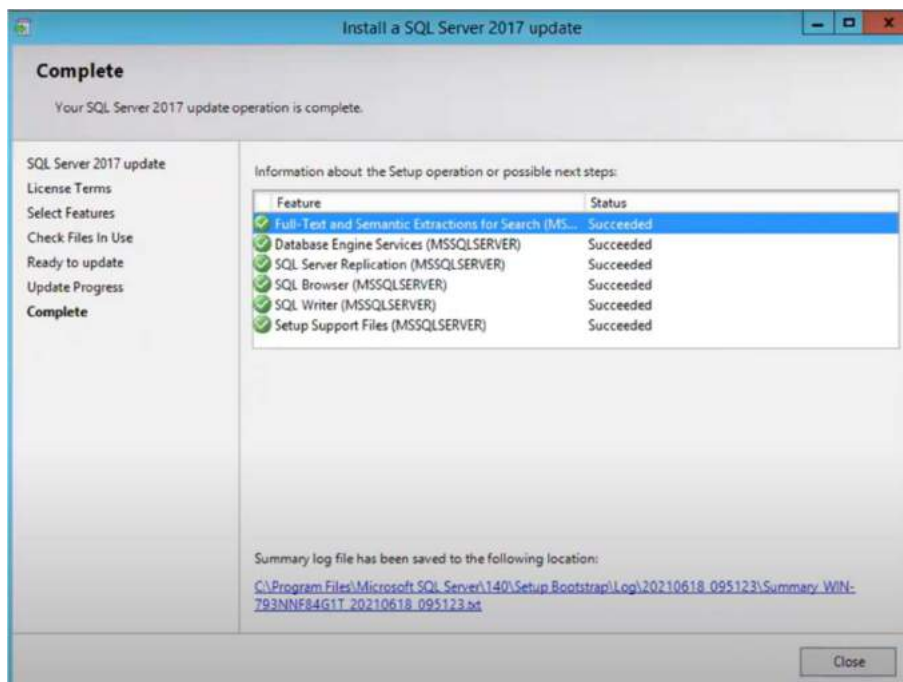
- e. Now it will check files in use. Like below, click next



- f. Ready to update screen will come like below, click next.

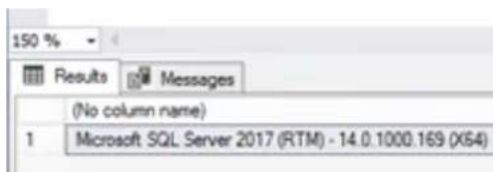


- g. Update completed screen and make sure all the features in this screen succeeded. If any feature is failed, Patching is failed.

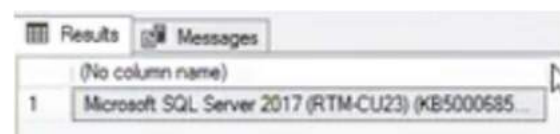


- h. Now check the versions of build no's

Before patching version screenshot



After patching version below



- i. Now follow the precautions after installation discussed in the before pages.

50. How to Remove patches of SQL Server?

Ans. Organization very rarely ask to remove the patches, when patches are not installed successfully like the object explorer build no. and @@Version build no. are not same. At this time, we have to remove the patch again we have to reinstall the patch. One more case also is there like, Applications are not compatible with the latest service packs or CU, then also we have to remove patches.

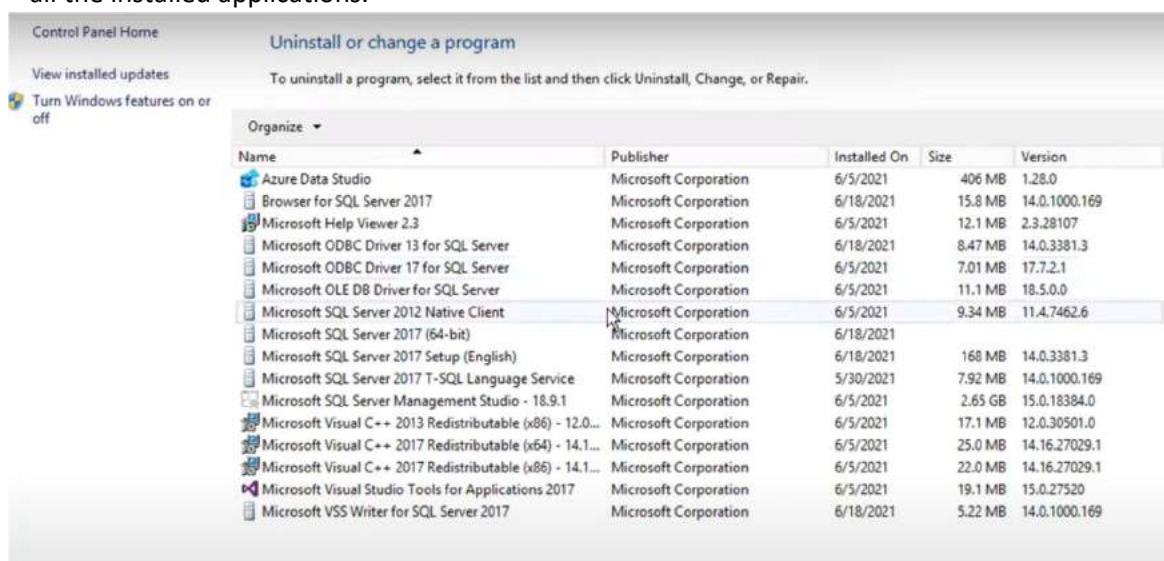
For compatibility issues, one way we can ask the application team to upgrade the application or else We have to remove the patch.

When we remove the CU, SQL Server goes to previous installed CU. If there is no install CU installation happened previous, then it will to RTM.

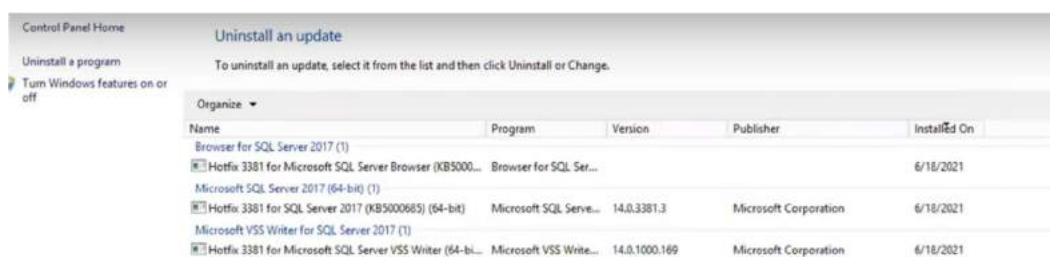
To remove patches also need a change request, why because this is also a downtime activity.

Steps to remove Patch:

- Type Run command “appwiz.cpl” or open control panel and select add or remove programs. Here we can see all the installed applications.



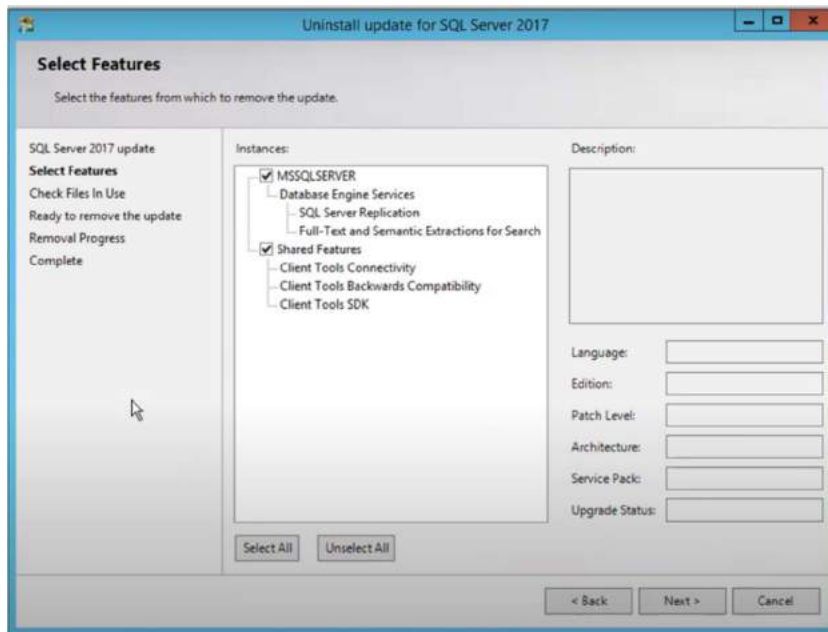
- Click on view installed updates, where we can see all the service packs installation and CU installations. There will be components updates and main sql server update also. We have to remove main sql server update, then automatically components update also removed.



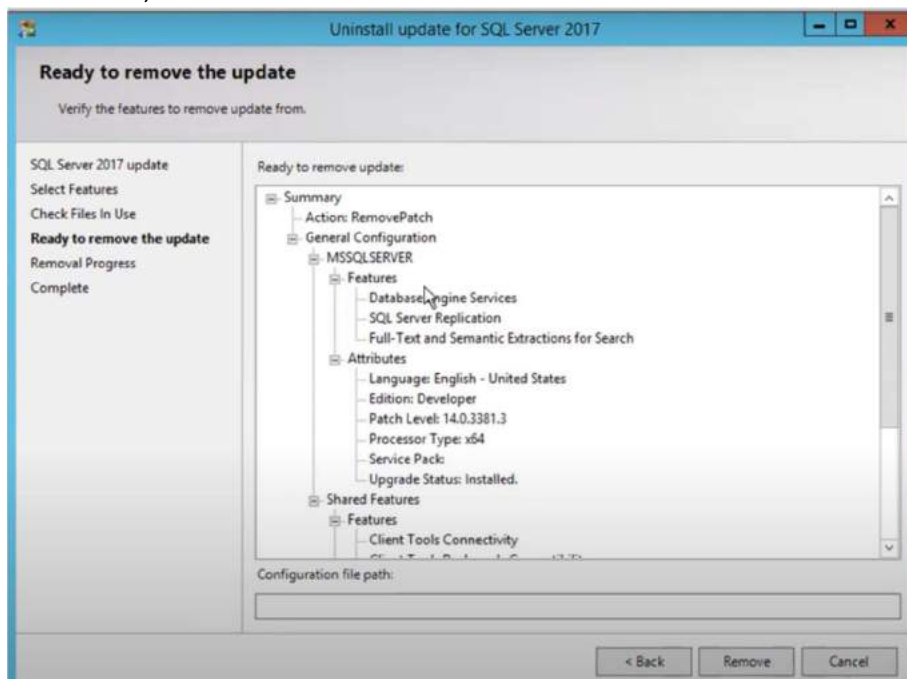
- Select the Main sql server update and click on uninstall. Now uninstall process starts.



- d. After that, uninstall window will go to features tab, where we have to select, which sql server instance update you want remove. After that click next.

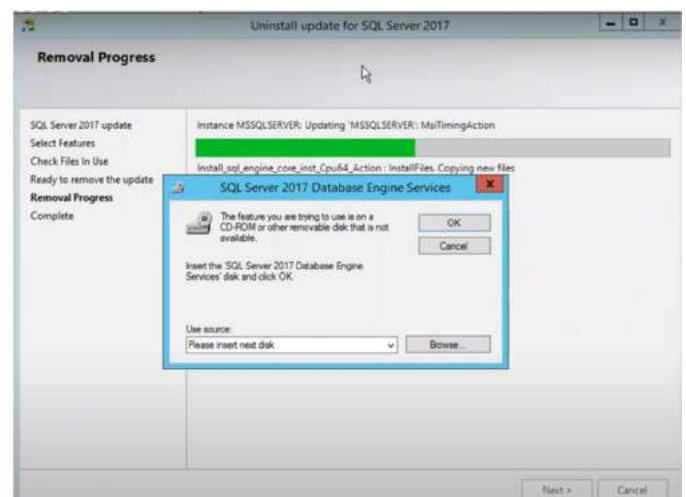


- e. After that, Click remove.

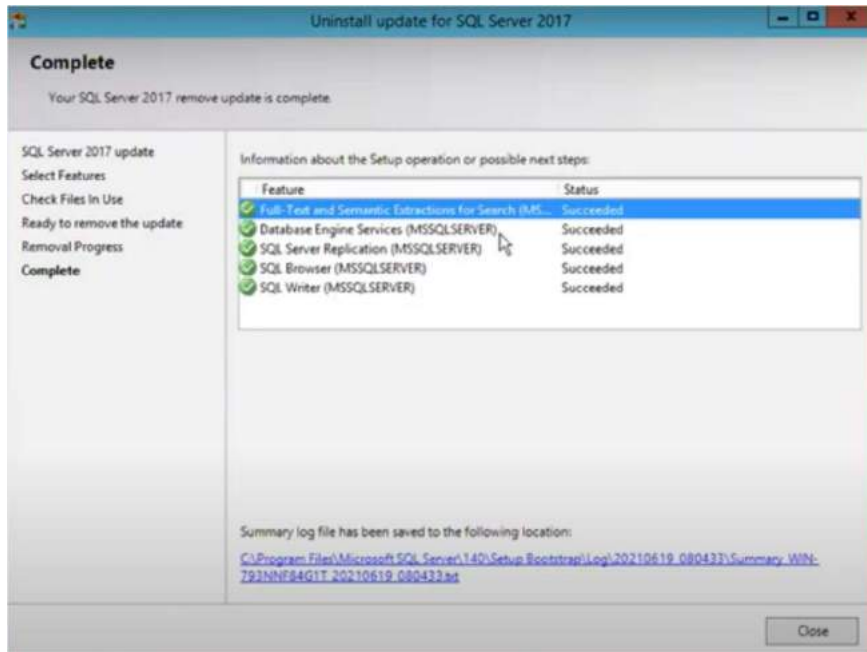


- f. Some times while doing the uninstalling, Sql server will asks for Main RTM Software CD like below. Give the address where the RTM Copy installation setup is there and click ok.

If the same error is kept on coming after given the path of RTM Copy, Then Microsoft introduced one tool called "Fix msi missing tool". Download it and install in the windows server.



g. Uninstall complete.



51. What are the terminologies used in organizations for different types of teams?

Ans. Windows team are called WIN OPS (Windows operators)
Virtual server team are called V OPS (Virtual Operators)
DBA Team are called DB OPS (Data base operators)
Network Operators are called NET Ops (Network Operators)

52. What are the terminologies in the SQL Server security?

Ans. Mainly we have 4 terminologies,

- Authorization: having access on objects or databases is called authorization.
- Authentication: The process of verifying the authorization is called authentication. For example: giving username and password to enter a database, after that authentication process starts on verifying the login, whether this login is there or not in the database. If it's there, then user can login otherwise can't.
- Principle: Those who are accessing the resources or database is called principle.
- Securable: What resource you are accessing, those resources are called securable.

53. What are the authentication types of SQL Server?

Ans. There are 2 types,

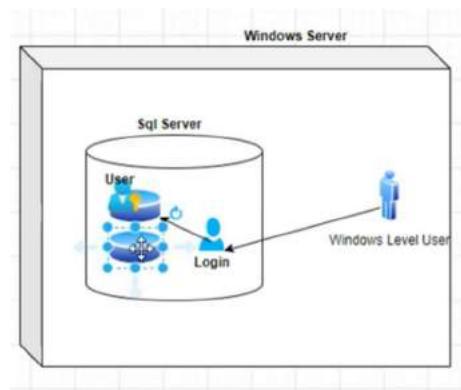
- Windows authentication: This mode not required any username and password.it reads the user name from windows and login in to sql server.
- Mixed mode or SQL server authentication: This mode required username and password for the users, which are created by the DBA.

54. What is the difference between login and user?

Ans. To access the SQL Server, Logins are created. Once login created, to access the database dba will give permissions, now a user will be created in the database. By simply creating login, user can't enter in to any database. A user will be created in the database, then only login can access to that particular database.

Instance level access is a login

Database level access is a user



55. How to create a SQL Server login via windows authentication mode?

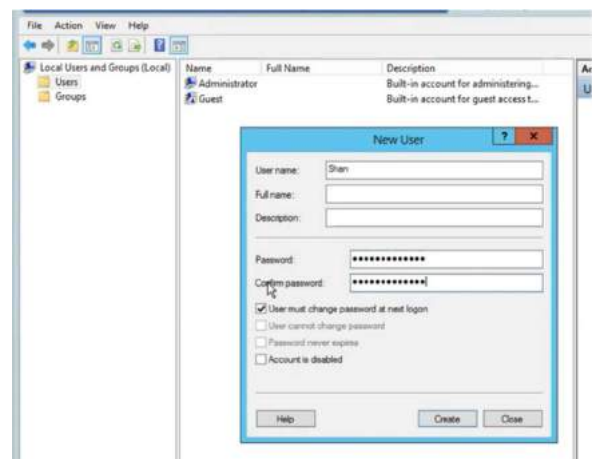
Ans. There is prerequisite before creating a windows login, that is user name should be there in the windows level or in the active directory level. In the organization, users will be created in the active directory level.

Creating windows level user:(This is not DBA job. Windows people will create the windows level user)

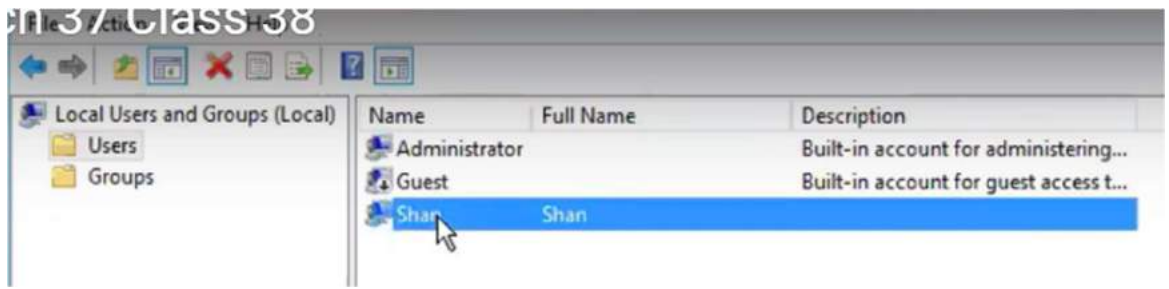
- First login in to windows server.
- Type run command "lusrmgr.msc" (Local user manager)



- Right click on the user, select new user, Fill the columns username, password etc. Click create.

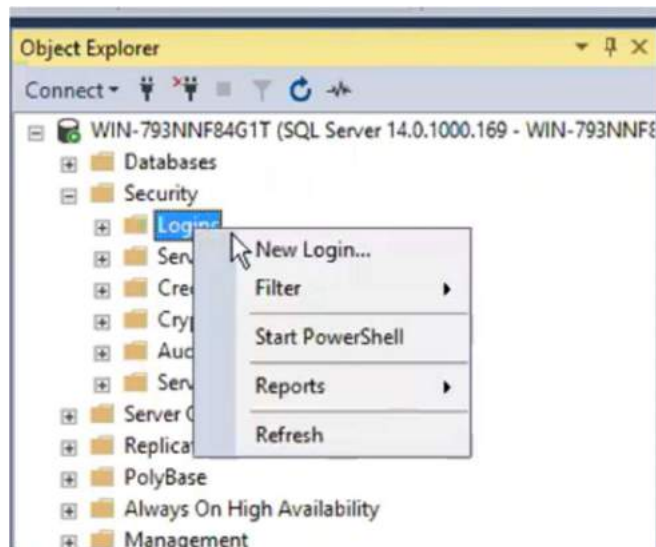


d) Now user got created

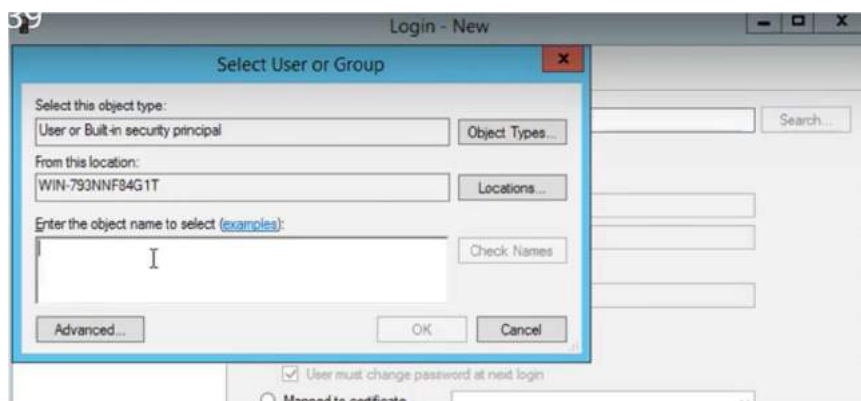


Windows user level account has been created. Now we have to add the user in the sql server via Windows mode Authentication.

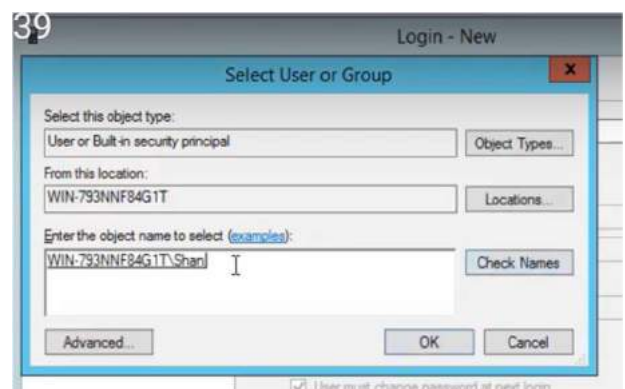
A. Open SQL server, Expand Security tab, Right click on logins, select new login.



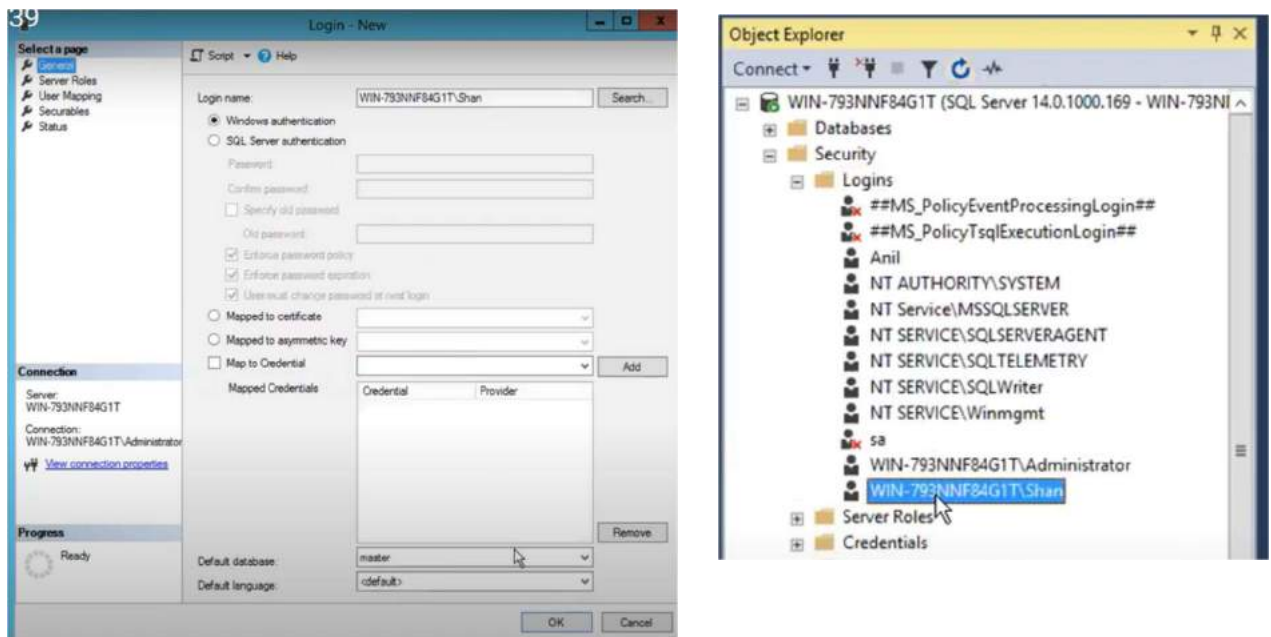
B. Select the Login type as windows authentication and click on the search button beside to the login name.



C. Give the created windows lever user name in the box, click check names. Now it will check for the username. Once it finds the windows level user, SQL Server will add the user. Click Ok.



- D. No need to give any password here, because while creating windows level user. We can give the password. The same will be used by SQL Server.
- E. Click ok. Login created. Now we can see that user name in SQL Server.

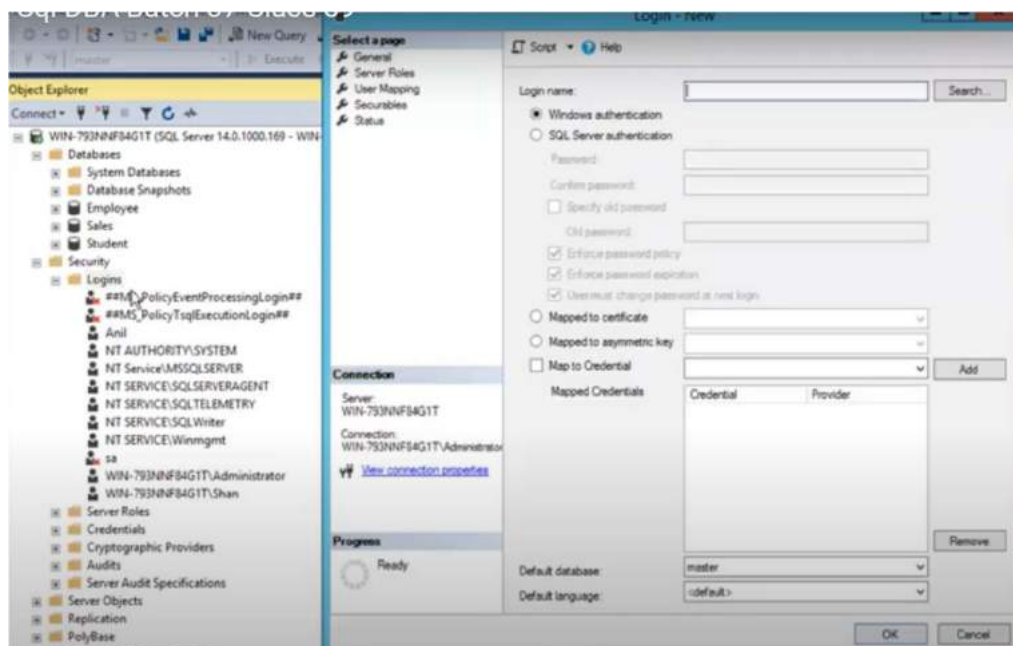


- F. Now user can access Sql server. But not able to access any Databases.

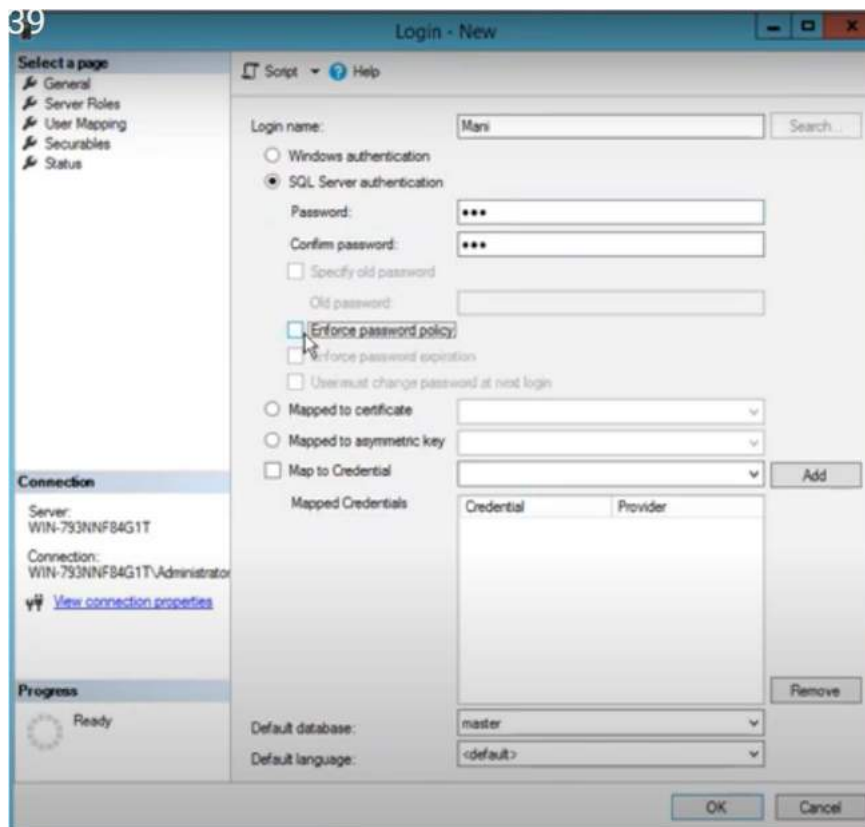
56. How to create a SQL Server login via windows authentication mode or SQL Server authentication mode?

Ans. There is no prerequisite to create a SQL Login via Mixed Mode.

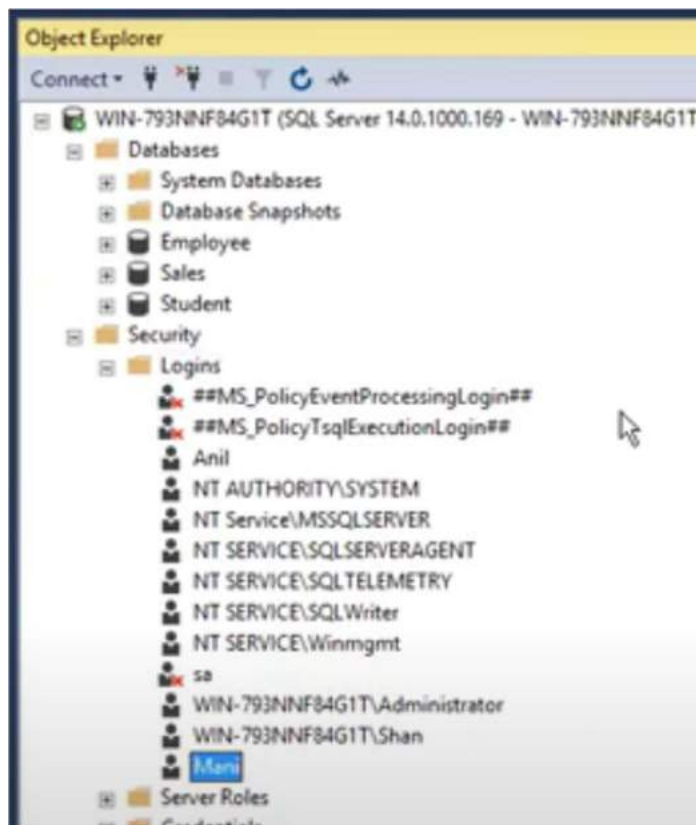
- a. Open SQL Server, expand security, Right click on Logins, Select New user.



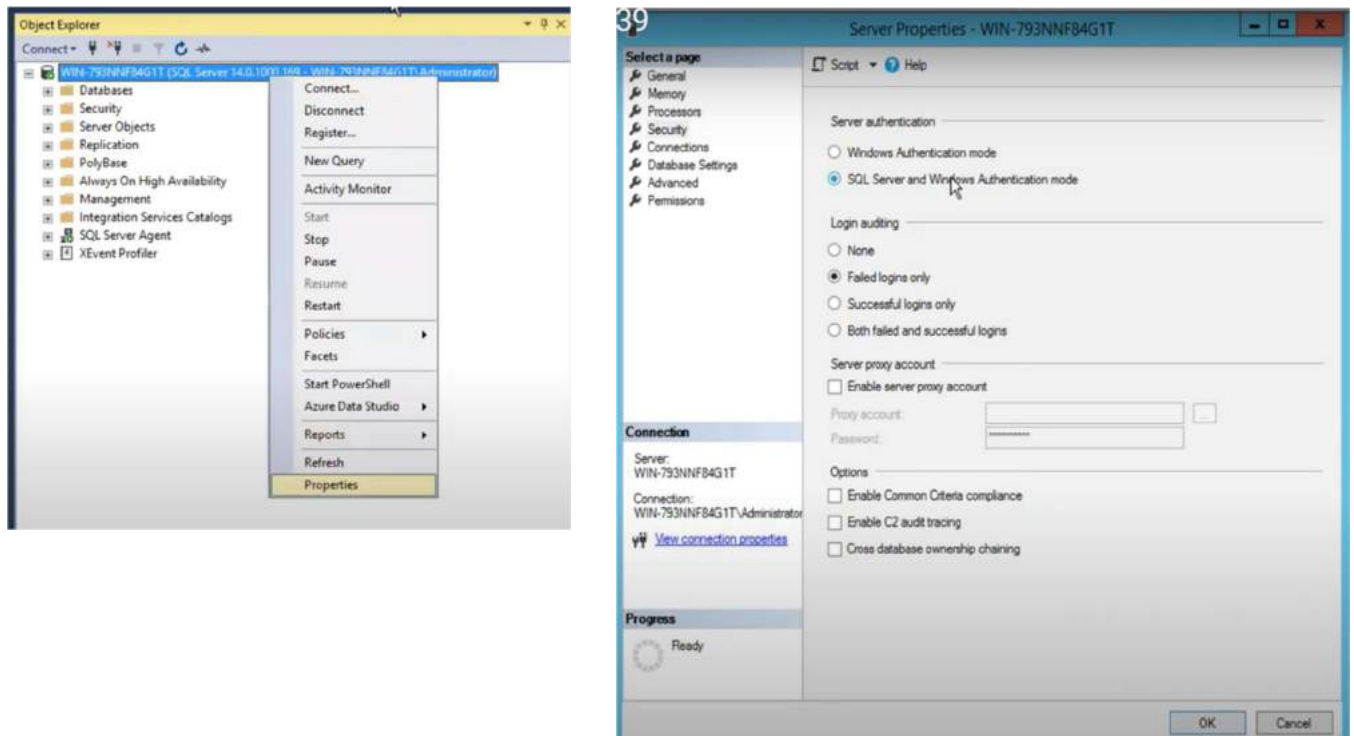
- b. Select SQL Server authentication, Give the login name, give the password, Click ok



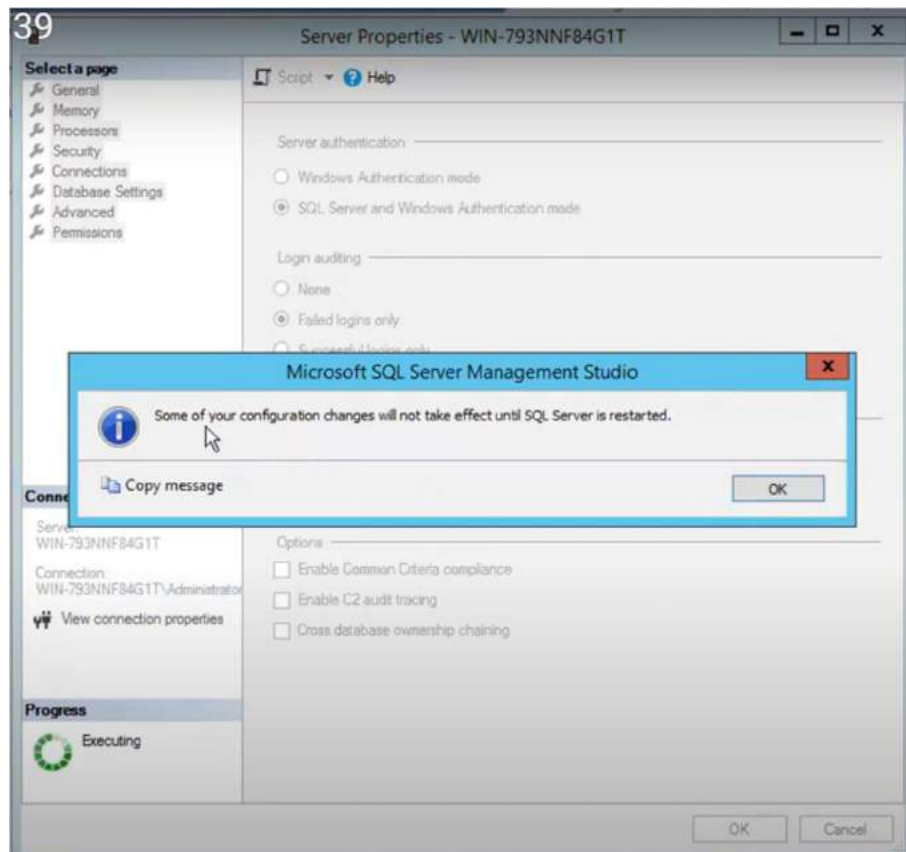
- c. Login successfully created.



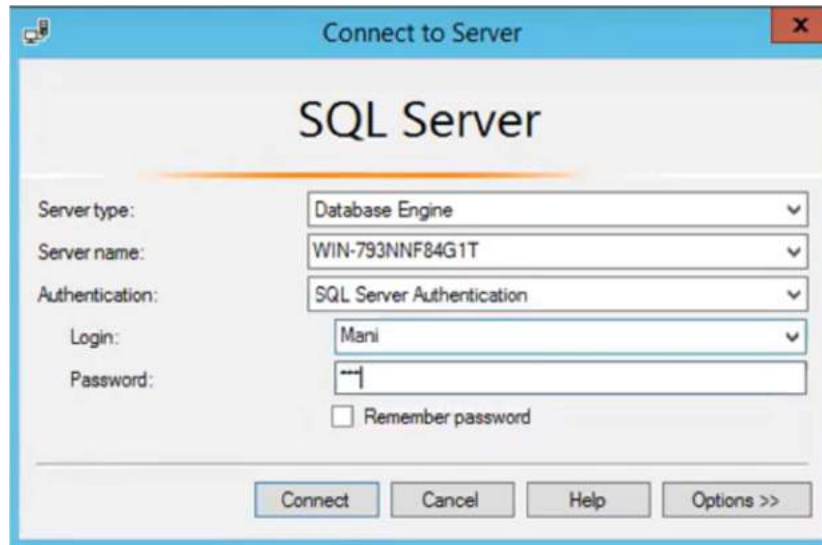
- d. Check the properties of SQL Server and keep the security type of server authentication mode to Mixed mode. Otherwise Sql Mode logins not able to enter SQL Server, we will get error.



- e. After changing the server authentication mode, it will ask to restart the server. Restarting the SQL Server is down time activity. While installing the SQL Server it self , We have to select Mixed mode . So in future, This kind of restart issues wont come while creating the logins.



- f. After restarting the SQL Server, now login to the SQL Server with the SQL user name and password. Like below.



57. How many users or logins or databases anything we can create in a single SQL Server.

Ans. 32767

58. What is a group login?

Ans. When a group from same team have the same permissions to access the database, that time we can create a group login. Instead of creating individual login for all the team members, we can simply create single group login and give access to a team. Even in future any new user joined to the team, we can simply add the user name in that existed group login. Now new user can access the database. All the organizations prefer mostly group logins instead of individual logins. All the users part of a group login has the same permissions.

Group logins, we can create only for the windows authentication mode. In SQL Server authentication mode, Group logins not possible to create.

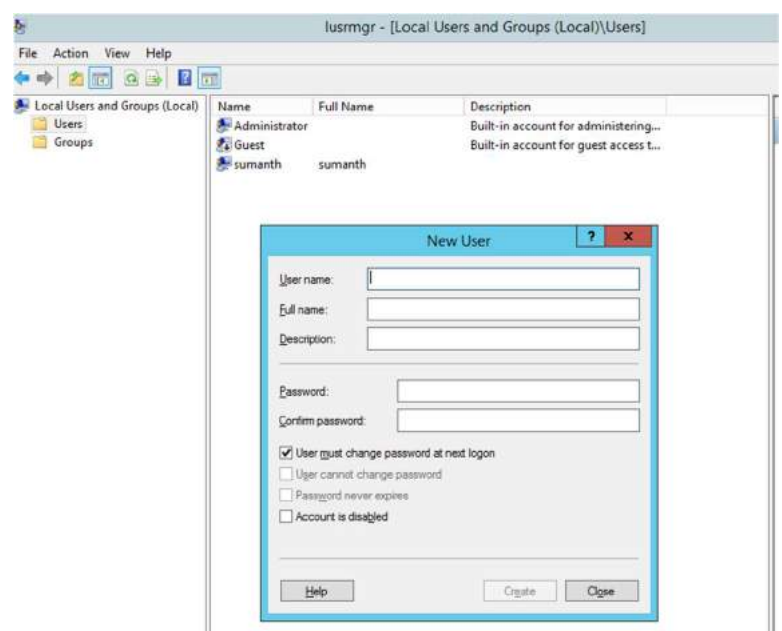
Prerequisites to create the group login is windows level logins or Active directory user logins

These windows level logins or Active directory user logins will be created by windows team not the dba.

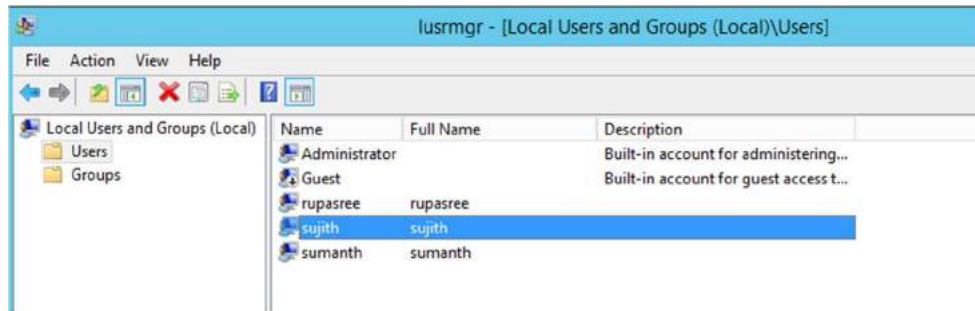
Once the group users created by Windows team, then we have to create group login in sql server.

How to create a Windows group login: (it's not dba job)

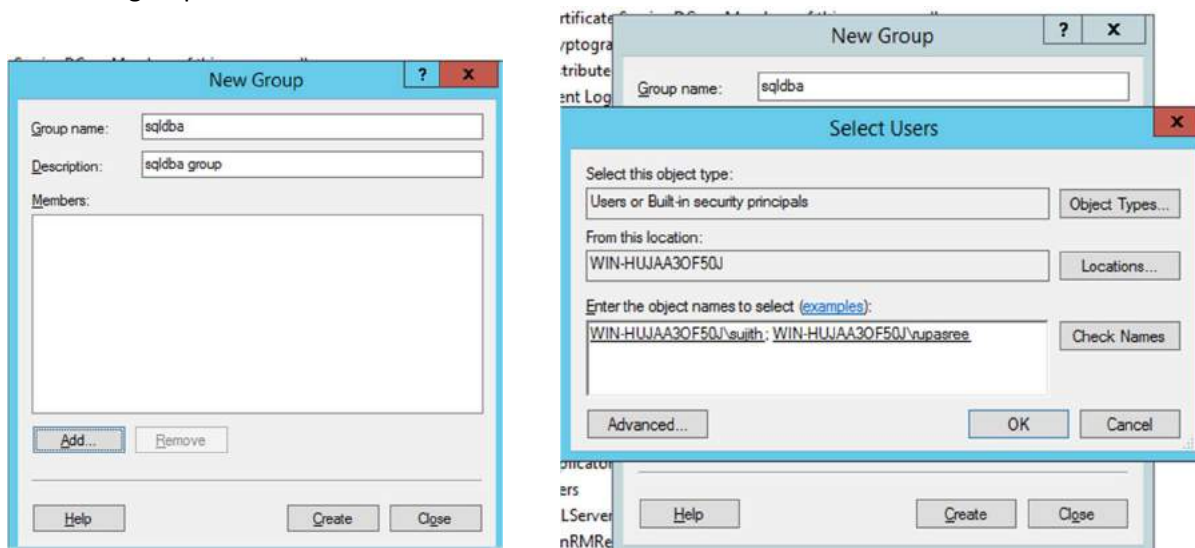
- Open run type "lusrmgr.msc" and enter
- Create the list of users, by right clicking on users option and select new.
- Give the username and password



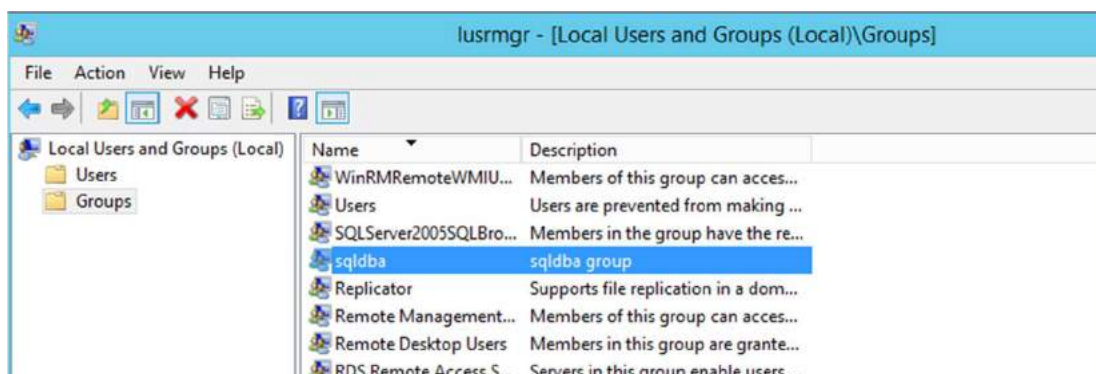
C. After created the list of users, you can see like below.



D. Right click on Groups folder and select the new group. Enter the Group name and add the users to that group. Click ok.



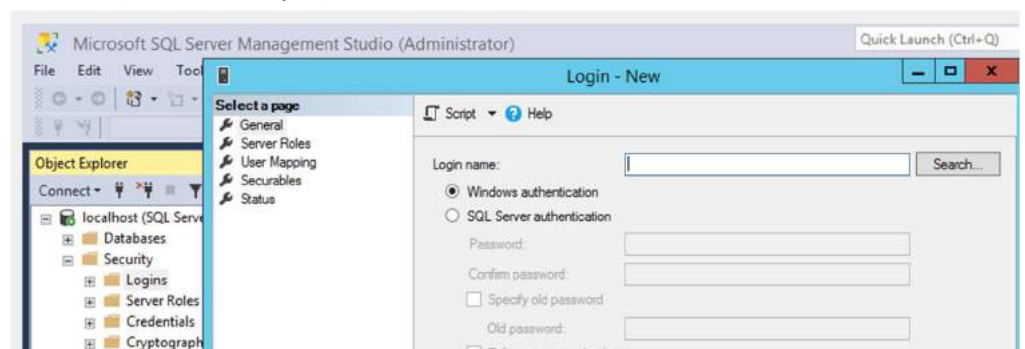
d. You can see that group name in the list, like below.



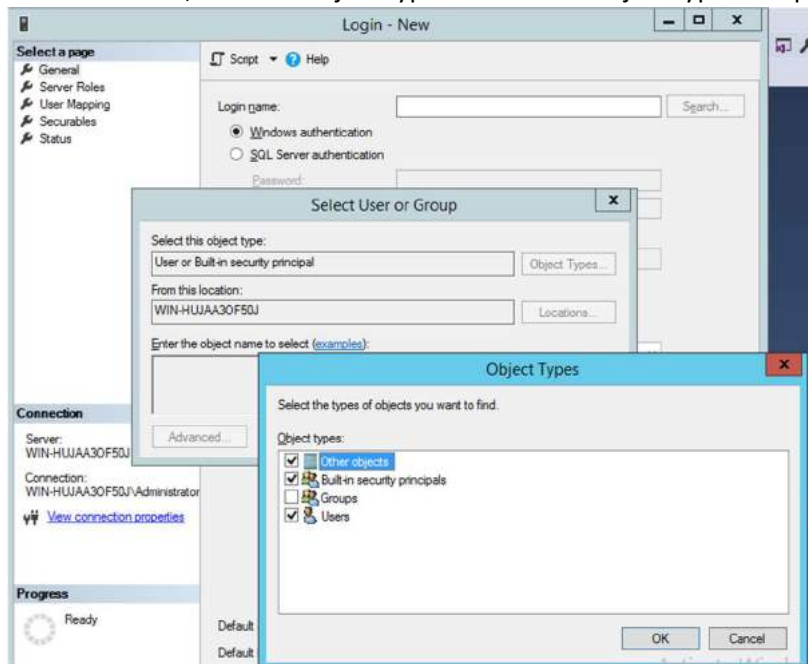
e. We can add the users to the group, if any new users joined in the same team.

Creating group login in the SQL Server: (This is a DBA job)

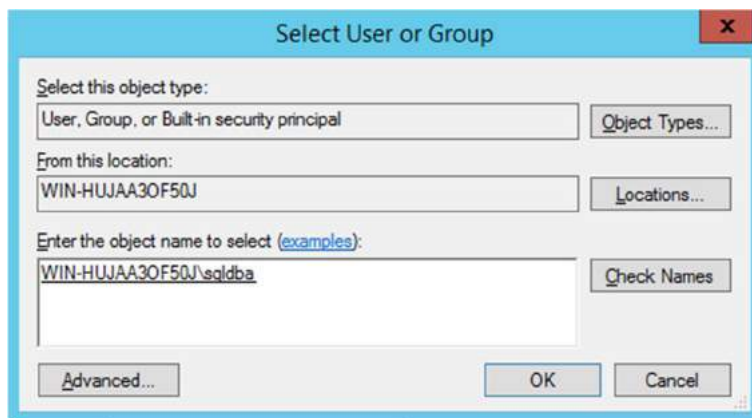
A) Open SQL Server,
Expand Security, Right
click on logins, Create
new.



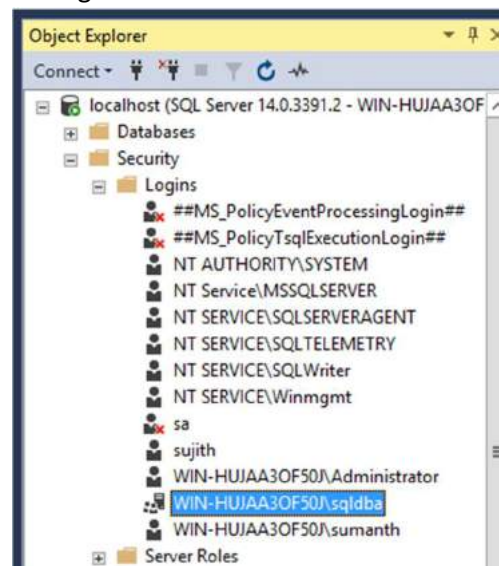
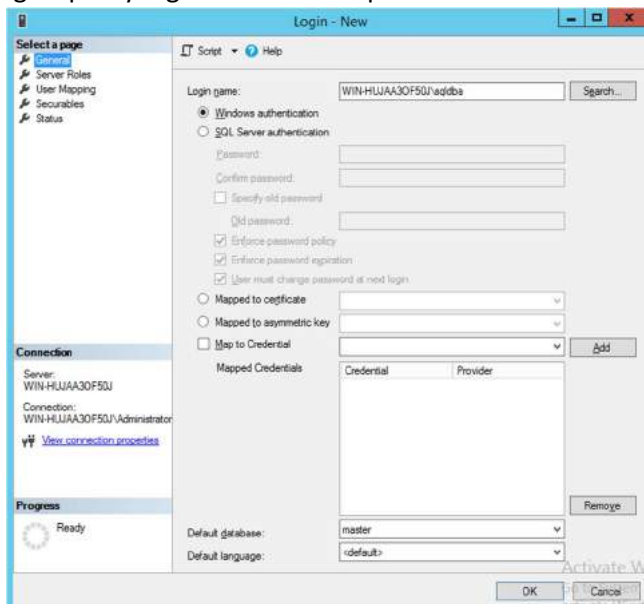
- B) Select Windows authentication mode only for Group logins.
- C) Click on search button, click the object type and tick the object type Groups like below. Click ok.



- D) Type the group data base name and click on check names. Click Ok.



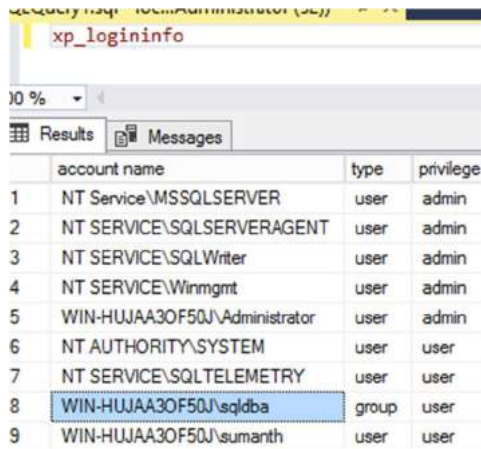
- E) Click ok. Group login created. By seeing the icons of the logins, We can findout, Which is individual login and Group login. All windows authentication logins starts with computername only, whether it's a individual or group. Any login with out computer name those are sql server logins.



F) If any new user joined in the same team, DBA no need to do anything. Windows team create the Windows user and they will add in the SQL DBA group. Now new user can access the SQL Server.

59. What is the query to see all the logins list in the SQL Server?

Ans. "xp_logininfo"



	account name	type	privilege
1	NT Service\MSSQLSERVER	user	admin
2	NT SERVICE\SQLSERVERAGENT	user	admin
3	NT SERVICE\SQLWriter	user	admin
4	NT SERVICE\Winmgmt	user	admin
5	WIN-HUJAA30F50J\Administrator	user	admin
6	NT AUTHORITY\SYSTEM	user	user
7	NT SERVICE\SQLTELEMETRY	user	user
8	WIN-HUJAA30F50J\sqldba	group	user
9	WIN-HUJAA30F50J\sumanth	user	user

60. What is the query to see the list of users in group login?

Ans. "xp_logininfo groupname , members"



	account name	type	privilege	mapped login name
1	WIN-HUJAA30F50J\rupasree	user	user	WIN-HUJAA30F50J\rupasree
2	WIN-HUJAA30F50J\sujith	user	user	WIN-HUJAA30F50J\sujith

61. What are server roles?

Ans. There are 2 types of server roles.

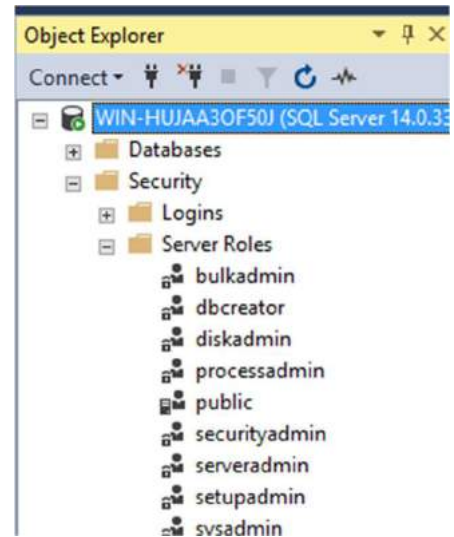
- Server – level – roles – It helps to manage the permission on server level(8+1 roles)
- Database – level – roles – it helps to manage the permissions in database level(9+1 roles)

62. What are server level roles?

Ans. Collection of permissions is called a role. Each role has a set of permissions. Based on the user requirements will allow the users to access the particular roles. These roles are fixed and we can't modify the permissions of the roles. To remember the below roles code BD2PS4

There are 8 + 1 roles in the server level, 8 roles are fixed and 1 role is not fixed. PUBLIC & SVS admin are not fixed roles.

1. Bulkadmin :
2. Dbcreator :
3. Diskadmin :
4. Processadmin :
5. Public :
6. Security admin :
7. Serveradmin :
8. Setupadmin :
9. Sysadmin :



“Add member “ permission in the all the server roles has the same kind of activity , if you have access to that server role, you can add other users also to that role.

To know permissions of each server role use the below query

“SP_SRVROLEPERMISSION”

When we want a particular server role permission, use the below query

OR

“SP_SRVROLEPERMISSION BULKADMIN”

SQLQuery1.sql - loc...Administrator (52))

sp_srvrolepermission

100 %

Results Messages

	ServerRole	Permission
1	bulkadmin	Add member to bulkadmin
2	bulkadmin	BULK INSERT
3	dbcreator	Add member to dbcreator
4	dbcreator	ALTER DATABASE
5	dbcreator	CREATE DATABASE
6	dbcreator	DROP DATABASE
7	dbcreator	Extend database
8	dbcreator	RESTORE DATABASE
9	dbcreator	RESTORE LOG
10	dbcreator	sp_renamedb
11	diskadmin	Add member to diskadmin
12	diskadmin	DISK INIT
13	diskadmin	sp_addumpdevice
14	diskadmin	sp_diskdefault
15	diskadmin	sp_dropdevice
16	processadmin	Add member to processadmin
17	processadmin	KILL
18	securityadmin	Add member to securityadmin
19	securityadmin	Grant/deny/revoke CREATE DATABASE
20	securityadmin	Read the error log
21	securityadmin	sp_addlinkedsevrlogin
22	securityadmin	sp_addlogin
23	securityadmin	sp_defaultdb

sp_srvrolepermission[bulkadmin]

100 %

Results Messages

	ServerRole	Permission
1	bulkadmin	Add member to bulkadmin
2	bulkadmin	BULK INSERT

1. Bulk admin : Bulk admin server role has 2 permissions.

- a. Add member to bulk admin:

If you are member in bulk admin, you can add another user to the bulk admin.

- b. Bulk insert

2. Data creator:

Bulk admin have 8 permissions.

A member of db creator can do all the below activities.

- a. ADD member to Dbcreator
- b. Alter database
- c. Create database
- d. Drop database
- e. Extend database
- f. Restore database
- g. Restore Log
- h. SP_renamedb

3. Diskadmin:

99% we won't this disk admin roles.

SQLQuery1.sql - loc...Administrator (52))* -> X

```
sp_srvrolepermission diskadmin
```

100 %

	ServerRole	Permission
1	diskadmin	Add member to diskadmin
2	diskadmin	DISK INIT
3	diskadmin	sp_addumpdevice
4	diskadmin	sp_diskdefault
5	diskadmin	sp_dropdevice

4. Process admin : A member of process admin can kill the sessions.

Below Q .no 53 explained about sessions and how to view sessions.

sp_srvrolepermission[bulkadmin]

100 %

	ServerRole	Permission
1	bulkadmin	Add member to bulkadmin
2	bulkadmin	BULK INSERT

SQLQuery1.sql - loc...Administrator (52))* -> X

```
sp_srvrolepermission dbcreator
```

100 %

	ServerRole	Permission
1	dbcreator	Add member to dbcreator
2	dbcreator	ALTER DATABASE
3	dbcreator	CREATE DATABASE
4	dbcreator	DROP DATABASE
5	dbcreator	Extend database
6	dbcreator	RESTORE DATABASE
7	dbcreator	RESTORE LOG
8	dbcreator	sp_renamedb

SQLQuery1.sql - loc...Administrator (52))* -> X

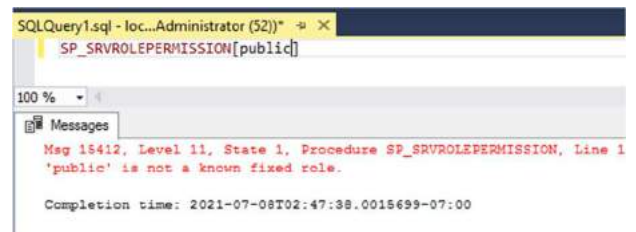
```
sp_srvrolepermission processadmin
```

100 %

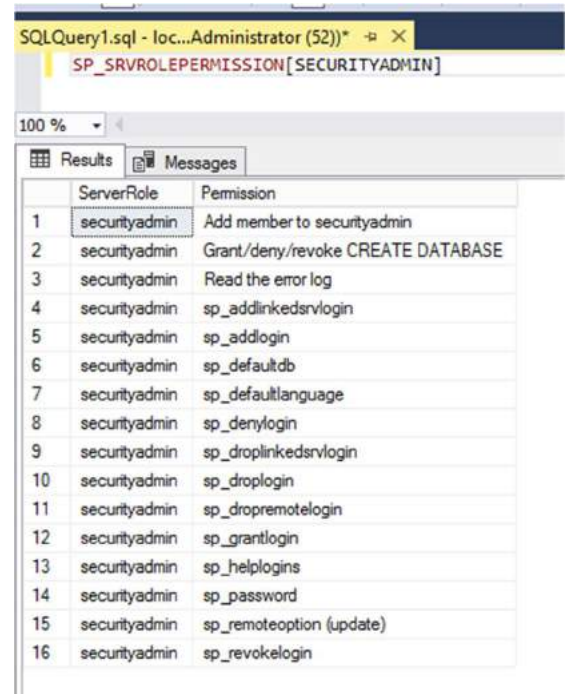
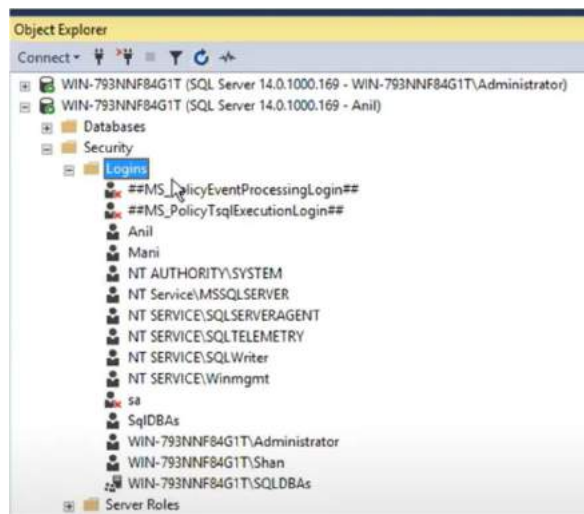
	ServerRole	Permission
1	processadmin	Add member to processadmin
2	processadmin	KILL

5. Public :

It is not a known fixed role. By default, each user got the public server role. This role doesn't have any permissions. We can't remove this role to any user. while creating the login itself, login created with default member.



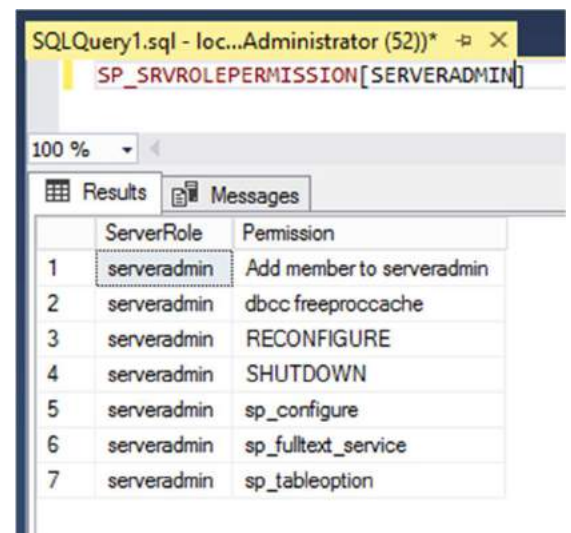
- ## 6. Security admin: A member of security admin has 16 permissions. Security admin can create logins, drop logins, changing the passwords of existing logins etc. Security admin can see all the logins in the security tab. They can also GRANT, DENY, and REVOKE database-level permissions



```
SQLQuery1.sql - loc...Administrator (52))* -> X
SP_SRVROLEPERMISSION[SECURITYADMIN]

100 %
Results
ServerRole  Permission
1 securityadmin  Add member to securityadmin
2 securityadmin  Grant/deny/revoke CREATE DATABASE
3 securityadmin  Read the error log
4 securityadmin  sp_addlinkedsvlogin
5 securityadmin  sp_addlogin
6 securityadmin  sp_defaultdb
7 securityadmin  sp_defaultlanguage
8 securityadmin  sp_denylogin
9 securityadmin  sp_droplinkedsvlogin
10 securityadmin  sp_droplogin
11 securityadmin  sp_dropremotelogin
12 securityadmin  sp_grantlogin
13 securityadmin  sp_helplogins
14 securityadmin  sp_password
15 securityadmin  sp_remoteoption (update)
16 securityadmin  sp_revokelogin
```

- ## 7. server admin: A member of server admin can shut down the SQL Server. Server admin can configure the sql server like changing the settings.

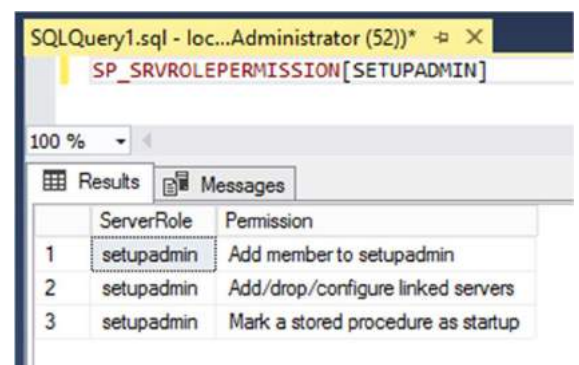


```
SQLQuery1.sql - loc...Administrator (52))* -> X
SP_SRVROLEPERMISSION[SERVERADMIN]

100 %
Results
ServerRole  Permission
1 serveradmin  Add member to serveradmin
2 serveradmin  dbcc freeproccache
3 serveradmin  RECONFIGURE
4 serveradmin  SHUTDOWN
5 serveradmin  sp_configure
6 serveradmin  sp_fulltext_service
7 serveradmin  sp_tableoption
```

8. Setupadmin:

Members of the Setupadmin can add and remove linked servers. By using the linked servers, we can create link between the other RDMS Software to the Sql Server. Like we can connect from sql Server to Oracle.



```
SQLQuery1.sql - loc...Administrator (52))* -> X
SP_SRVROLEPERMISSION[SETUPADMIN]

100 %
Results
ServerRole  Permission
1 setupadmin  Add member to setupadmin
2 setupadmin  Add/drop/configure linked servers
3 setupadmin  Mark a stored procedure as startup
```

9. Sys admin: Sys admin have all the roles. A member of sys admin can do anything. If a user is member of sysadmin, he/she got all the server role permissions. SYS admin can access any permission, even there is deny.

ServerRole	Permission
sysadmin	Add extended procedures
sysadmin	Add member to bulkadmin
sysadmin	Add member to dbcreator
sysadmin	Add member to diskadmin
sysadmin	Add member to processadmin
sysadmin	Add member to securityadmin
sysadmin	Add member to serveradmin
sysadmin	Add member to setupadmin
sysadmin	Add member to sysadmin
sysadmin	Add/drop to/from db_accessadmin
sysadmin	Add/drop to/from db_backupoperator
sysadmin	Add/drop to/from db_datareader
sysadmin	Add/drop to/from db_datawriter
sysadmin	Add/drop to/from db_dtladmin
sysadmin	Add/drop to/from db_denydatareader
sysadmin	Add/drop to/from db_denydatawriter
sysadmin	Add/drop to/from db_owner
sysadmin	Add/drop to/from db_securityadmin
sysadmin	Add/drop/configure linked servers
sysadmin	All DDL but GRANT, REVOKE, DE...
sysadmin	ALTER DATABASE
sysadmin	BACKUP DATABASE
sysadmin	BACKUP LOG
sysadmin	BULK INSERT

63. What is GRANT, WITH GRANT, DENY?

Ans. GRANT : Which is used to give permissions to a user

WITH GRANT : which is used to give permission to user and the user also can give permissions to other users.

DENY : Not giving permission to access the database. Once the deny applied to any user, he/she can't access the database till the DBA removed the DENY

Deny is the highest privileged permission. For a particular database, a user got grant and deny permission. Deny only will work. GRANT option won't work. Except sysadmin. Sysadmin can access even there is deny.

Sysadmin is the first privileged user & deny is the second privileged user in the server level roles.

Deny is highest privileged user in the data base level roles.

64. What is the query to see what are the processes running currently in the sql server?

Ans. "SP_WHO2" OR "SELECT * FROM SYS.SYSPROCESSES"

Any work done in the sql server in a session only.

It helps to find out who is using what database and to find out how many users currently using the sql server.

- 1 to 50 sessions are back ground sessions.
- From 51 session onwards, user sessions will start.

	SPID	Status	Login	HostName	BlkBy	DBName	Command
1	1	BACKGROUND	sa	.	.	NULL	XTP_CKPT_AGENT
2	2	BACKGROUND	sa	.	.	NULL	LOG WRITER
3	3	BACKGROUND	sa	.	.	NULL	RECOVERY WRITER
4	4	BACKGROUND	sa	.	.	NULL	LAZY WRITER
5	5	BACKGROUND	sa	.	.	NULL	LOCK MONITOR
6	6	BACKGROUND	sa	.	.	master	SIGNAL HANDLER
7	7	BACKGROUND	sa	.	.	NULL	XIO_RETRY_WORKER
8	8	BACKGROUND	sa	.	.	NULL	XIO_LEASE_RENEWAL
9	9	BACKGROUND	sa	.	.	master	BRKR TASK
10	10	sleeping	sa	.	.	master	TASK MANAGER

	spid	kpid	blocked	waittype	waittime	lastwaittype
1	1	3752	0	0x0393	24368774	WAIT_XTP_HOST_WAIT
2	2	1132	0	0x0000	0	LOGMGR_QUEUE
3	3	3384	0	0x033C	196	DIRTY_PAGE_POLL
4	4	2844	0	0x0060	20	LAZYWRITER_SLEEP
5	5	4052	0	0x0083	4301	REQUEST_FOR_DEADLOCK_SEARCH
6	6	4032	0	0x00A0	24368164	KSOURCE_WAKEUP
7	7	828	0	0x006C	588	SLEEP_TASK
8	8	3496	0	0x006C	588	SLEEP_TASK
9	9	1260	0	0x00AC	24368214	BROKER_TRANSMITTER
10	10	0	0	0x0000	0	MISCELLANEOUS

We can't kill any of the back ground sessions. We can kill only user sessions. To kill the session, user should be member of Process admin. This killing sessions processes used for Performance tuning. In the organizations, user asks the DBA to kill the session , where session is taking longer than expected time.

Maximum sessions will be opened 32767.

Remember 1 to 50 are background processes, we can't able to kill.

TO kill the session query is " KILL sessionid or SPID"

Before killing any session, check with the user and then kill it.

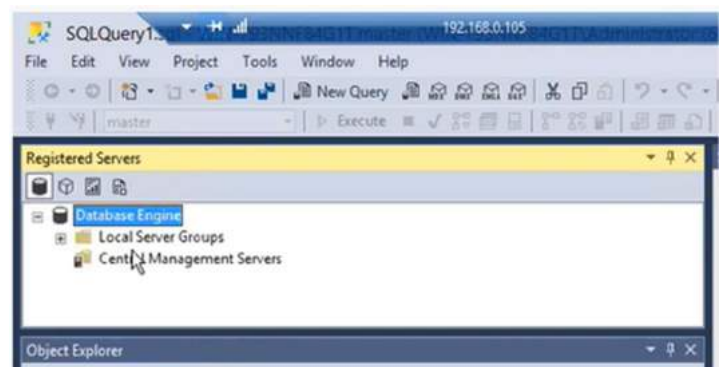
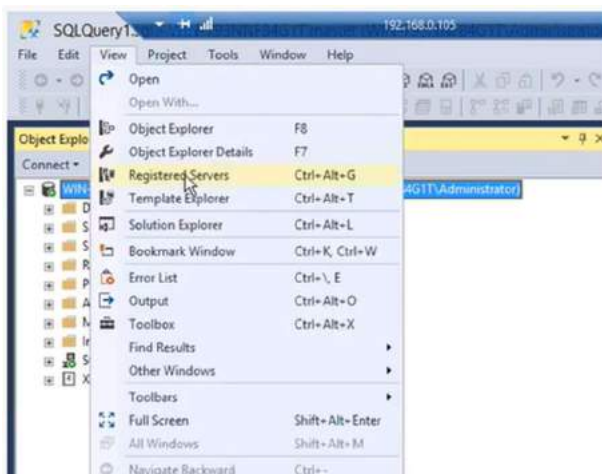
```
kill 56  
kill 42
```

65. How to run a query in multiple instances or multiple sql servers at a time?

Ans. By using the registered servers, we can achieve this.

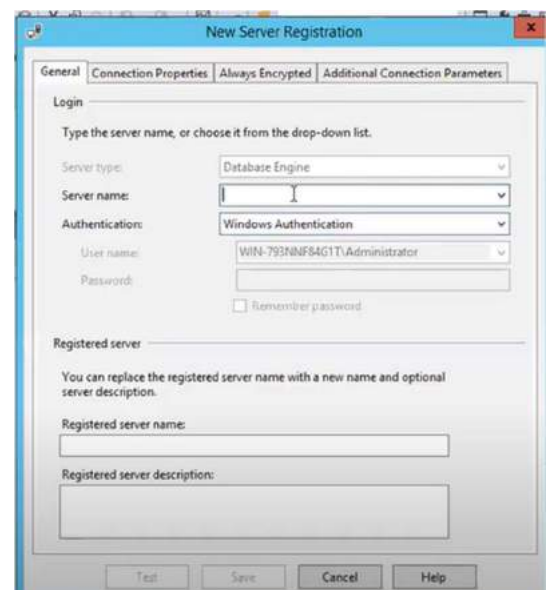
Below are steps,

A) OPEN SSMS, Go to VIEW, Select registered services, Now registered services window appear.



B) Right click on local server group, Select new server group or new server registration,

Give the SQL SERVER INSTANCE NAME IN THE SERVER NAME COLUMN, Click on Test. If the instance already created with the name. Then server will be added in the registered services.



66. What are customized server roles?

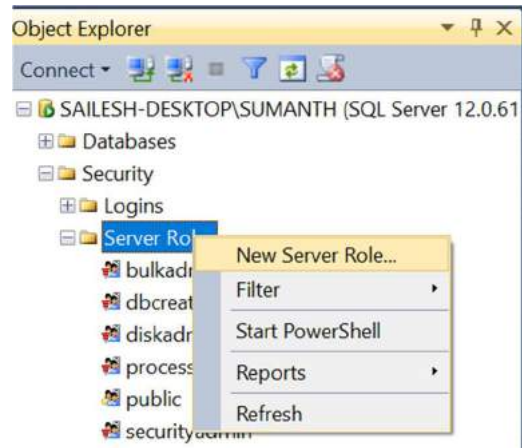
Ans. Creating our own server roles are called customized server roles. Collecting the Combination of permissions as per our requirement from the different fixed roles are assigning in to customized server roles. If we any login to

that customized server role, he/she are allowed to access the permissions of customized server roles. We can add or alter or delete the customized server roles, but fixed roles we can't. Customized server roles are introduced from SQL Server 2012 onwards.

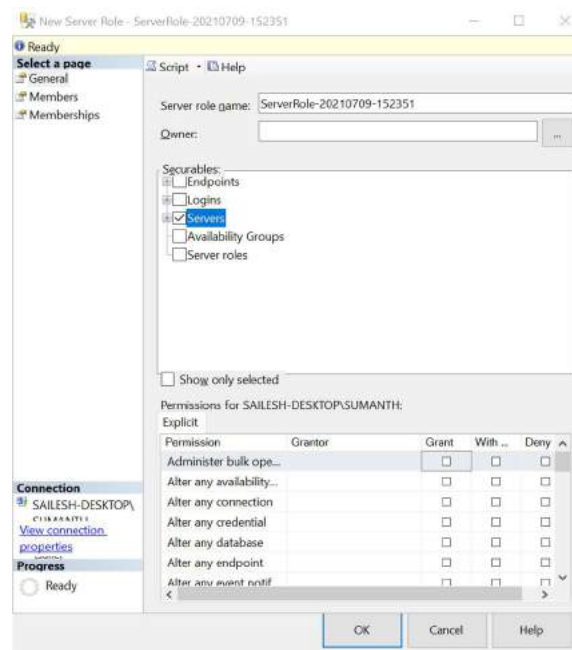
Below are the steps to create customized server roles,

Mostly DBA won't create Customized server roles. Developers write a query they ask us to run.

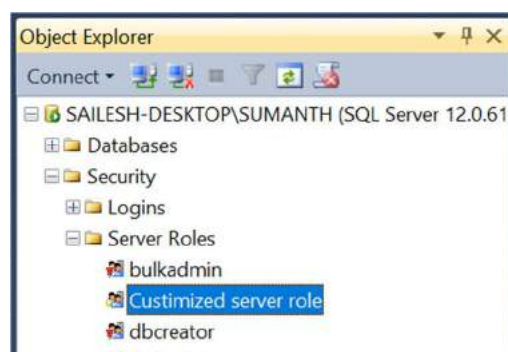
- a. Open SSMS, Expand security, Right click on server roles, select new role,



- b. Give the server role name, owner is sa, click on server and select the required permission, add the logins, who want to access, Click ok.



- c. Customized role created.



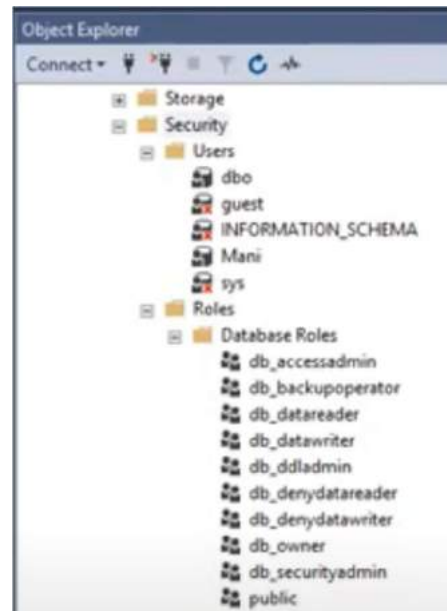
67. What are database level roles?

Ans. The database-level roles are a group of security principals that are used to manage the permissions within the databases more efficiently. There are two types of database-level roles. One is a predefined database-level role, and another is custom database-level roles. The custom database-level roles can be created by the database administrator or the member of the **db_owner** role. To see the permissions of database level roles

Use the query “ `sp_dbfixedrolepermission` ”

The SQL Server database has 9+1 fixed database roles. Here also public is not a fixed role and public is the default database level role for all the users.

- a. db_owner
- b. db_securityadmin
- c. db_accessadmin
- d. db_backupoperator
- e. db_ddladmin
- f. db_datareader
- g. db_datawriter
- h. db_denydatareader
- i. db_denydatawriter
- j. Public



- a. db_owner:

A member of db_owner can do anything. If deny applied to the db_owner, deny only work. A member of Db_owner can add a user to the other database level roles.

1 `sp_dbfixedrolepermission db_owner`

100 %

Results Messages

	DbFixedRole	Permission
1	db_owner	Add/drop to/from db_accessadmin
2	db_owner	Add/drop to/from db_backupoperator
3	db_owner	Add/drop to/from db_datareader
4	db_owner	Add/drop to/from db_datawriter
5	db_owner	Add/drop to/from db_ddladmin
6	db_owner	Add/drop to/from db_denydatareader
7	db_owner	Add/drop to/from db_denydatawriter
8	db_owner	Add/drop to/from db_owner
9	db_owner	Add/drop to/from db_securityadmin
10	db_owner	All DDL but GRANT, REVOKE, DENY

- b. db_securityadmin:

1 `sp_dbfixedrolepermission db_securityadmin`

100 %

Results Messages

	DbFixedRole	Permission
1	db_securityadmin	DENY
2	db_securityadmin	GRANT
3	db_securityadmin	REVOKE
4	db_securityadmin	sp_addapprole
5	db_securityadmin	sp_addrole
6	db_securityadmin	sp_addrolemember
7	db_securityadmin	sp_approlepassword
8	db_securityadmin	sp_changeobjectowner
9	db_securityadmin	sp_dropapprole
10	db_securityadmin	sp_droprole
11	db_securityadmin	sp_droprolemember

- c. db_accessadmin: Members of the **db_accessadmin** fixed database role can add or remove access to the database for Windows logins, Windows groups, and SQL Server logins.

The screenshot shows the 'sp_dbfixedrolepermission db_accessadmin' command executed in SQL Server Enterprise Manager. The 'Results' pane displays a table with the following data:

	DbFixedRole	Permission
1	db_accessadmin	sp_dropuser
2	db_accessadmin	sp_grantdbaccess
3	db_accessadmin	sp_revokedbaccess

- d. db_backupoperator: A member of backup operator can run the check point on a particular database.

The screenshot shows the 'sp_dbfixedrolepermission [db_backupoperator]' command executed in SQL Server Enterprise Manager. The 'Results' pane displays a table with the following data:

	DbFixedRole	Permission
1	db_backupoperator	BACKUP DATABASE
2	db_backupoperator	BACKUP LOG
3	db_backupoperator	CHECKPOINT

- e. db_ddladmin: A member of ddladmin can truncate ,rename a table.

The screenshot shows the 'sp_dbfixedrolepermission db_ddladmin' command executed in SQL Server Enterprise Manager. The 'Results' pane displays a table with the following data:

	DbFixedRole	Permission
1	db_ddladmin	All DDL but GRANT, REVOKE, DENY
2	db_ddladmin	dbcc cleantable
3	db_ddladmin	dbcc show_statistics
4	db_ddladmin	dbcc showcontig
5	db_ddladmin	REFERENCES permission on any table
6	db_ddladmin	sp_changeobjectowner
7	db_ddladmin	sp_fulltext_column
8	db_ddladmin	sp_fulltext_table
9	db_ddladmin	sp_recompile
10	db_ddladmin	sp_rename
11	db_ddladmin	sp_tableoption
12	db_ddladmin	TRUNCATE TABLE

- f. db_datareader: This role is very important ,why because regular we are going to give these permissions to the users. A member of data reader can read the data. Only select permissions are available in the data reader roles.

The screenshot shows the 'sp_dbfixedrolepermission [db_datareader]' command executed in SQL Server Enterprise Manager. The 'Results' pane displays a table with the following data:

	DbFixedRole	Permission
1	db_datareader	SELECT permission on any object

- g. db_datawriter: This role is very important ,why because regular we are going to give these permissions to the users. A member of data writer can have delete, insert & update permissions on a particular database.

The screenshot shows the 'sp_dbfixedrolepermission db_datawriter' command executed in SQL Server Enterprise Manager. The 'Results' pane displays a table with the following data:

	DbFixedRole	Permission
1	db_datawriter	DELETE permission on any object
2	db_datawriter	INSERT permission on any object
3	db_datawriter	UPDATE permission on any object

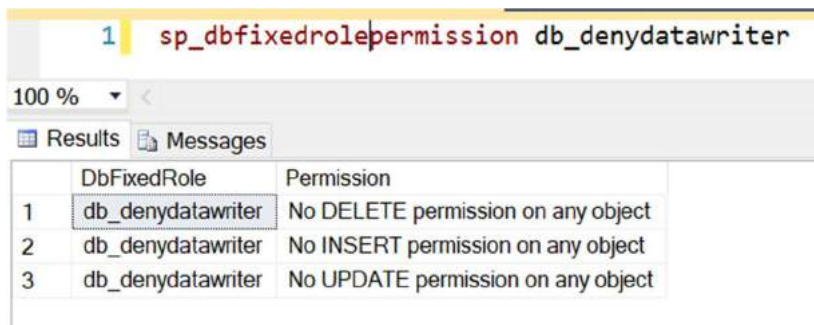
h. db_denydatareader:

A member of deny_datareader, user cant read the data of a particular database.



	DbFixedRole	Permission
1	db_denydatareader	No SELECT permission on any object

i. db_denydatawriter: A member of deny_datawriter, user cant insert, delete, update the data of a particular database.



	DbFixedRole	Permission
1	db_denydatawriter	No DELETE permission on any object
2	db_denydatawriter	No INSERT permission on any object
3	db_denydatawriter	No UPDATE permission on any object

j. Public: This is not fixed database level role

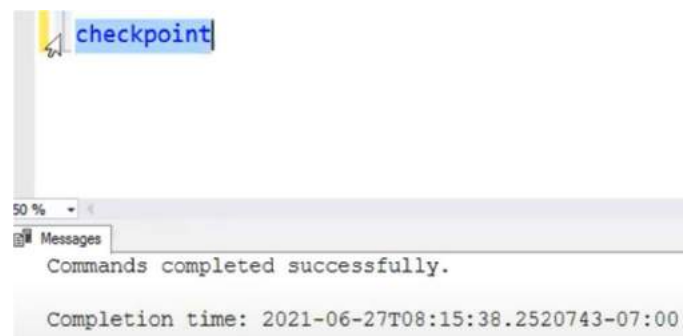
68. What are check points?

Ans. Check point helps to copy all the committed transactions from log file to datafile.

No one runs the check point. It's an automated process.

There are two types of check points,

- Automated checkpoint: Automatically every 60 seconds, committed transactions transferred from log file to Data file
- Manual Check point: it helps when you want to copy the committed transactions of log file to data file manually. NO NEED TO DO THIS MANUAL.
Query for manual check point is "CHECKPOINT"

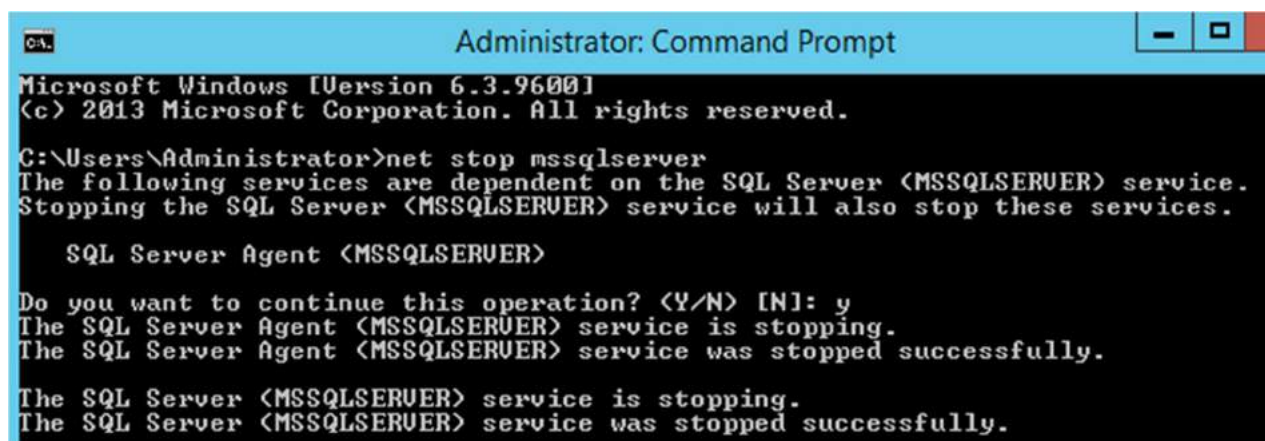


69. How to start or stop the sql server in Command prompt?

Ans. For Default Instance : Search command prompt, right click and select run as administrator

Enter the command to stop the sql server " net stop default instance name"

Enter the command to start the sql server “ net start default instance name”



```
Administrator: Command Prompt
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>net stop mssqlserver
The following services are dependent on the SQL Server (MSSQLSERVER) service.
Stopping the SQL Server (MSSQLSERVER) service will also stop these services.

    SQL Server Agent (MSSQLSERVER)

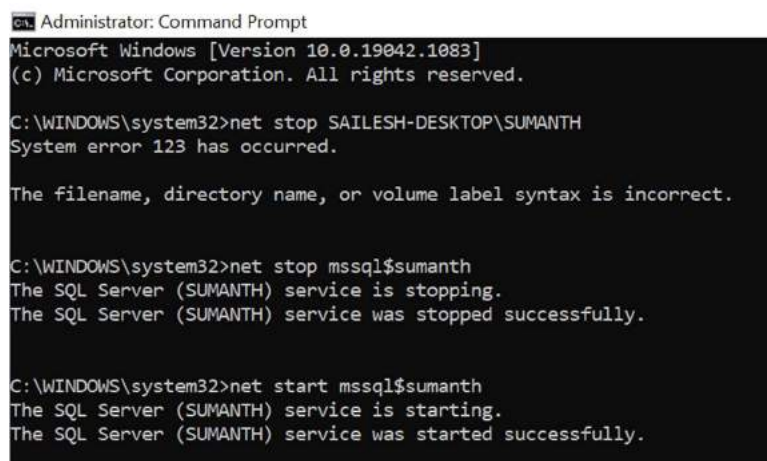
Do you want to continue this operation? (Y/N) [N]: y
The SQL Server Agent (MSSQLSERVER) service is stopping.
The SQL Server Agent (MSSQLSERVER) service was stopped successfully.

The SQL Server (MSSQLSERVER) service is stopping.
The SQL Server (MSSQLSERVER) service was stopped successfully.
```

For named instance : Search command prompt, right click and select run as administrator

Enter the command to stop the sql server “ net stop mssql\$named instance name”

Enter the command to start the sql server “net start mssql\$named instance name”



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19042.1083]
(c) Microsoft Corporation. All rights reserved.

C:\WINDOWS\system32>net stop SAILESH-DESKTOP\SUMANTH
System error 123 has occurred.

The filename, directory name, or volume label syntax is incorrect.

C:\WINDOWS\system32>net stop mssql$sumanth
The SQL Server (SUMANTH) service is stopping.
The SQL Server (SUMANTH) service was stopped successfully.

C:\WINDOWS\system32>net start mssql$sumanth
The SQL Server (SUMANTH) service is starting.
The SQL Server (SUMANTH) service was started successfully.
```

70. What is orphan user & how to find orphan users?

Ans. user doesn't have login, that user is a orphan.

When we delete any login with out removing the users of the database level, at that time users are orphans.

One more case, when we restore the database from production to non-production or from one server to another server, users of that database are also orphans in the nonproduction or restored database server.

To track the orphans in a database, use the bellow query, Select the database on which the below stored procedure you want to run

“sp_change_users_login report”

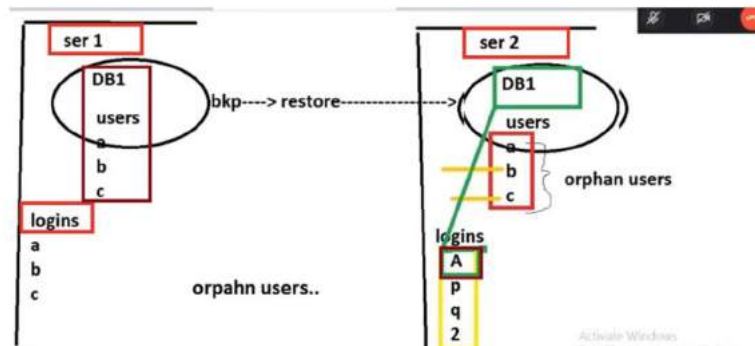


71. How to create login for orphan users?

Ans. by using usersid, we can recreate the login. To get the usersid (user security identification number) run the query, “sp_change_users_login report”.

ORPAHN USERS..

A user without any login is called as ORPHAN user. it will come when we move db from one server to another server.



We have four methods to recreate the login by using existed orphan user

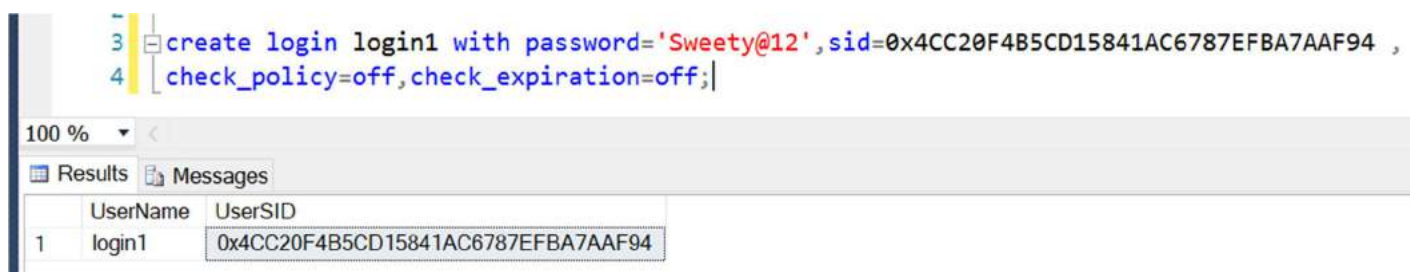
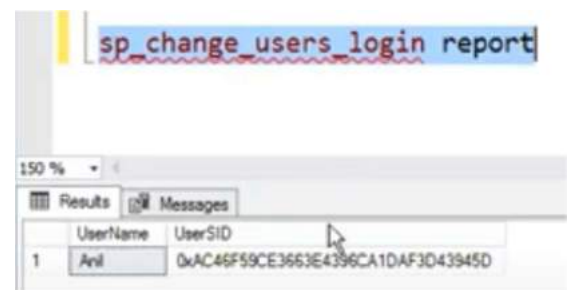
Method :1

Copy the user usersid, and create the new login by using this sid.

“Create login Anil with password=N’Sweety@12’ ,

Sid=paste the usersid , check_policy=off ,

check_expiration=off”



Method 2: If you have a login and user map both login and user by doing AUTOFIX METHOD.

EXEC SP_CHANGE_USERS_LOGIN ‘AUTOFIX’ , ‘USER NAME’

Method 3 : if you want to create a new login and password for the user ,

EXEC SP_CHANGE_USERS_LOGIN ‘AUTOFIX’ , ‘LOGIN NAME’ , NULL , ‘PASSWORD’

AGNET ROLES IN SQL SERVER

71. How to give SQL Server agent permissions to a user?

Ans. By selecting the MSDB database and choosing the SQL Server agent roles.

ID	Role	Description
1	SQLAgentUserRole	Ability to manage Jobs that they own
2	SQLAgentReaderRole	- All of the SQLAgentUserRole rights - The ability to review multiserver jobs, their configurations and history
3	SQLAgentOperatorRole	- All of the SQLAgentReaderRole rights - The ability to review operators, proxies and alerts - Execute, stop or start all local jobs - Delete the job history for any local job - Enable or disable all local jobs and schedules

72. What is Locks?

ZAns. A Lock will be created when any transaction performed on the table .

Below are the types of Locks.

Shared Locks (S): These locks acquired by readers during read operations such as "SELECT". Who ever writing the select transaction on the table shared locks will be created.

EX: select * from emp where id=121

Exclusive Lock(X) : These locks acquired by writers during data modification operators such as Insert , Update & Delete. These Locks prevent one object to be modified by the other sessions. These locks are always acquired and held till end of transaction. Exclusive access means complete access. Till the end of the modify transaction, others cant access the table.

Update Locks(U) : These locks are the mixture of shared and exclusive locks. when you run a modification transaction, Till the searching of the columns sql server keeps shared lock . once column found, it will convert shared lock to exclusive lock.

Ex : Update Mytable set column1=0 where column1 is null.

Now SQL Server put the shared lock on the table till searching for column1 is null. When eligible row found, SQL Serve converts S to X.

Intent Lock(IS,IX,IU,ETC) : Intent Lock not a table level lock . It's a range level lock. It wont keep the lock for the entire table. It will keep the lock only to the particular range of the table. Remaining range can be accessed by the other users.

Shared , Exclusive , update are table level locks. Where Intent lock is range level lock.

Ex : update emp set sal=sal+1000 where id=121.

Now SQL Server keep the lock for particular range i.e; is 1-1000 rows lock.

Remaining columns can be accessed by other users.

72. What is the difference between LOCK Vs Block Vs Dead Lock?

Ans. Lock : DB Engine locks the rows/page/table to access the data when user runs a query.

Blocking : When once person blocks the resources of another process then blocking happens. Below is the example, User 1 executing the query on emp and still its running. Now user2 is trying to access the same table emp. Now user1 query will block the user 2 query.

When ever blockings created we have to remove the blockings manually.

First we have to find blockings by using below query,

SP_WHO or SP_WHO2 or

Select * from sys.dm_exec_requests where blocking_sessions <> 0

Or

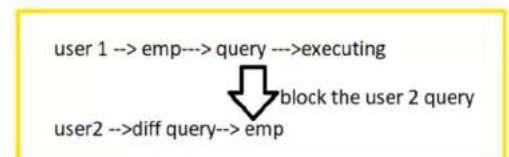
Select * from master..sysprocesses where blocked <> 0

Now we have to kill the blocked the session by using below query,

KILL Spid;(Server process identifier)

1 to 50 spids are server spids. We cant kill.

>50 spids are user spids. we can kill.



if blocking occurs , we have to manually identify blocking and release the blocking.

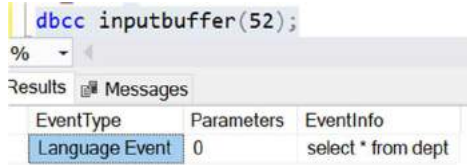
```
SELECT * FROM sys.dm_exec_requests where
blocking_session_id <> 0 [or]
SELECT * FROM master..sysprocesses where blocked <>
0 [or] sp_who [or] sp_who2
```

SP_WHO2				
% ▾				
Results Messages				
SPID	Status	Login	HostName	BlkBy
52	SUSPENDED	SAI...	SALES...	70

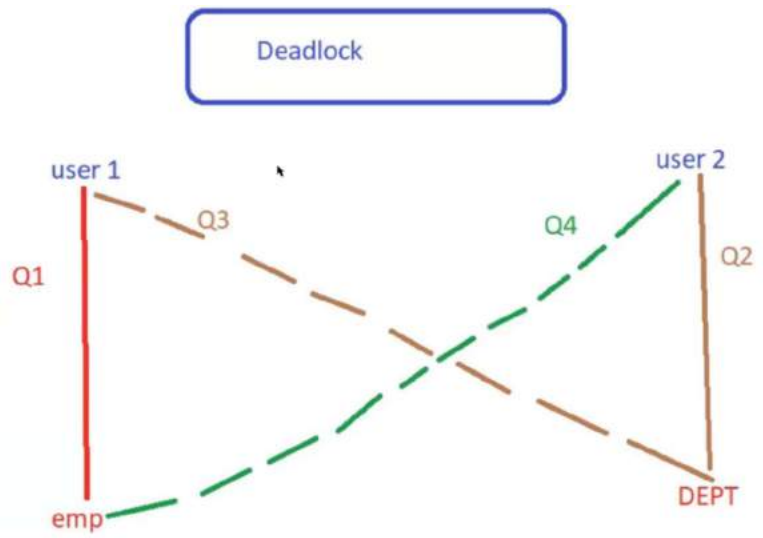
Always check which one is good to kill and then take the user approval. Then kill the session.

To see what is the query running on the sessions, we use DBCC,

DBCC inputbuffer(sessionid);



Dead Lock : If 2 users having lock on 2 different tables , each user trying to access the other user table, Then the dead lock will occur. In this , USER will not release the lock. When this happens, the sql server ends the dead lock automatically choose the one query to run queries continuously and other user abort the transaction. SQL server abort the user transactions, which is having least amount of resources to roll back the transaction.



Dead Lock Victim : The query which cancelled by sql server because of deadlock. That user is a dead lock victim.

this happens, the SQL Server ends the deadlock by automatically BY CANCELLING ONCE TRANSACTION AND all the other process to continue. Generally, the transaction that requires the least amount of overhead to rollback i transaction that is aborted.

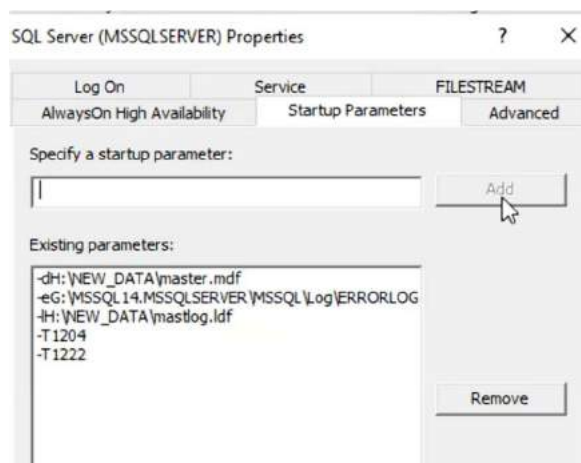
If you want to capture deadlock information in the SQL Server Error Log you need to enable below trace flags IN STARTUP PARAMETERS.

-T1204 - this provides information about the nodes involved in the deadlock

-T1222 - returns deadlock information in an XML format

Capturing dead locks : To capture the dead lock information in sql server error log, We need add two trace flags in the start-up parameters.

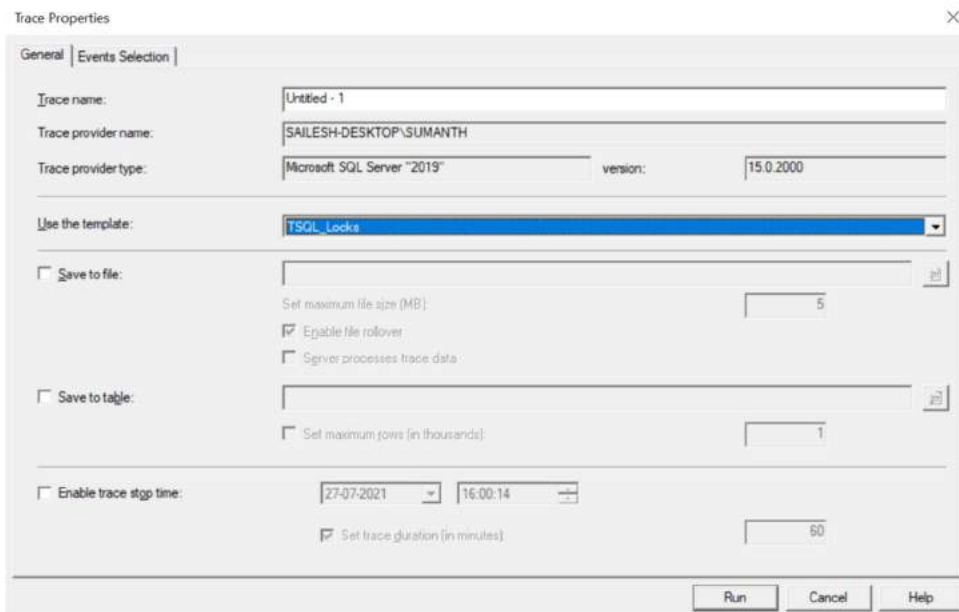
Those are -T-1204 & -T1222



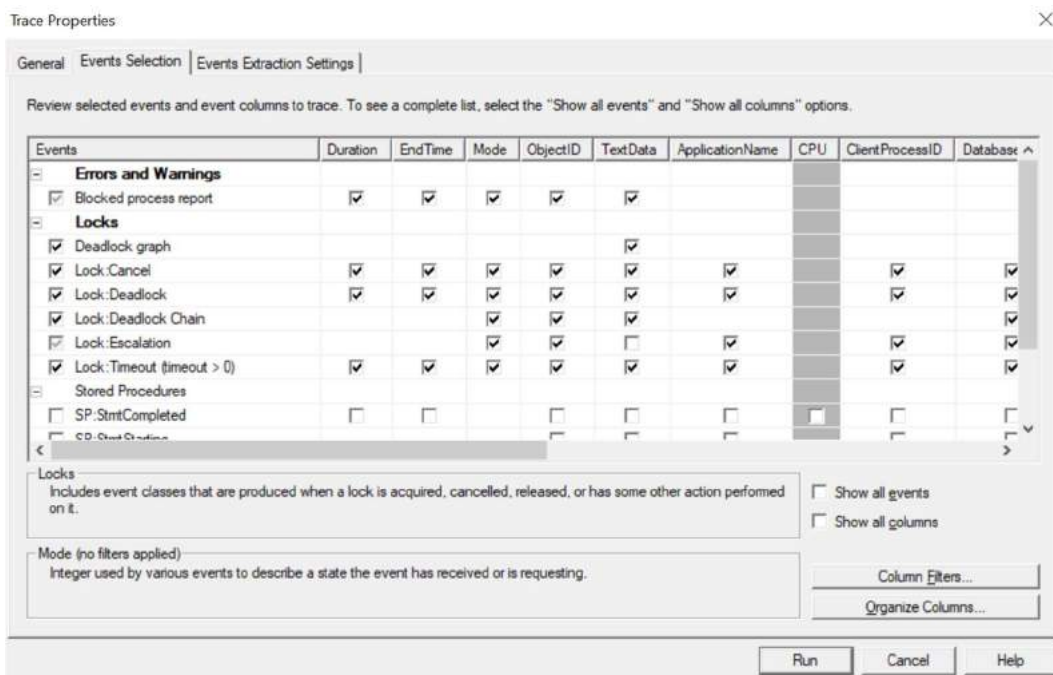
We can capture the dead lock errors by using profiler also.

Profiler : SQL Server Profiler is a tracing tool provided by Microsoft since SQL Server 2000 version. It is used to trace activities and operations executed on a specific SQL Server database engine or Analysis Service to be analysed later.

Open SSMS, Go to Tools, Select Profiler, Login to the instance, enter the name select the template type "TSQL_LOCKS", You can save to table or file, You can enter the end time to stop tracing the locks.



Now go to Events selection tab, Tick the errors and warnings, Locks. Click Run. Profiler start to trace the locks.



73. What is the query to see the list of logins?

Ans. select * from sys.sysusers

74. What are the different types of backups?

Ans.

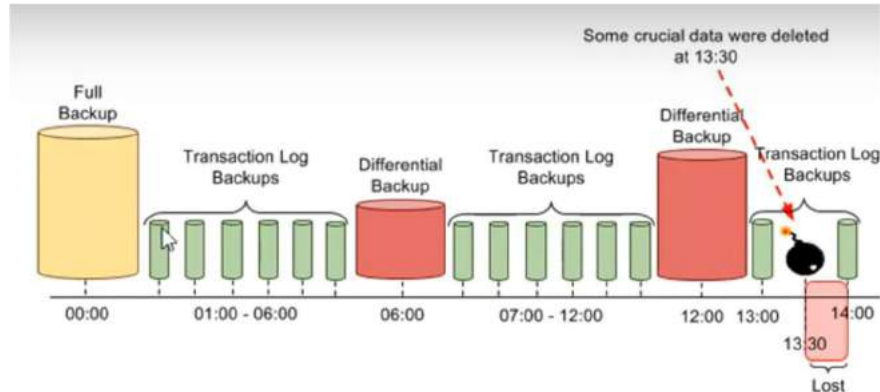
- a. Full backup : Full back up can take entire database backup that includes data file & log file. Once full backup completed, It resets the DCM Pages.

- b. Differential backup : Will take the backup of all the extents that has changed since last full backup, which is having datafile and log file. With the DCM(Differential change map) page only , differential back up can identifies the what are all the changed extents.

Interview question : how differential back up knows what are the extents changed since last backup?

Ans. DCM Page helps to track the extents, that are changed since last backup.

- c. Log Backup : It takes backup of the log file or transactions including committed and uncommitted transactions. First log backup starts from last full backup and second log backup onwards it starts from last log backup. It's a kind of chain system. Each log backup has a unique LSN number(Log sequence number) to restore the log backups. If we miss any of the Log file, LSN number mismatched will occur. Then we cant restore log backup



- d. Copy_only backup : While we taking backup, if we choose copy_only_backup, DCM Pages

wont be reset. When you are taking the any backups with out automation, always choose the copy only backup. Why because, if someone took the manual full backup without copy only option , the automated differential will start from the manual full backup onwards. It breaks the LSN, it leads to Chain break and restoration failed. So, When you are taking the manual backups, always choose the copy only backup.

- e. Tail log backup : When ever database is corrupted immediately, we have to take Tail log backup. The first action we have to do after the database corrupt, take the tail log up. We can take tail log up only with query,

“Backup log database name to disk='path.trn' with no_truncate “

- f. File or File group backup
- g. Stripped backup/Split backup : When we don't have free space in single drive, to split the backup in to multiple drives with the help of split backup. One backup can be taken to different locations. If any one of the split back is missed, we cant restore the database.
- h. Mirror backup: it creates the one more copy of backup for safety. If the original backup file is missed, we can use the this mirror backup copy.
Mirrored backups cannot reside with other nonmirrored backup sets, hence with format option is mandatory to mention.

Syn : backup database DBname TO disk='path.bak' mirror TO disk='path.bak' with format

- i. Partial backup : partial backups are useful whenever we want to exclude read only file groups . A partial does not contain all the file groups.

Syn : BACKUP DATABASE DBname READ_WRITE_FILEGROUPS,FG1,FG2 TO DISK='path.bak'

- j. Partial Differential backup: Is a differential backup for partial backup.

```
backup database DBName READ_WRITE_FILEGROUPS,READONLY1,READONLY2 to
disk=N'F:\DBName_PBKP.bak' WITH DIFFERENTIAL
```

- k. Cold backups : if backup is being taken for once specific database then

```
ALTER DATABASE DBName SET OFFLINE
WITH ROLLBACK IMMEDIATE
```

Copy MDF & LDF files of this database in Data Directory. This files backup taken can be attached to another database if required to restore.

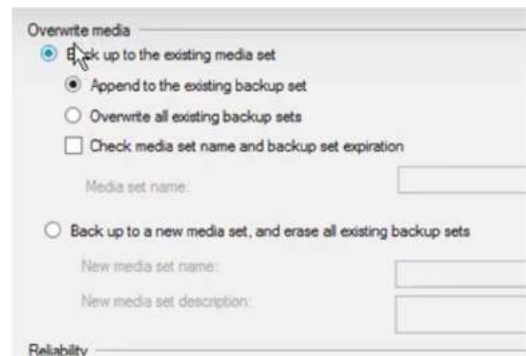
75. What are the media options in the backup?

Ans. There are two types,

- Backup set : A single backup is called Backup set.
- Media set : Collection of backup sets in to a single backup file is called Media Set.

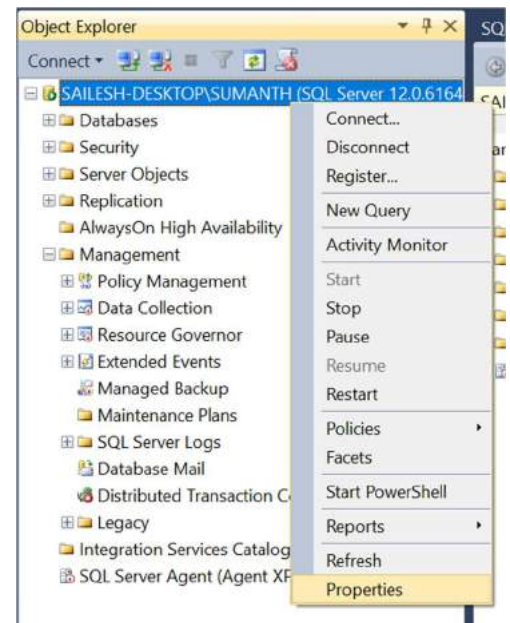
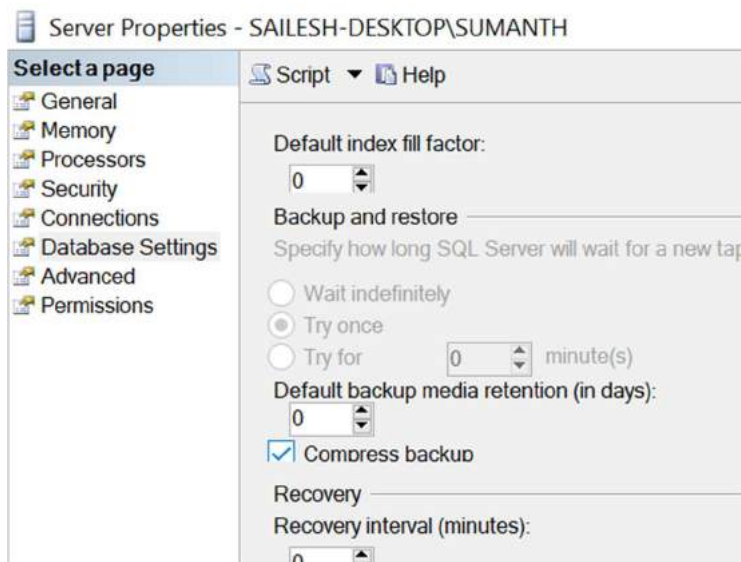
When we are first time taking backup, that backup file is backup set.

If you are taking second time backup the same database with the same name in the same location and selected the option append , now the backup file contains 2 backups. Now the same backup file called as media set. While restoring it will select the recent full backup from the backup file only.

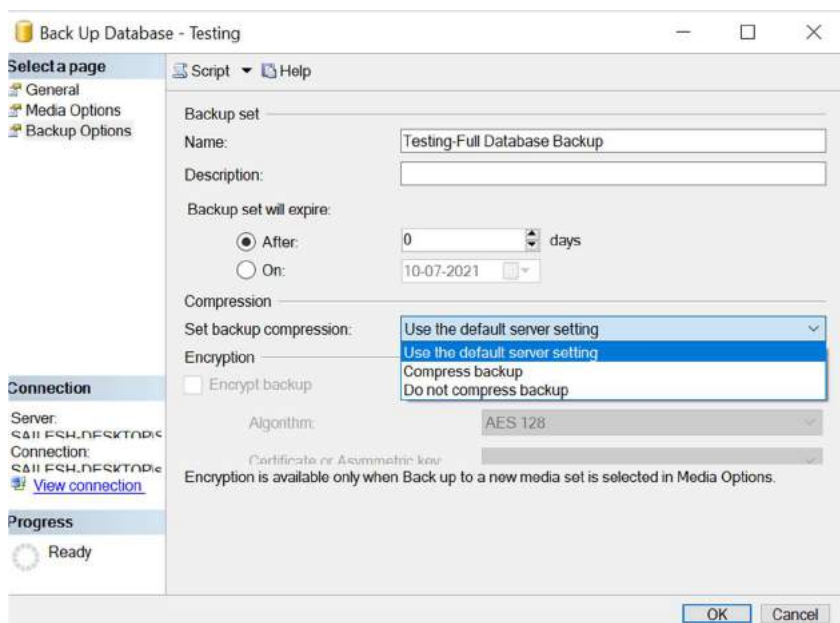


76. How to set default server setting for compression of backup?

Ans. Right click on server, go to properties, go to database settings, Tick on compress backup. Click ok.



Now all the database back up default setting will be compressed. With out changing the set backup compression type to compress backup, it will take the default setting as compressed.



Note : While running the backup, the transactions between the backup also part of full backup.

Ex : If a backup is started at 8 PM and going to complete at 10 PM. The transactions in between the 8 to 10 pm also now part the backup.

77. What is recovery ,types of recovery,?

Ans. Recovery means how the database is recovering.

There are 2 types of recovery,

- a. **Restart recovery :** Every time an instance is restarted/started the consistency of all the databases including master, model, msdb and TempDB is checked. This process is an internal operation and initiated just to keep the entire instance clean and with integrity.
Taking database offline/online involves restart recovery for the database.
Simply database engine checks the all the databases, whenever restarted the server or keeping a database from offline to online. That is called restart recovery.
- b. **Restore recovery :** As per backup strategy whenever a restore is started and recovery is done per backup sequence. This entire process of recovery initiated manually is called as restore recovery.
Simply Whenever we restored a database, Database engine checks the integrity of restored database.

78. What are the types of recovery models?

Ans. Recovery models are designed to control transaction log maintenance . recovery model control the behaviour of the log file. Using the recovery models , we can the behaviour of the log file.

There are 3 types.

- a. **Full recovery model:** When a database in full recovery model, it tracks all the 108 attributes of the log file completely. So, We can take the log backups and we can achieve point in time recovery.
When a database in full recovery model, we can take full backup , differential , log backups & tail log backups.
- b. **Bulk logged recovery model:** it acts as a full recovery model, when there is no bulk inserts or bulk transactions in the log file. If bulk operations are there, then all bulk transactions are minimally logged. In case If you know bulk operations are going to happen on a database, keep the recovery model as Bulk logged. If there is no bulk operations, its possible to restore point in time and if there is a bulk operations, not possible to restore point in time in bulk logged recovery model.
When a database in bulk logged model, we can take full backup , differential , log backups & tail log backups.

Interview question : how sql server know, which are bulk operations?

Ans. BCM(Bulk changed map) tracks all the extents, those changed by bulk operations.

- c. **Simple recovery model:** With the help of simple recovery model, we can able to take only full & differential backup. Log backup is not possible. Every 60 seconds log file will be deleted in the simple recovery model by the help of check point in regular intervals.

In the organizations, mostly use full recovery model in the database.

When ever we change the recovery models of the database, we have to take full backup of that database after changing the recovery model. Its must and should.

79. What is dirty page?

Ans. The changed or modified pages in the memory is called dirty page.

68. What is pseudo simple recovery model?

Ans. After changing the recover model, if we don't take the full database back up – the database behave like old recovery model. That behaviour of database is "Pseudo simple recovery model"

80. What are the scenarios , where restore the databases?

Ans.

- When data base is corrupted (Very rarely happens)
- Data base refresh tasks.(Developers want to test their logic in the non – production environment, at that time , DBA restore the production databases to non-production environment . Simply restoring databases from prod to non-prod environment)

81. What are the recovery states?

Ans.

- Restore with recovery : In case you have selected all the backup files to restore and there is no more backup files is pending, then you can choose this recovery option. For the last or final back up , we have to select this recovery option. If you choose this option, after completion of restore, database comes to online and we cant restore further backups.
- Restore with no recovery : till last backup to restore, we have to select this option. In this recovery state, database wont comes to online. Once the last back up selected and changed the state to with recovery, data base comes to online after restore.
- Restore with standby : if we restore the database with stand by, It only allowed to read operations on a database.

82. How to get the data base online , if you have restored a database with no recovery?

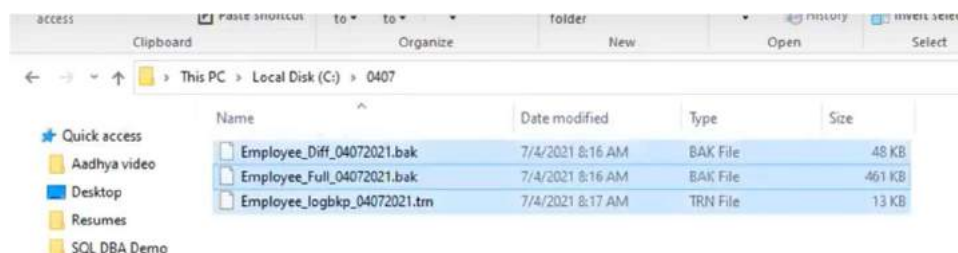
Ans. use the query " restore database db name with recovery"

Note : We can restore a database from lower sql version to higher sql version and same version to same version. But, we cant restore a database from higher version to lower version.

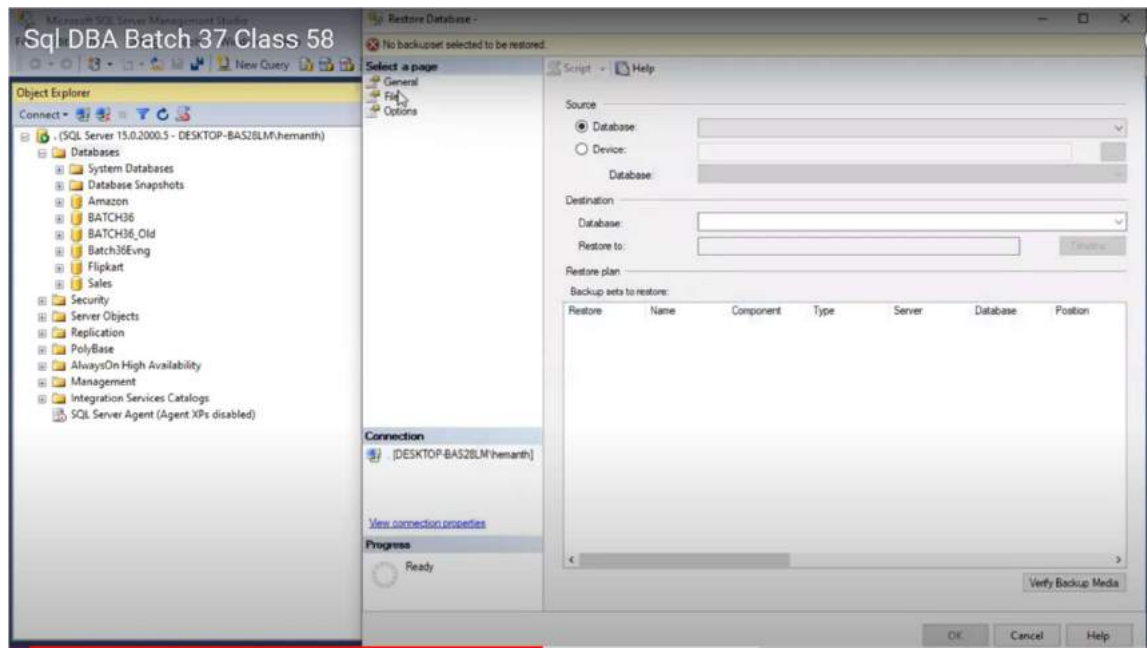
83. How to restore a database from one sql server to another sql server or production to non production or once instance to another instance?

Ans. We can restore a database from lower sql version to higher sql version and same version to same version. But, we cant restore a database from higher version to lower version.

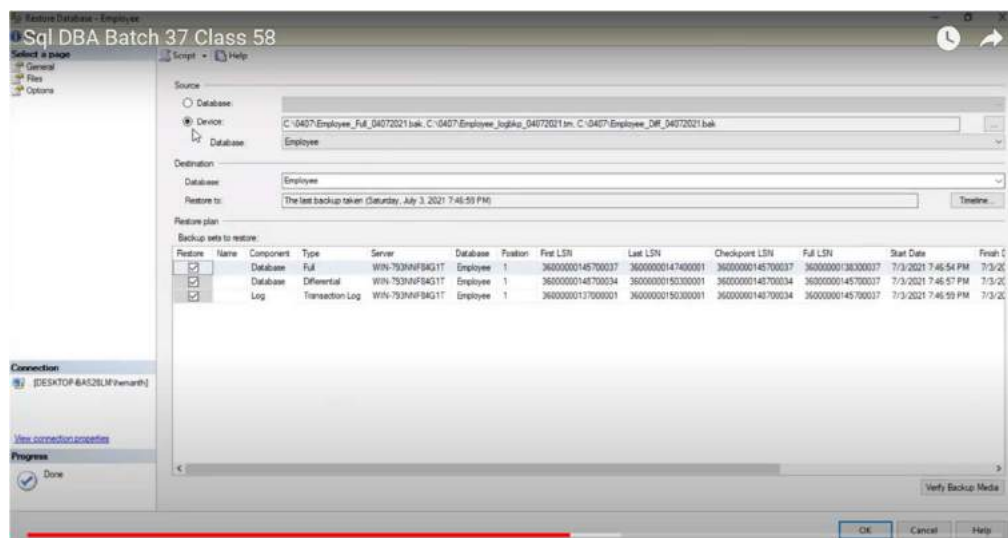
- First copy the backup files from the production server and copy in to non-production server.
- In the organizations, we will be having shared paths, where we can keep these backup files to access in the production server or else Ports should be open between production and non-production servers, then we can copy and paste the backup files directly. To open ports, we have to request network team.



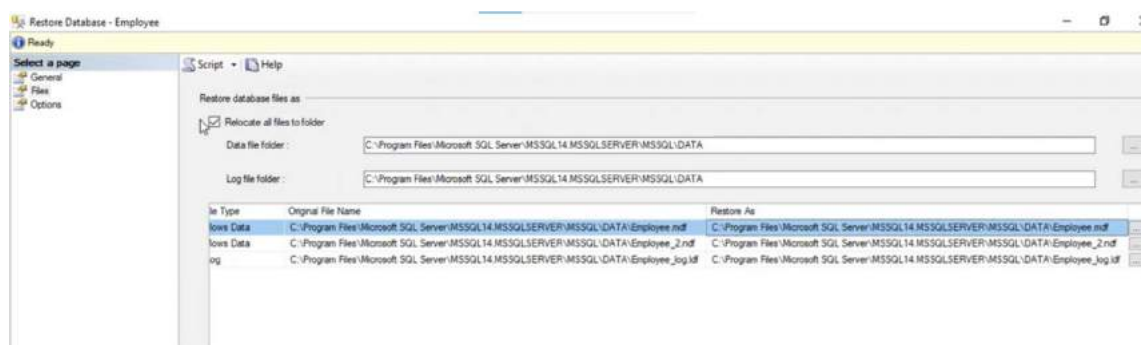
- c. Open SSMS in non production environment, right click on databases, go to tasks, select restore.



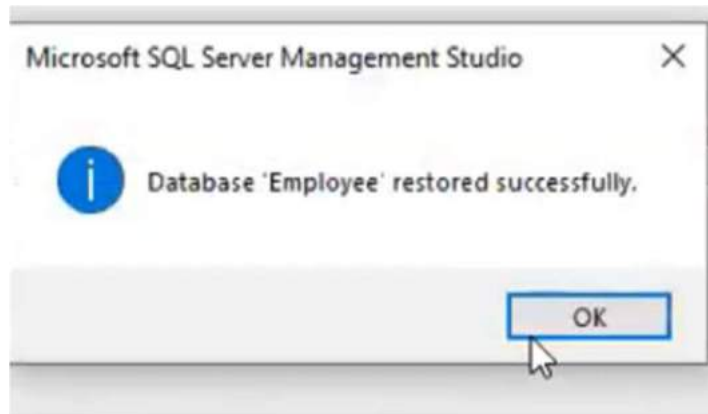
- d. Select the device , choose the path of the backup files in the database column, give the database name.



- e. Go to files tab, here we have to change the path of the restore as for mdf, ndf & ldf. Why because those paths , which are from the production environment . We don't have those paths in non-production environment. Now give the path , in which path we want store the data files and log files after the restore or else tick the relocate option , where sql server will create a new path.



- f. Go to options tabs, select recovery state. Click ok.



84. Is it possible to take full, differential , log backups on system data bases?

Ans. Recovery model for TEMP DB is always simple only. We cant change it.

System data bases	Full backup	Differential	Log backup
Master	Yes	NO	NO
Model	Yes	Yes	Yes
MSDB	Yes	Yes	Yes
TempDB	NO	NO	NO

85. What is job in SQL Server?

Ans. Job is an automation process. A job is a collection of steps. Scheduling a script to run. A job always depends in the script.

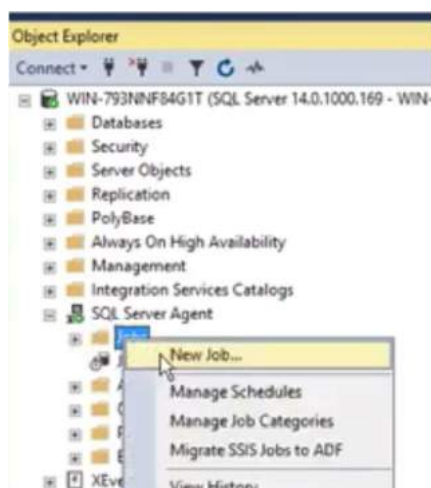
Mostly we have 5 types of jobs , we can automate.

- a. Backups
- b. Restore (Very rarely)
- c. DBCC
- d. Rebuild indexes
- e. Reorganizing indexes

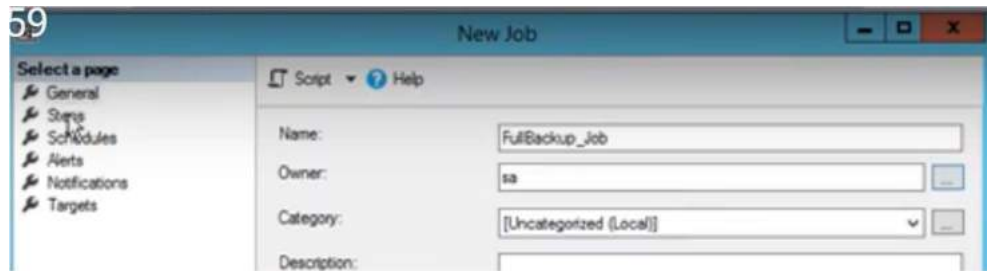
86. How to configure a full backup job?

Ans.

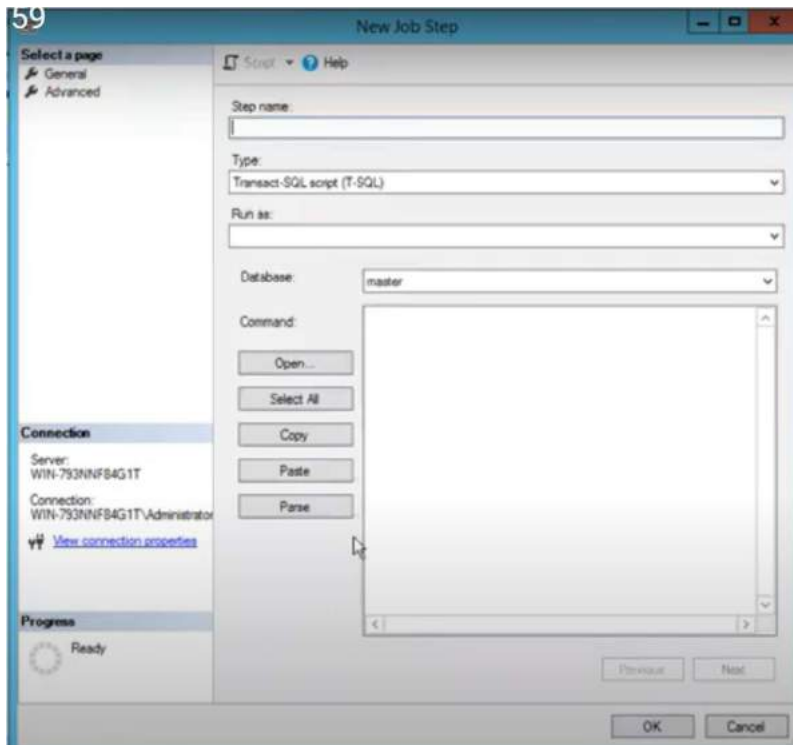
- a. Open ssms, expand SQL Server agent , Right click on Jobs, select new job,



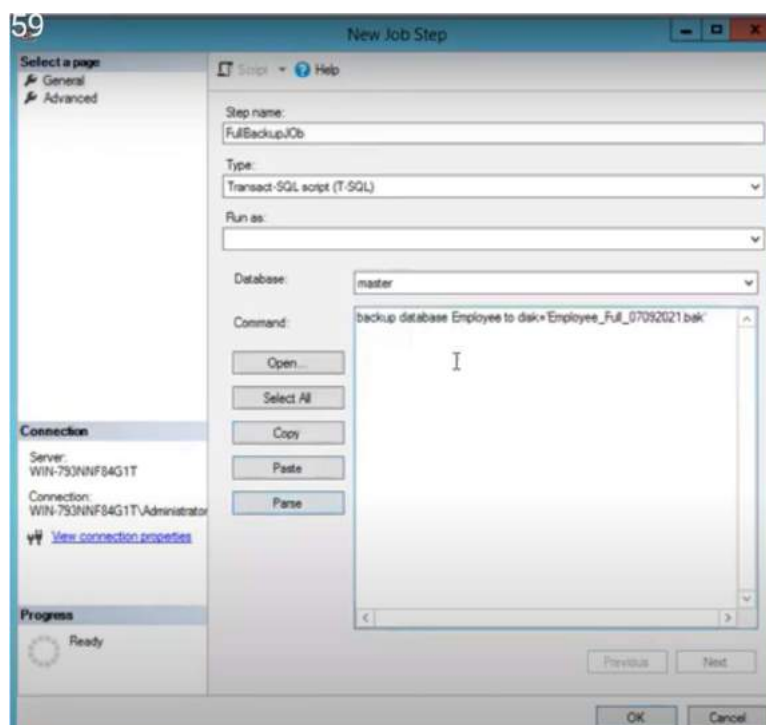
- b. Give the Job name, Give the owner as "sa"



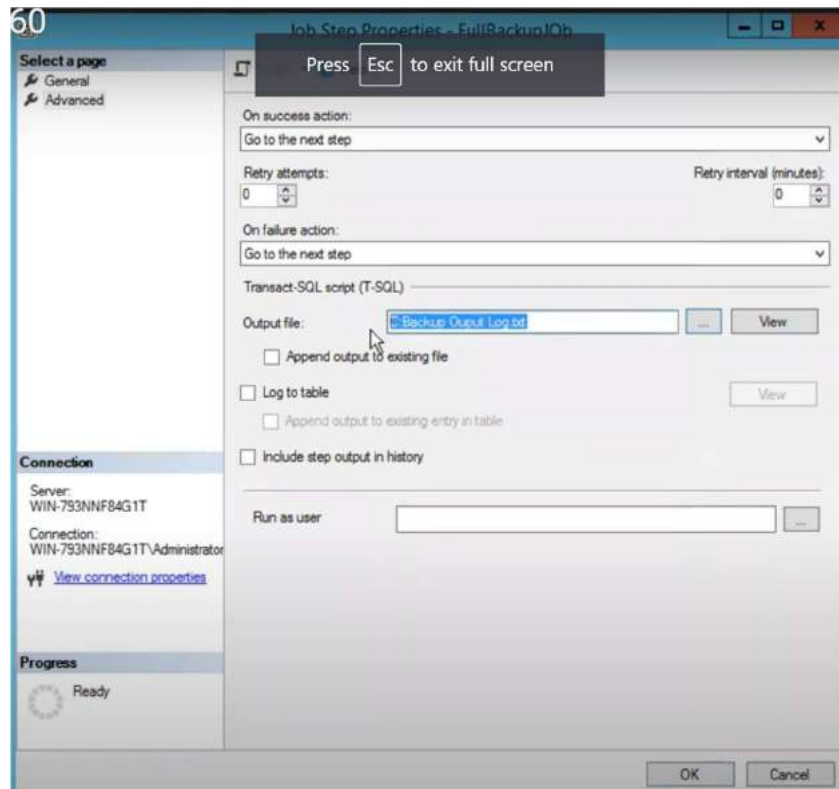
- c. Go to steps tab, Create new,



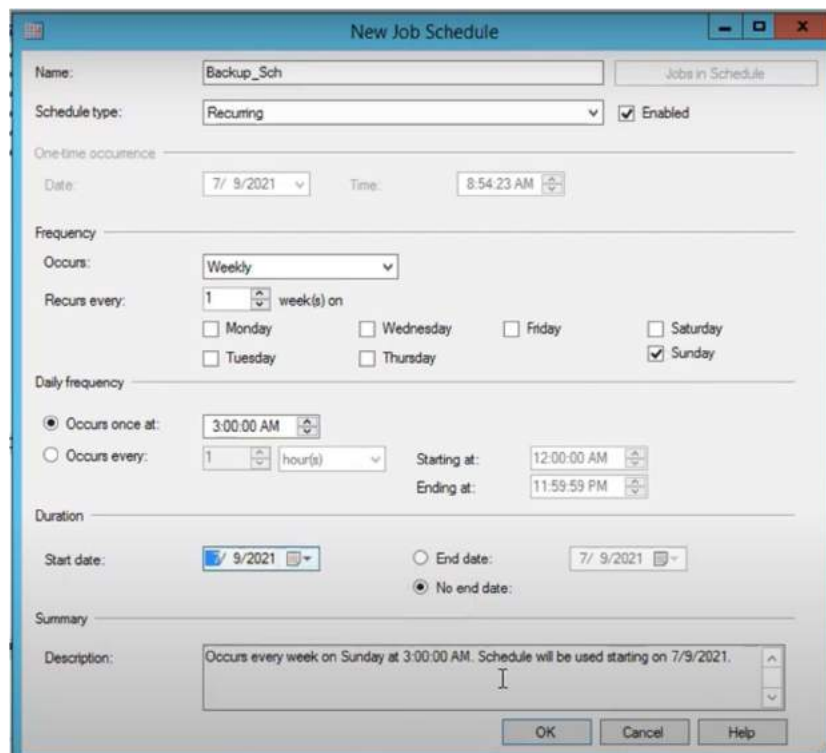
- d. Enter the step name, choose the database as master, enter the full backup query in the empty space, "backup database dbname to disk='path.bak' with init,format"
Now click on the parse to find out any syntax errors.



- e. Go to advanced tab, choose on success action (If step 1 succeeded what to do)- mostly choose go to next step , choose in failure action(if step 1 failed what to do)- mostly choose the quit the job reporting failure. Give the retry attempts and interval(not mandatory) , create notepad and save it . give this notepad path in the output file column. This can be helpful , if job fails at a certain point of time- we can open that note pad and check the error. Click ok. If you want to add one more step to take one more backup , do the same process.



- f. Go to schedule tab, create new, give the name , select the schedule type, tick enabled, click ok. Job created.

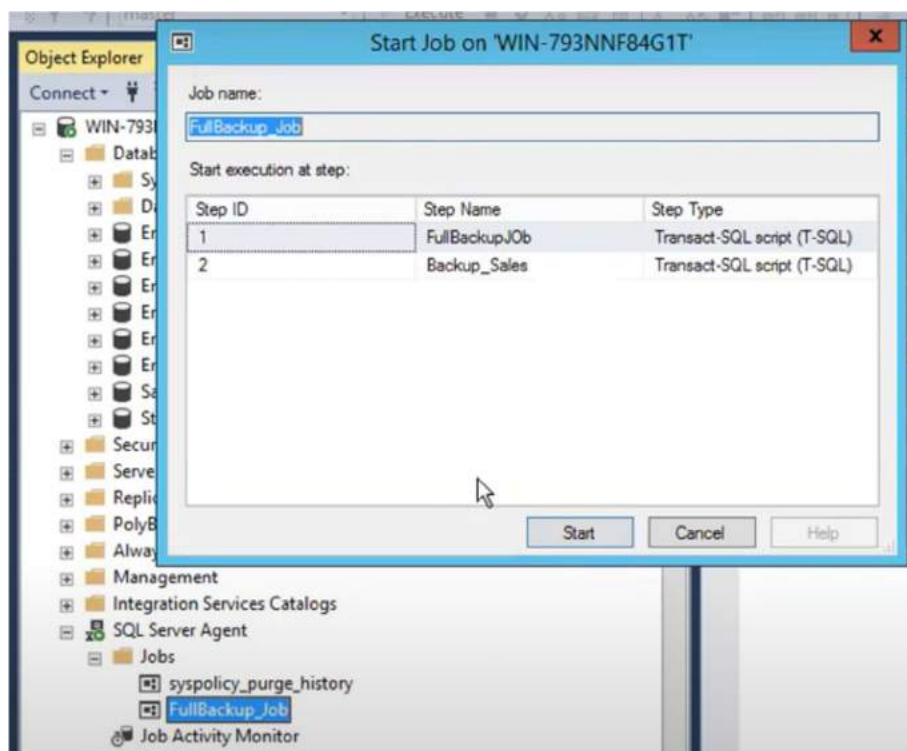
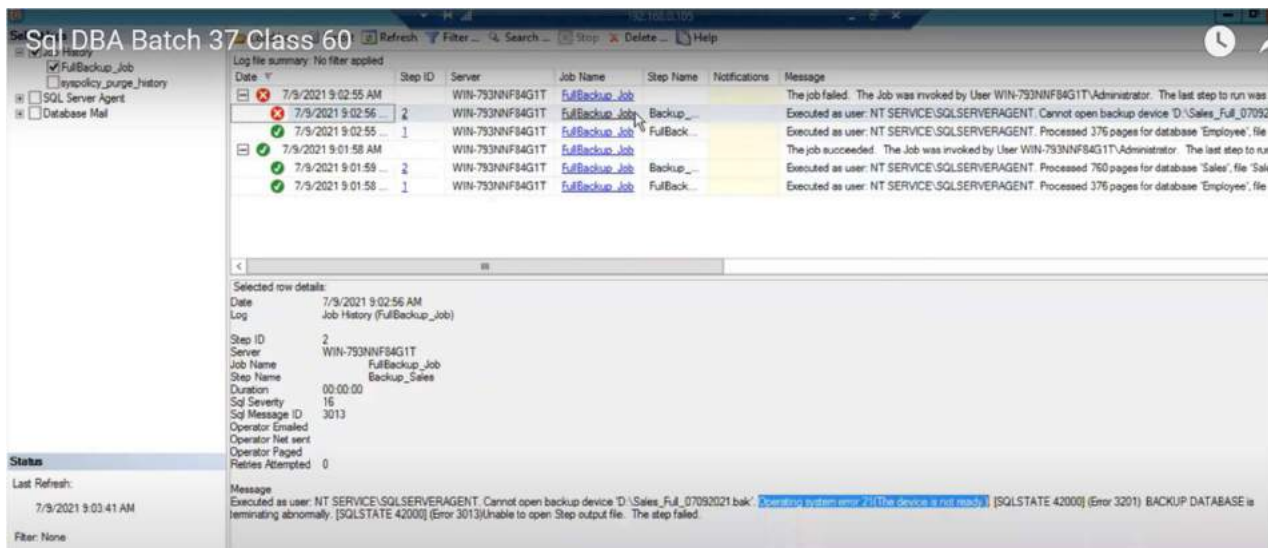


When ever you want restart the job , check the duration of the job completed. It its taking more time to complete the job like 4 to 5 hrs inform to client, why because it may slow down the server. Performance issues.

Note : if any step is failed in the job, after rectifying the error- don't start the job from beginning. Start from the failed step. Like below.

Right click on created job, click on start job at step, Select the step id where job has failed and click ok.

To find out the step id of failed job, right click on created job, go to view history , expand the line where step is in “X” mark. Below to that error will be displayed. Once rectified the error, start the job from that particular step.



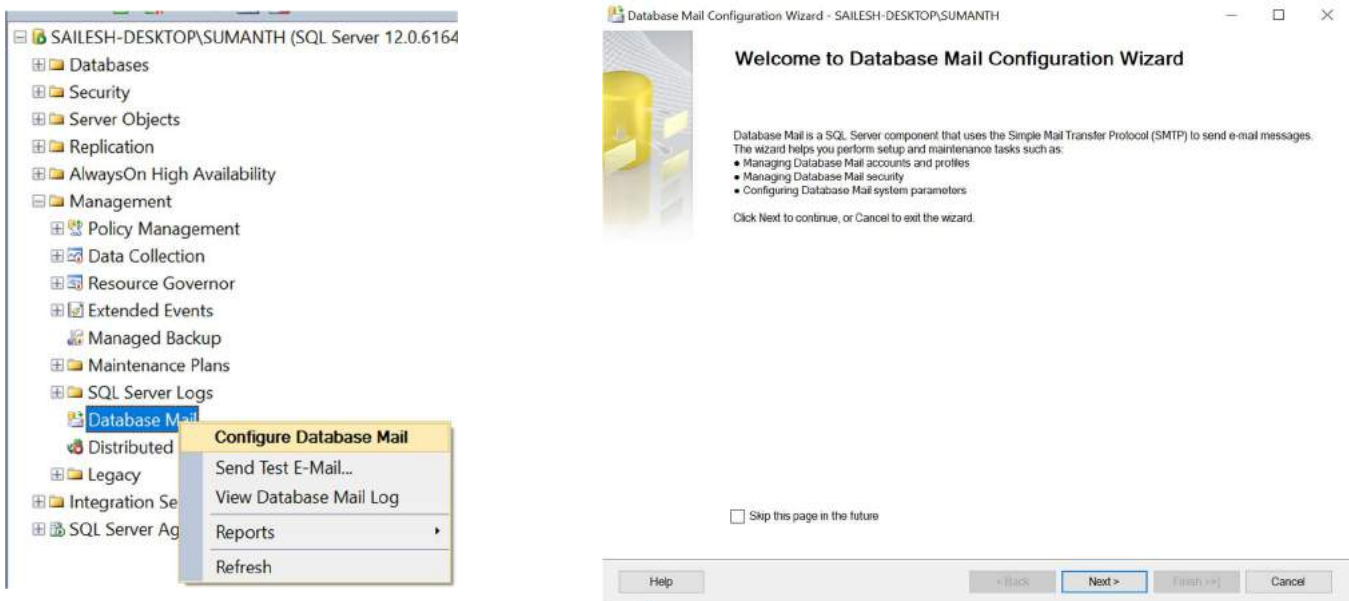
87. What is an alert in sql server agent?

Ans. You can use the **Database Alert** Monitor to identify memory problems in good time, to check the security of your **database** system, and to monitor performance. If data collection methods are periodically scheduled, **alerts** are automatically updated, and forwarded to the monitoring tree.

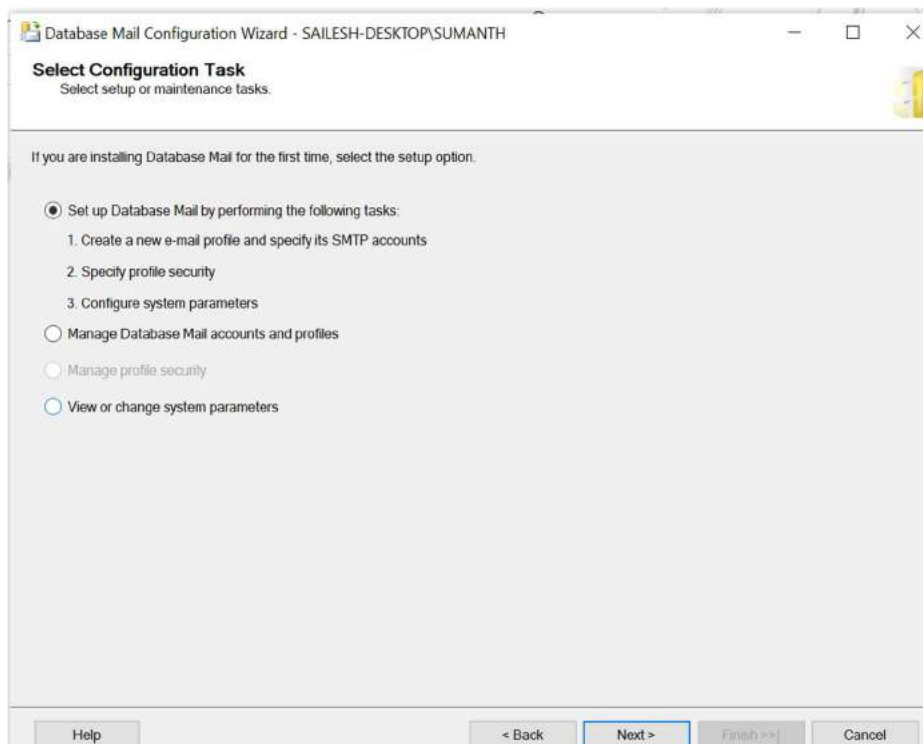
88. How to create a database mail?

Ans. To get the alerts from the sql server agent to the user, we need a database mail. As of now we don't have outlook mail. SO, below example for gmail.

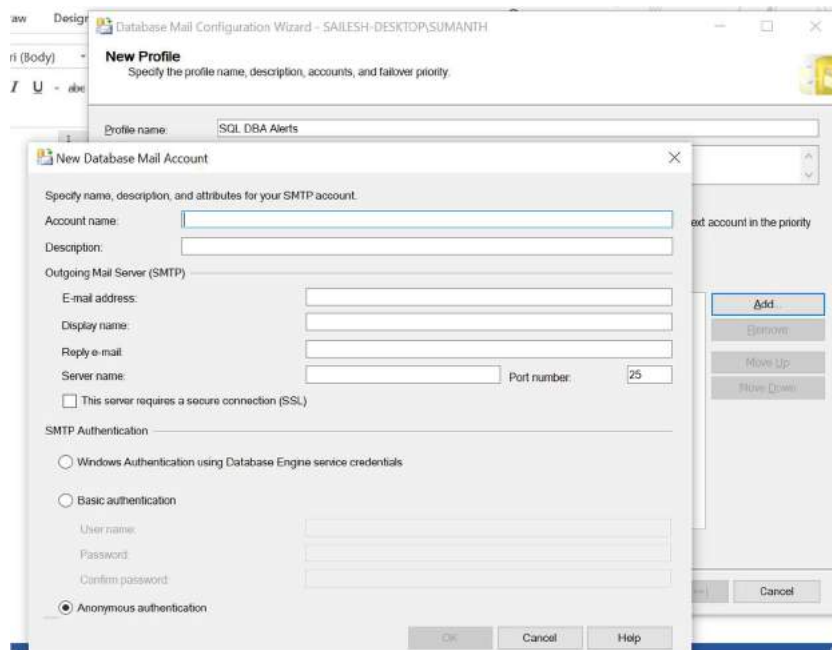
- a. Open SSMS , Expand management , right click on database mail, select configure database mail,



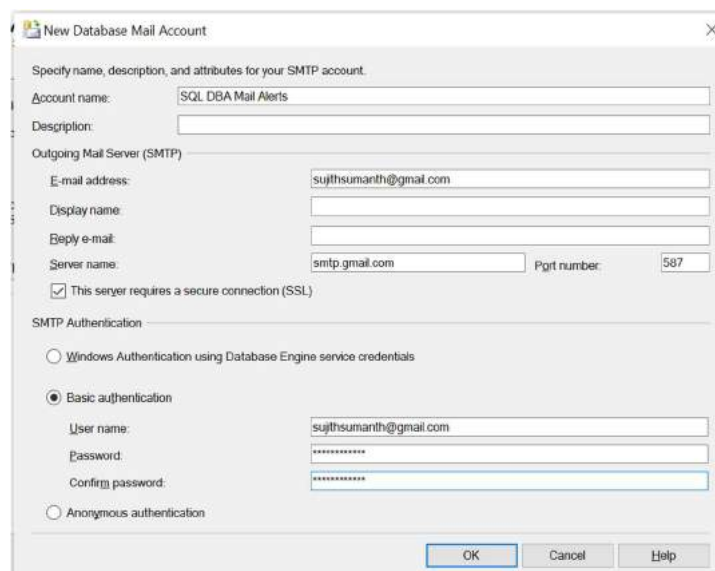
- b. Click next, select setup database mail , click next,



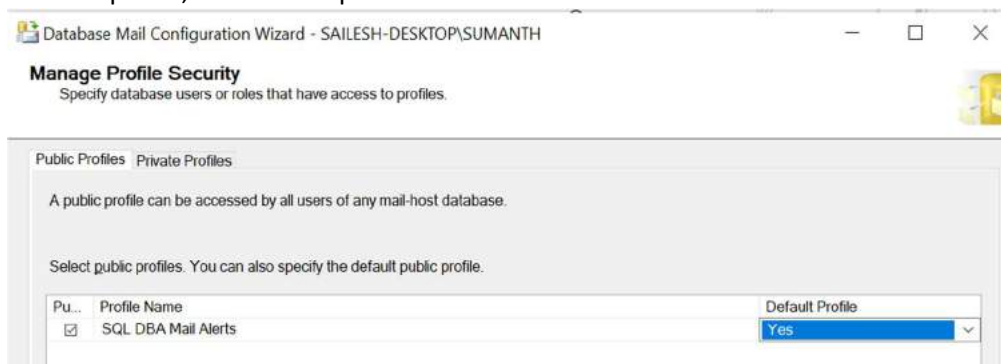
- c. Give the profile name , click on add button,



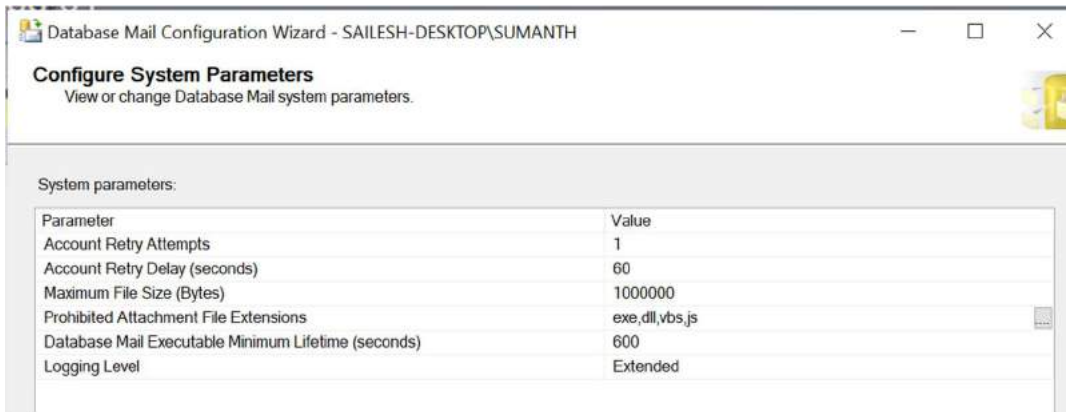
- d. Enter the account name , give the mail id, where you want the alerts, give the smtp(simple mail transfer protocol) every organization has smtp server for mail purpose. Gmail smtp is “smtp.gmail.com” and give port number 587 for gmail. In the organization , we give outlook details only. Outlook port number is 25. In the organizations, we will be having scripts to create the database mail. If we don't have the scripts, then we have to go like this.
- Tick – this server requires secure connection. Select basic authentication, give user name and password. From this basic authentication only mails will be send. Click ok. Click next.



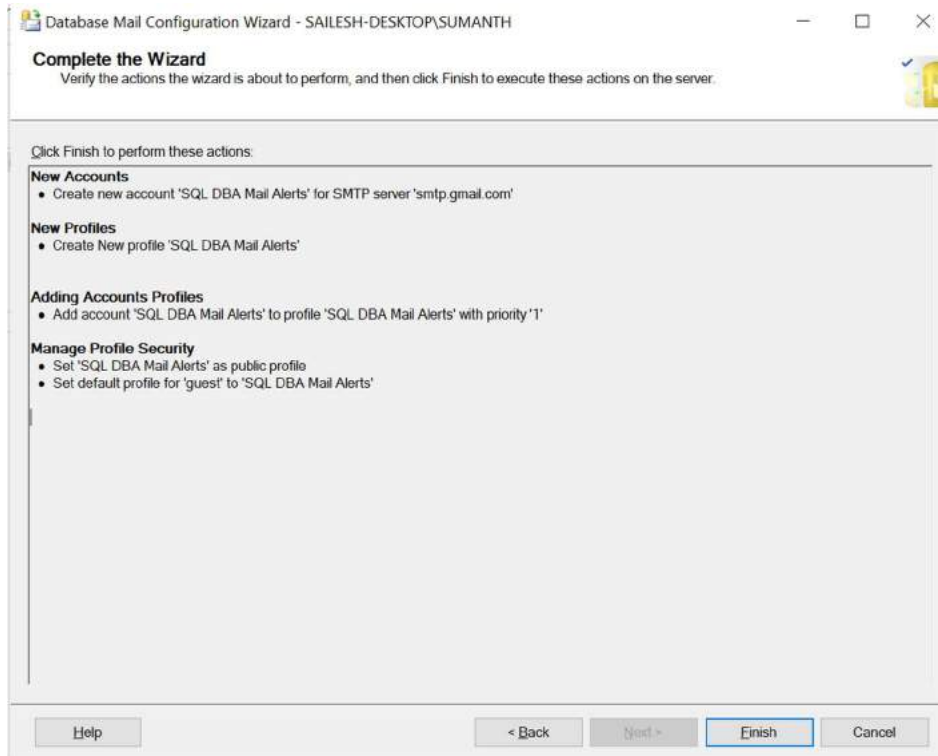
- e. Tick the public, and default profile YES. Click next.



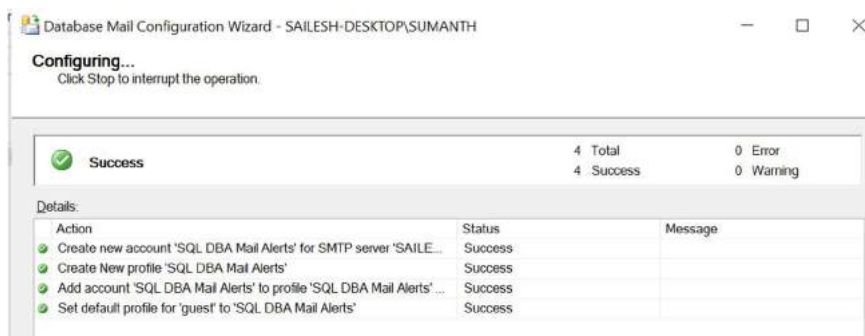
f. Configure the parameters. Click next



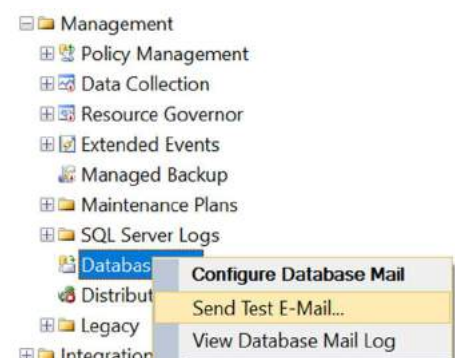
g. Click finish.



h. Success message



i. Now we can check the database mail is working or not by using the Test email option in the management.



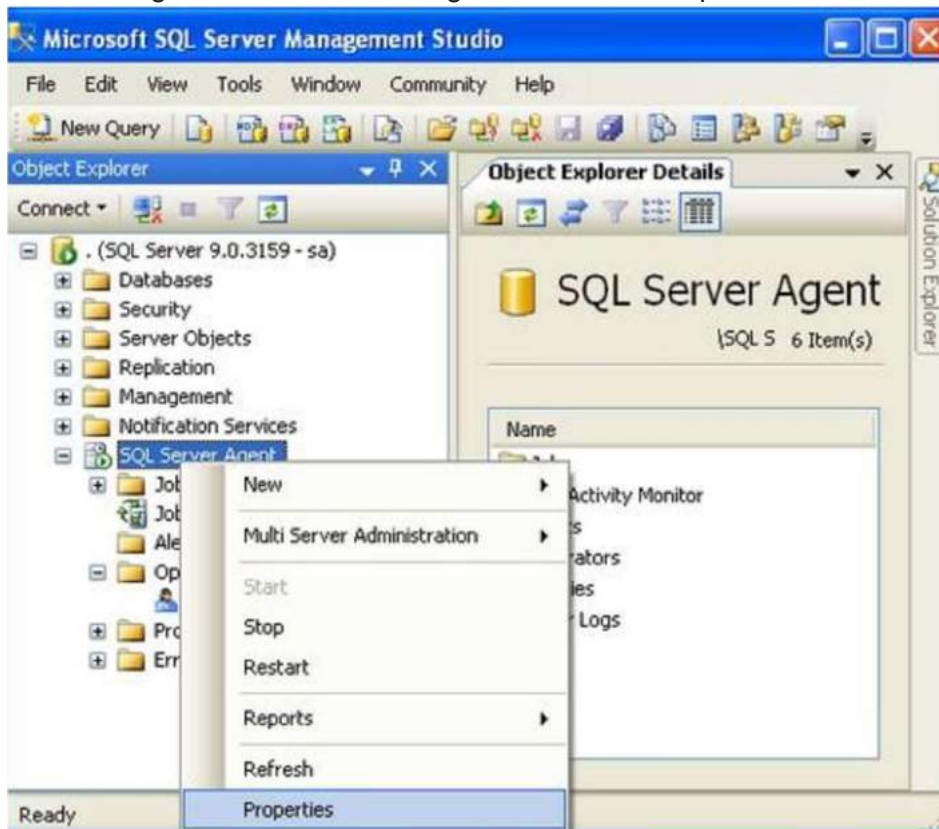
- j. Gmail won't allow the messages from unsecured emails. So, we have to turn off one security in the Gmail that is "Less secure app access". We have to turn it on.

← Less secure app access

Some apps and devices use less secure sign-in technology, which makes your account vulnerable. You can turn off access for these apps, which we recommend, or turn it on if you want to use them despite the risks. Google will automatically turn this setting OFF if it's not being used. [Learn more](#)

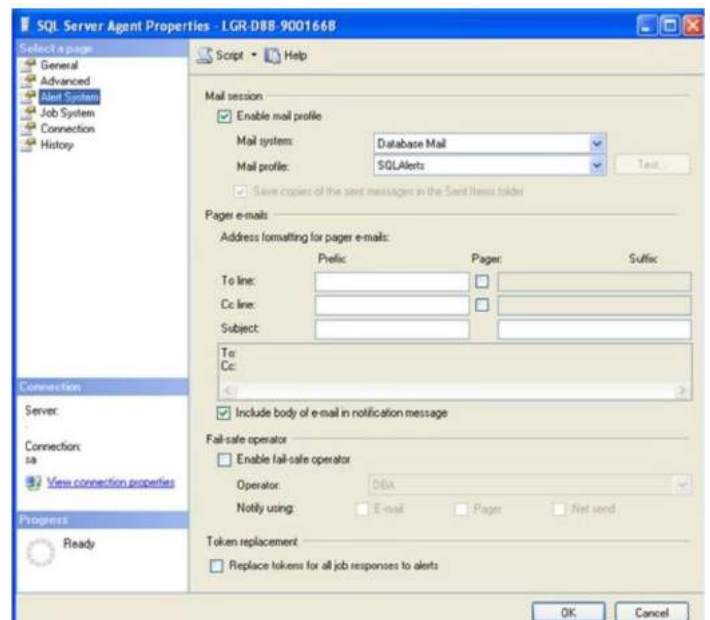
Allow less secure apps: ON

- i. Right click SQL Server Agent and select Properties



- j. Select Alert System in the left pane and do the following:

- Check **Enable mail profile**
- Verify Mail system is **Database Mail**
- Verify Mail profile is **SQLAlerts** that we discussed above
- Check **Include body of e-mail in the notification message**
- Click **OK** to save settings.
- Restart SQL Server Agent to Activate Settings



89. What kind of monitoring tool used in your current organization? (interview Question)

Ans. We are using SQL server alerting system and SCOM alerts for windows level.

We can use sql server alerting system or else we can use third party tools like Spot light , Fog Light , Idera etc.

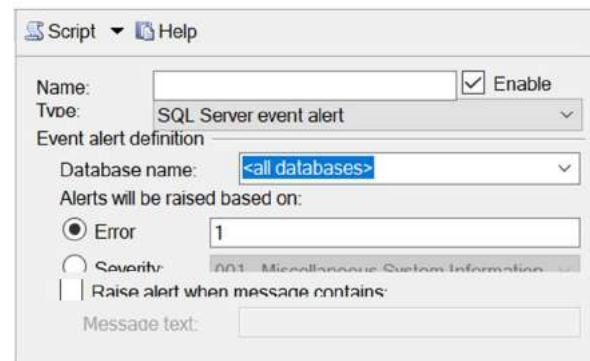
We can configure sql server alert system by using database mail configuration

90. What is an alert?

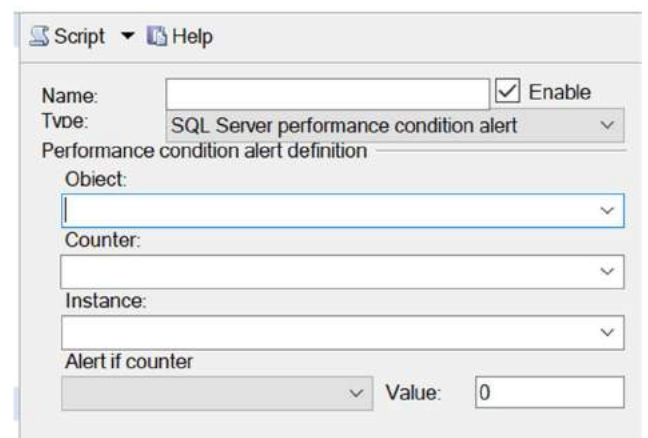
Ans. There are 3 types of alerts.

- a. SQL Server event alert : When we know the exact error number, we can configure event alert. Below are few error numbers.

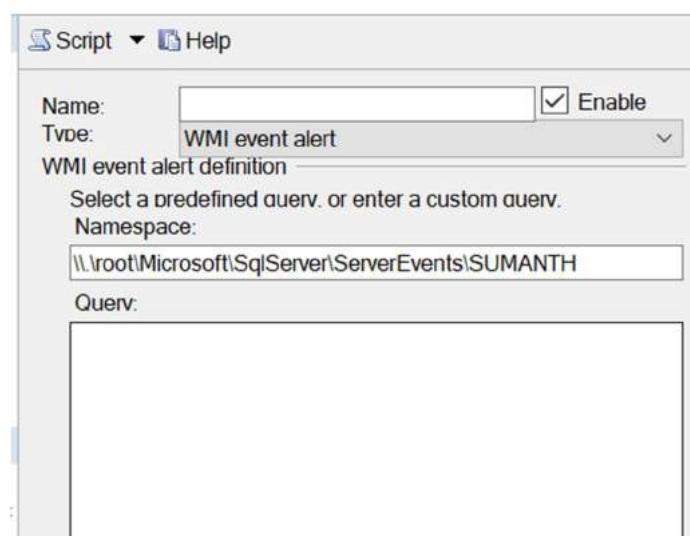
Error	Description
18456	Login failure
1105	Data File full
9002	Log file full



- b. SQL Server performance condition alert :
Actually its not an event or error, Its going to alert us before the error comes. Based on the threshold limit , we can set the condition alert. For Example , Creating an alert for data file full – When datafile reaches to 90% of size its going to alert us. One more example, When CPU Usage reaches 75% alert will come. We will fix the issue after the immediate alert. Based on our threshold limit, its going to work.



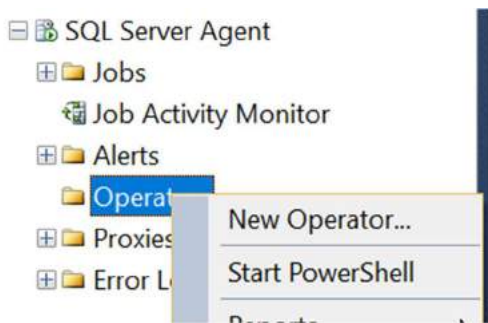
- c. SQL Server WMI Event alert : WMI(Windows management instrumentation) . It Tracks all the DDL Commands. Mostly we wont use this alert. This alert we can enable based on the query. For example. Who deleted the an object like that we can track and we receive the email.



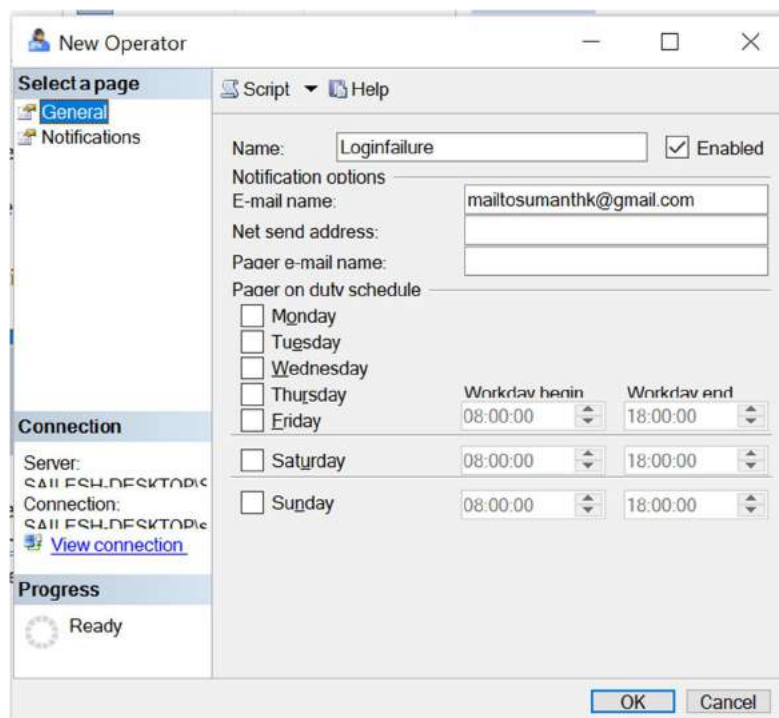
91. How to create an alert system?

Ans.

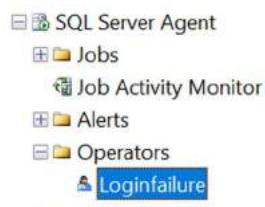
- a. Open SSMS, Expand SQL Server agent, Right click on operator, create new operator.



- b. Operator is a recipient of the error. Operators we can create with out mail configuration also. But we wont receive mails. So, After the successful mail configuration only alerts can send the operators.
- c. Enter the name of the operator , Enter mail address of recipient in the Email Name , Click ok.



- d. Operator created.



- e. Now right click on alerts, select new alert.

Setting Type SQL Server event alert:

Enter the alert name, Choose the type of event, select the database, enter the error number, enter the message text when error comes,

The 'New Alert' dialog box is shown with the 'General' tab selected. The 'Name' field is 'Loginfailure' and the 'Type' is 'SQL Server event alert'. The 'Event alert definition' section shows 'Database name' as '<all databases>', 'Alerts will be raised based on' as 'Error' with the value '18456', and 'Raise alert when message contains' as 'login failed'. The 'Connection' section shows the server name 'SAIL FISH DESKTOP'.

Setting Type SQL Server performance condition alert :

Enter the name , choose the type , choose the object , choose the counter , Choose the instance(_Total – selecting all the databases) , Counter type , value,

The 'New Alert' dialog box is shown with the 'General' tab selected. The 'Name' field is 'datafilesize' and the 'Type' is 'SQL Server performance condition alert'. The 'Performance condition alert definition' section shows 'Object' as 'Databases', 'Counter' as 'Data File(s) Size (KB)', 'Instance' as 'Total', and 'Alert if counter' as 'rises above' with a 'Value' of '90'. The 'Connection' section shows the server name 'SAIL FISH DESKTOP'.

- f. Go to Response tab, Click on notify operators, Select the operator, Tick Email

The 'New Alert' dialog box is shown with the 'Response' tab selected. The 'Notify operators' checkbox is checked. The 'Operator list' table shows the operator 'Loginfailure' with the 'E-mail' checkbox checked and the 'Pager' checkbox unchecked.

Operator	E-mail	Pager
Loginfailure	☒	☐

- g. Go to Options , Tick Email , Enter the text in empty space, Click ok.

New Alert

Select a page: General, Response, Options

Script Help

Include alert error text in:
☒ E-mail
☐ Pager
☐ Net send

Additional notification message to send:

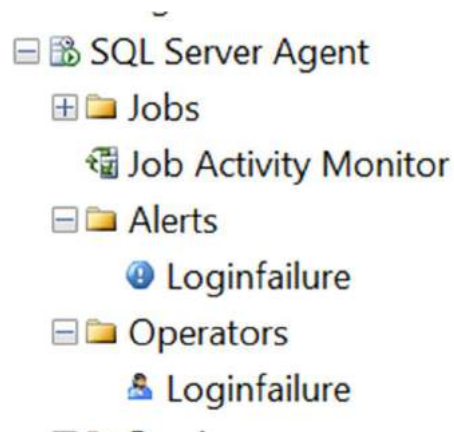
Delay between responses:
0 minutes 0 seconds

Connection
Server: S411 FSH.DFSKTOP16
Connection: S411 FSH.DFSKTOP16
[View connection](#)

Progress
Ready

OK Cancel

- d. Alert Created for login failure successfully.



Few examples :

Creating alert for CPU usage reaches 90-%

New Alert

Select a page: General, Response, Options

Script Help

Name: CPU90% ☒ Enable

Type: SQL Server performance condition alert

Performance condition alert definition

Object: Workload Group Stats

Counter: CPU usage %

Instance: internal

Alert if counter: rises above Value: 90

Creating alert for Datafile size reaches 90%

New Alert

Select a page: General, Response, Options

Script Help

Name: ☒ Enable

Type:

Performance condition alert definition

Object:

Counter:

Instance:

Alert if counter: Value:

Creating alert when blockings morethan 4

New Alert

Name: ☒ Enable

Type:

Performance condition alert definition

Object:

Counter:

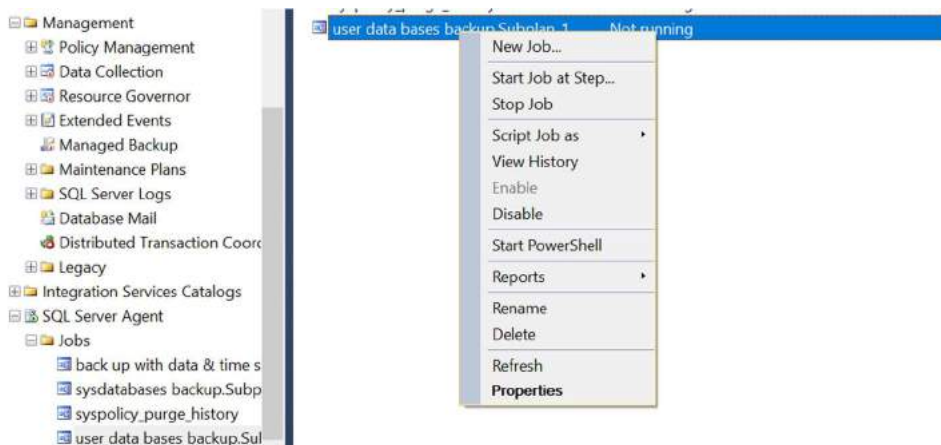
Instance:

Alert if counter: Value:

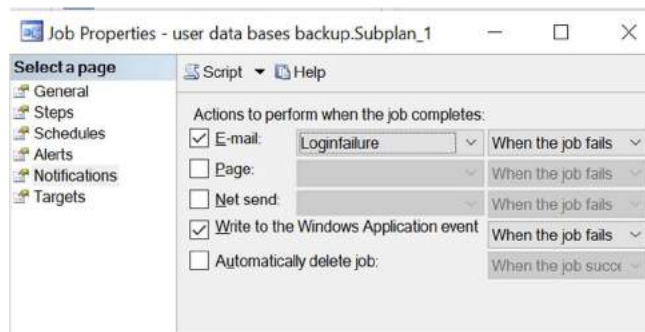
92. How to setup Email alert for a backup Job?

Ans.

- Expand SQL Server Agent ,Expand Jobs , Select and right click the already created job and go to properties,



- Go to notifications, Tick Email, Choose the operator, Choose the reason, Click ok.



93. What are general alerts , which is created in your current organization?

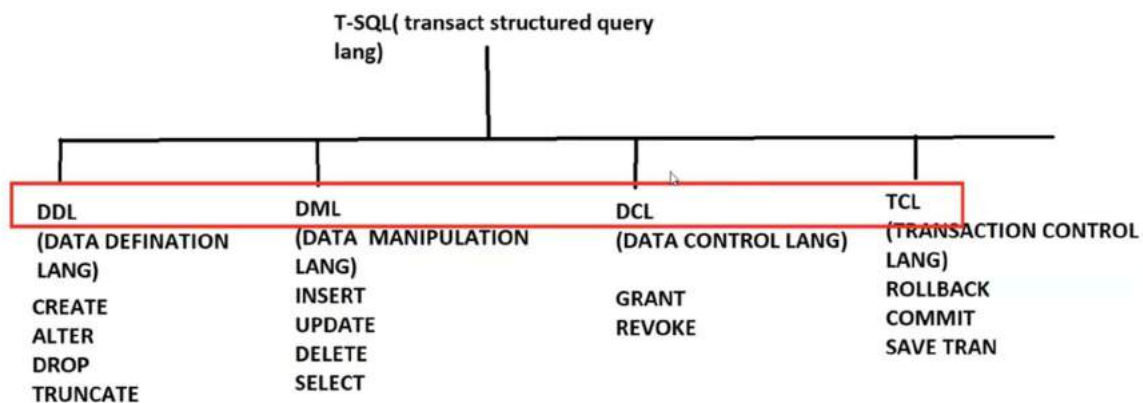
Ans.

- a. CPU alert
- b. Memory Alert
- c. Data file size alert
- d. Log file size alert
- e. Blocking alerts(VV IMP)

94. What is T-SQL?

Ans. T-SQL is Transact structured query language. By using T-SQL, We can modify, delete, select & update the data in the database. There are four languages in T-SQL.

- a. DDL(Data Definition language) :
- b. DML(Data manipulation language)
- c. DCL(Data Control Language)
- d. TCL(Transaction Control language)



95. What are the data types in SQL Serer?

Ans. The type of data which users provided to a specific column is called a Data Type.

There are 2 types of data types.

- a. System defined data types : SQL Server already contains some data types called System defined data types or predefined data types or built-in data types. System defined data types are again defined in to 4 types,
 - a) Numeric data types : used to provide numeric information for the columns. These includes

<u>Data Type</u>	<u>Size</u>
BIT	0 or 1
TINYINT	1 BYTE
SMALLINT	2 BYTES
* INT	4BYTES
BIGINT	8BYTES
REAL	4BYTES
FLOAT	8BYTES
DECIMAL (P, S)	5-17 BYTES

TINYINT, SMALLINT, INT, BIGINT are called **Exact Numerics(1101234)** where as REAL, FLOAT are called **Approximate Numerics.(1101234.456)**

b) String data types : used to provide Characters for the columns, These includes,

<u>Data Type</u>	<u>Size</u>
CHAR (N)]	1BYTE
* VARCHAR (N)]	1BYTE
TEXT	16BYTES
<u>VARCHAR(MAX)</u>	16 GB

- CHAR (N)]: It is a fixed length data type, which occupies by default 1 byte memory. When we N value then it occupies N bytes of memory where N is an integer. It follows Static memory allocation process.

VARCHAR (N)]: It is a variable length data type, which occupies by default 1 byte memory. When we N value then it occupies N bytes of memory where N is an integer. It follows dynamic memory allocation process.

Note : The Maximum limit for N is 8000, if it is more than 8000 characters we will use TEXT or VARCHAR(MAX)

The difference between Char and Varchar is Char is a fixed length data type where Varchar is a Variable length data type.

Ex:

```
char(20)
SQL--> 17 Char's. wasted.

varchar(20)
SQL--> 17 char's--> store other data.
```

c) Data time data types : used to provide date & time oriented information for the columns . These includes,

<u>Data Type</u>	<u>Size</u>	<u>Range</u>
SMALL DATETIME	2 BYTES	1900-01-01 TO 2079-06-06
DATETIME	4BYTES	1753-01-01 TO 9999-12-31

d) Special data types : used to provide miscellaneous information to the columns . These includes,

<u>Data Type</u>	<u>Size</u>
SMALLMONEY	4 BYTE
MONEY	8 BYTES
IMAGE	16 BYTES
VARBINARY (MAX)	Unlimited

b. User defined data types:

When user create a data type then that data type is called user defined data type

Syntax:

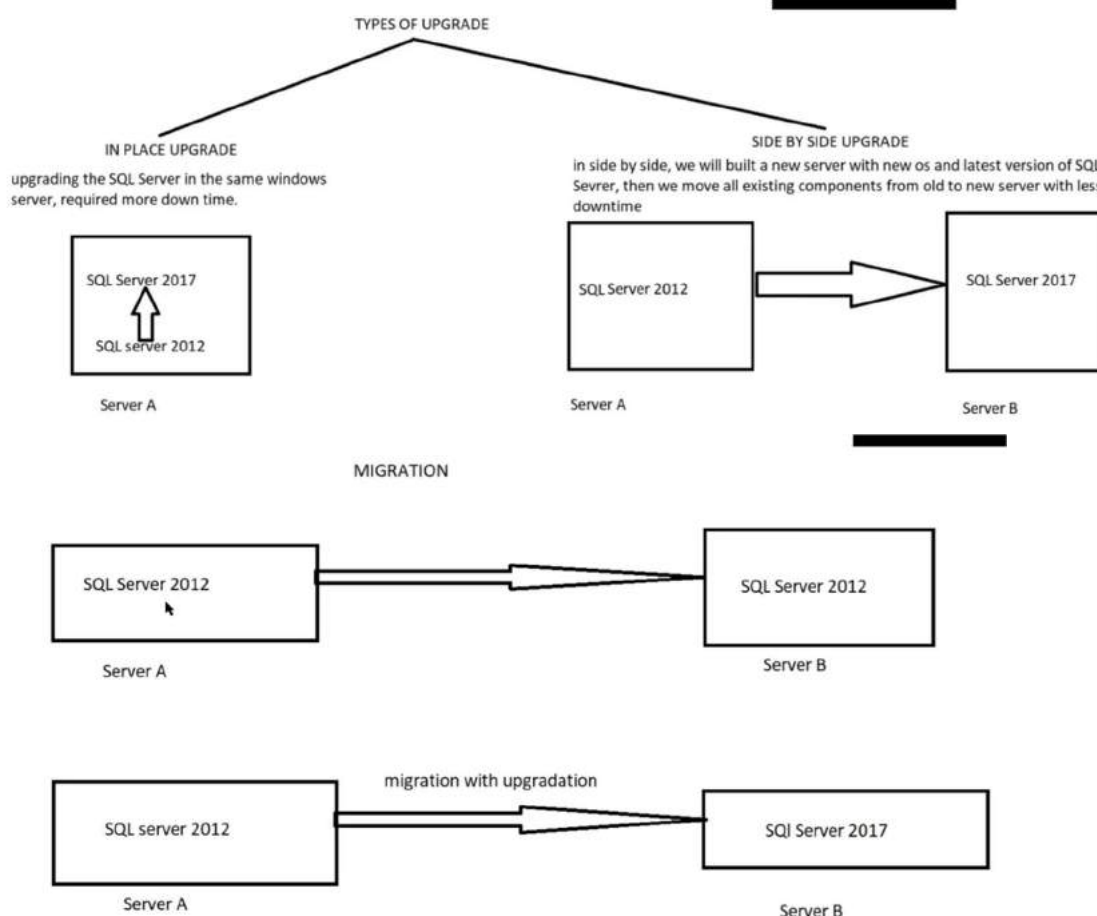
```
CREATE TYPE USER_DEFINED_DATATYPE FROM  
SYS_DEFINED_DATATYPE
```

Ex:

```
CREATE TYPE MYINT FROM INT  
CREATE TYPE MYFLOAT FROM FLOAT  
CREATE TYPE CASH from money
```

96. What is upgrade?

Ans. Upgradade means simply upgrading the SQL Server or instance from lower version to higher version. Mostly people go for migration only not upgradation in the organization because of lot of challenges in upgradation. Before doing the upgradation or migration, we have to run “Database migration assessment tool” to asses – what kind of issues will face. After that we can start upgrade.



We have two types of upgrade :

1. IN Place upgrade(Upgradation)
2. Side by Side upgrade(Migration)

1. In Place upgradation: IN Place upgradation lot of challenges is there for upgradation but its easy. In Place upgradation overwrites the existing SQL server or instance. We have to choose IN Place upgradation , only when the gap between lower version instance to higher instance is 7 years.

The challenges in IN Place upgradations are below

- a. We cant directly upgrade from very lower version to very higher version in a single shot. It means, The gap between lower version & higher is more than 8 years, direct upgrade is not possible. We have to upgrade split the upgradation.

For Example : if the client is requested upgrade my 2008 Instance to 2019 version First , we have to upgrade 2008 instance to 2012 and then from 2012 to 2019. Or else 2008 to 2016 and then 2016 to 2019. So, Its not possible upgrade from very lower version to very higher version in a single shot.

- b. One more challenge is, Currently 2008 instance running on windows 2003 server. Customer is requested to upgrade the 2008 instance to 2019 version. As we already discussed, Each sql server version can support -7 to +7 windows server machines. So, In this case- Upgradation of 2008 to 2019 version is difficult to upgrade because of windows server is 2003 version. So, first we have to upgrade windows server. Windows server also we cant upgrade in a single shot from very lower version to higher version. We have to split in to 2 parts. After the windows server upgradation only we can start sql server upgradation.
- c. After the upgradation, we cant go back to the lower version if you are facing any compatibility issues or any other. If you like it or don't like it, we have to use the upgraded server only. We cant down grade.
- d. Down time of instance is also high in upgradation
- e. We can upgrade to the same edition only. Like Developer edition to developer only like that. We cant upgrade from developer to any other. Same way for all the editions. For example, Customer is requesting to upgrade 2017 SQL server developer edition to 2019 enterprise edition. First we have to upgrade edition from developer to enterprise and then we have to upgrade the instance or else we can upgrade the 2017 version to 2019 version and then we have to upgrade the edition from developer to enterprise.

In this we have another 2 upgrades,

- a. Version Upgradation : Upgrading the versions of SQL Server is called Version Upgradation
- b. Edition Upgradation : Upgrading of Editions of SQL Server is called Edition Upgradation.

96. What is the use database migration assistant?

Ans. We can check what are issues will face before if we do any migration.

Below are the steps to use migration assistant,

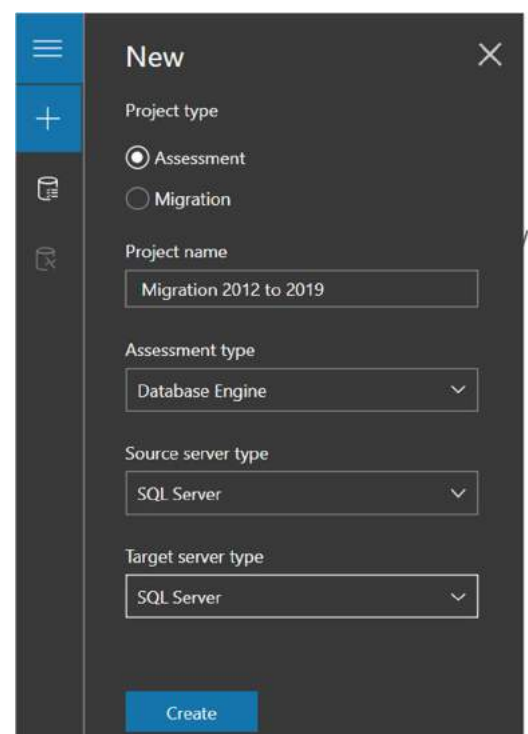
Step 1 : Open database migration assistant application

Step 2 : Click on new , Choose assistant, Give the name , Choose assessment type database engine, Select the source server type and Target server type. Click create.

Step 3 : Add the instance name and login Like

'Computername\instancename'

Click assessment. Now we can see if there any compatibility issues.

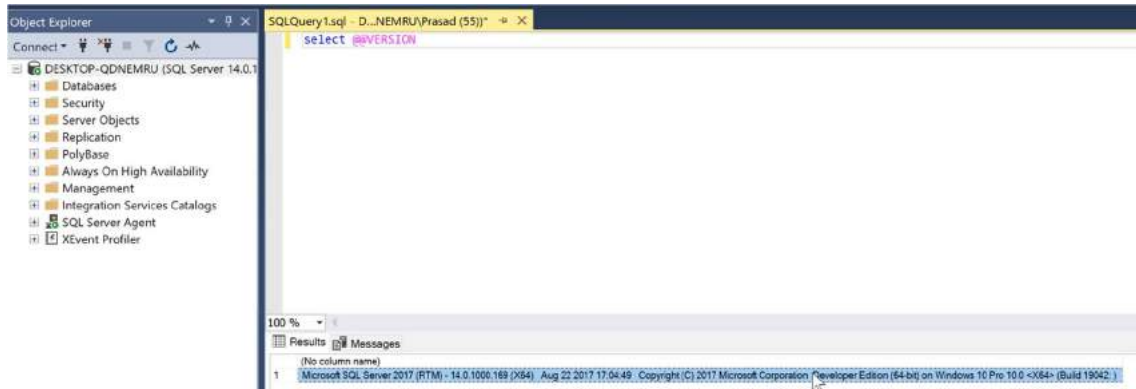


97. How to upgrade SQL Server?

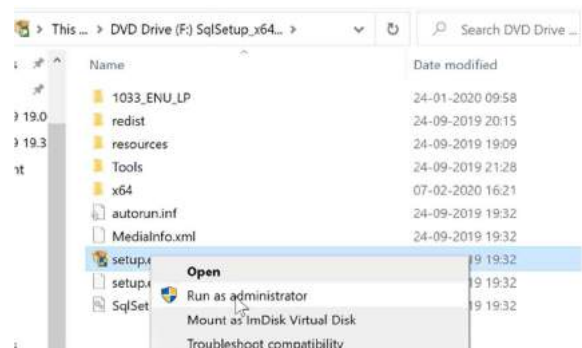
Ans. Below is the Example of upgrading SQL Server 2017 developer edition to SQL Server 2019 developer edition.

Whatever precautions we took while doing the patching, same precautions we have to take.

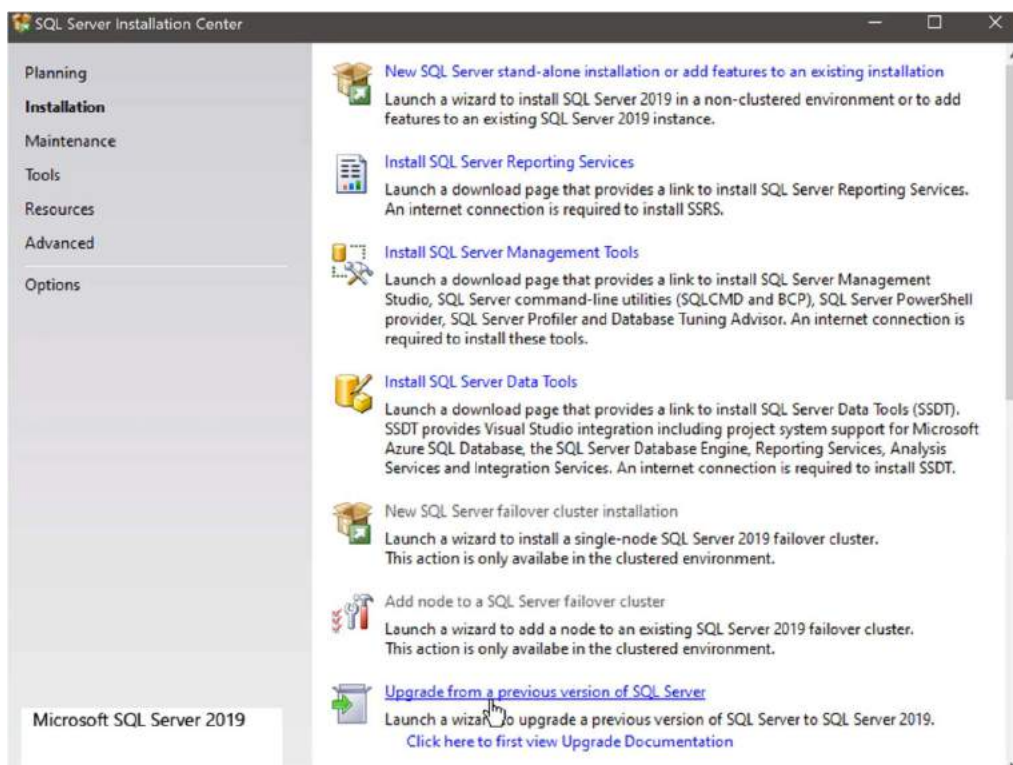
- a. First Check the what is the version and edition of the installed instance by using the query “select @@Version” – Below is the 2017 version , build no 14 and developer edition



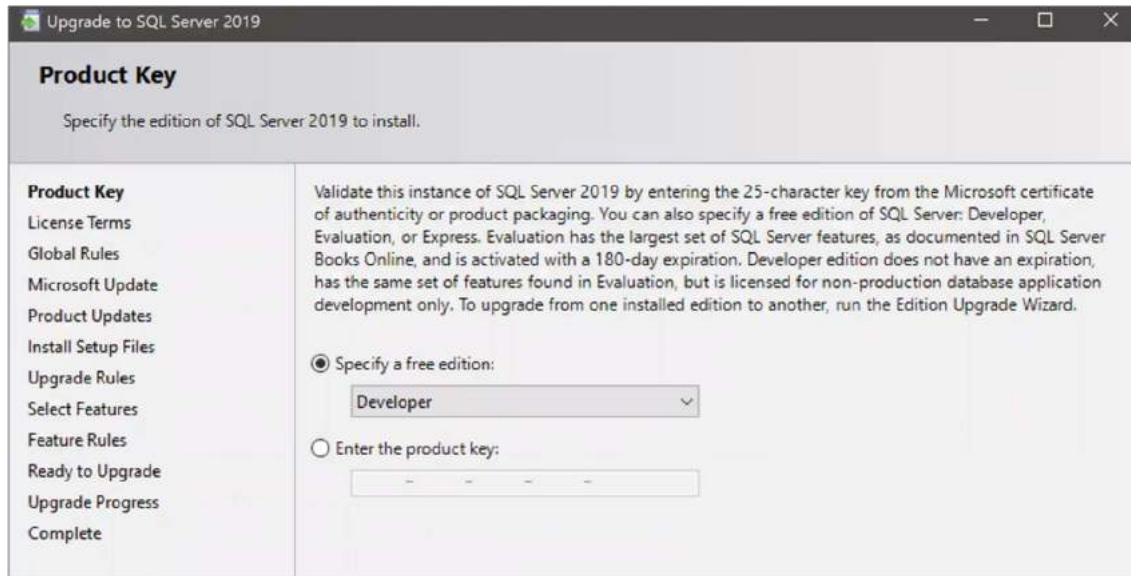
- b. Open SQL Server SQL Server 2019 developer edition setup folder , Right Click on setup.exe, RUN as administrator.



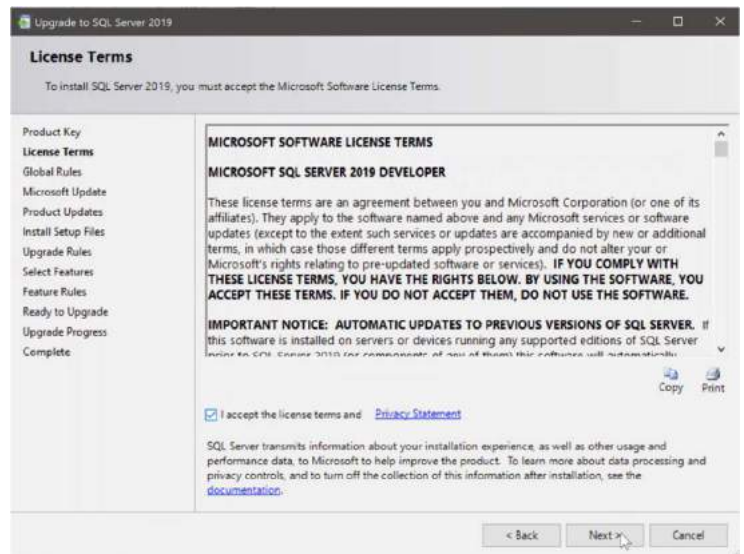
- c. Go to installation tab, Choose the upgrade from a previous version of sql server.



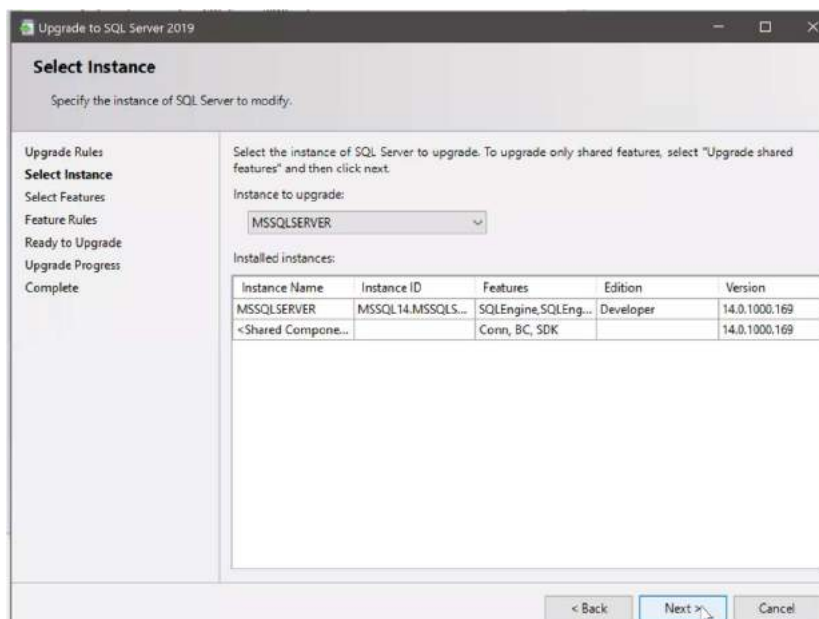
- d. Specify the developer edition , click next.



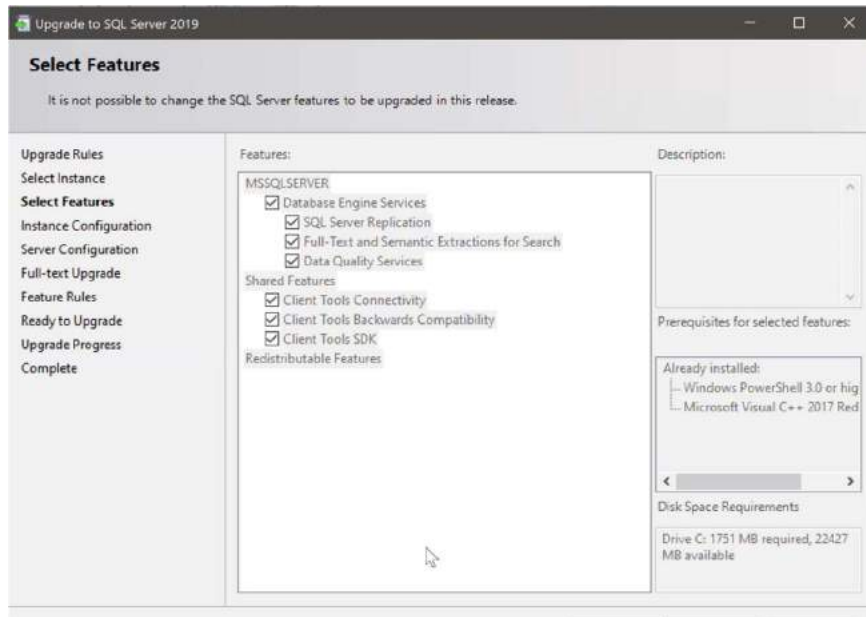
- e. Tick the Licence agreement , click next.



- f. After 3 times next, Instance selection window will come like below.
- g. Choose the instance, which you want to upgrade and click next. Below I have selected MSSQLSERVER instance.



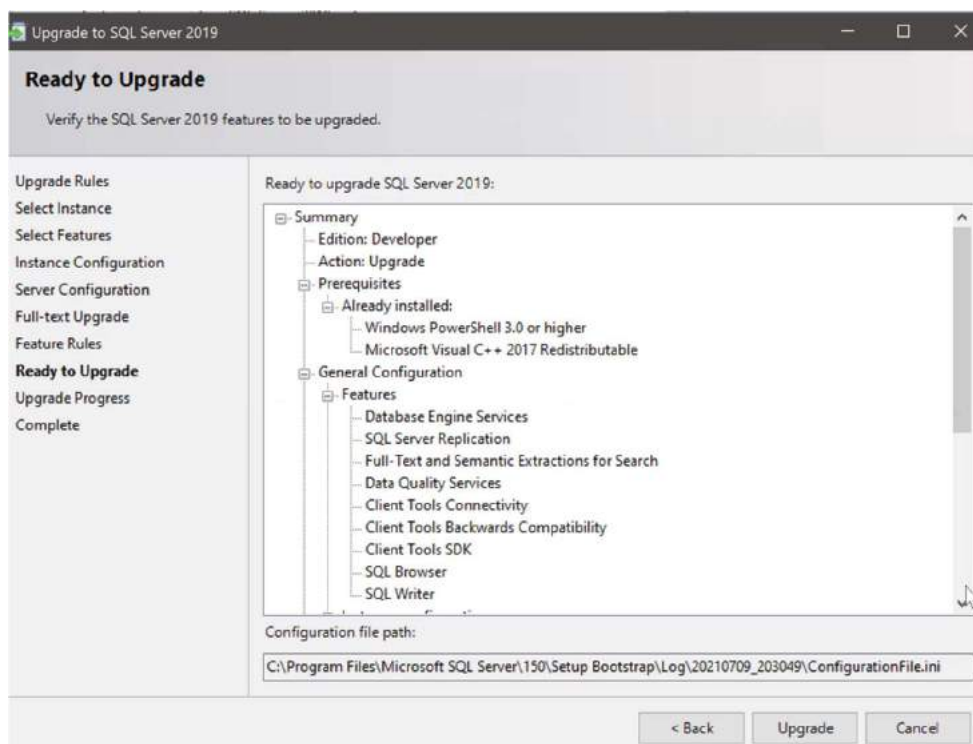
- h. Features selection window will come. No need to select any features. By default, what all are the features selected in the 2017 version, the same will be selected and same features will be upgraded to 2019. We cant even uncheck the features also. Click next.



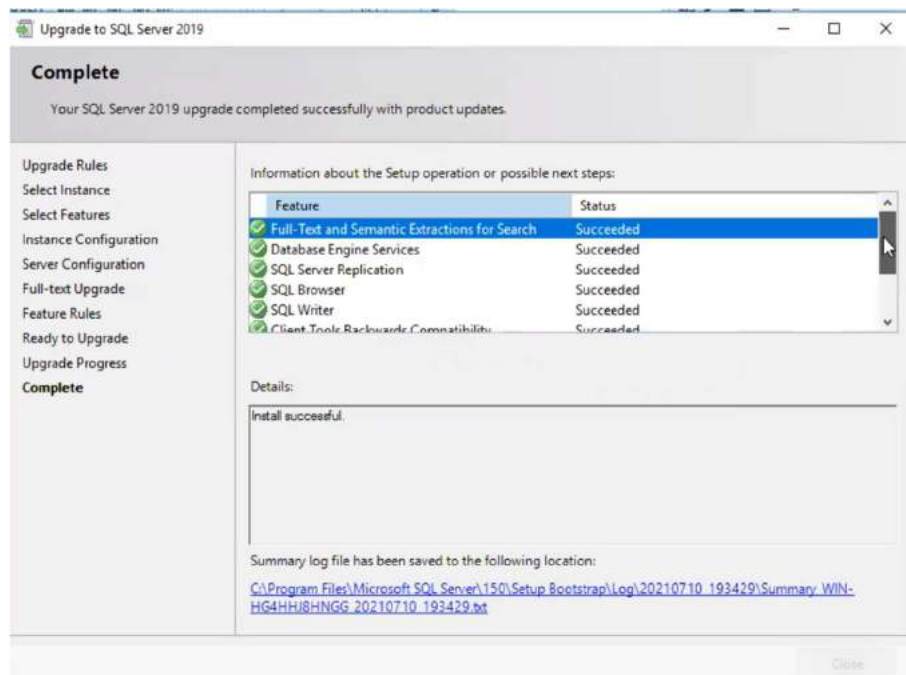
- i. Full text upgrade window, choose Import option , click next.



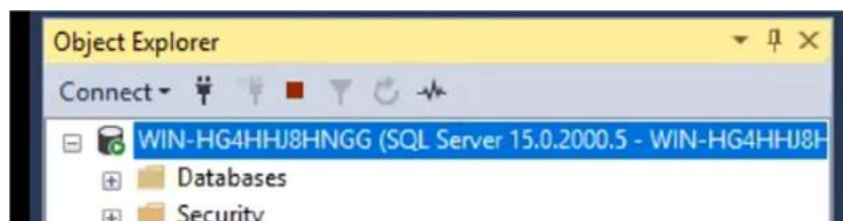
- j. Click on upgrade.



k. Upgradation completed.



l. We can check upgraded version number in the SSMS.



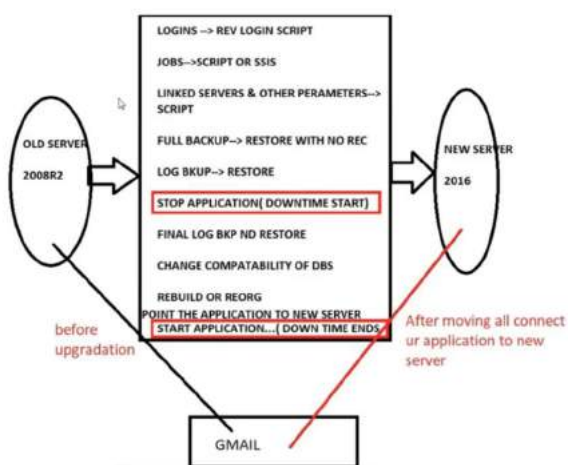
98. What is migration?

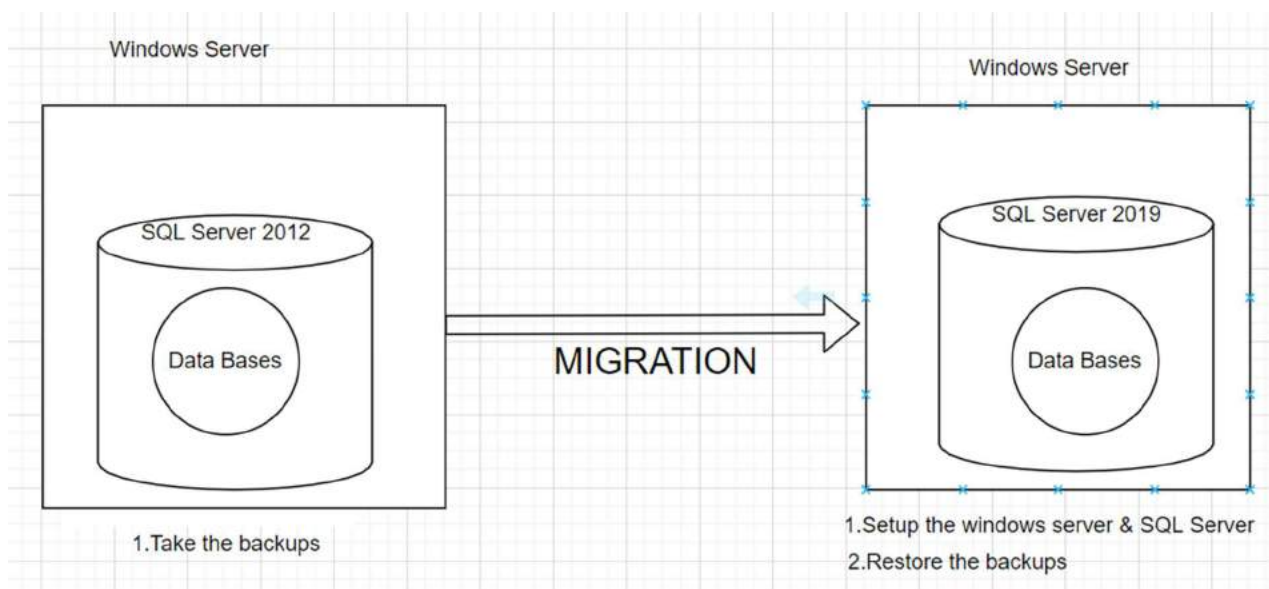
Ans. Migration means creating the same environment of already existed instance. For migration there should be a target and destination. Migration is a very long process but its safe process. We normally do the migration, when there is any database refresh tasks & upgrading the sql server from one version to another version. In Migration, We can upgrade SQL Server to any version & any edition.

Migration is simply 1st step : Creating the same environment like target in the destination(Windows server , SQL Server of required version)

2nd Step : Take the database backups of target

3rd step : Restore the backups in the destination

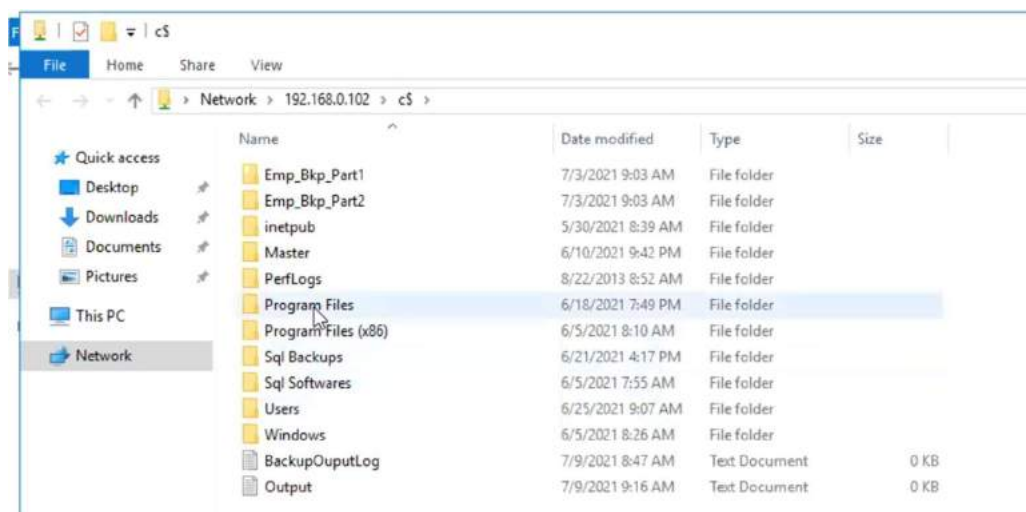
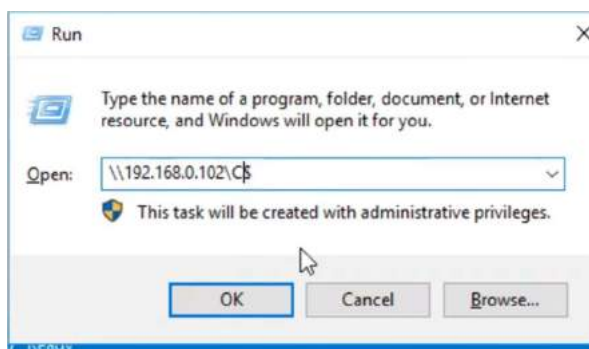




99. How to open another windows server drives in the current windows server?

Ans. Open run , type the command [\\IPADDRESS of server\drive letter\\$](#)

Example : [\\192.168.0.102\c\\$](#)



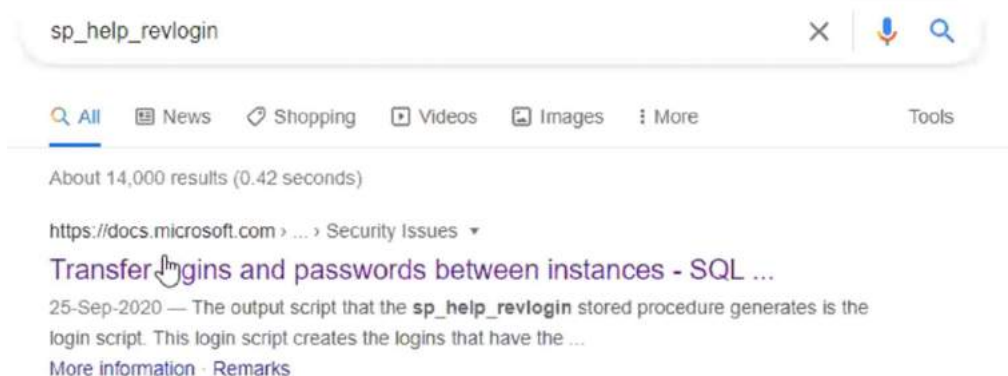
100. What are steps for migration?

Ans. Below is the example of migration from SQL Server 2017 to SQL Server 2019.

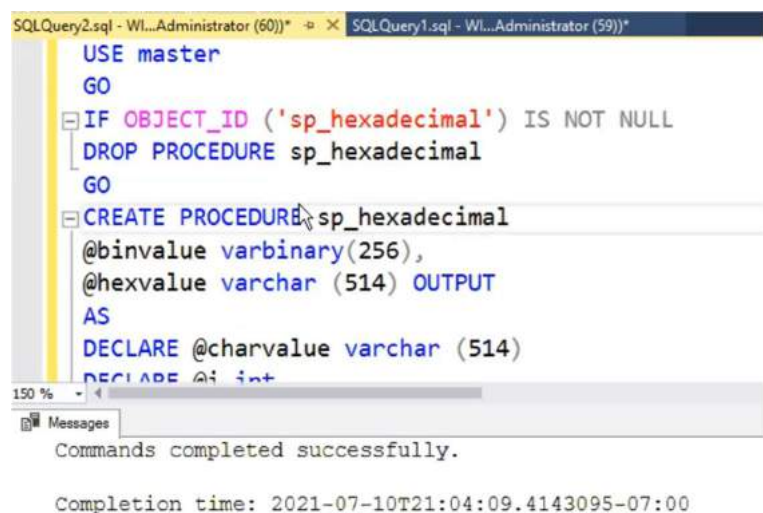
- a. Install the SQL Server 2019 in the destination.
- b. Took the backup of all the user databases from the Target sql server 2017 by using the maintenance plan.
- c. Copy the backup plans from target machine to destination machine.
- d. Restore the backups in the SQL Server 2019
- e. Now to export the logins from target to destination, that can be achieved by creating a stored procedure "SP_HELP_REVLOGIN" in the target server sql server 2017.



- f. Search the script in the google for "SP_HELP_REVLOGIN". This stored procedure script out all the logins with password. By using the script create the stored procedure and run it. Don't copy from third party websites , use Microsoft. Don't blindly copy every thing, read the script and check whether there is any links or anything.



- g. Paste the script in the target server and run



SQL



```
USE master
GO
IF OBJECT_ID ('sp_hexadecimal') IS NOT NULL
DROP PROCEDURE sp_hexadecimal
GO
CREATE PROCEDURE sp_hexadecimal
@binvalue varbinary(256),
@hexvalue varchar (514) OUTPUT
AS
DECLARE @charvalue varchar (514)
DECLARE @i int
DECLARE @length int
DECLARE @hexstring char(16)
SELECT @charvalue = '0x'
SELECT @i = 1
SELECT @length = DATALENGTH (@binvalue)
SELECT @hexstring = '0123456789ABCDEF'
WHILE (@i <= @length)
BEGIN
    DECLARE @tempint
    DECLARE @firstint
    DECLARE @secondint
    SELECT @tempint = SUBSTRING (@binvalue, @i, 2)
    SELECT @firstint = CAST (@tempint AS int)
    SELECT @secondint = CAST (@tempint AS int)
    SELECT @charvalue = @charvalue + RIGHT (@hexstring + @hexstring, 2 - LEN(@charvalue))
    SELECT @i = @i + 2
END
```

h. After run the script, it will create 2 stored procedures

i. Sp_hexadecimal : Converts the passwords in to encryption

ii. Sp_help_revlogin

i. Execute the STORED Procedure in the target,

sp_help_revlogin

150 %

```
Messages
CREATE LOGIN [WIN-793NNF84G1T\SQLDBAs] FROM WINDOWS WITH DEFAULT_DATABASE = [mas
-- Login: SqlDBAs
CREATE LOGIN [SqlDBAs] WITH PASSWORD = 0x020018C8AEC1F194EE4156A5FC7D5324F0160BC
-- Login: Ankam
CREATE LOGIN [Ankam] WITH PASSWORD = 0x0200BC69D6EB9F5C6242B5778348E312B0437D97F
-- Login: Anil
CREATE LOGIN [Anil] WITH PASSWORD = 0x02003354F433439762F2FC15E0F3558655A8F628C7
```

- j. Copy the output message and paste it in the query window of destination server. Run the query.

```
SQLQuery2.sql - WI_Administrator (SQI)*  SQLQuery1.sql - WI_Administrator (SQI)*
-- Login: WIN-793NNF84G1T\Shan
CREATE LOGIN [WIN-793NNF84G1T\Shan] FROM WINDOWS WITH DEFAULT_DATABASE = [master]
-- Login: Mani
CREATE LOGIN [Mani] WITH PASSWORD = 0x020018308F3141C004D300D81FF1ABAB1886C15B121A
-- Login: WIN-793NNF84G1T\SQLDBAs
CREATE LOGIN [WIN-793NNF84G1T\SQLDBAs] FROM WINDOWS WITH DEFAULT_DATABASE = [mas
-- Login: SqlDBAs
CREATE LOGIN [SqlDBAs] WITH PASSWORD = 0x020018C8AEC1F194EE4156A5FC7D5324F0160BC0E
-- Login: Ankam
CREATE LOGIN [Ankam] WITH PASSWORD = 0x0200BC69D6EB9F5C6242B5778348E312B0437D97F8E
-- Login: Anil
CREATE LOGIN [Anil] WITH PASSWORD = 0x02003354F433439762F2FC15E0F3558655A8F628C74E
Completion time: 2021-07-10T21:04:20.8521756-07:00
```

- k. We will get few errors, because already default logins available in the destination server. if you don't want the errors, delete the default login queries.

```

-- Login: WIN-793NNF84G1T\Shan
CREATE LOGIN [WIN-793NNF84G1T\Shan] FROM WINDOWS WITH DEFAULT_DAT...

-- Login: Mani
CREATE LOGIN [Mani] WITH PASSWORD = 0x020018308F3141C004D300D81FF...

-- Login: WIN-793NNF84G1T\SQLDBAs
CREATE LOGIN [WIN-793NNF84G1T\SQLDBAs] FROM WINDOWS WITH DEFAULT_...

-- Login: SqlDBAs

```

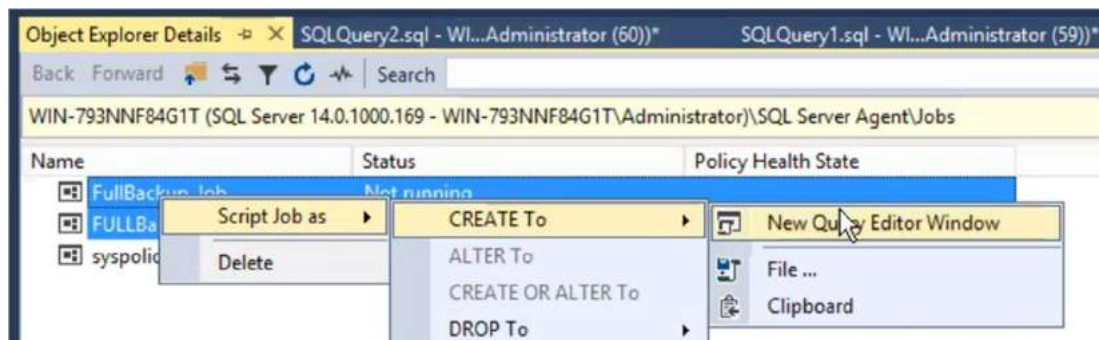
Messages

```

Msg 15025, Level 16, State 2, Line 12
The server principal 'NT SERVICE\SQLWriter' already exists.
Msg 15025, Level 16, State 2, Line 15
The server principal 'NT SERVICE\Winmgmt' already exists.
Msg 15025, Level 16, State 2, Line 18
The server principal 'NT Service\MSSQLSERVER' already exists.
Msg 15025, Level 16, State 2, Line 21
The server principal 'NT AUTHORITY\SYSTEM' already exists.
Msg 15025, Level 16, State 2, Line 24
The server principal 'NT SERVICE\SQLSERVERAGENT' already exists.

```

- l. After this step, same logins will be created in the destination.
- m. Now we have to copy the jobs from source to destination.
- n. Select the jobs, right click , choose script job as, select create to , New query window editor



- o. Query window will open like below, Copy this script and paste the script in the query window of destination server.

```

USE [msdb]
GO

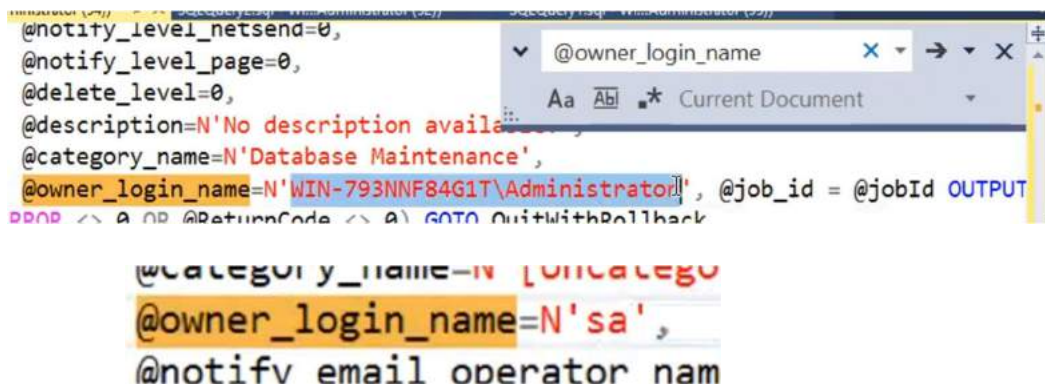
/***** Object: Job [FullBackup_Job]    Script Date: 7/10/2021 9:11:42 PM
BEGIN TRANSACTION
DECLARE @ReturnCode INT
SELECT @ReturnCode = 0
/***** Object: JobCategory [[Uncategorized (Local)]]    Script Date: 7/10/2021 9:11:42 PM
IF NOT EXISTS (SELECT name FROM msdb.dbo.syscategories WHERE name=N'Uncategorized')
BEGIN
EXEC @ReturnCode = msdb.dbo.sp_add_category @class=N'JOB', @type=N'LOCAL', @name=N'Uncategorized'
IF (@@ERROR <> 0 OR @ReturnCode <> 0) GOTO QuitWithRollback

END

DECLARE @jobId BINARY(16)
EXEC @ReturnCode =  msdb.dbo.sp_add_job @job_name=N'FullBackup_Job',
      @enabled=1,
      @notify_level_eventlog=0,
      @notify_level_email=2,
      @notify_level_netsend=0,

```

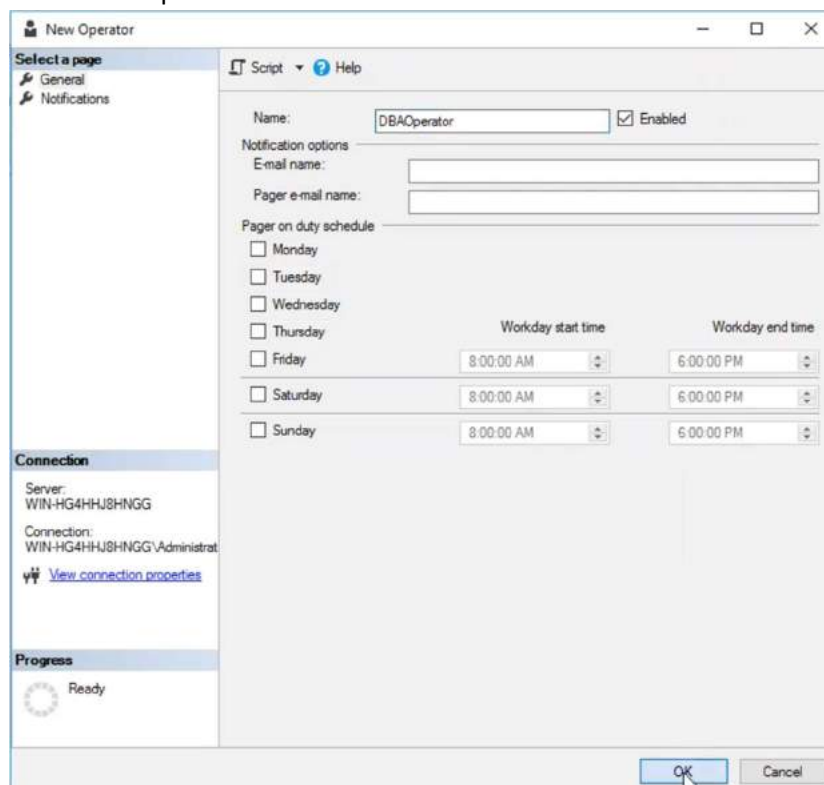
- p. After paste the script, Change the owner name in the script to "sa"



- q. If you have created any job with alerts, you have to create the same notify email operator in the destination server. In the below screenshot, there is a dba operator.

```
@notify_email_operator_name=N'DBAOperator', @job_id = @jobId OUTPUT
:ERROR <> 0 OR (@ReturnCode <> 0) GOTO QuitWithRollback
```

- r. Create the same operator in the destination. Expand sql server agent , right click operators, select new, Create new operator.

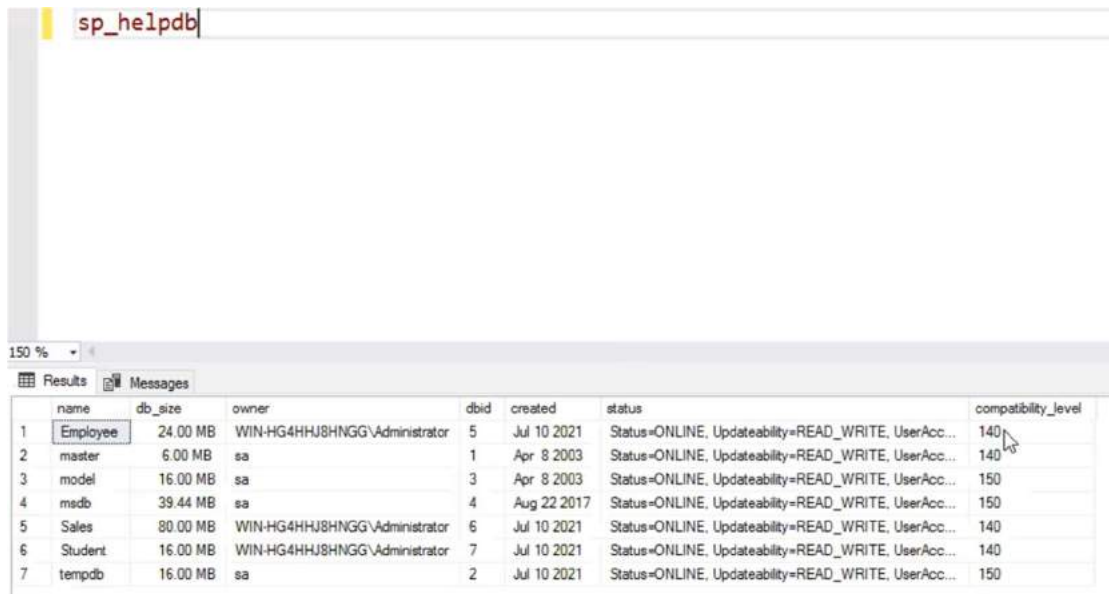


- s. Jobs created in destination server



- t. Now we have to change the compatibility of all the databases in the destination server. Run the stored procedure “sp_helpdb”. It will show the compatibility of databases. Here its showing the compatibility of databases in destination server as 140. This is source compatibility .

sp_helpdb



	name	db_size	owner	dbid	created	status	compatibility_level
1	Employee	24.00 MB	WIN-HG4HHJ8HNGG\Administrator	5	Jul 10 2021	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	140
2	master	6.00 MB	sa	1	Apr 8 2003	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	140
3	model	16.00 MB	sa	3	Apr 8 2003	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	150
4	msdb	39.44 MB	sa	4	Aug 22 2017	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	150
5	Sales	80.00 MB	WIN-HG4HHJ8HNGG\Administrator	6	Jul 10 2021	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	140
6	Student	16.00 MB	WIN-HG4HHJ8HNGG\Administrator	7	Jul 10 2021	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	140
7	tempdb	16.00 MB	sa	2	Jul 10 2021	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	150

- u. Now we have to change the compatibility of databases manually. By using the query

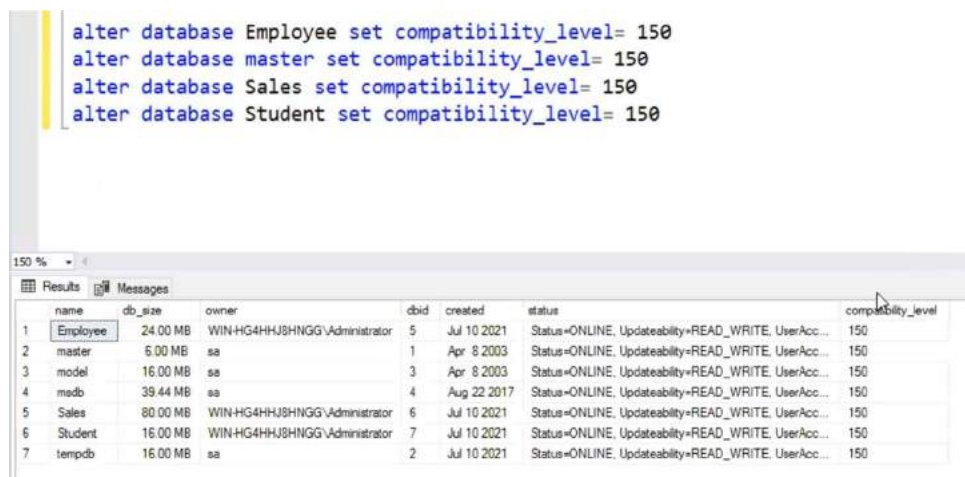
“ALTER DATABASE dbname SET COMPATIBILITY_LEVEL=COMPATIBILITY NUMBER”

Ex:

```
sp_helpdb  
  
alter database Employee set compatibility_level= 150
```

- v. Run the queries and compatibility will be changed.

```
alter database Employee set compatibility_level= 150  
alter database master set compatibility_level= 150  
alter database Sales set compatibility_level= 150  
alter database Student set compatibility_level= 150
```



	name	db_size	owner	dbid	created	status	compatibility_level
1	Employee	24.00 MB	WIN-HG4HHJ8HNGG\Administrator	5	Jul 10 2021	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	150
2	master	6.00 MB	sa	1	Apr 8 2003	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	150
3	model	16.00 MB	sa	3	Apr 8 2003	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	150
4	msdb	39.44 MB	sa	4	Aug 22 2017	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	150
5	Sales	80.00 MB	WIN-HG4HHJ8HNGG\Administrator	6	Jul 10 2021	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	150
6	Student	16.00 MB	WIN-HG4HHJ8HNGG\Administrator	7	Jul 10 2021	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	150
7	tempdb	16.00 MB	sa	2	Jul 10 2021	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	150

- w. Migration completed.
- x. Don't Decommission of old server Immediately. Wait for 2 to 3 days, if there is no issues we can decommission.

101. How to setup Active Directory in the windows server?

Ans. Active directory setup is windows team work. DBA won't create any active directory.

To configure the HADR , We need a domain, So, we are creating the Active directory setup.

Every user in the organization is the part of Active Directory only. If any user want to access the server, Logins will be created in the active directory. In the active directory , we have different groups. Each group has a group policy. Policy nothing but permissions.

When a new person joined in a company, a login will be created in the active directory setup and they will set the permissions to access the servers, which they will work. With the help of active directory, A person can access all the servers with out created any new logins.

Based on the groups in the Active directory, user can access to the only those particular servers.

Setting the Active directory setup n my local machine to practice:

Step 1 : Click on add roles and features

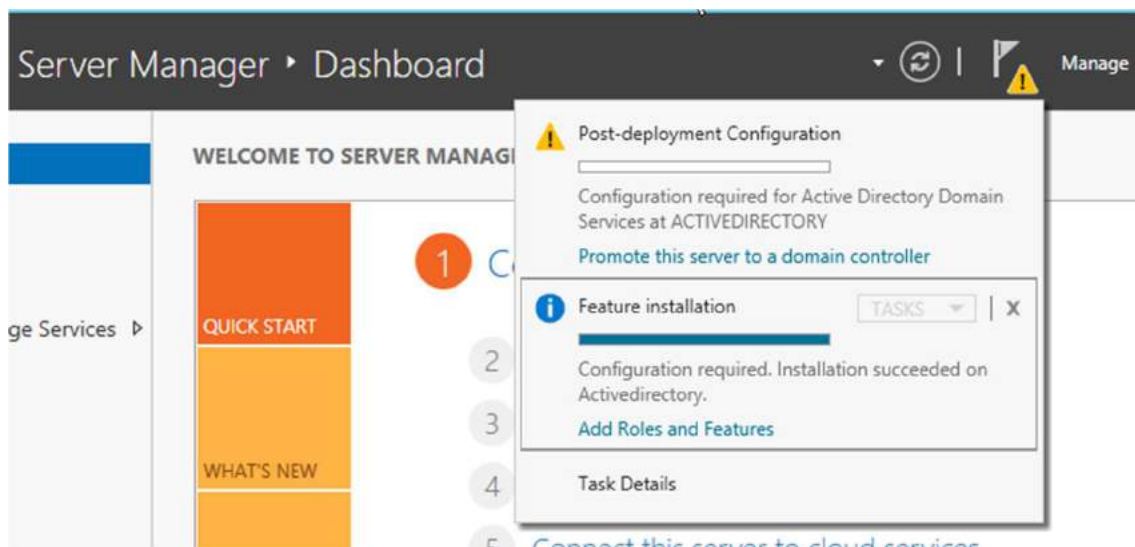
Step 2 :Choose role based , click next

Step 3 : select active directory domain services, Click next

Step 4 : Click install.

Step 5 : Now we have to configure active directory setup after the successful installation.

Step 6 :Click on the FLAG symbol and promote the server to a domain controller



Step 7: After this , select add a new forest and give the name of the domain like IBM.COM, WIPRO.COM etc, CLICK NEXT

Select the deployment operation

☐ Add a domain controller to an existing domain

☐ Add a new domain to an existing forest

☒ Add a new forest

Specify the domain information for this operation

Root domain name:

Step 8 : Enter the new domain password, Click Next.

Step 9 : enter the net bios name – What ever we mentioned in the root domain name in the above screenshot, Enter the same name with out '.com' Like IBM. It will pick automatically. Click Next.

Step 10 : Install

102. What are the steps to follow to create continuous connections between Active directory , NODEs?

Ans.

Step 1 : Turn off the Windows Firewall in the server

Step 2 : Create a Manual IP Address . Open network connections by using RUN " ncpa.cpl". Right click on Ethernet, go to properties, Select IPV4 , click properties, In the general tab- Choose 'use the following IP Address", and enter the IPADDRESS like below,

Give different IP address for Active Directory and Nodes

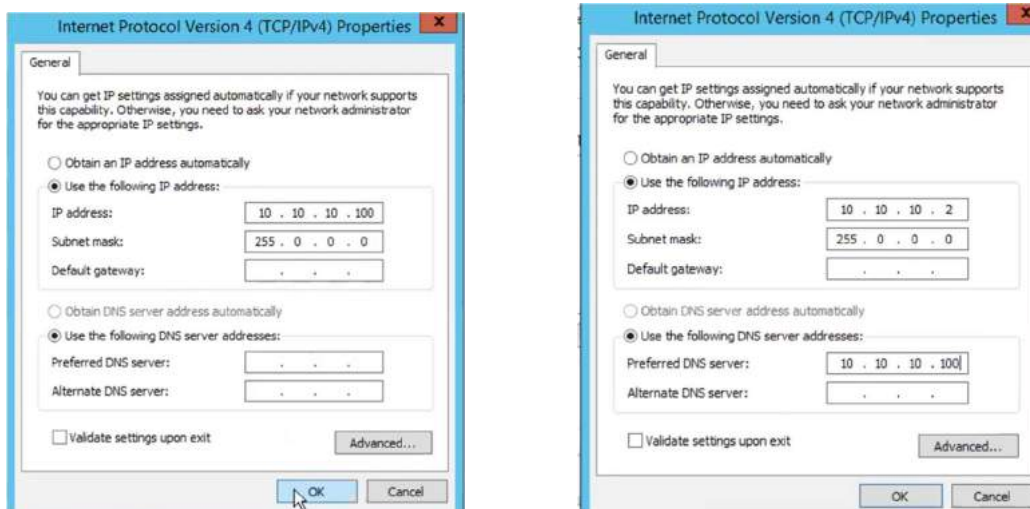
Ex : Active Directory : 10.10.10.100

Node1: 10.10.10.101 , Node 2 : 10.10.10.102

Once we fixed the IP address, then servers can communicate always each other.

We have to fill preferred DNS server as "Active directory IP address" for all NODES.

No need for Active directory.



Step 3 : Change the computer name as "ActiveDirectory" , "NODE1" , "NODE2". In the Workgroup column of NODE1 and NODE2 give the IBM.COM as group and click ok. Now NODE 1 & NODE2 will be added in the Active Directory.

103. How to change the location of user database in SQL Server?

Ans. To change the location of a database, we have to move datafile & log file of a database.

Step 1 : Run the query "select * from Sys.sysaltfiles" to know the name & location of the datafile, Logfile of each database.

Step 2 : copy the path of datafile & logfile to copy files from old location to new location in the last step.

Step 3 : Write the Alter Query for moving data file , "ALTER DATABASE dbname MODIFY FILE (NAME='DATAFILENAME' , FILENAME='NEW PATH.MDF')

Write the Alter Query for moving LOG file , "ALTER DATABASE dbname MODIFY FILE (NAME='LOGFILENAME' , FILENAME='NEW PATH.LDF')

Execute.

Step 3: Now path of the database will be changed and we can check in "SELECT * FROM SYS.SYSAULTFILES"

Step 4 : Now keep the database offline.

Step 5 : Now copy the files physically from old location to New location.

Step 6 : Now bring the database online

104. How to setup log shipping?

Ans. In log shipping , A database is going synchronize between the servers. Log shipping can be done up to the Database level only. We have to maintain same data between the servers.

Simply Log shipping means, Shipping of the LOG backups from primary server to secondary server.

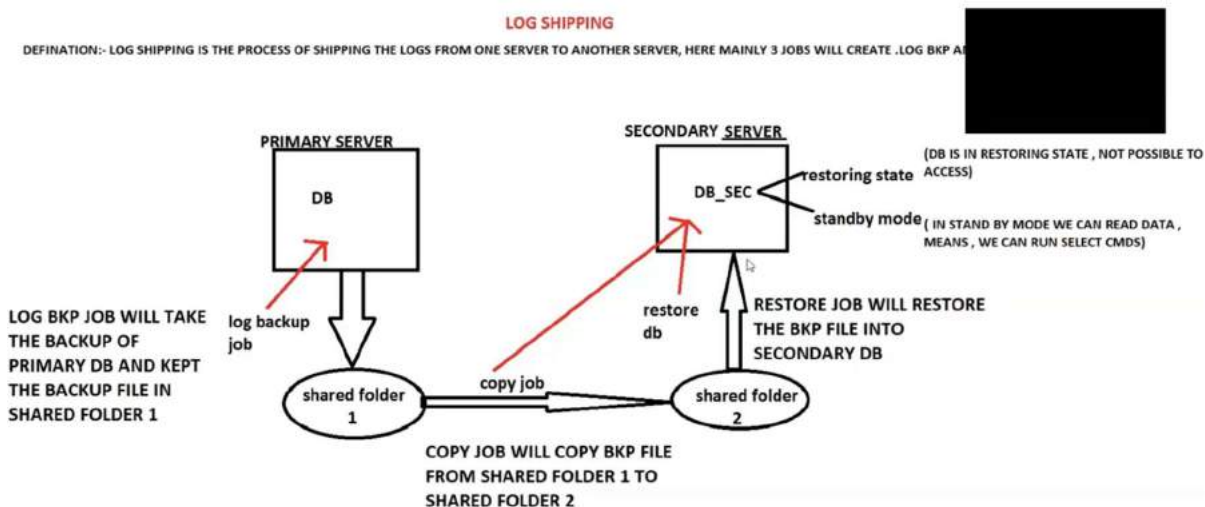
Log shipping works based on the backups. We can keep 4 secondary servers for a single database.

In Log shipping , We can restore the backup in the secondary server with No Recovery and Standby mode.

With No recovery, we cant access the database. With Stand BY, We can read the data of the database

We will have 3 jobs in the log shipping:

- Backup job in Primary server
- Copy Job in secondary server
- Restore job in secondary server

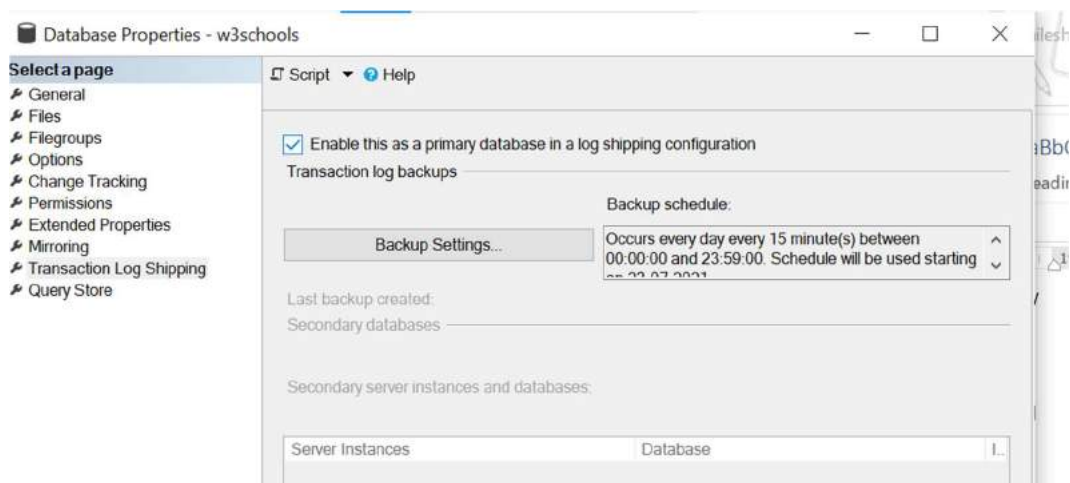


To setup the Log Shipping, We need the Shared folder between the servers. One is in Primary and one is in secondary server. Below are the steps to create shared folder,

Step 1 : Create a folder, Right click properties, go to share tab, Click Share, Type everyone, Click add, permission level change to Read/Write, Click ok. To know the shared folder lists, We can use the RUN command “\\Computename

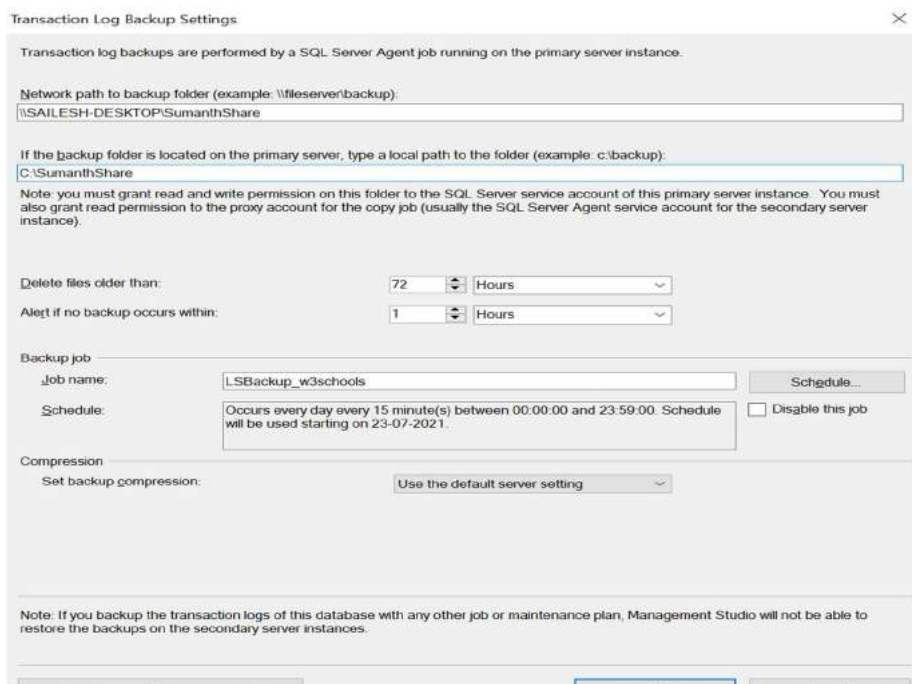
Below are the steps to setup log shipping,

Step 1 : Open the primary server, Select the database which you want to log shipping, Right click, go to properties, select Transact log shipping option, Enable this as primary database in log shipping configuration.



Step 2 : Click on backup settings, Enter the Shared folder address in the network path , Enter the local path Shared folder address in the second column. Setting the automatic deletion of older backups. Schedule the back job time. Click ok.

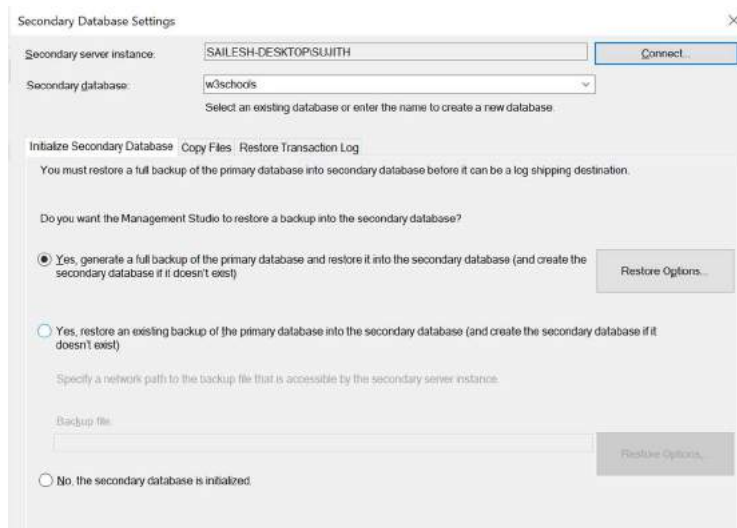
Every job name in log shipping starts with “LS” only Like LSBATCHUP_Databasename



The 'Transaction Log Backup Settings' dialog box is shown. It contains the following fields and options:

- Network path to backup folder (example: \\filesrvr\backup):** \\SAILESH-DESKTOP\SumanthShare
- If the backup folder is located on the primary server, type a local path to the folder (example: c:\backup):** C:\SumanthShare
- Note:** you must grant read and write permission on this folder to the SQL Server service account of this primary server instance. You must also grant read permission to the proxy account for the copy job (usually the SQL Server Agent service account for the secondary server instance).
- Delete files older than:** 72 Hours
- Alert if no backup occurs within:** 1 Hours
- Backup job:**
 - Job name:** LSBATCHup_w3schools
 - Schedule:** Occurs every day every 15 minute(s) between 00:00:00 and 23:59:00. Schedule will be used starting on 23-07-2021.
 - Schedule...** button
 - Disable this job** checkbox (unchecked)
- Compression:** Set backup compression: Use the default server setting
- Note:** If you backup the transaction logs of this database with any other job or maintenance plan, Management Studio will not be able to restore the backups on the secondary server instances.

Step 3 : Add the secondary server instance by clicking add option, Click connect, select the secondary server, login



The 'Secondary Database Settings' dialog box is shown. It contains the following fields and options:

- Secondary server instance:** SAILESH-DESKTOP\SQLITH
- Connect...** button
- Secondary database:** w3schools
- Select an existing database or enter the name to create a new database.**
- Initialize Secondary Database / Copy Files / Restore Transaction Log** tabs
- You must restore a full backup of the primary database into secondary database before it can be a log shipping destination.**
- Do you want the Management Studio to restore a backup into the secondary database?**
 - ☒ Yes, generate a full backup of the primary database and restore it into the secondary database (and create the secondary database if it doesn't exist)
 - ☐ Yes, restore an existing backup of the primary database into the secondary database (and create the secondary database if it doesn't exist)
- Restore Options...** button
- Specify a network path to the backup file that is accessible by the secondary server instance.**
- Backup file:**
- Restore Options...** button
- ☐ No, the secondary database is initialized.

Step 4 : Choose the option based in requirement,

- If we don't have created the same database in the secondary server and we don't have full back up of primary server choose the 1st option
- If we don't have created the same database in the secondary server and we have full back up of primary server choose the 2nd option

c. If we already restored the database in the secondary server, choose the 3rd option.

The screenshot shows the 'Secondary Database Settings' dialog box with the 'Initialize Secondary Database' tab selected. The 'Secondary server instance' is 'SAILESH-DESKTOP\SUJITH' and the 'Secondary database' is 'w3schools'. The 'Restore Options...' button is visible. The 'Yes, generate a full backup of the primary database and restore it into the secondary database (and create the secondary database if it doesn't exist)' option is selected. A message states: 'You must restore a full backup of the primary database into secondary database before it can be a log shipping destination.' Below this, it asks 'Do you want the Management Studio to restore a backup into the secondary database?' with the same 'Yes' option selected. A 'Backup file' field is present with a 'Restore Options...' button. At the bottom, the 'No, the secondary database is initialized.' option is also visible.

Step 5 :go to copy files tab, and give the shared folder address or local secondary shared address folder where we want to copy the backups. Choose the time period to delete the old backup files. Schedule the Job timing.

The screenshot shows the 'Secondary Database Settings' dialog box with the 'Copy Files' tab selected. The 'Destination folder for copied files' is set to '\\SAILESH-DESKTOP\Sujithshare'. The 'Delete copied files after' is set to 72 hours. The 'Copy job' section shows a job name 'LSCopy_SAILESH-DESKTOP\SUMANTH_w3schools' and a schedule 'Occurs every day every 15 minute(s) between 00:00:00 and 23:59:00. Schedule will be used starting on 23-07-2021.' The 'Schedule...' button is visible. The 'OK' button is at the bottom right.

We can give [\\SAILESH-DEKSTOP\Sujithshare](#) or

C:\Sujithshare

Step 6 : Go to restore transaction log Tab, Select the restoring mode whether No recovery or Stand by. Tick the Disconnect users when selected stand by. Schedule the restore job. Click ok.i

Secondary Database Settings

Secondary server instance: SAILESH-DESKTOP\SUJITH Connect...

Secondary database: w3schools
Select an existing database or enter the name to create a new database.

Initialize Secondary Database Copy Files **Restore Transaction Log**

Files are restored from the destination folder by a SQL Server Agent job running on the secondary server instance.

Database state when restoring backups:

☒ No recovery mode

☐ Standby mode

☐ Disconnect users in the database when restoring backups

Delay restoring backups at least: 0 Minutes

Alert if no restore occurs within: 45 Minutes

Restore job

Job name: LSRestore_SAILESH-DESKTOP\SUMANTH_w3schools Schedule...

Schedule: Occurs every day every 15 minute(s) between 00:00:00 and 23:59:00. Schedule will be used starting on 23-07-2021. ☐ Disable this job

Step 7 : If we have another server, we can choose monitor server also.

Step 8 : Click ok. Log shipping will be created with 3 jobs – 1. backup Job in primary server, 2. Copy & Restore Job in Secondary server.

Database Properties - w3schools

Select a page: General Files Filegroups Options Change Tracking Permissions Extended Properties Mirroring **Transaction Log Shipping** Query Store

Script Help

☒ Enable this as a primary database in a log shipping configuration

Transaction log backups

Backup schedule: Occurs every day every 15 minute(s) between 00:00:00 and 23:59:00. Schedule will be used starting on 23-07-2021.

Last backup created:

Secondary databases

Secondary server instances and databases:

Server Instances	Database
SAILESH-DESKTOP\SUJITH	w3schools

Add... Remove...

Monitor server instance

☐ Use a monitor server instance

Monitor server instance: Settings

This action will script the entire log shipping configuration. Script Configuration

OK Cancel

Other important information of log shipping:

The .TUF & .WRK files in Log Shipping

.TUF File

.TUF file is the Transaction Undo File, which is created when performing log shipping to a server in Standby mode.

When the database is in Standby mode the database recovery is done when the log is restored; and this mode also creates a file on destination server with .TUF extension which is the transaction undo file.

This file contains information on all the modifications performed at the time backup is taken.

The file plays a important role in Standby mode... the reason being very obvious while restoring the log backup all uncommitted transactions are recorded to the undo file with only committed transactions written to disk which enables the users to read the database. So when we restore next transaction log backup; SQL server will fetch all the uncommitted transactions from undo file and check with the new transaction log backup whether committed or not.

If found to be committed the transactions will be written to disk else it will be stored in undo file until it gets committed or rolledback.

.WRK File

The .wrk files are produced when the transaction log backups are copied from the backup location to the secondary server (by the agent job on the secondary). The files are named .wrk during the copy operation; when they have been completely copied they are renamed to have the .trn extension. The temporary naming using the .wrk extension ensures that the files are not picked up by the restore job until successfully copied.

So, if the .wrk file you are seeing is there all the time it is likely to be an old copy that failed for some reason. So for normal operation, you'll see a .wrk file when each transaction log backup file is copied and then that will be renamed to a .trn file.

104. What are steps to failover the Log Shipping?

Ans. Fail over means , When primary server is down bringing the secondary server online is called failover.

Below are steps to failover the log shipping,

Step 1 : Took the Tail log backup of primary server database

Step 2 : Stop the restore Job in the secondary server

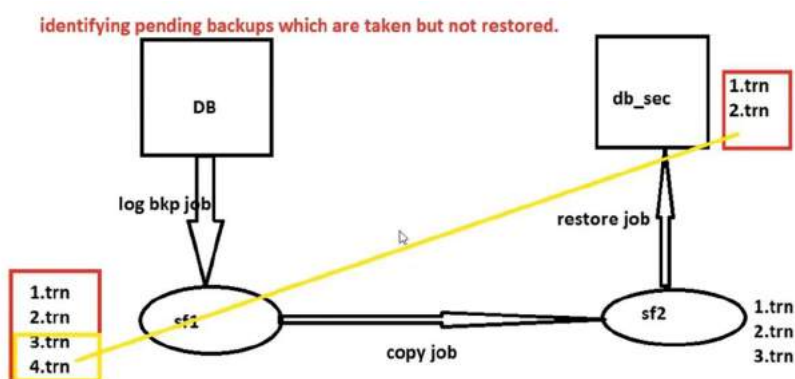
Step 3 : Check what are the backups pending to restore in the secondary server from primary server

Step 4 : Restore the pending backups

Step 5 : Restore the tail log backup with Recovery.

Step 5 : Secondary server database will come online.

Step 6 : Now will do reverse log shipping- Secondary server will acts as primary, Primary server acts a secondary



failover steps or process to bring sec online

1. take tail log bkp.
--> stop restore job..
2. we have to manually restore the backups, which are taken and but not restored on the sec side.
3. finally restore the tail log backup with rec.

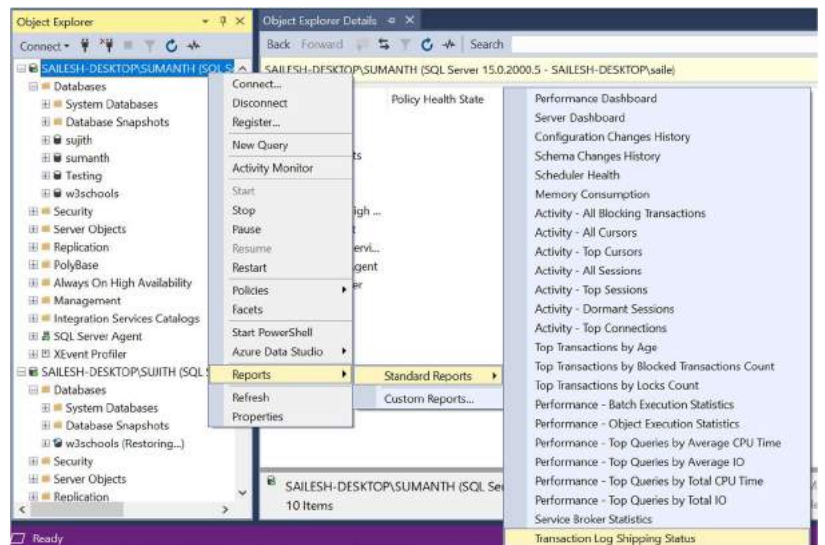
105. How to check the status of the Log Shipping ?

Ans. Right Click on the server , Go to rep orts, Choose Standard Reports, select Transaction log Shipping status.

If we select primary server log shipping status it will show only Backup job status.

If we select Secondary server log shipping status it will show copy job and restore job status.

Transaction Log Shipping Status				
SAILESH-DESKTOP\SUMANTH				
This report shows the status of log shipping configurations for which this server instance is a primary, secondary, or monitor.				
Status	Primary Database -- Secondary Database	Time Since Last	Threshold	Alert Enabled
Good	[SAILESH-DESKTOP\SUMANTH] [w3schools]	14 min	60 min	True
	-- [SAILESH-DESKTOP\SUWITH] [w3schools]			



106. How to connect multiple instances of different windows server machines in a single windows server?

Ans. By activating the SQL Server Browser services, we can connect.

Below are the steps,

Step 1 : Open SQL Server configuration manager

Step 2 : Go to SQL Server network configuration, Enable the TCP/IP.

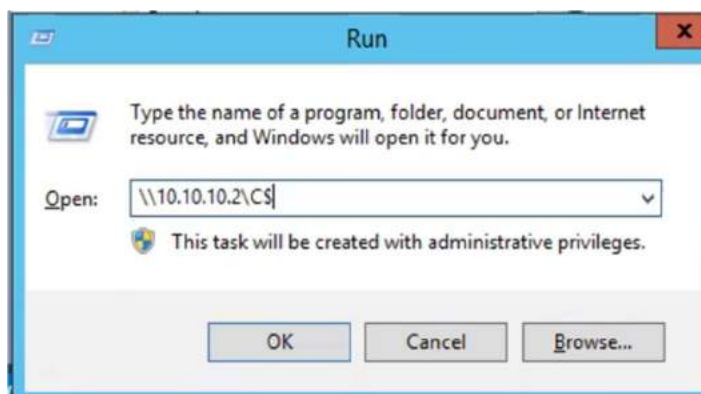
Step 3 : GO to SQL Server Services , Start the SQL Server browsing services.

107. In which edition , we don't have sql server agent?

Ans. Express edition. Log shipping also not possible.

108. How to access a folder of another windows server in our computer?

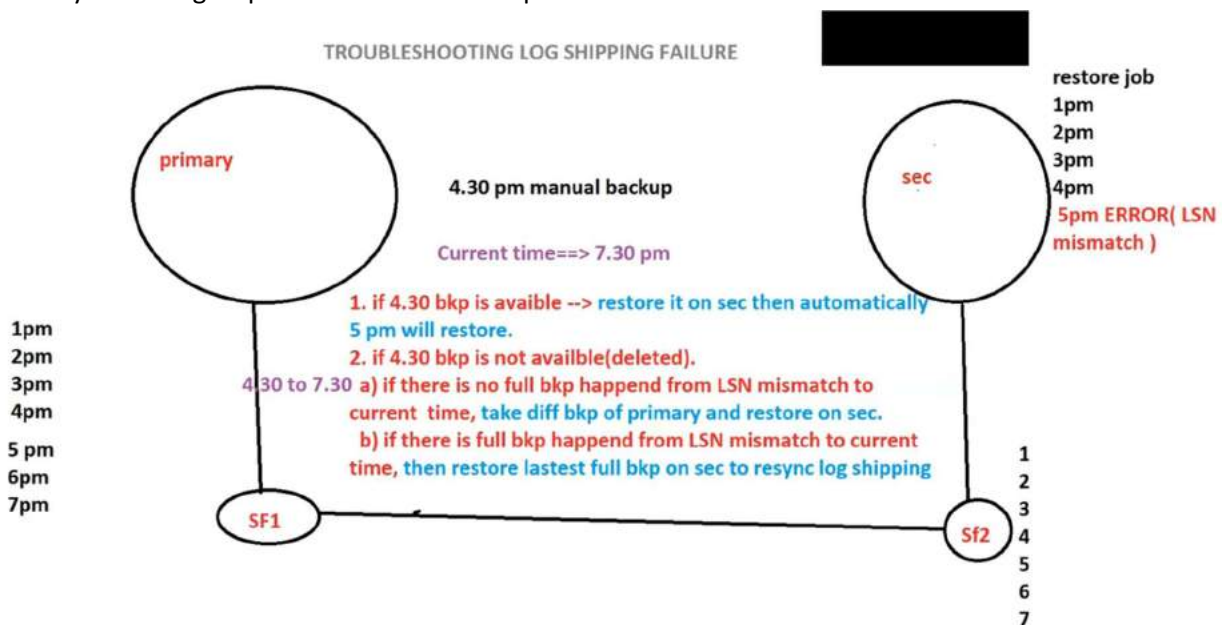
Ans. By using RUN command \\IP Address\drive letter\$ or \\computername\drive letter\$



109. How to Troubleshoot log shipping failure?

Ans. In what cases, Log shipping will break.

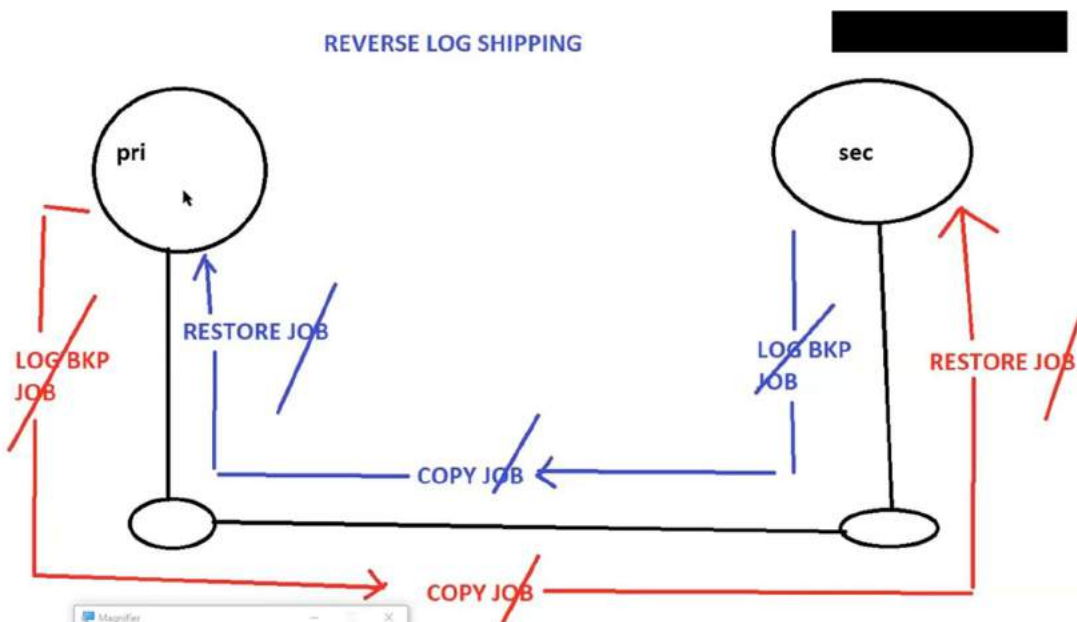
- When the LSN number mismatch.
- When any new file group created with manual path



110.

What is reverse log shipping?

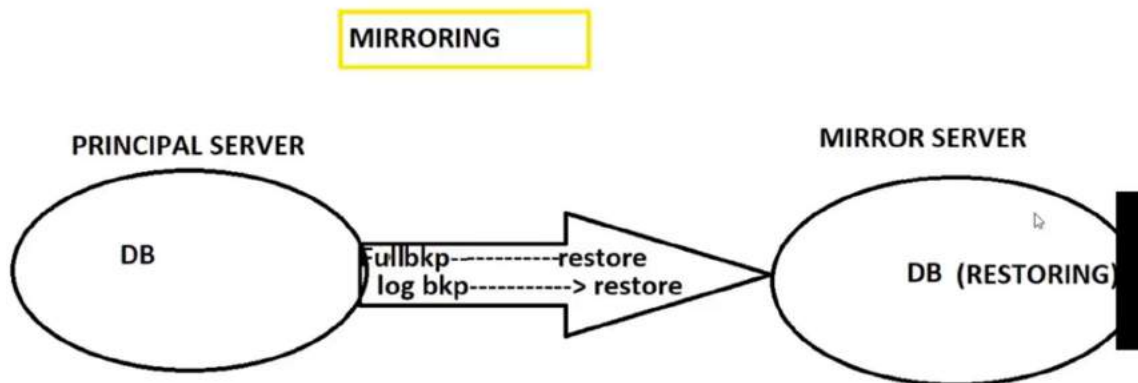
Ans. Changing the Secondary server to primary and primary server to secondary. Reverse log shipping will do when there is failover.



111. What is Mirroring?

Ans. Maintaining the same copy of database in another service is called Mirroring. To configure the Mirroring, version and edition should be same in both the servers. To configure the mirroring, Database should be in Full recovery model. Servers should run in the same service account. We can use different service accounts but using same service account for both the servers is good practice. We have to keep same database name in the both the servers. In Mirroring, we have only NO Recovery mode for restoring. We don't have stand by mode used in Log shipping. So, We cant able to read the data of the secondary server. In Mirroring, Primary server called as principal and Secondary server called as Mirror. Mirroring can be created only to the 10 databases of a sql server.

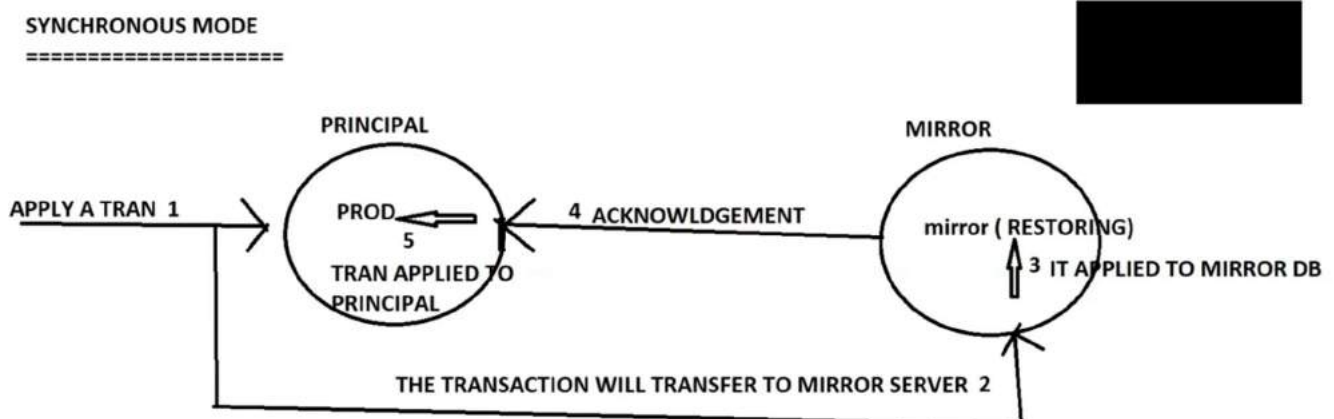
We don't have jobs in mirroring, Data will be transferred to another server by using 2 modes. Synchronous mode and Asynchronous mode.



Initial data will transfer with the help of full and log bkp...when we modify data the later data will transfer by using the mode of the mirroring.

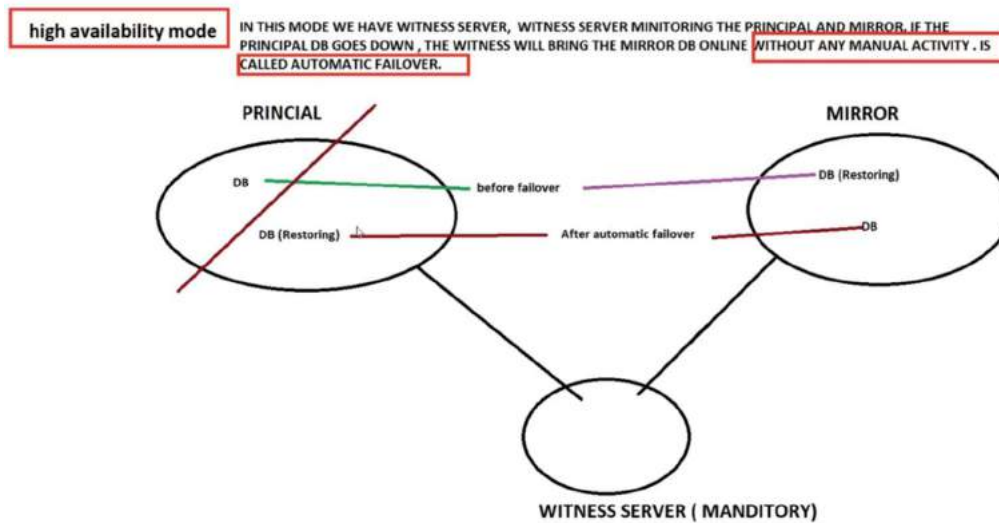
1. Synhronous mirroring.
2. Asynchronous mirroring.

a. Synchronous Mode : If we run any transaction in the principal server, first the transaction will go to mirror server, mirror server sends back the acknowledgement and then the same transaction will apply on principal. Automatic fail over can be done in synchronous mode only. Always both databases sync is possible.

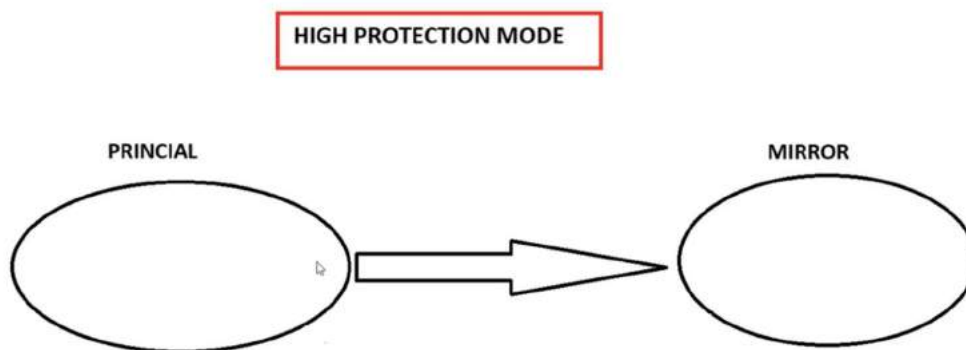


In Synchronous mode , we have 2 modes.

1) High Availability mode : In this we have 3rd server called witness server, which helps to automatic fail over of servers. Witness servers always monitor the principal and Mirror, if principal goes down- witness brings the mirror server online with out any manual activity. This automatic process is called Automatic failover.



2) High protection mode : In this we don't have witness server, So automatic failover not possible. When principal server goes down, We have to fail over manually from Principal to Mirror.



THERE IS NO WITNESS SERVER, MEANS THERE IS NO AUTOMATIC FAILOVER.
IN THIS MODE IF THE PRINCIPAL IS GOES DOWN, WE HAVE TO MANUALLY BRING THE MIRROR ONLINE

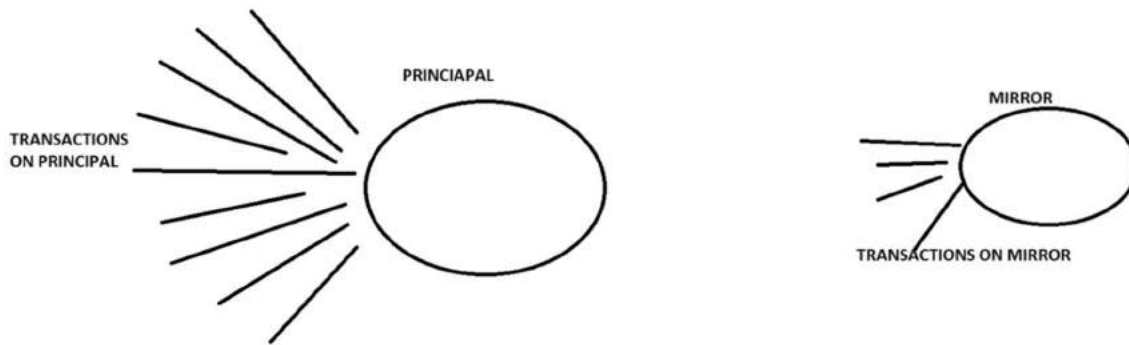
Asynchronous Mode : If we run any transaction in the principal server ,it apply on the principal first and then transaction applied on the mirror server. Automatic fail over not possible in asynchronous mode. In this case, with out verification, whether transaction applied in the mirror or not – Transactions will be applied continuously on the principal. In this mode, there is data difference. Chances data loss will be there. Asynchronous mode is faster than Synchronous mode.

In Asynchronous mode, we have only High performance mode.

1) High performance mode : In this mode, we don't have automatic fail over .

ASYNCHRONOUS MIRRORING

THE PRINCIPAL WILL KEEP ON ACCEPTING THE TRANSACTIONS... WITHOUT VERIFYING THE TRANSACTION ARE APPLIED TO MIRROR DB, IN THIS MODE THERE IS A DATA DIFFERENCE, MEANS THERE IS POSSIBLE DATA LOSS.



112. how to setup Mirroring?

Ans. Below are the steps ,

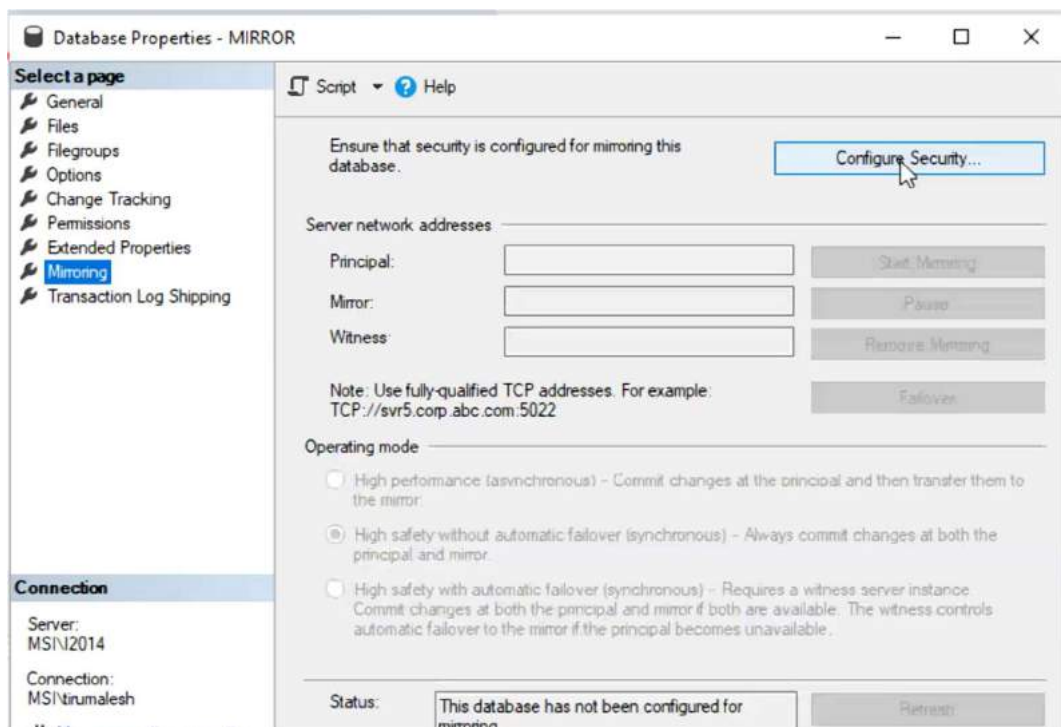
Step 1 : Take the Full and log backup of Principal server

Step 2 : Keep the same service account for principal , Mirror & Witness servers.

Step 3 : Give the SYSADMIN permissions to the service account in the all the 3 servers.

Step 3 : Restore Principal server database backups in Mirror server with NO recovery.

Step 4 : Right click on the principal server database, go to properties, select mirroring.



Step 5 : Click on Configure security, Click next, if client is requested for automatic failover, then only select yes for witness server, Click next.

Include Witness Server
Specify whether to include a witness server in the security configuration.

To operate database mirroring in synchronous mode with automatic failover, you must configure a witness server instance to monitor the status of the principal and mirror server instances and control the failover.

Do you want to configure security to include a witness server instance?

☒ Yes
☐ No

Help < Back **Next >** Finish >>| Cancel

Step 6 : It will ask for the port number, The default port number for mirroring is 5022. This number comes automatically no need to change or update. Click Next.

Principal Server Instance
Specify information about the server instance where the database was originally located.

Principal server instance:
SAILESH-DESKTOP\SUMANTH

Specify the properties of the endpoint through which the principal server instance will accept connections from the mirror and witness server instances:

Listener port: ☒ Encrypt data sent through this endpoint

Endpoint name:

NOTE: If the principal, mirror or witness are instances on the same server, their endpoints must use different ports.

Help < Back **Next >** Finish >>| Cancel

Step 7 : Now connect to mirror instance, here the port is 5023. Click Next.

The screenshot shows the 'Mirror Server Instance' step of the 'Configure Database Mirroring Security Wizard'. The window title is 'Configure Database Mirroring Security Wizard'. The main heading is 'Mirror Server Instance' with a subtext 'Specify information about the server instance where the mirror copy of the database will be'. Below this, there is a dropdown menu for 'Mirror server instance:' showing 'SAILESH-DESKTOP\SUJITH' and a 'Connect...' button. A section titled 'Specify the properties of the endpoint through which the mirror server instance will accept connections from the principal and witness server instances:' contains a 'Listener port:' field with '5023', an 'Endpoint name:' field with 'Mirroring', and a checked checkbox for 'Encrypt data sent through this endpoint'. A note states: 'NOTE: If the principal, mirror or witness are instances on the same server, their endpoints must use different ports.' At the bottom are buttons for 'Help', '< Back', 'Next >', 'Finish >>|', and 'Cancel'.

Step 8 : Now connect to Witness instance, here the port is 5024. Click Next.

The screenshot shows the 'Witness Server Instance' step of the 'Configure Database Mirroring Security Wizard'. The window title is 'Configure Database Mirroring Security Wizard'. The main heading is 'Witness Server Instance' with a subtext 'Specify the server instance that monitors the status of the principal and mirror server instances'. Below this, there is a dropdown menu for 'Witness server instance:' showing 'MSI\I2014' and a 'Connect...' button. A section titled 'Specify the properties of the endpoint through which the witness server instance will accept connections from the principal and mirror server instances:' contains a 'Listener port:' field with '5022', an 'Endpoint name:' field with 'Mirroring', and a checked checkbox for 'Encrypt data sent through this endpoint'. A note states: 'NOTE: If the principal, mirror or witness are instances on the same server, their endpoints must use different ports.' At the bottom, there is a yellow warning box with a triangle icon and the text: 'The principal and witness server instances cannot be the same instance of SQL Server. Select another instance as the witness server instance.' Below the warning box are buttons for 'Help', '< Back', 'Next', 'Finish >>|', and 'Cancel'. A mouse cursor is pointing at the 'Next' button.

Step 9 : It will ask for the service account names for the all the 3 servers. If we using same service account for all the 3 servers at the beginning, no need to give any thing here. If different service accounts, then only we have to give the name of the service account. Click next.

The screenshot shows the 'Configure Database Mirroring Security Wizard' window at the 'Service Accounts' step. The title bar reads 'Configure Database Mirroring Security Wizard'. The main heading is 'Service Accounts' with the instruction 'Specify the service accounts of the server instances.' Below this, a note states: 'For SQL Server accounts in the same domain or trusted domains, specify the service accounts below. If the accounts are non-domain accounts or the accounts are in untrusted domains, leave the textboxes empty.' The section 'Service accounts for the following instances:' contains three textboxes: 'Principal:', 'Witness:', and 'Mirror:'. At the bottom, there are buttons for 'Help', '< Back', 'Next >' (which is highlighted with a mouse cursor), 'Finish >>|', and 'Cancel'. A small padlock icon is visible in the top right corner of the wizard area.

Step 10 : Click Finish .

The screenshot shows the 'Configure Database Mirroring Security Wizard' window at the 'Configuring Endpoints' step. The title bar reads 'Configure Database Mirroring Security Wizard'. The main heading is 'Configuring Endpoints' with the instruction 'Click Stop to interrupt the operation.' Below this, a progress bar shows the operation is complete. A summary table indicates: 2 Total, 0 Error, 2 Success, 0 Warning. A 'Details:' section contains a table with the following data:

Action	Status	Message
Configuring endpoint on principal server(SAIL...	Success	
Configuring endpoint on mirror server(SAILES...	Success	

At the bottom of the wizard area, there are buttons for 'Stop' and 'Report'. A 'Close' button is located at the bottom right of the window.

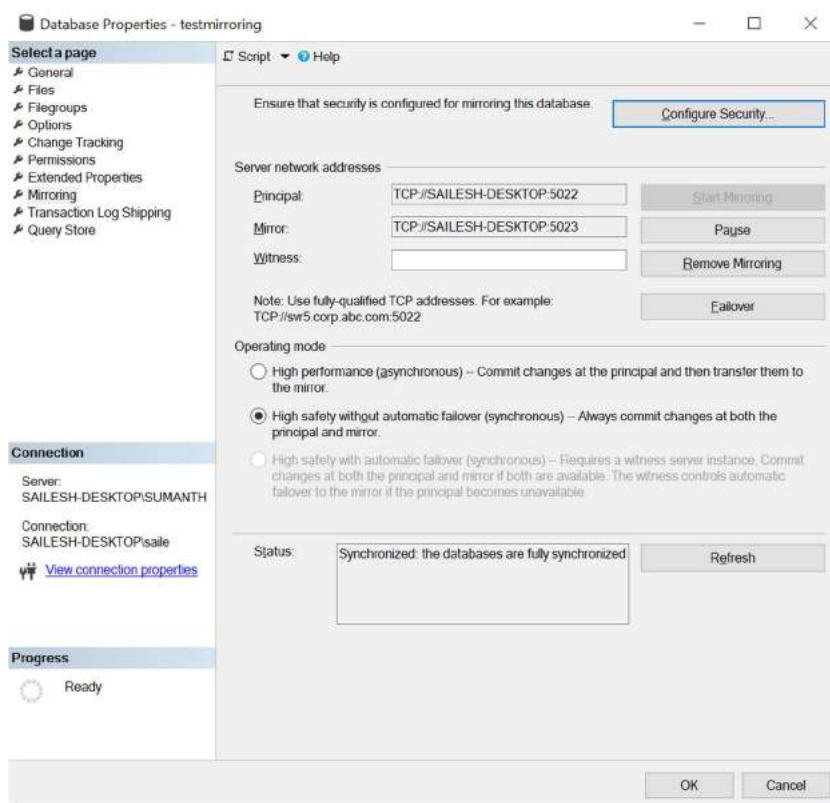
113. How to fail over the Mirroring?

Ans. When the mirroring created the witness server, we need to do failover. Witness server automatically failover the mirroring with the help of synchronous high availability mode.

If there is no witness server, we have to manually failover the mirroring. Below are the steps,

Step 1 : Right Click on principal database, go to properties, select mirroring.

Step 2 : Click on failover

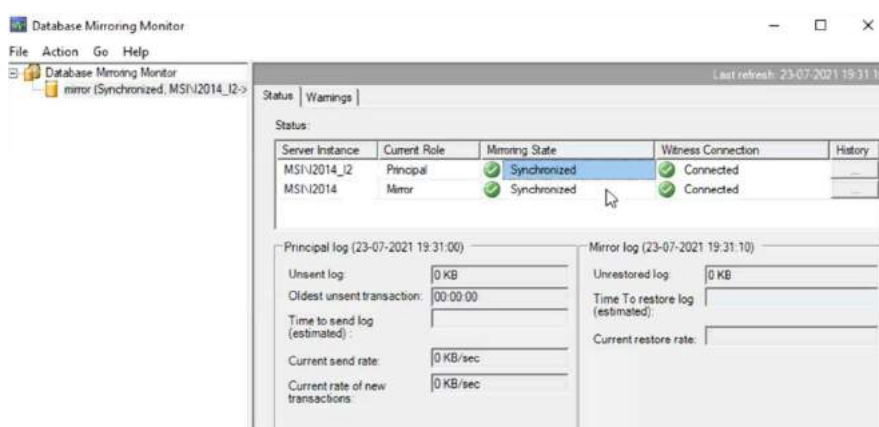


114. How to monitor the Mirroring?

Ans. Right Click on Principal Database, go to tasks, select launch database mirroring

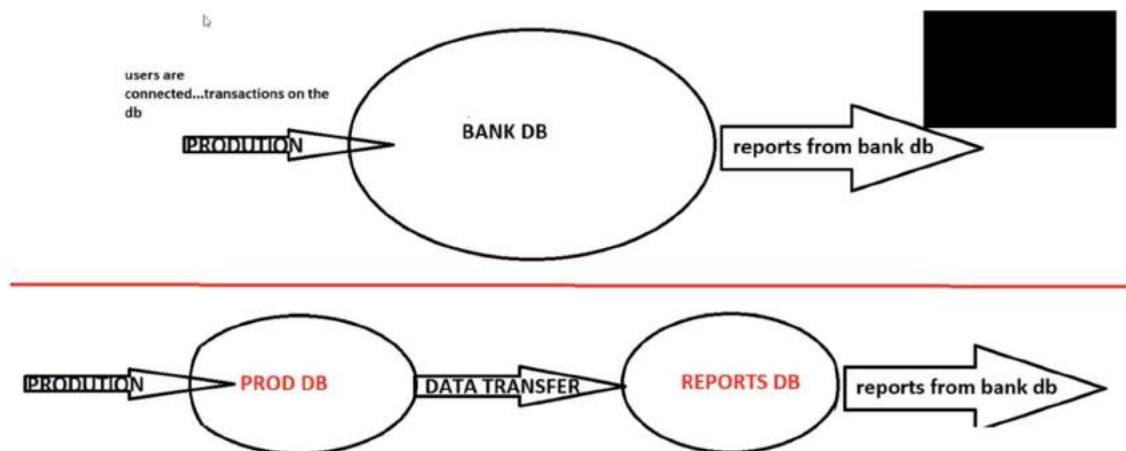


Below is the mirroring monitor



115. What is Replication?

Ans. Replication is the concept of transferring the data from one server to multiple servers. It mainly used for reporting purpose. In replication, we can transfer all tables or only few tables or only few columns from few tables. Replication is a object level concept. Replication increase the performance of the database. Replication creates a same database in destination server, Once server used for production and one server used for reports. Below is the example.



In Log Shipping, With the help of Stand by mode- We can take the reports but not possible to take continuously.

In mirroring, There is no concept of taking reports. Not at all possible.

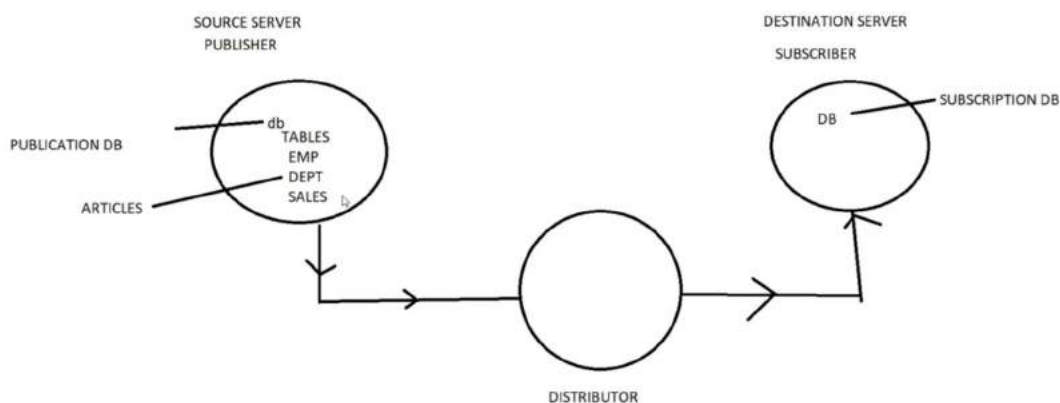
In Replication, we can take the reports continuously.

Log shipping & Mirroring is database level concept, Where replication is table or object level concept.

Replication is not a high availability concept. Why because, we wont transfer all the tables or objects of the data to the secondary server and we won't use replication for High availability. Here, we cant use secondary server for failover. Why because, We don't have all the data of primary server in the secondary server.

In replication source server called as Publisher and destination server called as Subscriber. In between the primary and secondary, We use another server called Distributor.

- a. Publisher – Source server- The database which we used for transfer the data is called Publication database.
 - b. Distributor – Acts as a mediator b/w Publisher and Subscriber- it receives the data from publisher and transferred to Subscriber.
 - c. Subscriber – Destination Server- The database which we used to receives the data is called subscription database.
- In Replication, We called tables as Articles.



In Log shipping & mirroring, we took the backup of primary server database and restored in secondary server to initiate the process. Where in replication By using snapshot process the initial data from publisher will be transferred to Subscriber.

Snap shot process scripts all the objects from publisher and will be executed the scripts at subscriber. By using BCP(Bulk Copy Process) the data will transfer from publisher to subscriber.

Snapshot will create tables in the subscriber without data. BCP transfers the data to the tables.

Snapshot process will create only the empty tables in the subscriber and with help of BCP data will be transferred to the tables.

116. What are types of Replications?

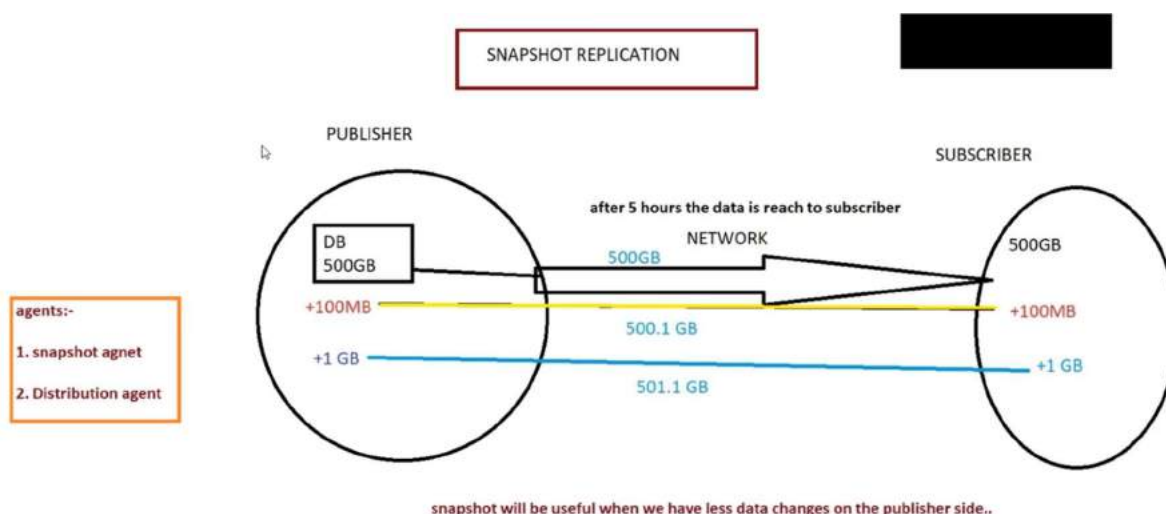
Ans. There are 4 types of replications ,

- a. Snapshot Replication
- b. Merge Replication
- c. Transaction Replication

a. Snapshot Replication : it will take entire data as an image and then it will transfer from publisher to subscriber. This will be used when we have less data changes in publisher side. In snapshot replication, Data will be transferred via Network(Internet).

Below is the example :

A database which contains 500 gb data. We took the snapshot replication and transferred the data to subscriber with the network help. Another 100 MB of data has been added to the publisher. Now again we took the snapshot replication, Now snapshot replication will take again the entire database image instead of 100MB. This time , 1 GB data has been added to the publisher. We run the snapshot replication, it took the entire database image again 501.1GB and transferred to subscriber. Here , we have an disadvantage with the snap shot replication is every time snap shot replication took entire database as image when ever changes in the publisher. It took all the resources of network every time. The other process will be slow because of this. That's why, We use the Snapshot Replication only for the databases- where changes is very less on the publisher. For example, Gold price changes only once in a day. Here we can use snapshot replication, why because only one-time changes in the publisher. We run snapshot replication once in a day.

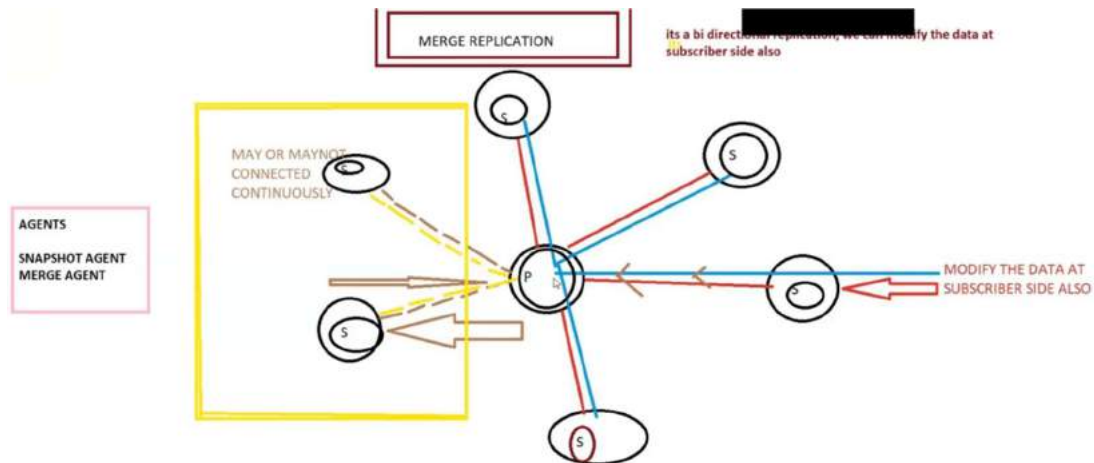


DEFINITION: IT WILL TAKE THE ENTIRE DATA AS A IMAGE AND WILL TRANSFER FROM PUBLISHER TO SUBSCRIBER

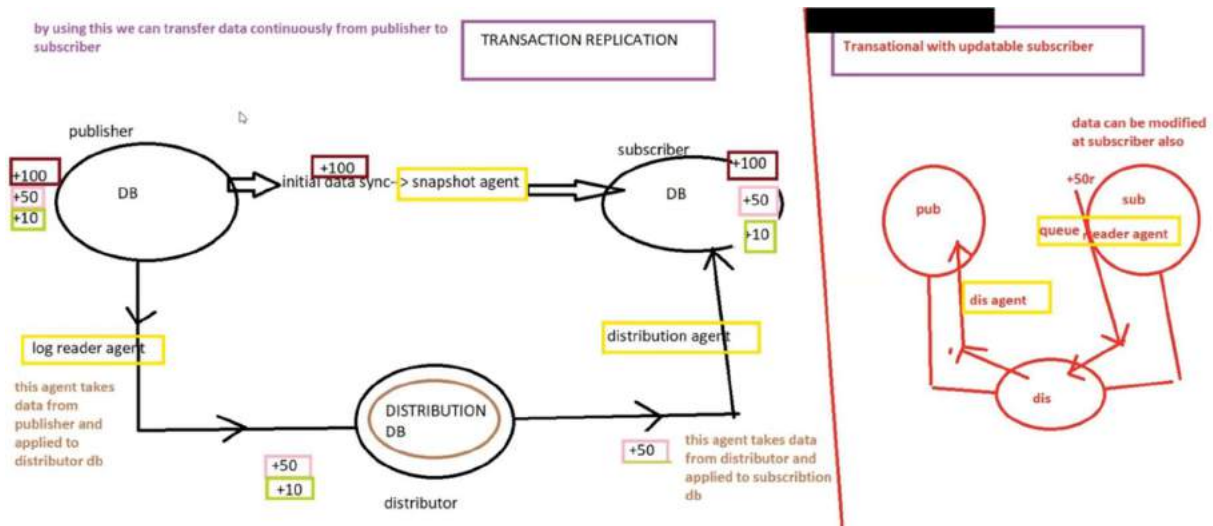
b. Merge Replication: It's a bi directional replication. Bi directional means data can be modified in the subscriber level also. What ever data modified in the publisher will be transferred to Subscriber and at the same time, What ever modified in the subscriber also will be transferred to subscriber.

In merge replication, couple of subscribers may or may not connected continuously to the publisher. But whenever they connected to publisher, the data will be interchanged.

The disadvantage in Merge replication is continuous data transferring is not possible. Why because couple of subscribers may or may not connected to publisher.



C. Transaction Replication : By using Transaction replication , we can transfer data continuously from publisher to subscriber. What ever records will be there in the publisher the same data will be in the subscriber.



To initiate the Transaction replication , there are 3 agents,

- 1) Snapshot agent : To initiate the replication. Initial data of publisher will be transferred to subscriber.
- 2) Log reader agent : This agent takes the data from publisher and applied to the distributor db
- 3) Distribution Agent : this agent takes the data from the distributor and applied to subscription db.

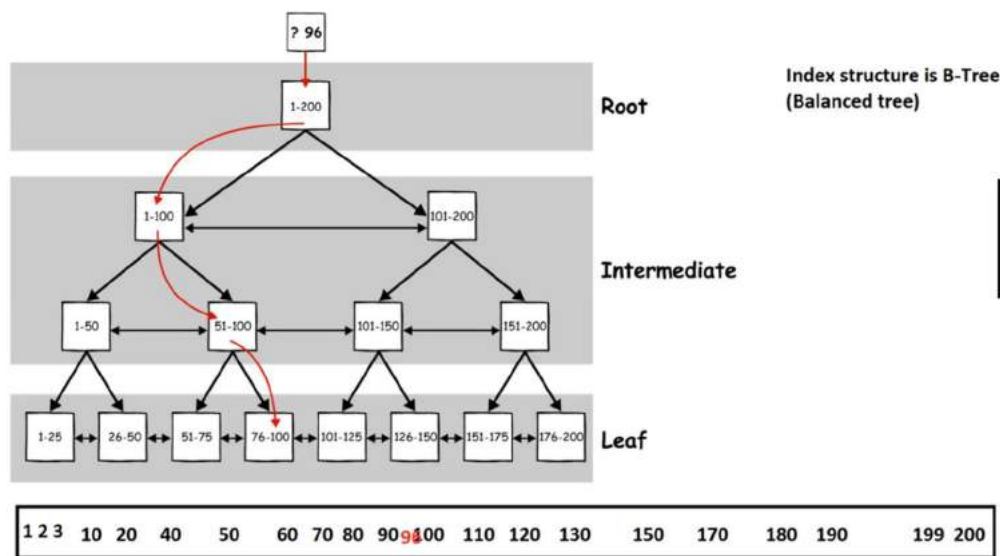
snapshot agent will initiate the transaction replication, after that what ever modified will be transferred to subscriber with the help of log reader & distribution agent . Log reader agent and Distribution agent runs continuously to transfer the data from publisher to subscriber.

We have another concept “ **Transactional with updatable subscriber** ” in the transaction replication. With this we can modify the data in the subscriber. we have agent called “**Queue reader agent**”. This agent takes the modified data from subscriber and sends to distributor db. Now distribution agent sends the data from distributor db to publisher.

117. What are Indexes?

Ans. Mainly Indexes are used to improve the query performance. Index Structure is like a balanced Tree Like below.

When there is no index, SQL server search each and every page to get the data. With the help of INDEX, SQL Server directly identifies the required page with out any delay.



Scanning : Searching each and every page to get the data is called scanning.

Seek : Directly pointing the page , where requested data is available without searching all the pages called Seek.

There are 2 types of indexes,

a. **Clustered Index** : Clustered Index will be created when we place a primary key in the column. Primary key can be placed only to one column of the table. In Clustered index the leaf level pages contain data. That’s why, when we run the query, we get results quickly. We can able to create only one clustered for one table.

Primary Key : Primary key does not accept duplicate values and null. So, when we place primary key to a column, The data will be unique.

b. **Non-Clustered Index** : Non clustered index will be created when we place unique key in a column. We can able create 999 non clustered indexes in a single table. In Non-clustered index, the leaf level does not contain any pages. Non clustered index has a pointer to Clustered index for getting data. If there is no clustered index, Non clustered index depends on the data pages available in the disk. Non clustered Index will improve the performances of other columns.

Unique key : It won’t accept duplicate values and accept only one null value for a column.

In a table, We can create one clustered index for one column and rest of the columns we can create non clustered indexes.

When there is no Clustered index on a table then there is no use of creating NON Clustered indexes.

118. How to see the performance of a query?

Ans. Optimizer creates a plan to execute the query in an optimal manner. That plan is called Execution Plan.

There are 2 ways to find out the execution plan,

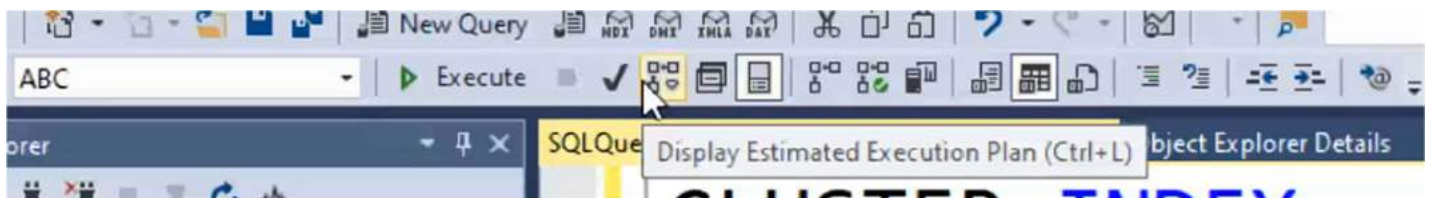
a. Actual Execution Plan : It will execute the query and it will provide the plan of executed Query.

To enable the Actual Execution Plan, We can use Short cut Ctrl+M or we click on the icon “Display Actual execution plan”

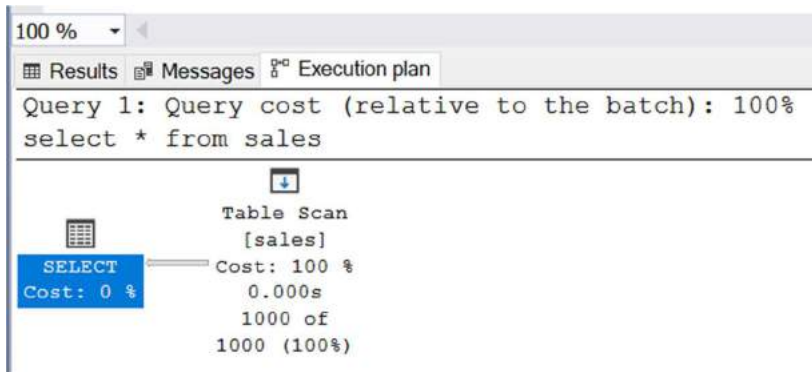


b. Estimated Execution plan : With out Executing the query , it will provide the plan.

To enable the Estimated Execution Plan, We can use Short cut Ctrl+L or we click on the icon “Display Estimated execution plan”



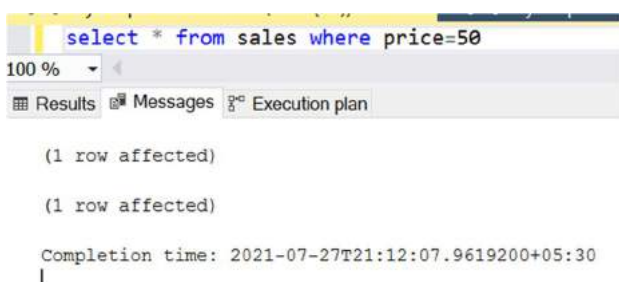
After enabling the execution plan, Execution plan tab will come like this. We can click on that tab and we can see the plan of the Execution of a query.



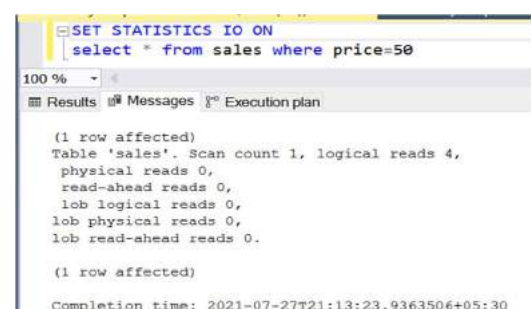
If you want to see how many pages a query reads to retrieve the data, we have to enable “SET STATISTICS IO ON” .

Now we can see in the messages tab

Before



After SET Statistics IO ON



We have to read execution plan from right to left.

To see all the costs of the query, just put the cursor on the table.

Physical Operation	Table Scan
Logical Operation	Table Scan
Actual Execution Mode	Row
Estimated Execution Mode	Row
Storage	RowStore
Number of Rows Read	200000
Actual Number of Rows for All Executions	1
Actual Number of Batches	0
Estimated I/O Cost	0.838759
Estimated Operator Cost	1.05884 (100%)
Estimated CPU Cost	0.220079
Estimated Subtree Cost	1.05884
Number of Executions	1
Estimated Number of Executions	1
Estimated Number of Rows Per Execution	1.02748
Estimated Number of Rows to be Read	200000
Estimated Row Size	41 B
Actual Rebinds	0
Actual Rewinds	0
Ordered	False
Node ID	0

It's used SCAN process because there is no index on the table.

We will see mostly Estimated operator cost only. Estimated operator cost is a combination of CPU, IO & Memory.

In the above image, the operator cost is 1.05884 and rows read is 200000 and actual number of rows is 1.

It means to find 1 row, SQL Server reads all the 200000 rows. Due to this, operator cost is increased.

Now to reduce the Operator cost and Rows read, we create a Clustered index on a unique column of a table.

Syn: `CREATE CLUSTERED INDEX name ON tablename(column name);`

Ex: `CREATE CLUSTERED INDEX CI_SALES ON sales(id);`

Now we can see SQL Server used SEEK process because we created the index.

Physical Operation	Clustered Index Seek (Clustered)
Logical Operation <td>Clustered Index Seek</td>	Clustered Index Seek
Actual Execution Mode <td>Row</td>	Row
Estimated Execution Mode <td>Row</td>	Row
Storage <td>RowStore</td>	RowStore
Number of Rows Read <td>1</td>	1
Actual Number of Rows for All Executions <td>1</td>	1
Actual Number of Batches <td>0</td>	0
Estimated I/O Cost <td>0.003125</td>	0.003125
Estimated Operator Cost <td>0.0032831 (100%)</td>	0.0032831 (100%)
Estimated CPU Cost <td>0.0001581</td>	0.0001581
Estimated Subtree Cost <td>0.0032831</td>	0.0032831
Number of Executions <td>1</td>	1
Estimated Number of Executions <td>1</td>	1
Estimated Number of Rows Per Execution <td>1</td>	1
Estimated Number of Rows to be Read <td>1</td>	1
Estimated Row Size <td>41 B</td>	41 B
Actual Rebinds <td>0</td>	0
Node ID <td>0</td>	0

Now we can see in the below screenshot the operator cost came to 0.00323 and number of rows read and execution is one. For to retrieve one row of a table, SQL Server reads only one page and executed one page by using Clustered index.

Physical Operation	Clustered Index Seek (Clustered)
Logical Operation <td>Clustered Index Seek</td>	Clustered Index Seek
Actual Execution Mode <td>Row</td>	Row
Estimated Execution Mode <td>Row</td>	Row
Storage <td>RowStore</td>	RowStore
Number of Rows Read <td>1</td>	1
Actual Number of Rows for All Executions <td>1</td>	1
Actual Number of Batches <td>0</td>	0
Estimated I/O Cost <td>0.003125</td>	0.003125
Estimated Operator Cost <td>0.0032831 (100%)</td>	0.0032831 (100%)
Estimated CPU Cost <td>0.0001581</td>	0.0001581
Estimated Subtree Cost <td>0.0032831</td>	0.0032831
Number of Executions <td>1</td>	1
Estimated Number of Executions <td>1</td>	1
Estimated Number of Rows Per Execution <td>1</td>	1
Estimated Number of Rows to be Read <td>1</td>	1
Estimated Row Size <td>41 B</td>	41 B
Actual Rebinds <td>0</td>	0
Node ID <td>0</td>	0

Normally Indexes will be created on the columns , where column name used more after the WHERE condition.

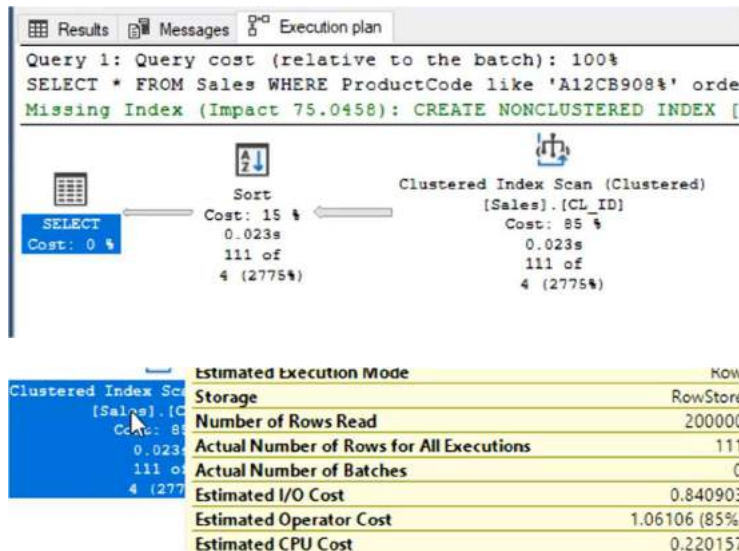
For Example, In the below Query PRICE column used after where condition. So, We will create Index on the PRICE Column.

Select * from sales where PRICE=50;

We reduced the Operator cost of query by using Clustered index. In case, The user runs the query by using another column instead of Clustered index column. Then, Again operator cost and logical reads will increase. To avoid these, we will create NON-Clustered indexes on the other columns.

Below is the example , where Query executed with NON Index column and used the SCAN process to retrieve the data.

Operator cost is 0.84 and reads 20000 rows to retrieve 1 row.

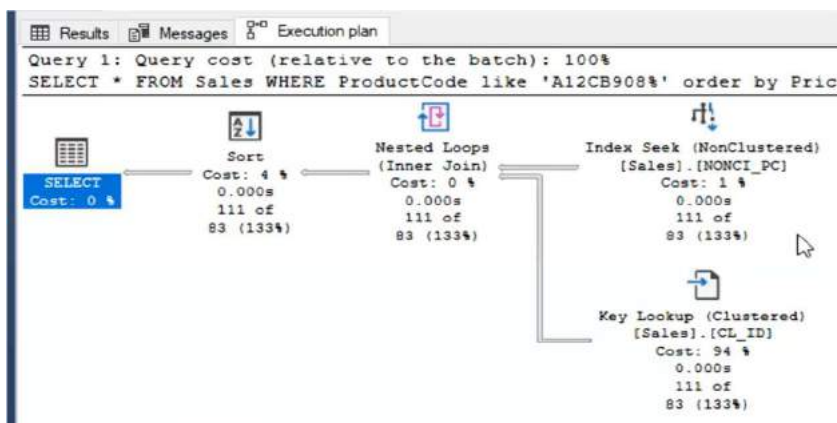


To avoid this, Now we will create NONClustered index on the Executed column.

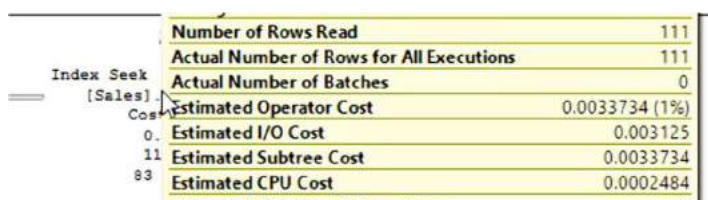
Syn : CREATE NONCLUSTERED INDEX name ON TABLE NAME(COLUMN NAME)

Ex: CREATE NONCLUSTERED INDEX NCI_SALES ON SALES(PRODUCT);

Now if we see operator changed from SCAN to SEEK process and operator cost came to 0.003 and rows read and executed 111 only. It means, What ever rows we requested the same pages has been SQL Server pointed directly and retrieve the data.



NON Clustered index pointed to clustered index to retrieve the data.



In the above screenshot , there is key lookup. Key lookup is not good sign, it takes more operator cost to perform. So, We have to eliminate the key lookup. Key look up will be created automatically , when any one of the column of a table don't have the index on the retrieved the data.

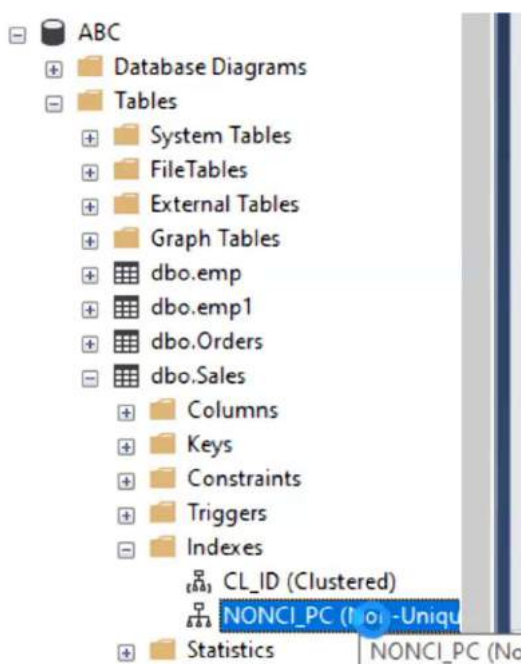
In the below query, Index not created on price and date transaction. In this case, Key lookup will be created.



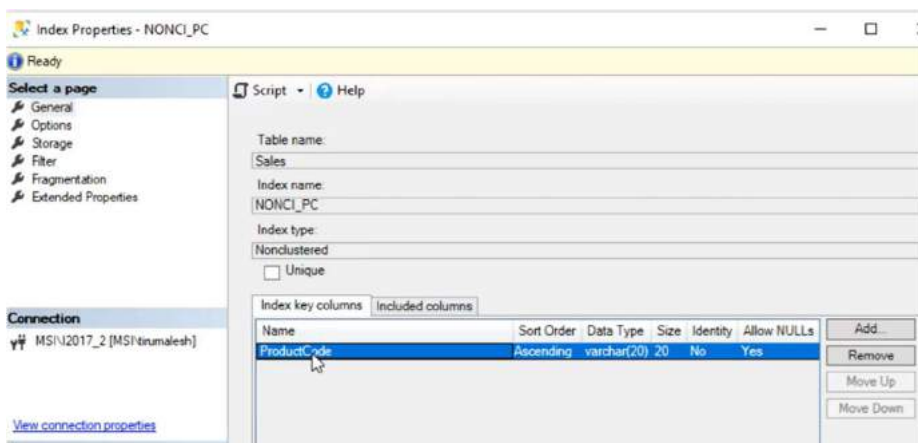
To eliminate the key lookup, we have to use **covering index**. Covering index means, We have to add these columns to the existing non-clustered index. Covering index will cover all the columns with one non clustered index.then key lookup will be eliminated automatically.

Below is the procedure of covering index,

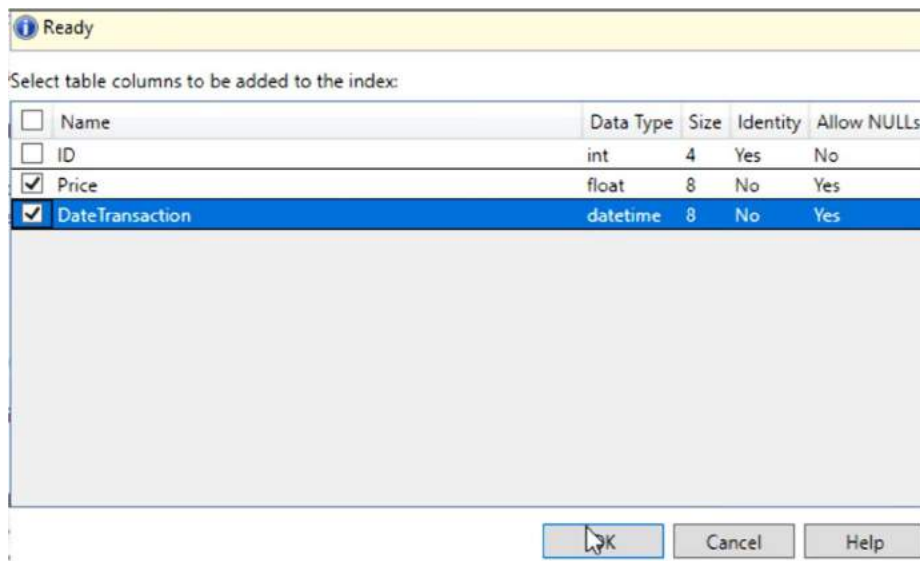
Step 1 : Expand the database, Expand tables, Expand the required table, Expand indexes , select non clustered index,



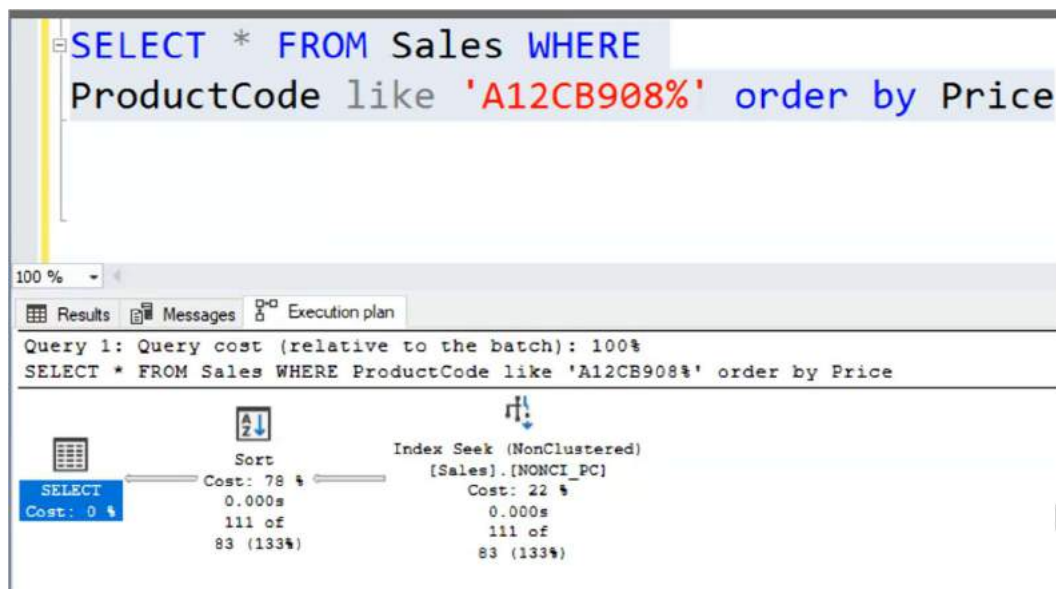
Step 2 : Right click properties, General tab, click add,



Step 3 : Tick the columns and click ok.



Now run the query and we can see in the below screen shot , Key lookup eliminated.



In the above scenario, we know on what columns clustered index & non clustered indexes required.

But in the organization , so many users runs different queries on different tables and columns.

We cant do manually like above. Why because, we don't have track on that.

Without knowing on which column the user is running the query, we cant create index.

To achieve this, we have **profiling and DTA(Database tuning advisor).**

Profiler : profiler capture/trace the queries run by the users. Extension of profiler is ".trc". trc means trace.

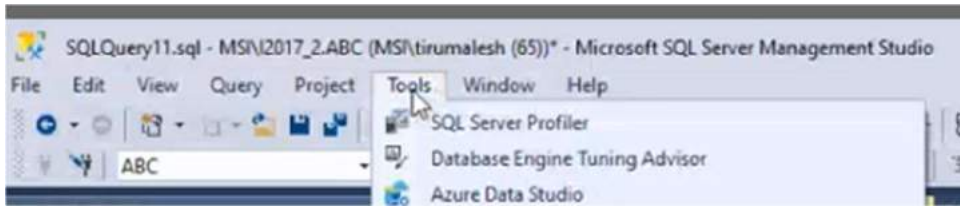
We have to give profiler output to DTA as input.

Profiler output = DTA Input

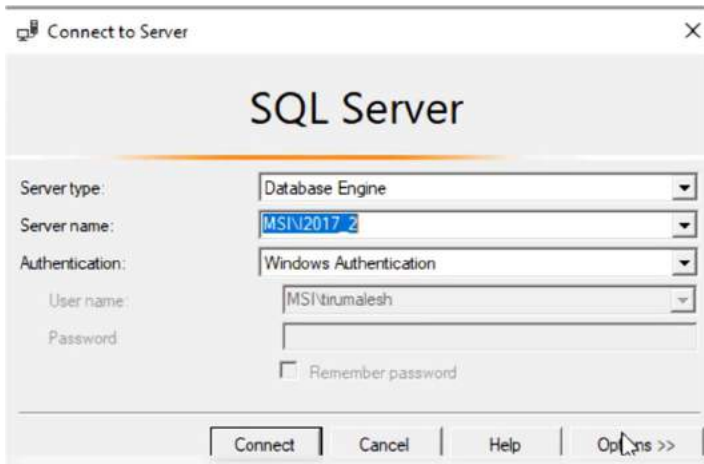
DTA : It will analyse the input and provide the recommendation of indexes.

Now will see practical of using profiler and DTA.

Step 1 : OPEN SSMS, Go to tools, Select SQL Server Profiler.

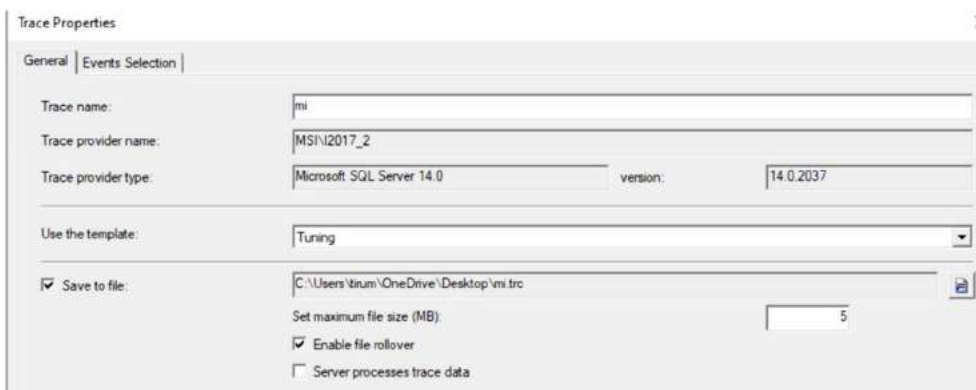


Step 2 : Login to the instance.

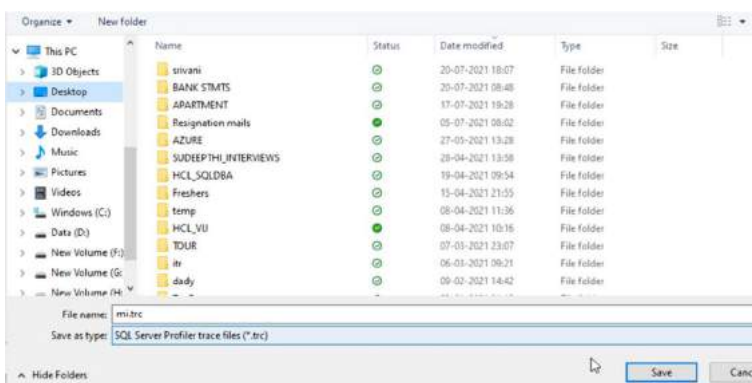


Step 3 : Enter the trace name, select the template from the list "Tuning", Set the maximum size and enable the file roll over option. Tick the SAVE file and give the location where you want save the output.

By enabling the file roll over option, Once the trn file reached limit which we set, automatically one more file will created. We don't know how much size the trace file. That's why we are enabling the file roll over. Keep maximum size starts from 50 MB.



Step 4 : File extension should be ".trc". Click save.



Step 5 : Profiler is dangerous tool. why because, It took more resources of the system like CPU, MEMORY etc. It may lead slow down the sql server. That's why, we have to run profiler for short time only.

To control this, we have to enable trace stop time. Here, we have to give the minutes to run the profiler. Its helps to stop profiler after that time period automatically.

The screenshot shows the 'Trace Properties' dialog box with the 'General' tab selected. The 'Trace name' is 'MI', 'Trace provider name' is 'MSIN2017_2', and 'Trace provider type' is 'Microsoft SQL Server 14.0'. The 'Use the template' dropdown is set to 'Tuning'. The 'Save to file' checkbox is checked, with the file path 'C:\Users\trun\OneDrive\Desktop\mi.trc' and a maximum file size of 500 MB. The 'Save to table' checkbox is also checked, with the table name 'MSIN2017_2 [master] [dbo] [MI]'. The 'Enable trace stop time' checkbox is checked, with the start time '20-07-2021' and end time '20:10:44'. The 'Set trace duration (in minutes)' checkbox is also checked. The 'Run', 'Cancel', and 'Help' buttons are at the bottom.

Step 6 : Go to event tab, Tick the required,

The screenshot shows the 'Trace Properties' dialog box with the 'Events Selection' tab selected. The 'General' tab is also visible. The 'Events Selection' tab shows a list of events to trace. The 'Stored Procedures' and 'TSQL' event categories are expanded. The 'RPC:Completed', 'SP:StmtCompleted', and 'SQL:BatchCompleted' events are selected. The 'TextData', 'Duration', 'SPID', 'DatabaseID', 'DatabaseName', 'Object Type', and 'LoginName' columns are also selected. The 'Show all events' and 'Show all columns' options are checked. The 'Run', 'Cancel', and 'Help' buttons are at the bottom.

Events	TextData	Duration	SPID	DatabaseID	DatabaseName	Object Type	LoginName
Stored Procedures							
☒ RPC:Completed	☒	☒	☒	☒	☒		☒
☒ SP:StmtCompleted	☒	☒	☒	☒	☒	☒	☒
TSQL							
☒ SQL:BatchCompleted	☒	☒	☒	☒	☒		☒

Step 7 : select the Column filters option in events selection, select database name , Click like, Enter the database name , click ok. Here if we wont choose the database, profiler trace the queries of all the databases and took more CPU. It lead to slow down the server.

The screenshot shows the 'Edit Filter' dialog box. The 'DatabaseName' filter is selected. The 'Like' radio button is selected, and the text 'ABC' is entered in the filter value field. The 'Exclude rows that do not contain values' checkbox is unchecked. The 'OK', 'Cancel', 'Show all events', and 'Show all columns' buttons are at the bottom. The 'Column Filters...' and 'Organize Columns...' buttons are also visible.

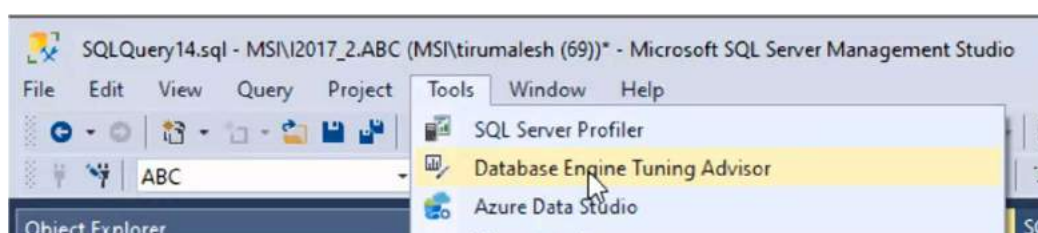
Step 8 : Click Run. Now it will starts capturing the queries. It wont trace already executed queries. It will trace after initiating the profiler, what ever queries runs.

RPC:Completed	exec sp_executesql N'SELECT SCHEMA_N...	1	71	7	ABC
SP:StmtCompleted	SELECT i.name AS [Name] FROM sys.tab...	0	71	7	ABC
RPC:Completed	exec sp_executesql N'SELECT i.name A...	0	71	7	ABC
SQL:BatchCompleted	declare @MasterPath nvarchar...	2	71	7	ABC
SP:StmtCompleted	SELECT CAST(serverproperty...	0	71	7	ABC
RPC:Completed	exec sp_executesql N'SELECT CAST(...	0	71	7	ABC
SP:StmtCompleted	SELECT CAST(serverproperty...	0	71	7	ABC
RPC:Completed	exec sp_executesql N'SELECT CAST(...	1	71	7	ABC
SP:StmtCompleted	SELECT CAST(serverproperty...	0	71	7	ABC
RPC:Completed	exec sp_executesql N'SELECT CAST(...	0	71	7	ABC
SP:StmtCompleted	SELECT i.name AS [Name], CAST(i.inde...	20	71	7	ABC
RPC:Completed	exec sp_executesql N'SELECT i.name A...	21	71	7	ABC
SP:StmtCompleted	CREATE TABLE #tmp_extended_remote_da...	1	71	7	ABC
SP:StmtCompleted	IF EXISTS(SELECT 1 FROM master.sys.s...	0	71	7	ABC
SP:StmtCompleted	INSERT INTO #tmp_extended_remote_dat...	0	71	7	ABC
SP:StmtCompleted	EXECUTE(N'INSERT INTO #tmp_extended_...	47	71	7	ABC
SP:StmtCompleted	SELECT tbl.name AS [Name], tbl.objec...	19	71	7	ABC
SP:StmtCompleted	DROP TABLE #tmp_extended_remote_data...	1	71	7	ABC
RPC:Completed	exec sp_executesql N' CREAT...	71	71	7	ABC
SQL:BatchCompleted	/***** Object: Index [NONCI_PC] ...	2	71	7	ABC
SQL:BatchCompleted	SELECT @@SPID;	0	68	7	ABC
SQL:BatchCompleted	SELECT * FROM Sales WHERE ID=189923 ...	388	68	7	ABC

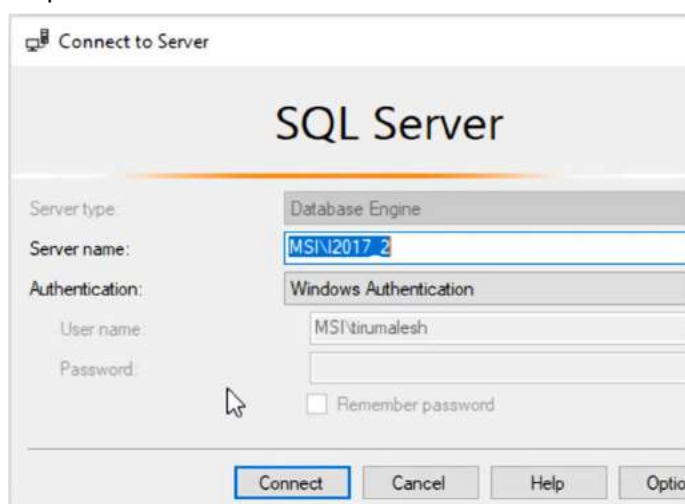
SELECT * FROM Sales WHERE ID=189923
SELECT * FROM Sales WHERE ID=9923
SELECT * FROM Sales WHERE ID=19923
SELECT * FROM Sales WHERE ID=123
SELECT * FROM Sales WHERE ID=3
SELECT * FROM Sales WHERE ID=89923
SELECT * FROM Sales WHERE ID=9923
SELECT * FROM Sales WHERE ID=23
SELECT * FROM Sales WHERE ID=999

If already indexes are there in the table, delete the indexes. Otherwise DTA wont recommend the indexes.

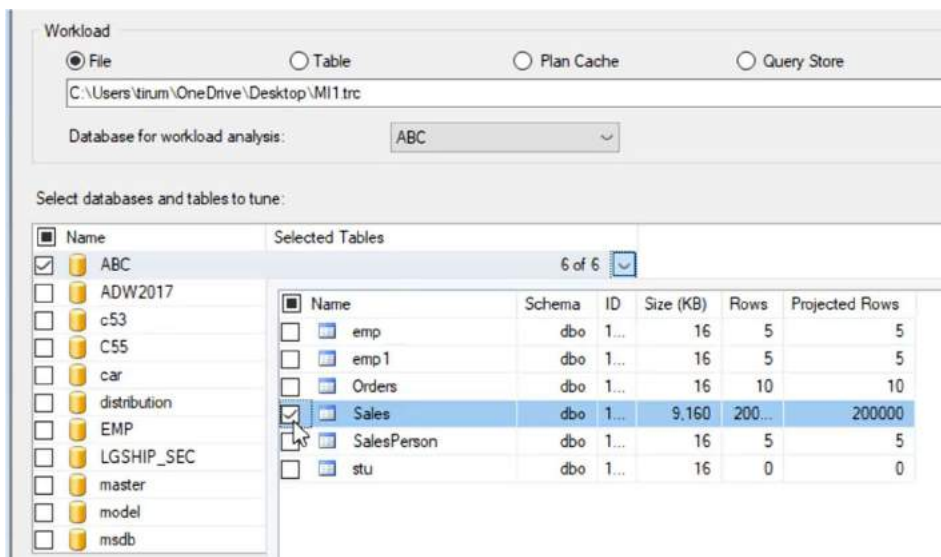
Step 9 : In SSMS, go to tools, select database engine tuning advisor.



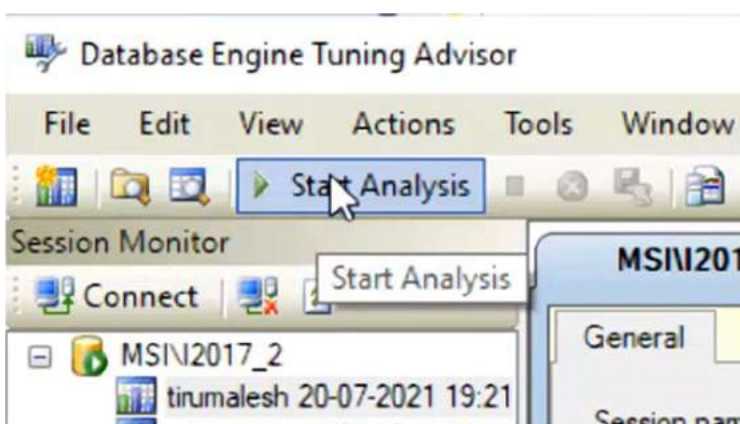
Step 10 : Connect to the instance.



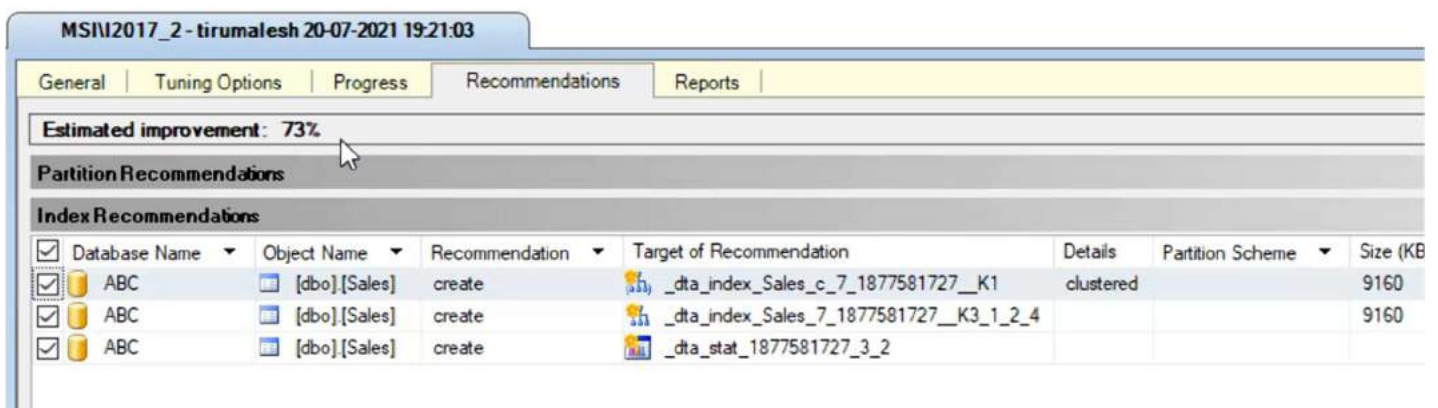
Step 11 : Enter the session name, Select the work load and choose the type of output, We have saved the profiler trace in file format. So, choose the file option, give the path of ".trc" file which we saved earlier. Choose the database, select the table ,



Step 12 : Click on start analysis.



Step 13 : now it will show the index recommendations and it will show the improvement of query performance if we create the indexes as per DTA recommendation.

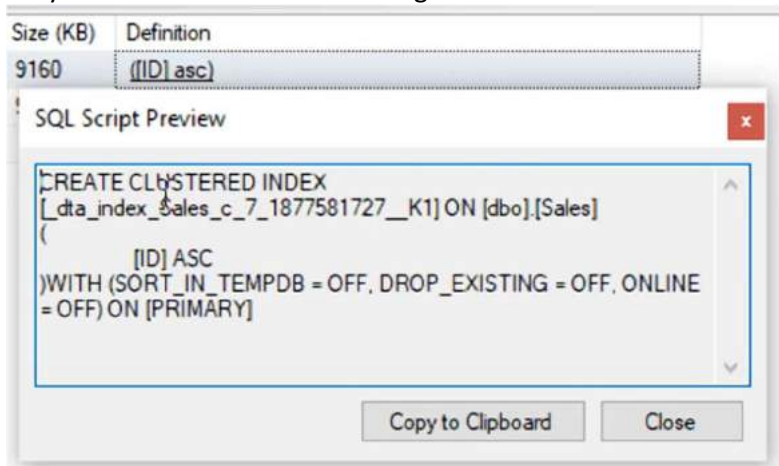


Step 14 : Go to definition column. Here we can see DTA suggested Clustered index on ID column and remaining two columns non clustered index.

Partition Scheme	Size (KB)	Definition
	9160	(ID asc)
	9160	(Price asc) include (ID, ProductCode, DateTransaction)
		(Price, ProductCode)

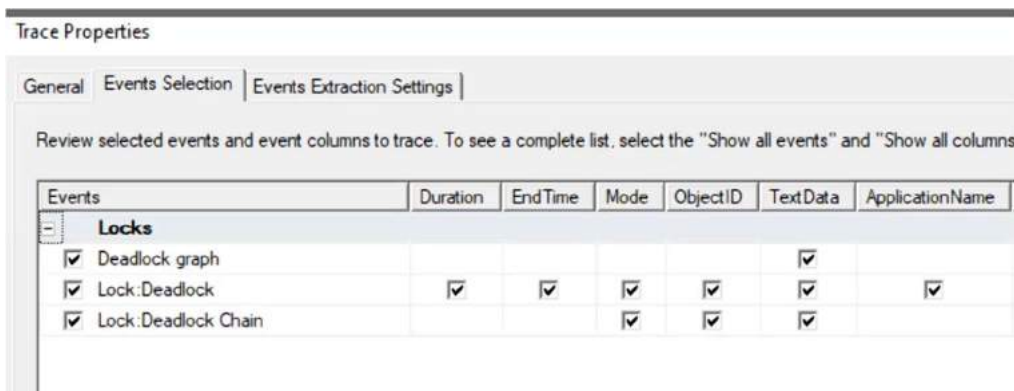
Step 15 : Put the cursor, and copy the script and execute on the sql server. Now index will be created.

Coy and execute for the remaining recommendations also.



119. How to capture the deadlocks by using the profiler?

Ans. Open Profiler, Select the template "TSQL Locks" , select the all the locks related to dead locks in events selection.

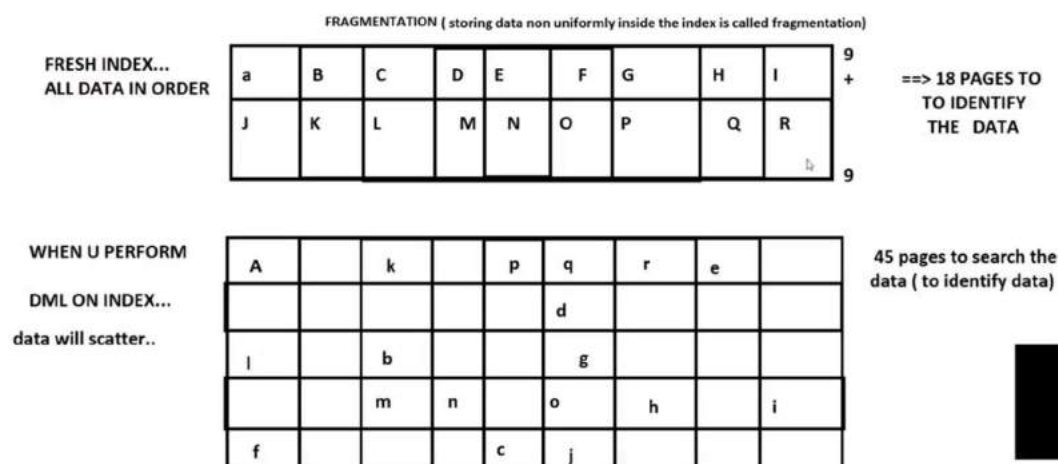


120. What is fragmentation?

Ans. When we create a fresh index, data will be organized in order. After performing DML in the data, data will scatter and order will be gone.

In the below example , if we see.

When fresh index created, data placed in order and took 18 pages. After the DML queries applied on the data, Data scattered and pages not in order. Now, Query took long time to execute even index is there because of DML queries performed.

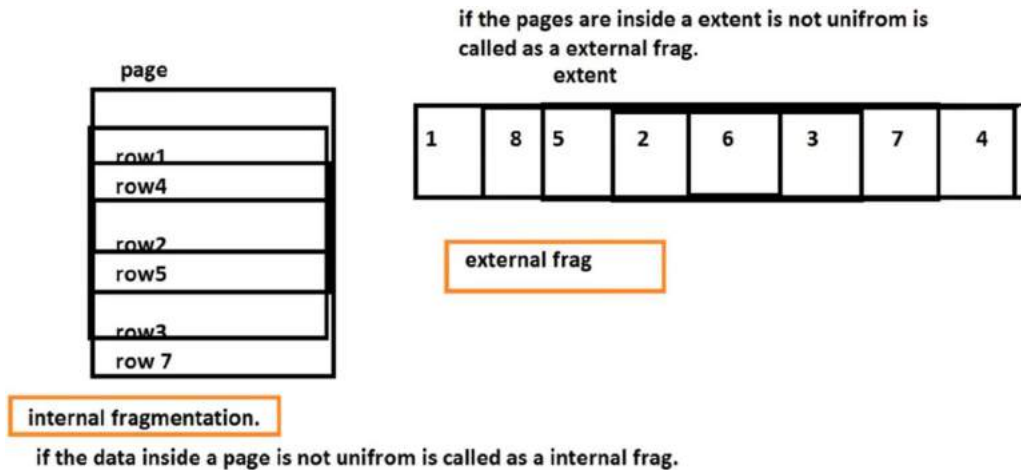


Storing data non uniformly inside the disk is called fragmentation. Simply, Data stored not in order is called fragmentation. Because of fragmentation , query runs slowly. Always we have to reduce the fragmentation.

Fragmentation occurs when we perform DML operations on the data.

There are 2 types of fragmentation,

- Internal Fragmentation : if the data is not in uniform inside a page is called internal fragmentation.
- External Fragmentation : if the pages are not in uniform inside a extent is called External Fragmentation.



INTERNAL VS EXTERNAL FRAGMENTATION

121. How to identify and remove the fragmentation?

Ans. Fragmentation can be identified in two ways,

One is “ DBCC SHOWCONTIG”

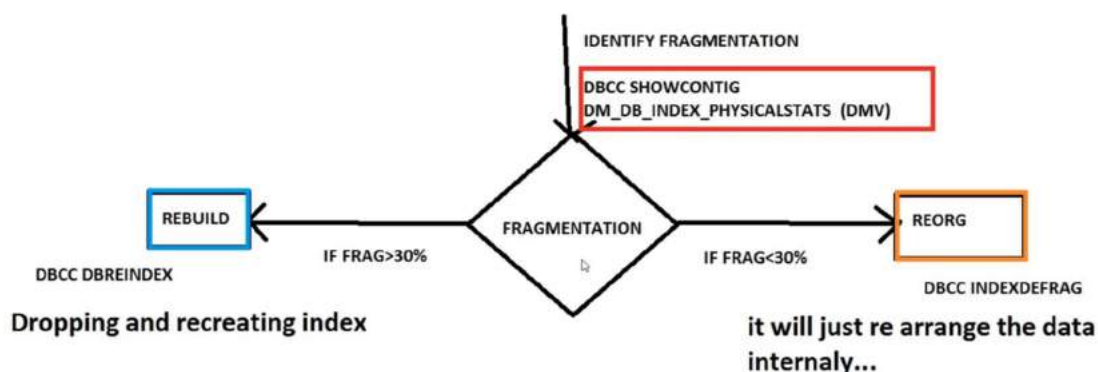
Second is

“DM_DB_INDEX_PHYSICALSTATS” (DMV)

How to detect Fragmentation: We can get both types of fragmentation using the DMV: sys.dm_db_index_physical_stats. For the screenshot given below, the query is as follows:

```
SELECT *
FROM sys.dm_db_index_physical_stats
(DB_ID('dbname'), NULL, NULL, NULL, 'SAMPLED')
ORDER BY avg_fragmentation_in_percent DESC
```

(No column name)	index_id	index_type_desc	index_level	avg_fragmentation_in_percent
1 DF_BIOFMaterials_StartDate	1	CLUSTERED INDEX	0	99.009900990099
2 DF_Address_ModifiedDate	1	CLUSTERED INDEX	0	97.0588235294118
3 DF_Contact_emailid	1	CLUSTERED INDEX	0	94.0000000000000



IDENTIFY FRAGMENTATION AND DIFF BETWEEN REBUILD VS REORG

To clear the Fragmentation, We have 2 options,

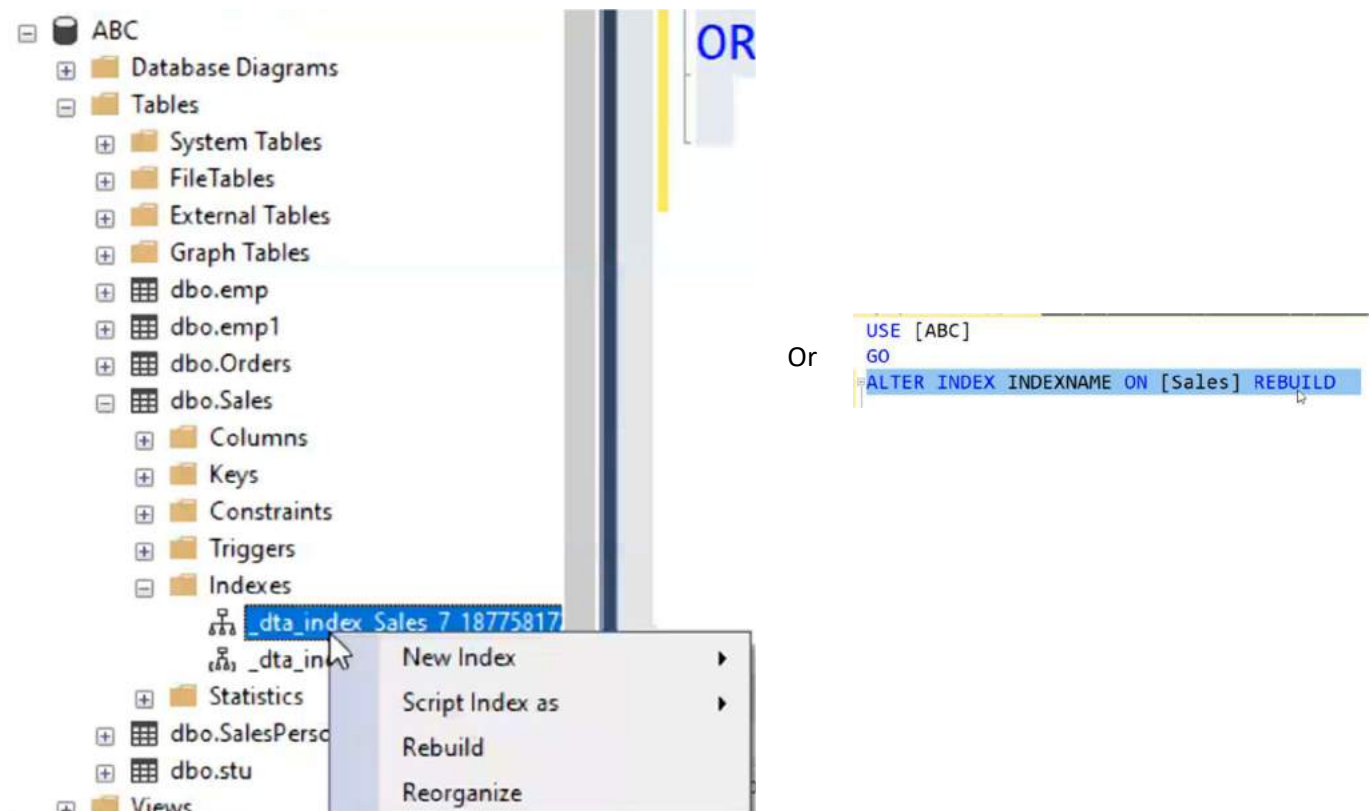
a. Rebuild Index : It will drop the existed index and create the new index. Whenever new index created, automatically data will places in order. Fragmentation will be eliminated.

b. Reorg Index :it wont drop any index, It will rearrange the data internally . Fragmentation will be controlled.

Rebuild is the best option .

Below are the steps to eliminate Fragmentation,

Select & expand the database, Expand tables, select the table and expand the table, Expand indexes, right click index , Choose the rebuild or reorganize.

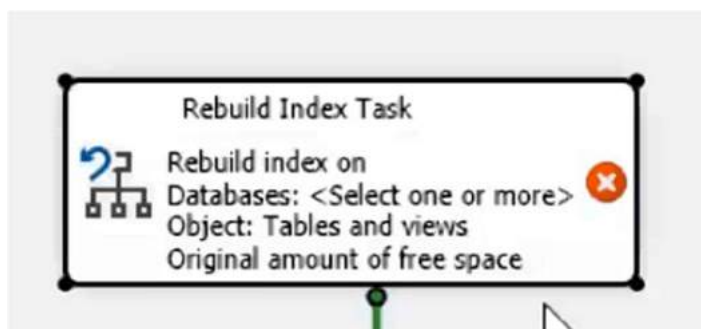


We can do the above process to eliminate the fragmentation for one or two tables like that. To do all the tables , it will take long time. So, We use maintenance plan to eliminate the fragmentation of the all the tables.

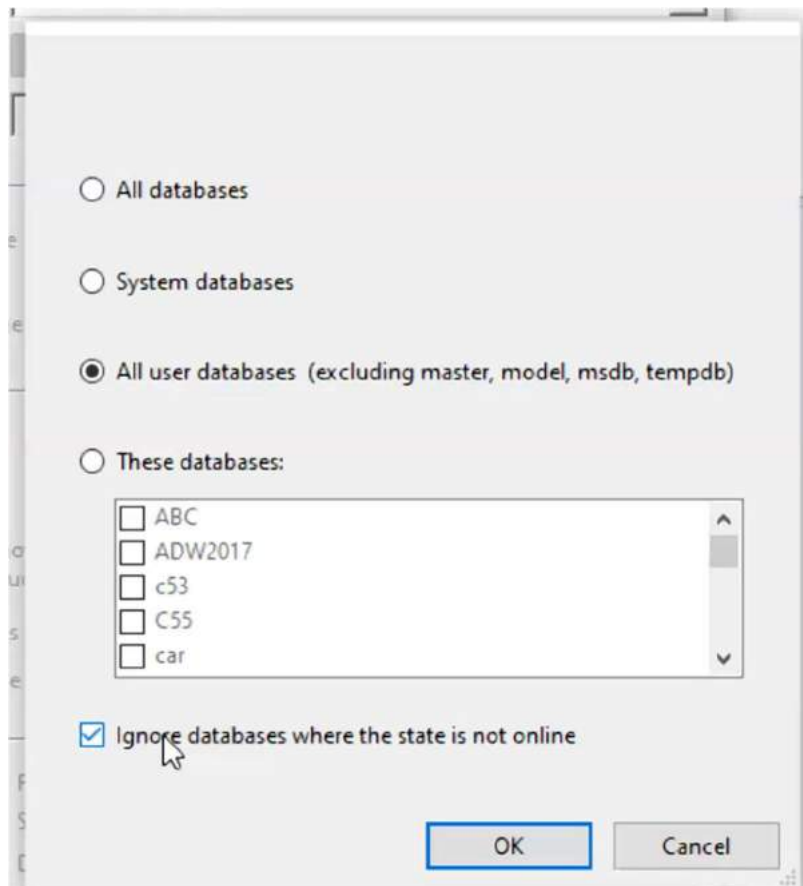
Creating maintenance plan:

Below are the steps,

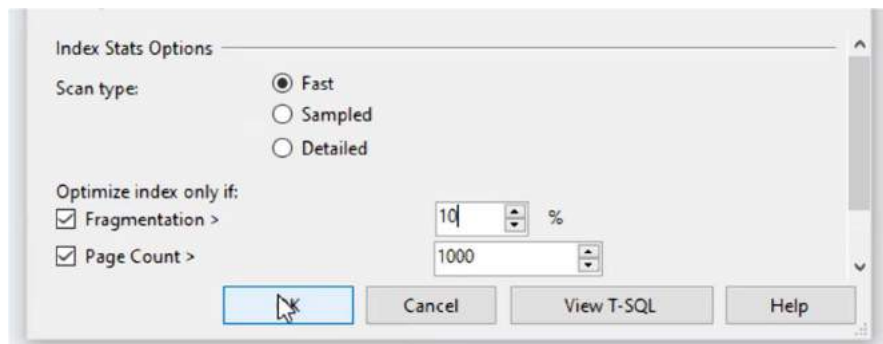
Step 1 : Select new maintenance plan, Drag the Rebuild or Reorg index task to the empty space.



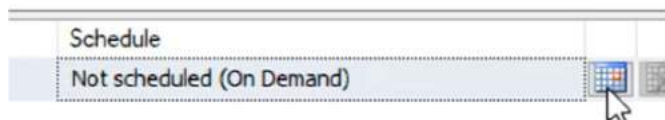
Step 2 : Select the database, Ignore the database where state is not online, Click ok.



Step 3 : go to optimize index , Choose the fragmentation percentage and page count, Click ok.



Step 4 : Schedule the maintenance plan and save the maintenance plan.



Step 5 : maintenance plan will be created to eliminate fragmentation as per schedule.

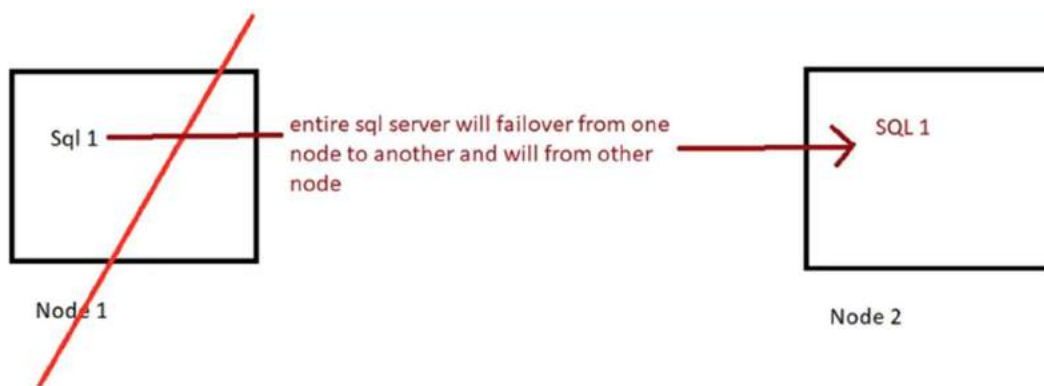
118. If Temp DB is full, How to solve the issue?

Ans. We have to shrink the TEMP DB.

120. What is Clustering?

Ans. Clustering is a instance level high availability. Instance level means, logins, db, jobs and everything will move from one server to another server in case any issue with the windows server.

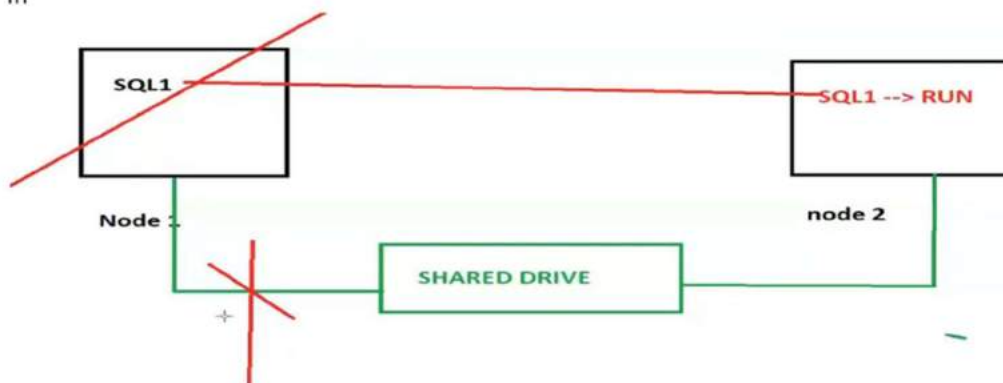
In clustering, we called windows servers as Nodes.



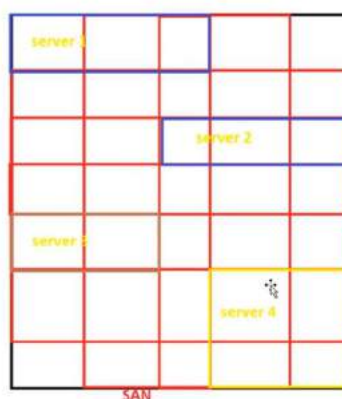
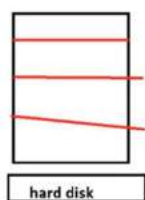
Cluster storage:

For Cluster configuration, we required shared storage. That is Storage area network(SAN).

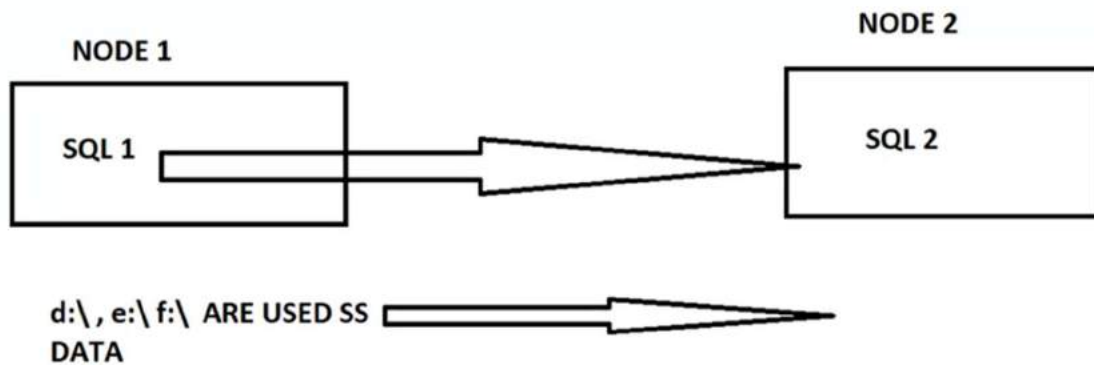
In



In real time, SAN will be considered as shared storage. (STORAGE AREA NETWORK)



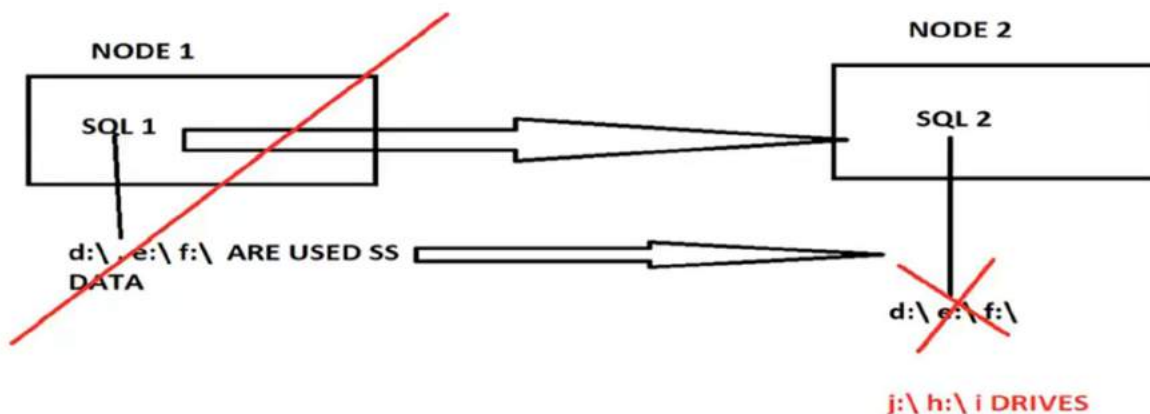
In cluster, When node 1 is down, The sql server will failover from node1 to node2. When sql server moves , the associate drives(data and log bkp) will be also failover from node 1 to node2.



We don't keep any data files and log files in the C: Drive . why because it contains windows files. So, we configure the sql server in the other drives except C drive.

One more imp thing is, We should not keep same drives in the both the nodes, Why because if the same drives already created in the node2 , Failover not possible. Already same drives exists in the node 2.

We have to take unique drive letters while doing the Clustering. If Node 1 has the D & E drives used for Clustering then in the NODE 2 , there should not be any drives with the same drives. Always take unique drive letters for each sql server.



Types of Clustering :

There are 2 types

a. Active – Active Cluster : Each Node contains a different sql servers and each node is capable of maintain 2 sql servers. Simply, Both the Nodes having the SQL Server. Here we called Node 1 as Active Node and Node 2 also called as Active node. Here, Each server should be capable of maintain the 2 servers.

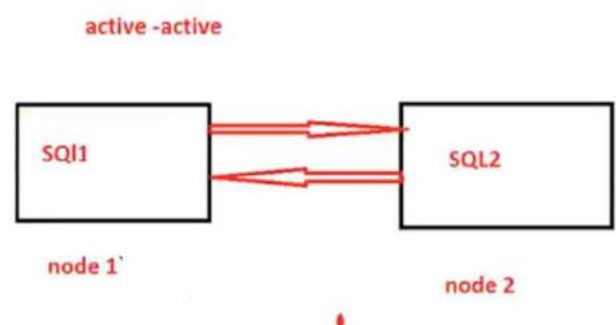
Active means SQL Server available in the windows server.

For example,

In Node 1, Flipkart SQL Server is running.

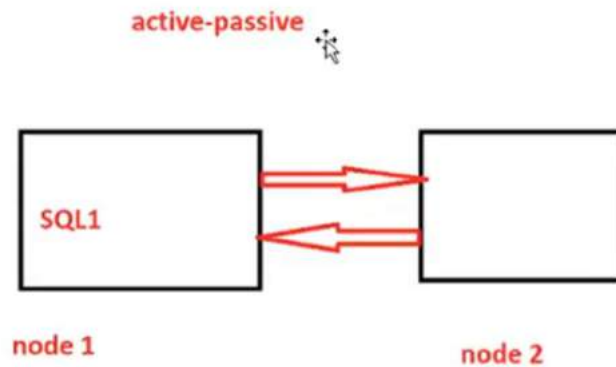
In Node 2, Amazon SQL Server is running.

When Node1 Windows server is down, Flipkart SQL Server will failover to NODE2 windows server. Now, Both the Flipkart and Amazon sql server running in the same NODE2 Windows server.



b. Active – Passive Cluster : Only one Node has the SQL Server and another NODE don't have any SQL Server. The Node, which has SQL Server is called ACTIVE Node. The Node, Which don't have the sql server called as Passive Node. Passive node is waiting for the failover of the active node.

Passive Means SQL server not available in the Windows server.



How Cluster Works: Cluster works through heart beat process. A signal will send from one server to another server to verify the NODE is LIVE or not. Heart beat helps to find out whether the other nodes are alive or not.

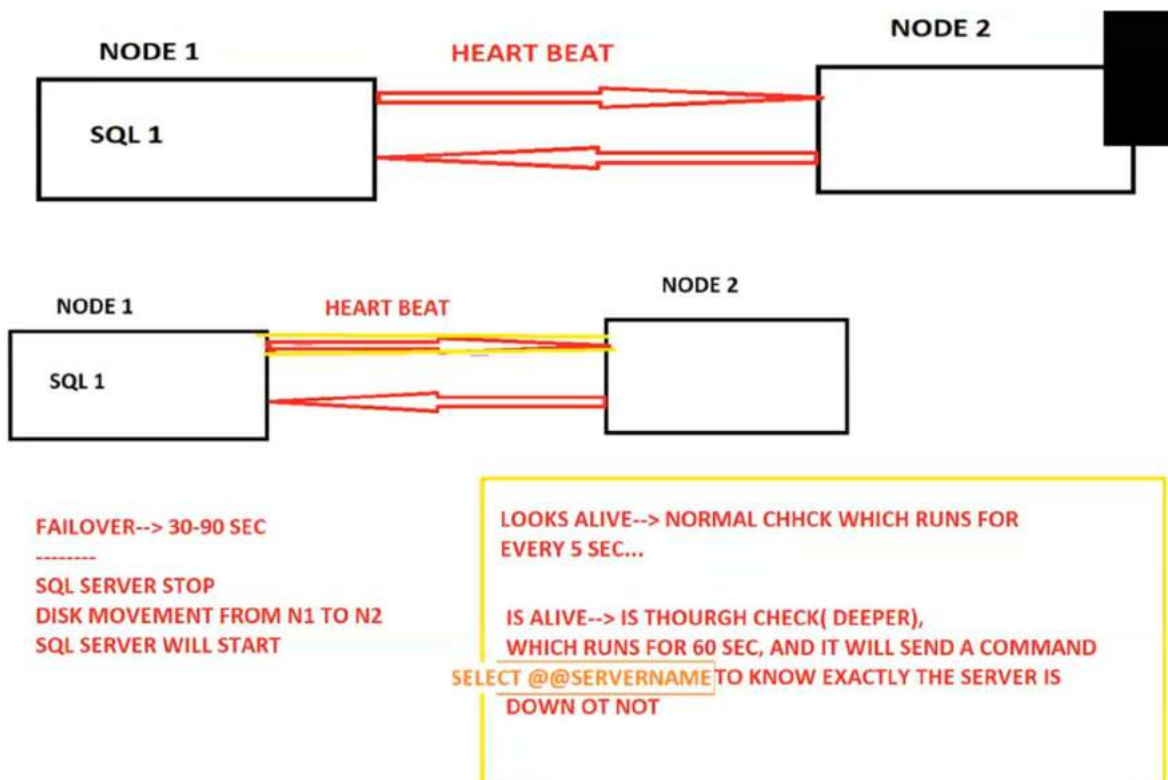
Heart beat setup will be done by using private network.

There are 2 types of heart beats, LOOKS ALIVE & IS ALIVE.

a. LOOKS ALIVE : It's a normal check , which runs for every 5 seconds.

b. IS ALIVE : it's a deeper check , Which runs for every 60 seconds. It will send a command “select @@Servername” to know exactly the server is down or not.

When there is a fail over in clustering, SQL Server will be in stopped state for 2 to 3 minutes.



When any of the reason the heart beat of the SQL Server is not working. Then Heart beat will do 2 checks.

1st it will send LOOKS ALIVE check – If receive back the signal from the server, heart beat will continue. If there is no signal again, then it will send IS ALIVE Check. IS ALIVE Check sends the Command signal to the server and runs for 60 Sec. If the receives the signal back, heart beat will continue. If there is no signal in this check also, SQL SERVER will fail over to another NODE.

Why 2 checks is , Some times due to network issues also signal wont come. That's why, SQL SERVER cross check once again with the LOOKS ALIVE and IS ALIVE. Then only sql server will failover to another windows server.

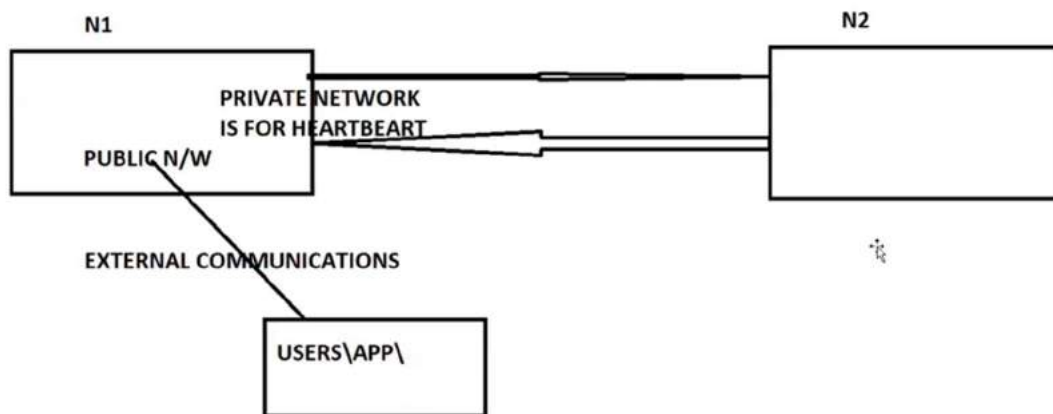
The total time to failover of a SQL Server is around 2 to 3 minutes in clustering. At this stage, SQL Server is in stopped state.

Types of network :

There are 2 types of networks,

- a. Private Network : used for Internal communication. Acts as a heart beat. To send the signals b/w the servers.
- b. Public Network : used for External communication.(users and applications).

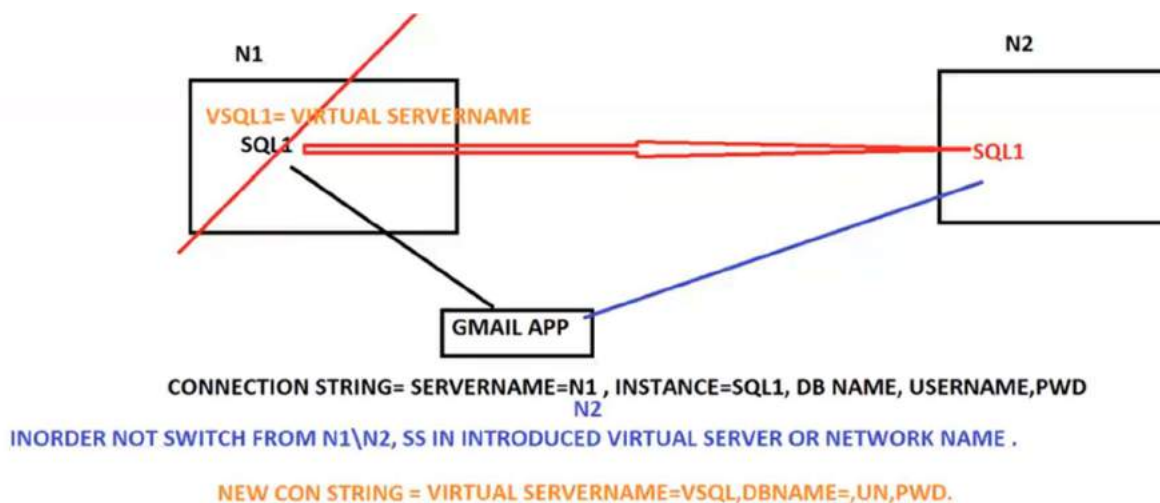
Every windows server has one public and one private network.



Virtual Server name or network name : By using virtual server name, the application will connect where ever the sql server is running. We don't need to change the connection string with network name whenever the failover happens.

Simply if SQL Server is failover from node 1 to node 2 , We need to ask application team to change the connections from node 1 to node 2. Automatically with the help of Virtual server name, Connection string also changed to Node 2.

After automatic failover also, Application connection string also automatically changed to where ever sql server is running.

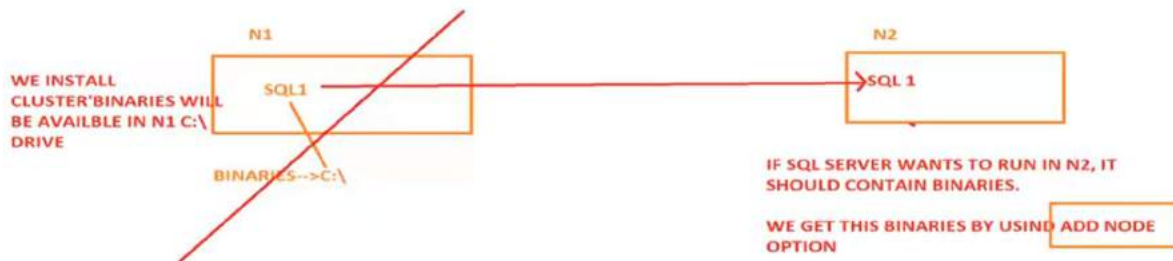


Adding Binaries : All the SQL Server installation related files are called binaries. Binaries are files, which helps to run the SQL Server.

In ACTIVE Passive Cluster, There is no sql server in the NODE2. After the failover to NODE2, To run the SQL Server – binaries are required.

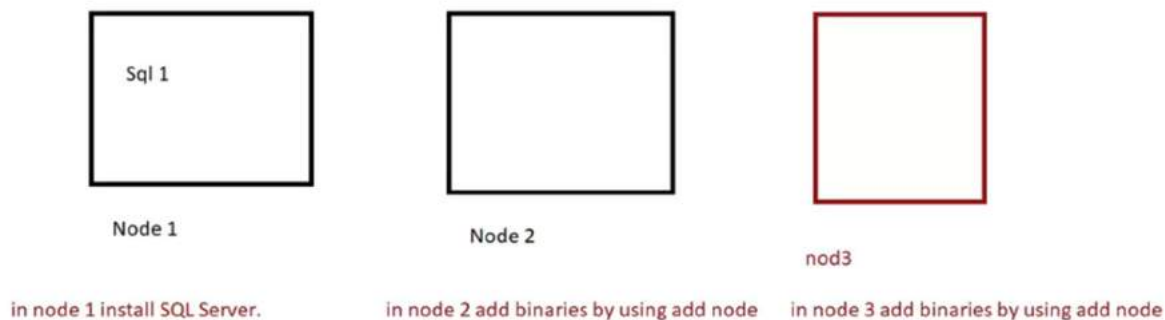
In ACTIVE ACTIVE Cluster , There is already a SQL Server installed and used by some other instance. After the failover from NODE1 to NODE 2- SQL Servers because already a SQL Server installed in NODE 2.

By using the “ADD NODE” option, we can get the binary files of SQL Server from NODE1 TO NODE2.



If we have installed the SQL Server NODE1, In node 2 we have to add binaries by using ADD NODE.

In Node 3 We have to add binaries by using ADD NODE.



SQL 1 installation

N1--> INSTALL CLUSTER (SQL1) IN N1.. And N2,N3,N4 --> ADD NODE SQL1

SQL 2 INSTALLTION

N2->INSTALL SQL2 and perform N1, N3 N4-> ADD NODE SQL2.

How many IP's required for Cluster Configuration ?

All the SQL Servers in the Clustering connected to Windows Cluster only.

For 2 Node Clustering, We need 7 IP's to configure ACTIVE – ACTIVE Clustering

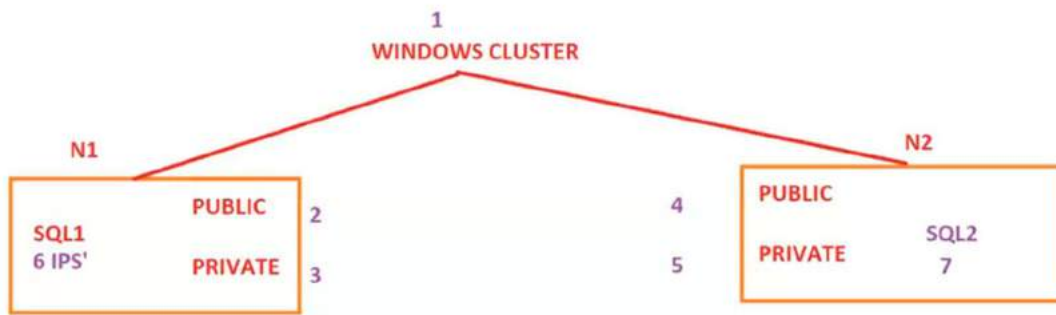
We need 6 IP's to configure ACTIVE – PASSIVE Clustering

If 3 node clustering, 8 IP's – A-A Clustering and 7 IP'S A-P Clustering.

To remember the how many IP's, use the formula – $2n+2$ for Active passive clustering

Use the formula – $2n+3$ for Active - Active clustering

Don't tell this formula in interview. Its just to remember only.



IP'S REQUIRED

2-NODE A-P --> 6

2-NODE A-A--> 7

3 -NODE--> A-A-A--> 10 (1 WIN CLU, 3 PUB, 3 PRI, 3 SQL SERVERS

3-NODE-->A-A-P-->9 (1 WIN CLU, 3 PUB, 3 PRI, 2 SQL S

QUORUM : IT contains QUORUM.LOG file , which contains all cluster configuration and will decide the sql server from one server to another server by using voting process. Which node gets the highest votes, SQL server failover to the highest voting node. Simply, QUORUM decides to which node sql server should failover.

Installation :

When we cluster installation request...

1. WIN TEAM WILL INSTALL NEW SERVERS AND THEN WILL INSTALL CLUSTERING SOFTWARE, BY ADDIGN ALL THE NODES.(called as windows cluster configuration)
2. STORAGE TEAM WILL CONFIGURE THE STORAGE.
3. NETWORK TEAM WILL CONFIGURE PUBLIC AND PRIVATE NETWORK.
4. THEN ONLY SQL TEAM WILL PERFORM INSTALLTION.
5. BEFORE SQL SERVER CLUSTER INSTALLTION WE HAVE TO **VALIDATE THE CLUSTER.** (VALIDATION MEANS CHECKING WINDOWS, DISKS, CLUSTER CONFIGURATION AND NETWORK)
6. THEN INSTALL SQL SERVER cluster IN NODE N1 AND THEN PERFORM ADD NODE option IN N2.

How to open Failover cluster Manager?

RUN , CLUADMIN.MSG

121. How to setup 2 node SQL Server Clustering?

Ans. SQL Server clustering installation also same as Stand alone installation only. Only the few changes will be there.

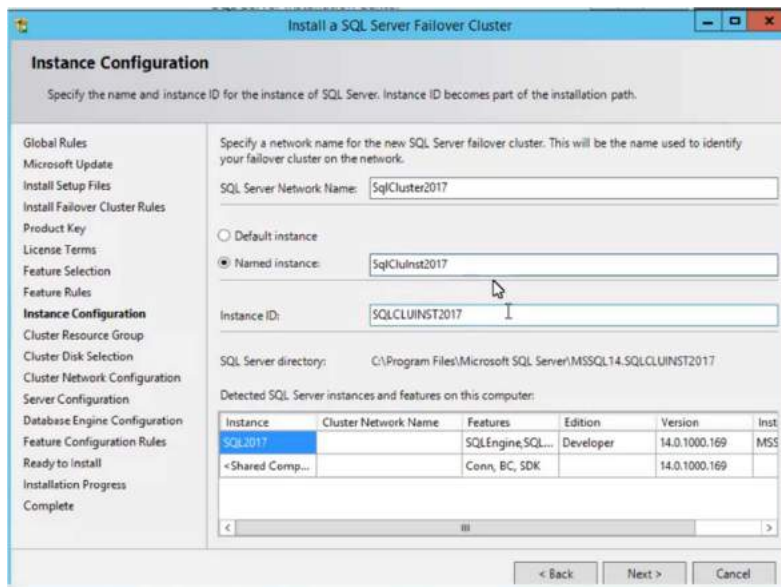
Step 1 : First Windows team will setup windows server failover clustering. After that we have to start.

Step 2 : Open the SQL Server setup file as administrator, Go to Installation Tab, Select New SQL Server failover installation.

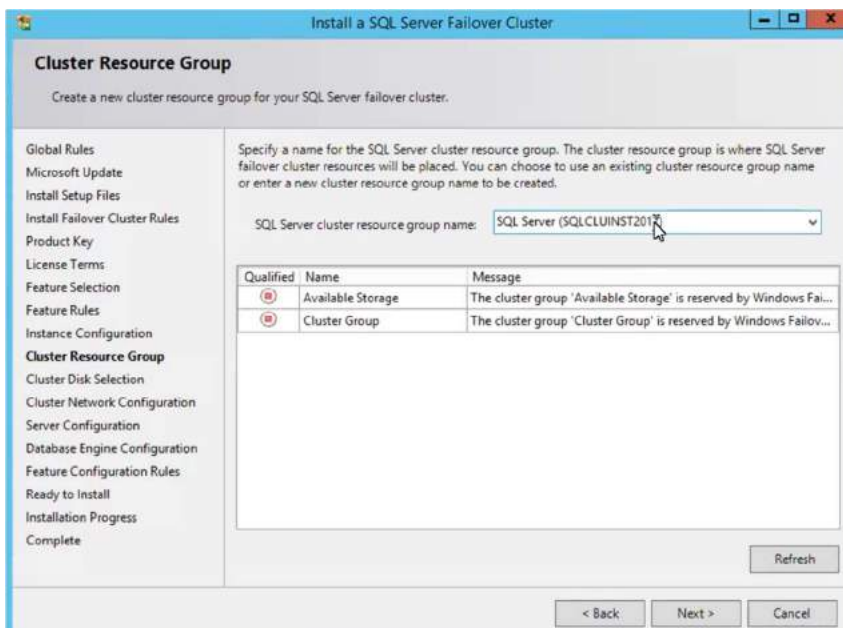


Step 3 : 1st difference – In the Instance configuration Setup page, We have to give SQL Server network name, Provide the Instance name . Here SQL Server network name acts as a computer name to login in to the sql server.

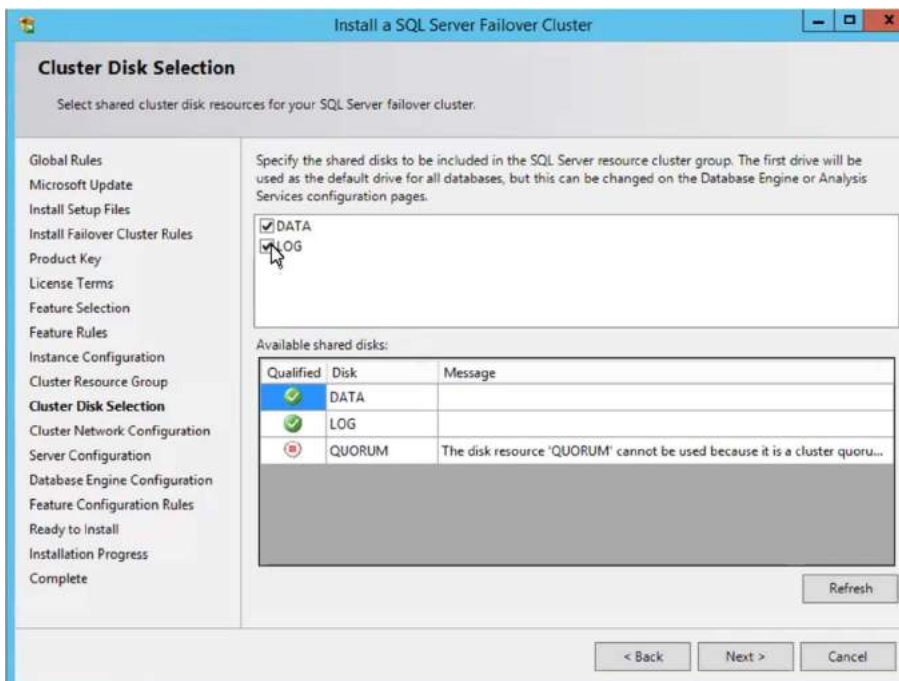
For example, To login in to a stand alone instance- user name is Computer name\Instance name. In the same way to login in to the Cluster SQL Server is “SQL Server network name\Instance name”.



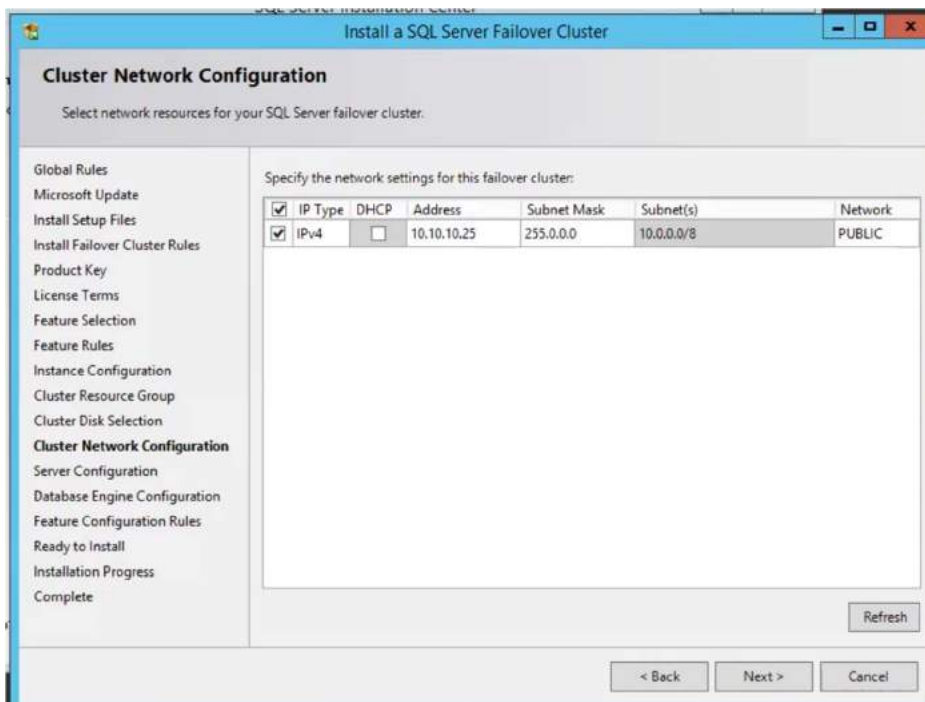
Step 4 : 2nd Difference – Choose the resource group. Resource group contains all the resources of the SQL Server.



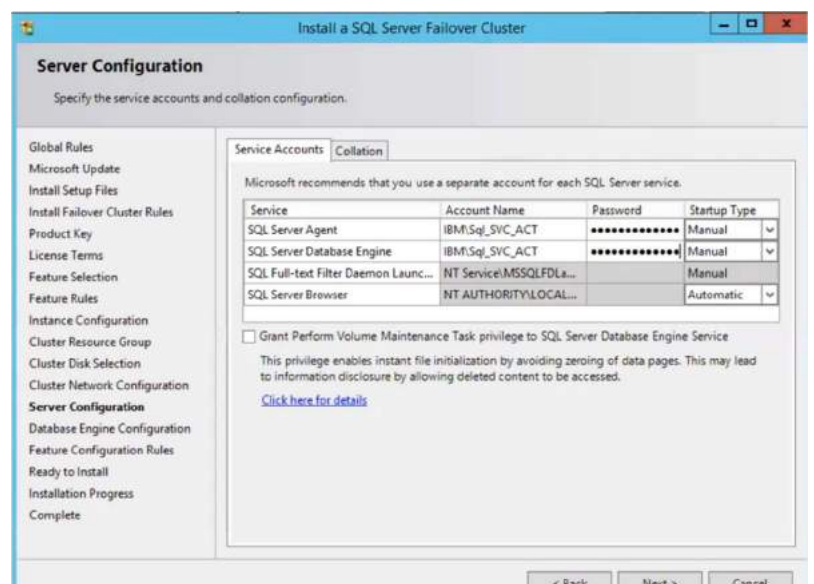
Step 5 : 3rd difference – choose the eligible drives, where we store data files, log files , backup& tempdb. We cant store any data on the QUORUM drive.



Step 6 : 4th difference : Now we have to IP address , which will be provided by network team. No need to select DHCP.



Step 7 : 5th difference – Choose the service account and give password of the service account. Always startup type should be manual in the SQL Server cluster. We don't get option also to change automatic.



Step 8 : 6th Difference – In the database configuration page, Go to data directories, Choose the drives where we want to store data, log, backup files.

The screenshot shows the 'Database Engine Configuration' window with the 'Data Directories' tab selected. The left sidebar lists various configuration steps, with 'Database Engine Configuration' highlighted. The main area contains fields for specifying data directories:

- Data root directory: E:\
- System database directory: E:\MSSQL14.SQLEXPRESS\MSSQL\Data
- User database directory: E:\MSSQL14.SQLEXPRESS\MSSQL\Data
- User database log directory: L:\MSSQL14.SQLEXPRESS\MSSQL\Log
- Backup directory: E:\MSSQL14.SQLEXPRESS\MSSQL\Backup

Navigation buttons at the bottom include '< Back', 'Next >', and 'Cancel'.

In the same page , go to TEMPDB tab , Choose the path where we want to store TEMPDB files.

Temp DB is separate configuration has been implemented in the sql server installation from 2017 onwards.

For performance wise, Multiple data files recommended for TEMP DB. Based in the Cores of the PC, Number of temp db files will be recommended. Max temp db files can be 8 files only.

The screenshot shows the 'Database Engine Configuration' window with the 'TempDB' tab selected. The left sidebar is the same as in the previous screenshot. The main area contains configuration options for TempDB:

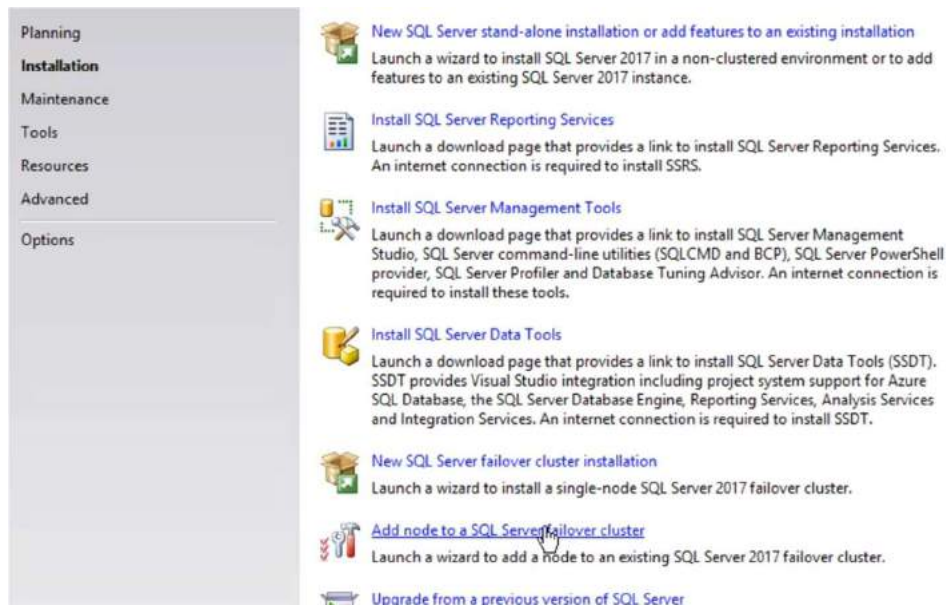
- TempDB data files: tempdb.mdf
- Number of files: 1 (with a dropdown arrow)
- Initial size (MB): 8
- Total initial size (MB): 8
- Autogrowth (MB): 64
- Total autogrowth (MB): 64
- Data directories: E:\MSSQL14.SQLEXPRESS\MSSQL\TempDB (with 'Add...' and 'Remove' buttons)
- TempDB log file: templog.ldf
- Initial size (MB): 8
- Autogrowth (MB): 64
- Log directory: E:\MSSQL14.SQLEXPRESS\MSSQL\TempDB

Navigation buttons at the bottom include '< Back', 'Next >', and 'Cancel'.

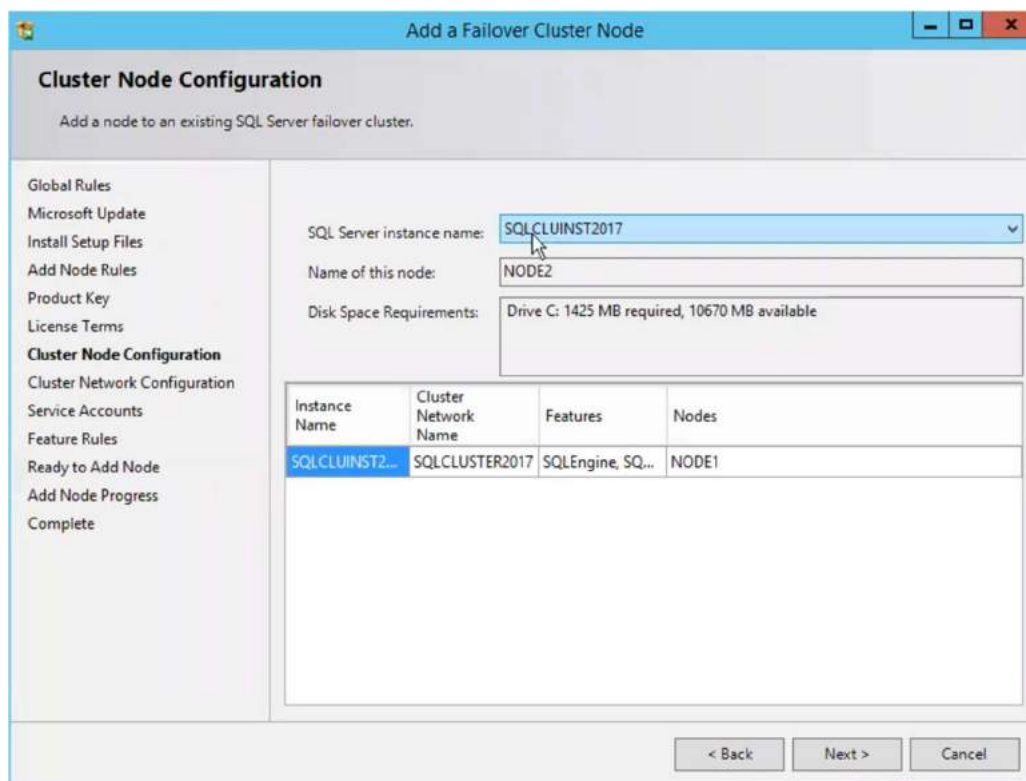
Step 9 : That's it. Install.

Step 10 : Now we have to perform ADD Node in the NODE 2 server.

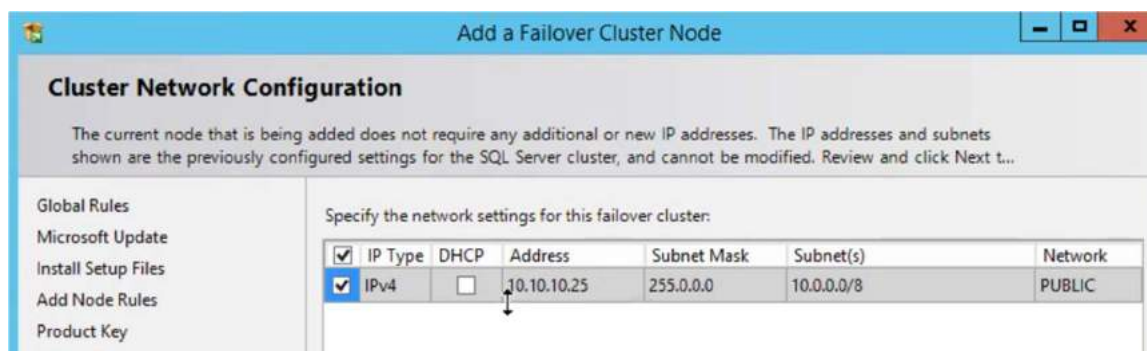
Step 11 : Go to NODE 2 server, Open SQL Server setup file, Run as administrator, Choose the option “ADD Node to a SQL Server failover cluster”



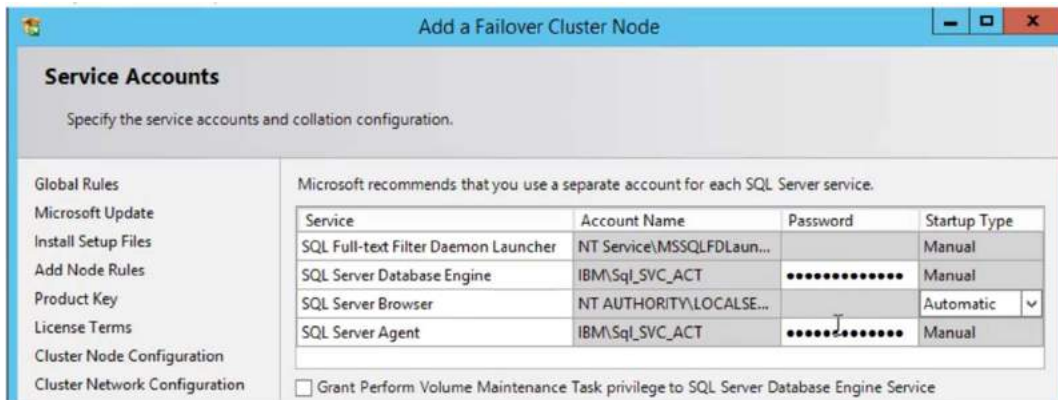
Step 12 : Choose the SQL Server instance name



Step 13 : Tick the IP Address,



Step 14 : Enter the password of service account



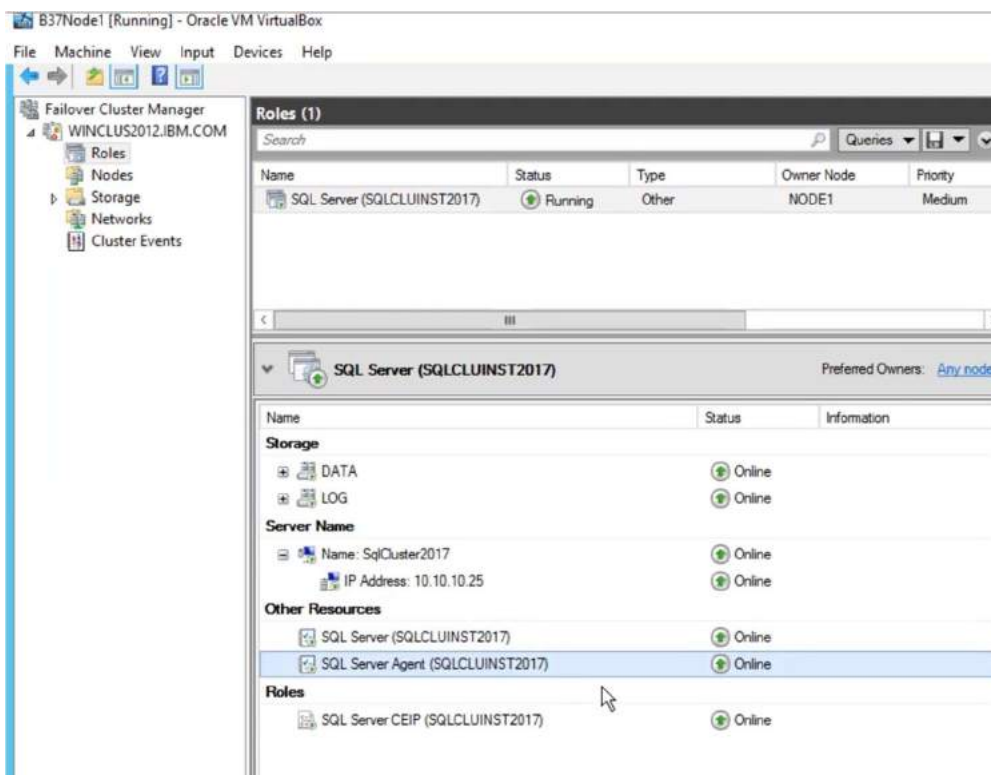
Step 15 : Install.

122. How to Manual failover the cluster?

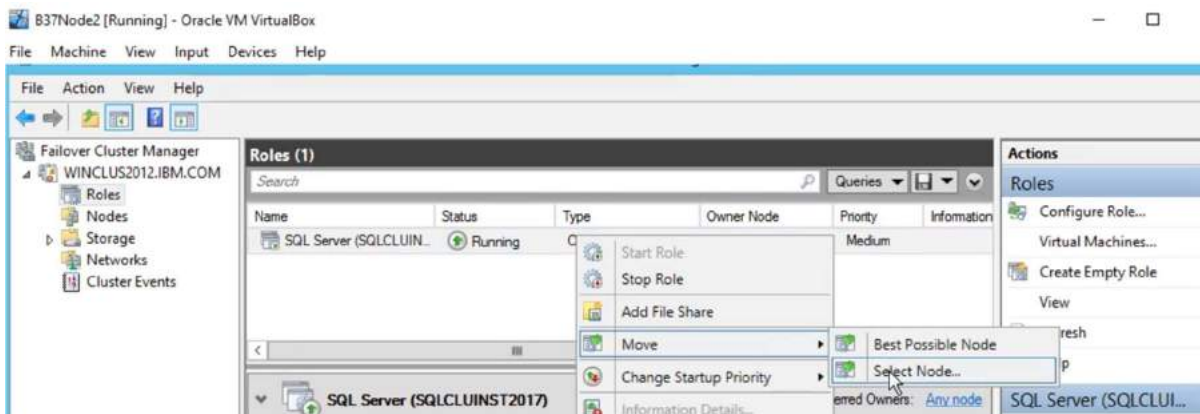
Ans.

Step 1 : open RUN command on any of the NODE, Type the command "CLUADMIN.MSG"

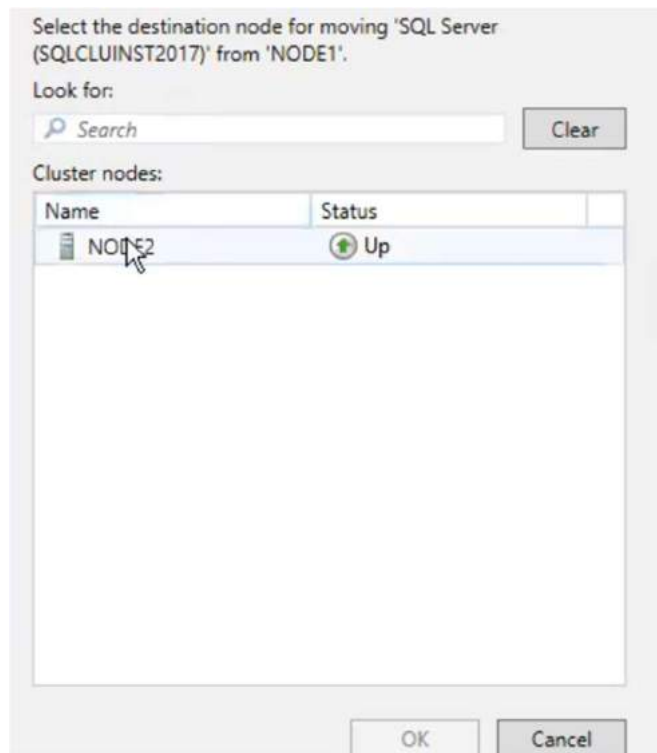
Step 2 : Go to roles, There we can see SQL Server cluster status.



Step 3 :Right click on the server , Choose the move option, select Node option,



Step 4 : Select the node , which we want to failover. Click ok.

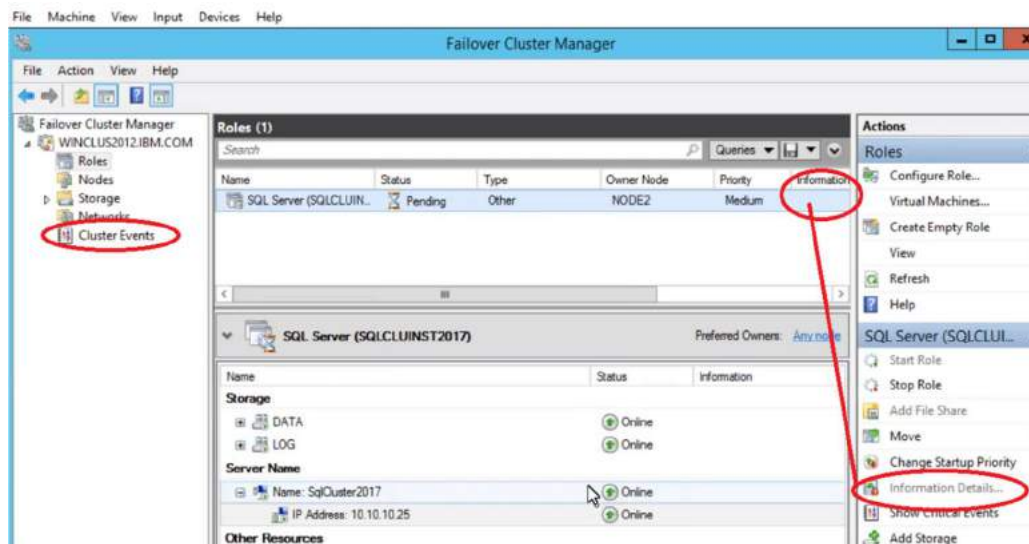


Step 5 : Now all the resource drives will move to NODE 2 and then SQL Server will come online.

Step 6 : Quorum drive wont move to Node 2. It will be in the NODE 1 only. If disaster happens , then automatic failover automatic failover happens. That time Quorum also moved to NODE2. Why because, Node1 server is fully down.

123. Where to check the cluster related errors?

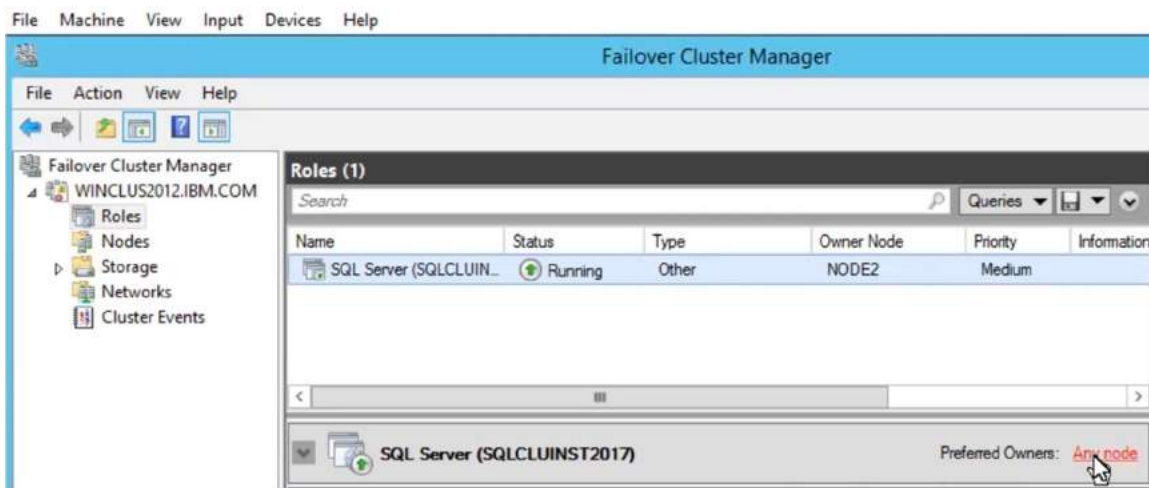
Ans. Few errors captured in Cluster events and few errors captured in Information details.



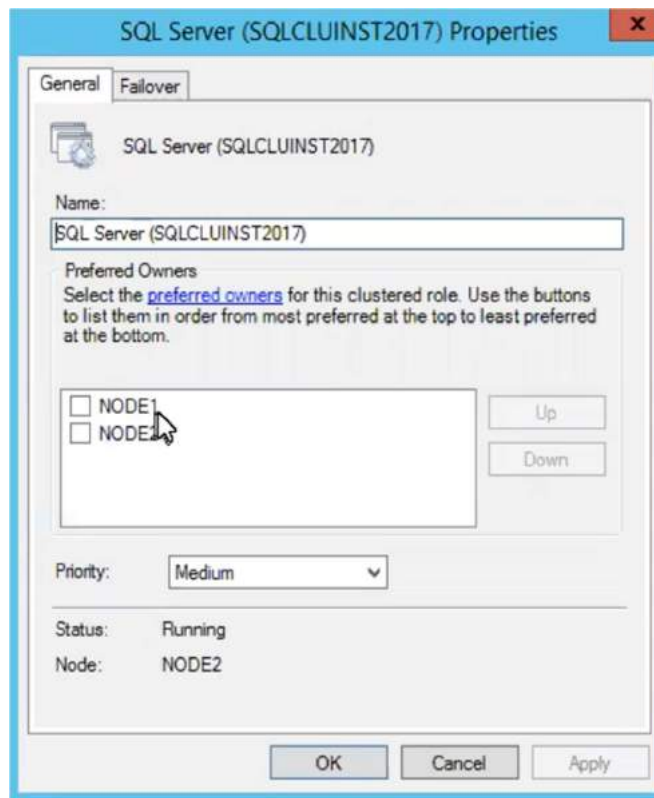
124. what are types of owners in Cluster?

Ans. There are 2 types ,

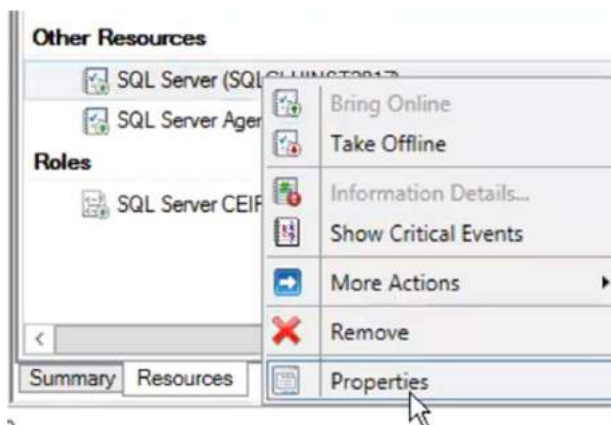
- a. Preferred owner : Which node shows first in the list, That server acts as a owner of the SQL Server.



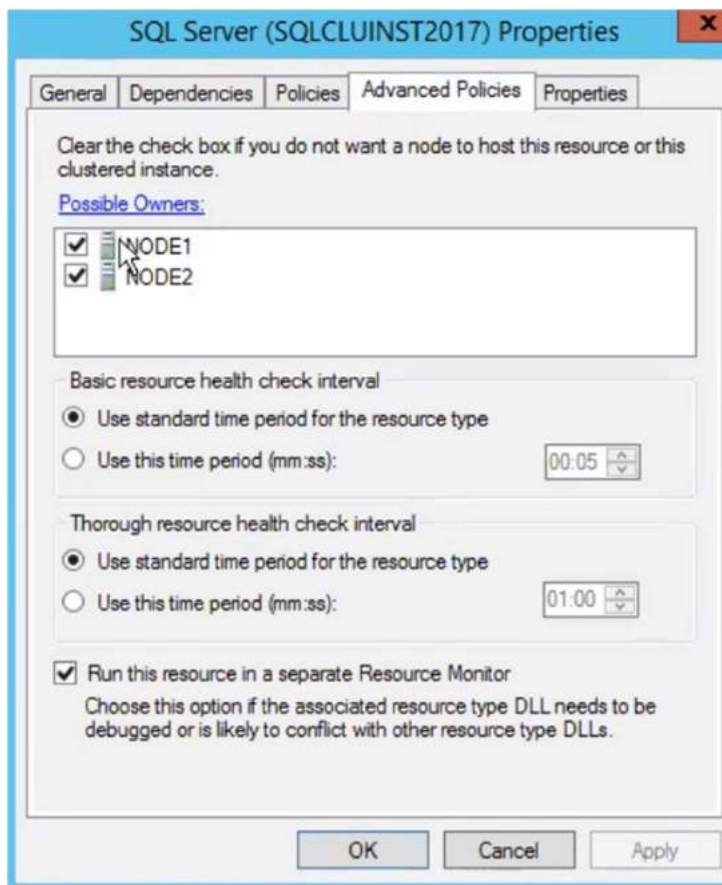
To change the preferred owner, simply select the node and click up and apply.



b. Possible owner : At the time failover, what are the possible nodes available. Those are possible owners.

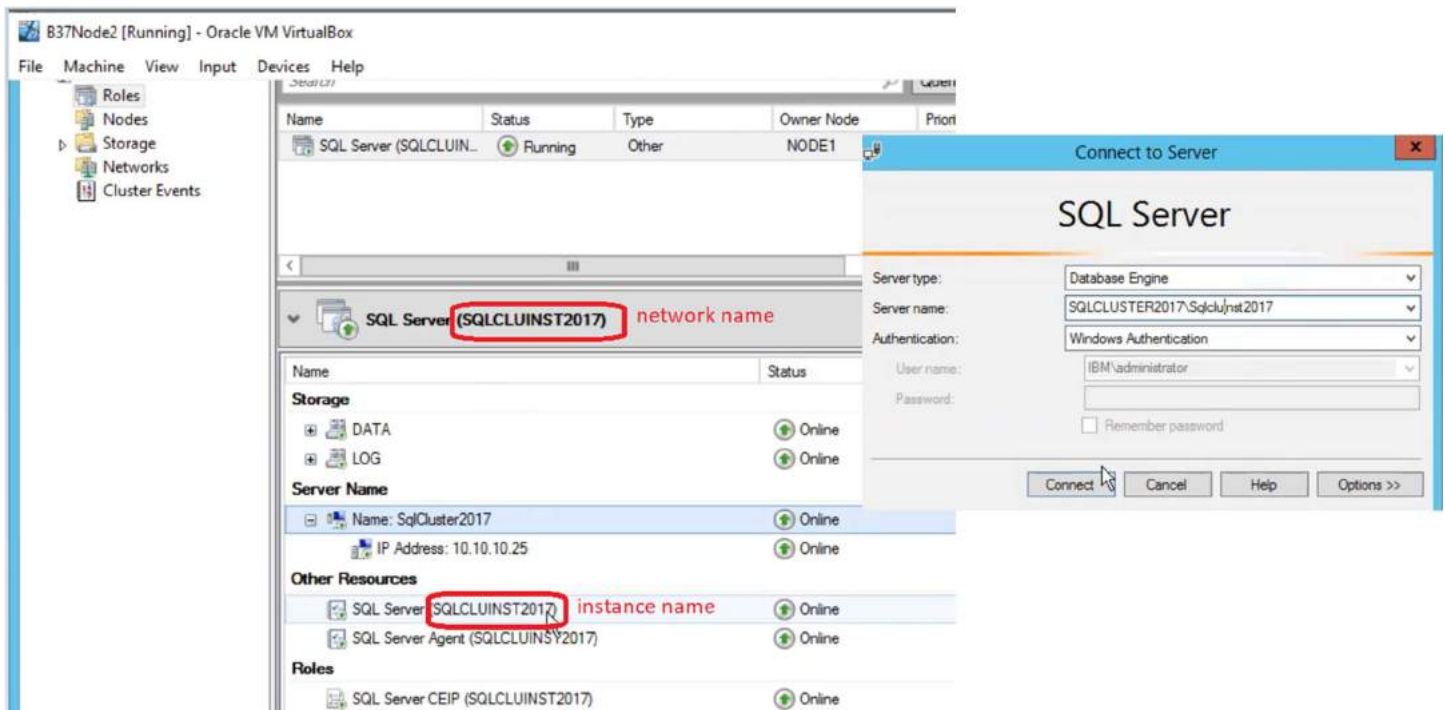


If we not tick the nodes in this tab, Failover not possible. Because there is not possible node.



125. How to connect SQL Server Cluster Instance?

Ans. Give the "SQL Server network name\Instance name". Both the names we can get in the Cluster manager.



126. how to patch a cluster setup?

Ans. First we have to check whether failover is working or not.

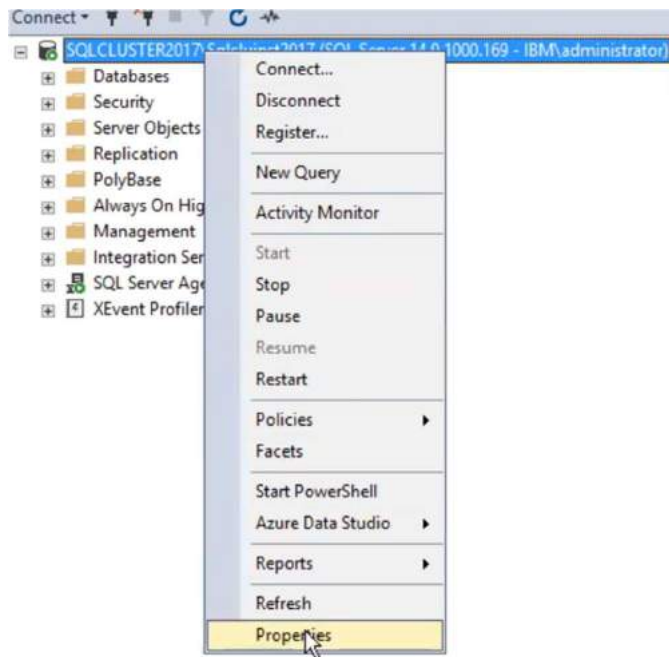
If its working fine. Then, Patch the passive Node first.

After successful patching, failover the server.

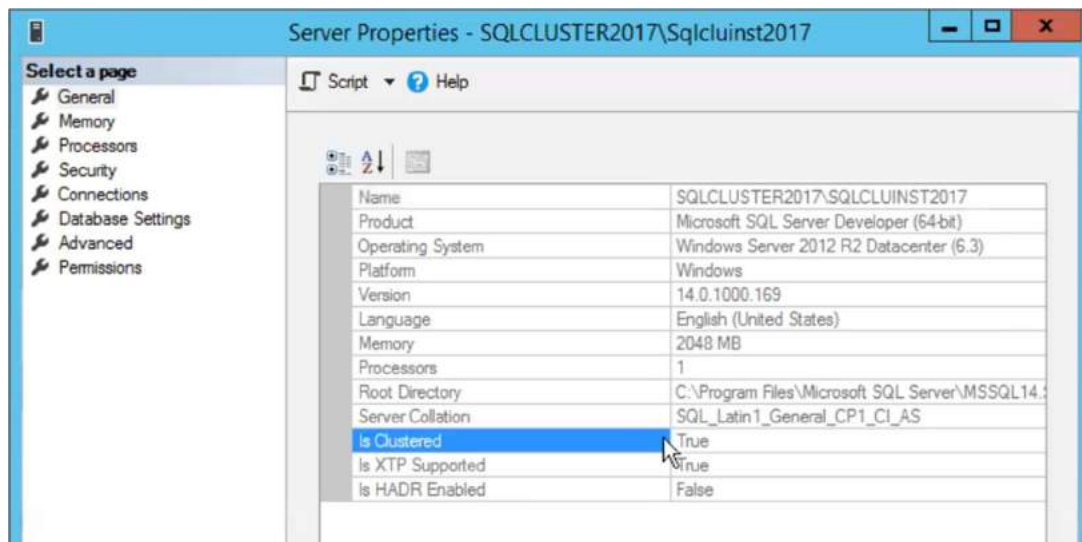
Do the patching for the remaining nodes.

127. how to check an instance is Cluster instance or not?

Ans. Right click on the server, go to properties,



In the general tab, Is Clustered is True then instance is a clustered.



128. How to shrink a log file on the real time environments?

Ans. Normally, When any log file size of the database is reached 90%. Then we will shrink the log file of the database.

The max log file size 2 TB only.

By using Disk level alerts, we will come to know , when there is disk space issues.

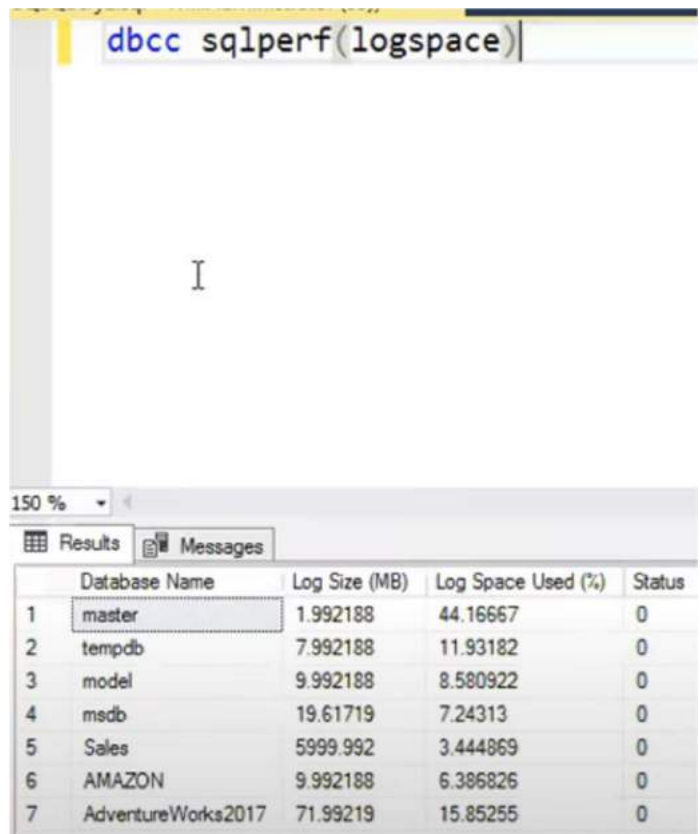
If we got the alert in the outlook, that log file drive is going to full. Then we have to take below steps,

Step 1 : Login in to the windows server.

Step 2 : Check that Logfile drive is really going to full or not in the my computer.

Step 3 : Login in to SQL Server, Type the command “dbcc sqlperf(logspace)”. This command shows how many log files are there in the sql server and what is the size of the log file.

In the screenshot, sales database log file occupied 5999MB = 6 GB. Now we will shrink the log file of sales database.



	Database Name	Log Size (MB)	Log Space Used (%)	Status
1	master	1.992188	44.16667	0
2	tempdb	7.992188	11.93182	0
3	model	9.992188	8.580922	0
4	msdb	19.61719	7.24313	0
5	Sales	5999.992	3.444869	0
6	AMAZON	9.992188	6.386826	0
7	AdventureWorks2017	71.99219	15.85255	0

Step 4 : Now we have to check why log file is growing and not reusable. To know that, we have to find the state of the log file. There are different states,

- Mainly are
- a. Nothing
 - b. Log backup
 - c. Active backup or restore
 - d. active transaction
 - e. availability replica
 - f. Check point
 - g. oldest page

Step 4 : To know the state of the log file use the query “ select name, log_reuse_wait_desc from sys.databases”

if the state of the Log file is ACTIVE_TRANSACTION , we have to wait till state to be changed to NOTHING. State of log file should be “NOTHING”. Then only we have to shrink the log file.

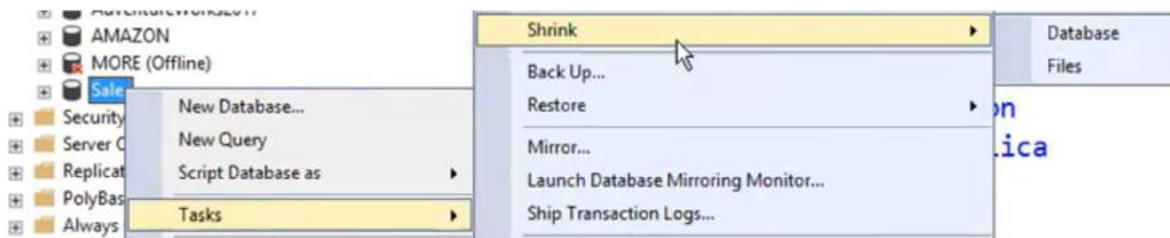
If the state is showing as “LOG BACKUP”, Then we have to take log up of the database and re run the same query , the state will be changed to NOTHING. Even some times, after taking the log backup also state wont be changed. So, until the state changed to nothing, we have to take log backups again and again. Here one more case, if the log backup file size is 2 TB and we don't have space in any of the drives to take backup. That we can use nul backup. Nul only not null.

“ backup log sales to disk='nul' ”. Here backup wont be stores any where. But our requirement full filled. State of the log file will be changed to NOTHING.

If the state of the log file is “ACTIVE BACKUP or RESORE”. Then we have to wait to change the status to nothing.

If the state is “OLDEST PAGE” , We have to run the CHECK POINT on the particular database. Then state will be changed to NOTHING.

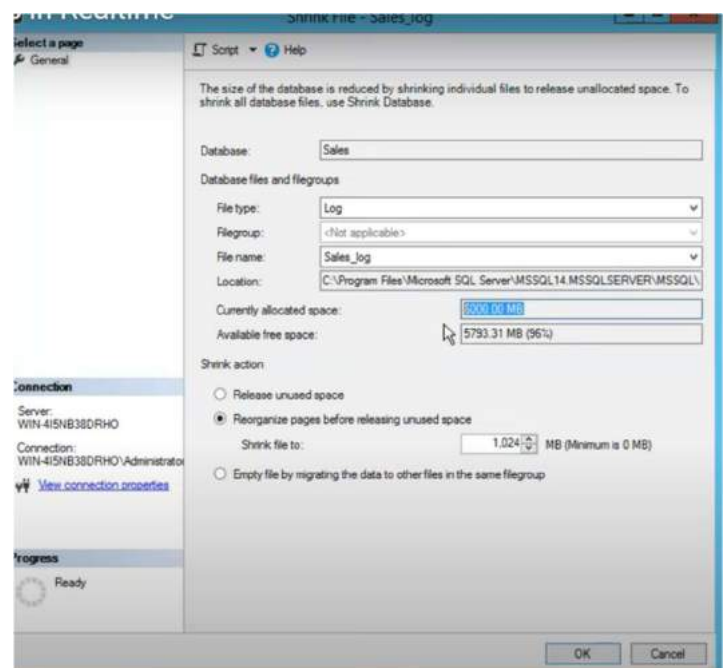
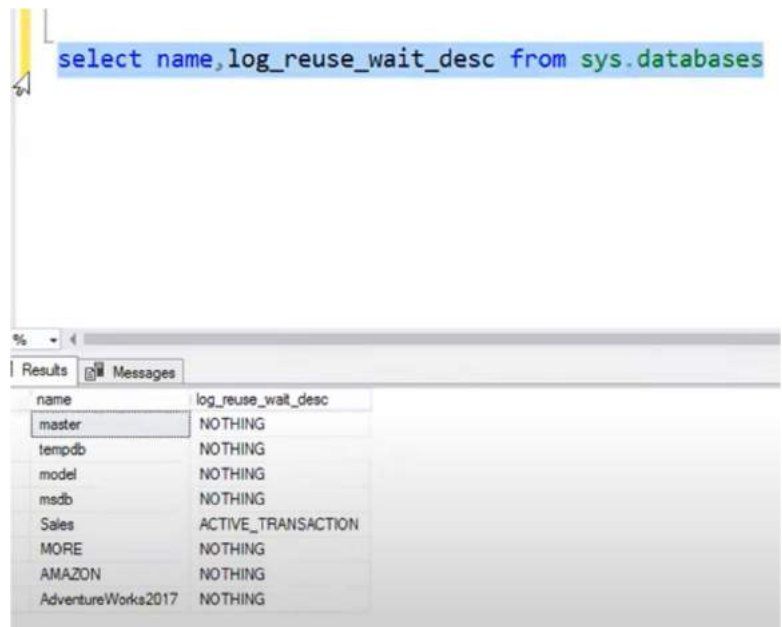
Step 5 : Right click on the database , go to tasks, Select shrink , Choose files. Never select the database to shrink. Always we have to shrink only log file not the data file. If we shrink the data file, fragmentation will be created. Which leads to performance issue. There is a another way for data file shrink.



Step 6 : Choose the file type as LOG, Cross check the Location & File name , Select the Reorganize pages before releasing unused space. In the organization ,, Shrink file size is 512 MB or 1024 MB. Normally we do shrinking of log file , which has more than 5 GB.

In the right screenshot, Log file size is 6000 MB. After clicking ok , The size will changed to what ever size we kept . In the screenshot we kept 1024 MB. Now log file size will be shrink from 6000 MB to 1000 MB.

End of the process.



129. What is ALWAYS ON?

Ans. Always on introduced from 2012 version onwards. ALWAYS ON is called as AOAG(ALWAYS ON AVAILABILITY GROUP). ALWAYS ON is a Stand alone instance. No need to instal cluster instance. But, we need Windows clustering to perform always on. ALWAYS ON is a advanced version of MIRRORING. In mirroring , we have to configure mirroring for each database for high availability. But, in ALWAYS ON , HIGH AVAILABILITY Group will be created. Simply, adding the databases to the group will be converted to HADR. ALWAYS ON also works with help of END POINTS only like Mirroring. Same synchronous and asynchronous modes available in ALWAYS ON.

Mirroring is a database level high availability, where ALWAYS ON is group level high availability.

Once we create the AG(availability group), we can add the other data bases also the group for high availability.

In ALWAYS ON, we called primary server as Primary Replica and secondary server called as Secondary Replica. Basically, servers called as Replica in always on.

In Mirroring , we can't read the data in the secondary server. Where as in always on, we can read the data in the secondary replica. Like select , back up queries.

Always ON helps to balance the Load of the sql server. For example, Always on applies the insert or update commands on the primary replica and select and backup commands will run the secondary replica. It will reduce the load on the same server. These load balance done by the **Listener** in the always on. Listener route the traffic between the servers.

Lister route the traffic based on the customer requirement.

When any of the database is offline in the always on group. Entire Group will be failover to secondary server.

We need use shared storage for ALWAYS ON. Only for SQL Server Clustering , we use shared storage.

In Always on Application connected to the listener, Whenever the failover happens connection strings of application also will be changed to active replica automatically with the help of listener.

In mirroring, we have to inform application team to change connection string after the failover.

Listener do 2 things; One is Traffic or load balance on the replicas and Change the connection strings automatically after the failover to the application. Listener knows which one primary replica and which one secondary replica.

We can configure always on with out listener but automatic failover not possible.

130. What are the prerequisites to configure always on?

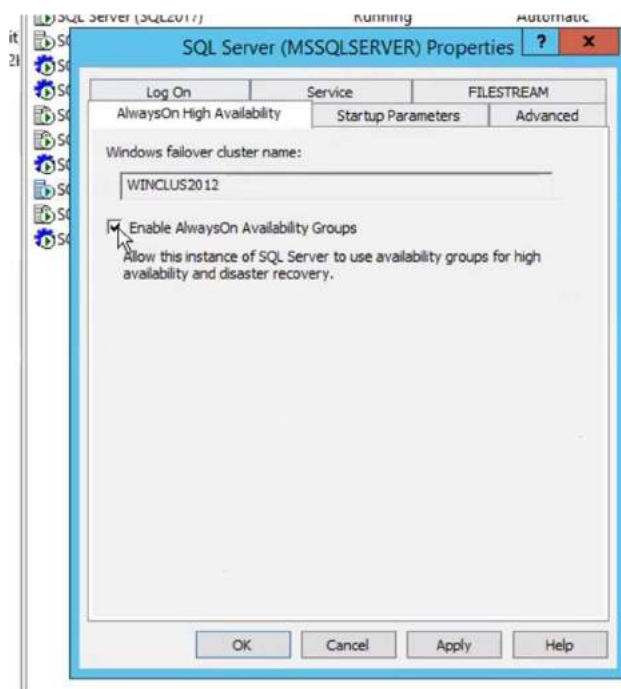
Ans.

- a. Windows level cluster setup
- b. IP address
- c. standalone sql server (Version should be 2012 or more)
- d. We have to enable the always on feature in configuration manager.
- e. database should be in FULL recovery mode.
- f. Service account sysadmin permissions.
- f. Take the backup of primary replica database and restore it in secondary replica with NO Recovery.

131. What are the steps to configure always on?

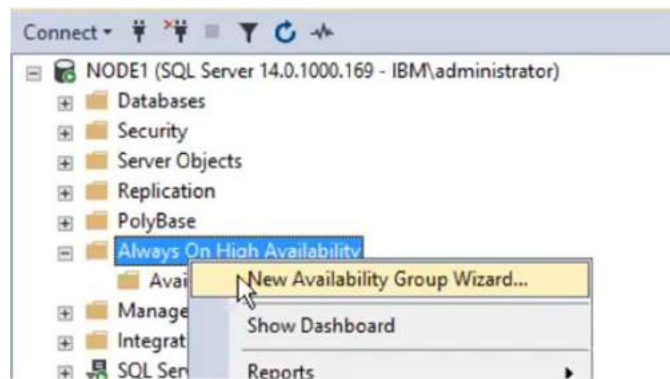
Ans. Below are the steps,

- a. Windows team will install the cluster setup.
- b. Install the Standalone sql server on the replicas(Startup type automatic only)
- c. Enable the Always on feature in the configuration manager for all the replicas.
- d. Right click on instance, go to properties, select always on high availability tab, Tick the “Enable Always on Availability group”, Click ok. Do the same process for all the replicas.



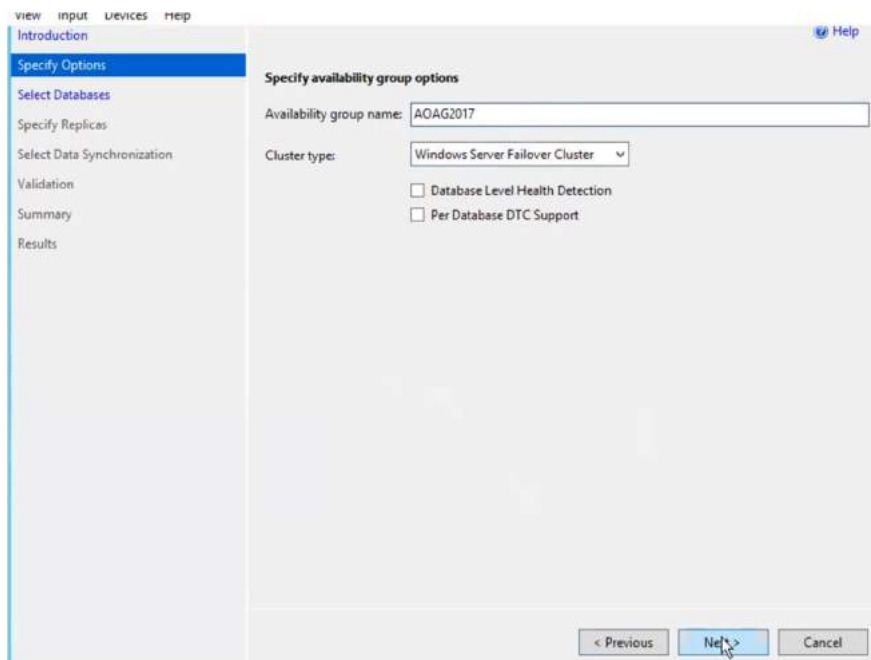
- e. Keep the database recovery model to FULL
- f. Take the database backup in the primary replica and restore it on the secondary replica with no recovery.

g. On the primary replica, select always on high availability, right click and select new availability group wizard,

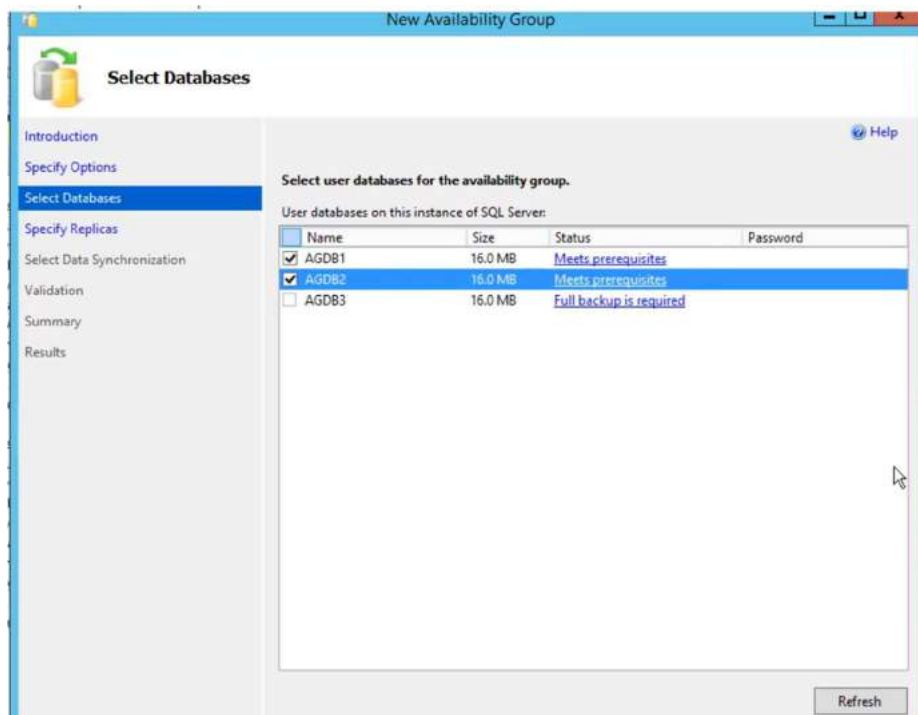


h. Give availability group name , choose the cluster "Windows level failover cluster"

type



i. Now it will show the all the databases of primary replica, choose the databases and it should meet the prerequisites.



j. Specify replicas window- This is the main window to setup always on.

Replicas Tab: Primary replica instance already available. Now add the secondary replica by using add option, Connect to sql server, Click Connect. Tick the automatic failover for all the replicas, choose the availability mode based on requirement. If automatic failover, synchronous mode like that.

Choose the readable secondary “Yes” for all the replicas. Then only read operations can be performed on the secondary replica.

In 2012 sql server, we can add 4 secondary replicas in AG.

In 2016 sql server , we can add 6 secondary replicas in AG.

In 2017 & 2019 server, We can add 8 Secondary replicas in AG.

Added the secondary replica and ticked the automatic failover.

Specify an instance of SQL Server to host a secondary replica.

Replicas | Endpoints | Backup Preferences | Listener | Read-Only Routing

Availability Replicas:

Server Instance	Initial Role	Automatic Failover (Up to 3)	Availability Mode	Readable Secondary
NODE1	Primary	☒	Synchronous commit	No
NODE2	Secondary	☒	Synchronous commit	No

Add Replica... Remove Replica

Summary for the replica hosted by NODE2

Replica mode: Synchronous commit with automatic failover
This replica will use synchronous-commit availability mode and will support both automatic failover and manual failover.

Readable secondary: No
In the secondary role, this availability replica will not allow any connections.

Required synchronized secondaries to commit 0

< Previous Next > Cancel

Changed readable to yes.

Specify an instance of SQL Server to host a secondary replica.

Replicas | Endpoints | Backup Preferences | Listener | Read-Only Routing

Availability Replicas:

Instance	Initial Role	Automatic Failover (Up to 3)	Availability Mode	Readable Secondary
	Primary	☒	Synchronous commit	Yes
	Secondary	☒	Synchronous commit	Yes

Add Replica... Remove Replica

Summary for the replica hosted by NODE2

Replica mode: Synchronous commit with automatic failover
This replica will use synchronous-commit availability mode and will support both automatic failover and manual failover.

Readable secondary: Yes
In the secondary role, this availability replica will allow all connections for read access, including connections running with older clients.

Required synchronized secondaries to commit 0

< Previous Next > Cancel

k. END POINTS TAB : Here we have to give end points port number. Default port number for mirroring or always on is 5022. If we have already configured mirroring 5022 will be already used. So here we have to use 5023. If 5023 also used for some other configuration, we can choose 5023. At this page , by default 5022 will come. We have to change the port number manually in the end point url and port number column.

The screenshot shows the 'Specify an instance of SQL Server to host a secondary replica' dialog box with the 'Endpoints' tab selected. The 'Endpoint values' table is as follows:

Endpoint URL	Port Number	Endpoint Name	Encrypt Data
TCP://NODE1.IBM.COM:5023	5023	Hadr_endpoint	☒
TCP://NODE2.IBM.COM:5023	5023	Hadr_endpoint	☒

Navigation buttons at the bottom include '< Previous', 'Next >', and 'Cancel'. A 'Refresh' button is also present.

l. backup preferences tab :

There are 4 options :

A. Prefer Secondary : Secondary replica will be used for backup. If secondary replica not available, Automatically backup will be taken from primary replica.

B. Secondary only : Backup will run only from the secondary replica.

C. Primary : backup will run only from the Primary Replica.

D. Any Replica : backups run from any replica based on the AG. By using Priority percentage or else ticking the Exclude replica option in the below ,we can exclude the replica for backup queries. Mostly don't use the Exclude replica option. When we did the failover, backup job may stop because of excluded the replica for the backup.

In the organization, mostly we prefer SECONDARY ONLY OPTION for backup.

The screenshot shows the 'Specify an instance of SQL Server to host a secondary replica' dialog box with the 'Backup Preferences' tab selected. The 'Where should backups occur?' section has the following options:

- ☐ **Prefer Secondary**
Automated backups for this availability group should occur on a secondary replica. If there is no secondary replica available, backups will be performed on the primary replica.
- ☒ **Secondary only**
All automated backups for this availability group must occur on a secondary replica.
- ☐ **Primary**
All automated backups for this availability group must occur on the current primary replica.
- ☐ **Any Replica**
Backups can occur on any replica in the availability group.

The 'Replica backup priorities' table is as follows:

Server Instance	Backup Priority (Lowest= 1, Highest=100)	Exclude Replica
NODE1	50	☐
NODE2	50	☐

Navigation buttons at the bottom include '< Previous', 'Next >', and 'Cancel'.

j. Listener Tab: Here if you want automatic failover, we have to create availability group listener. Even if we wont create the listener at this step, we can create after the AG setup also.

Give the Lister name and port number.

If you have selected the default instance as Primary replica then give the port number 1433.

If it's a named instance, check the port number of instance and then enter the same port number.

In the organization , After the SQL Server named instance installation, we will change the port numbers. There are two types of ports Static & Dynamic ports. Normally after installation a dynamic port will be allocated to sql server. In the organization, as a security precaution , they will change the port number manually. After changing the port number , we have to request network team to open the ports.

The screenshot shows the 'Specify an instance of SQL Server to host a secondary replica' dialog box with the 'Listener' tab selected. The 'Replicas' tab is also visible. The 'Specify your preference for an availability group listener that will provide a client connection point:' section has two radio buttons: 'Do not create an availability group listener now' and 'Create an availability group listener'. The 'Create an availability group listener' option is selected. Below this, the 'Specify your listener preferences for this availability group.' section contains the following fields: 'Listener DNS Name' (Sql2017Listner), 'Port' (1433), and 'Network Mode' (Static IP). At the bottom, there is a table with columns 'Subnet' and 'IP Address'.

Network mode keep it as Static IP address, Click add, give the IP address

The screenshot shows the 'Add IP Address' dialog box. It has the following fields: 'Subnet' (10.0.0.0/8), 'Address' (10.10.10.65), 'Subnet Mask' (255.0.0.0), and 'IPv6 Address'. At the bottom, there are 'OK', 'Cancel', and 'Help' buttons. The dialog box is overlaid on the 'Specify an instance of SQL Server to host a secondary replica' dialog box, which is partially visible in the background.

k. Read only Routing Tab : Not explained. Feature from 2016 version onwards.

Specify an instance of SQL Server to host a secondary replica.

Replicas Endpoints Backup Preferences Listener **Read-Only Routing**

Read-only routing summary

Server Instance	Read-Only Routing URL	Read-Only Routing List
NODE1		
NODE2		

- Replicas in the read-only routing list are separated by ";" and sorted by priority.
- Load-balanced replicas with same priority are grouped in "()".

Read-only routing list for: NODE1

Available replicas

Read-only routing list

Add

Remove

Move Up

Move Down

< Previous Next > Cancel

l. Select data synchronization page : This is a initialization setup page. Choose the join only option. Why because we have already initialized the database.

Introduction Specify Options Select Databases Specify Replicas **Select Data Synchronization** Validation Summary Results

Help

Select your data synchronization preference.

☐ Automatic seeding
SQL Server automatically creates databases for every selected secondary replica. Automatic seeding requires that the data and log file paths are the same on every SQL Server instance participating in the availability group.

☐ Full database and log backup
Starts data synchronization by performing full database and log backups for each selected database. These databases are restored to each secondary and joined to the availability group. Make sure the file share is accessible to all replicas and is mounted to the same directory on all Linux replicas.
Specify the file share path in Windows format:
 Browse...
Specify the file share location in Linux format:

☒ Join only
Starts data synchronization where you have already restored database and log backups to each secondary server. The selected databases are joined to the availability group on each secondary.

☐ Skip initial data synchronization
Choose this option if you want to perform your own database and log backups of each primary database.

< Previous Next > Cancel

M. Click Finish.

